Advanced Quantum Materials

Lecture Notes

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Session 1

Second Quantization, Mean-Field, and Spontaneous Symmetry Breaking

This is the first session of the course, so we start with the question that organizes the whole course. How do we go from a single atom, which is an object that we understand rather well, to a *quantum material*, where electrons that move together give rise to phenomena that no single atom can show? To answer this question we need a new mathematical language, which is second quantization, and a first tool to use it, which is mean-field theory. In this session we will use this tool three times, and we will obtain a charge density wave, a magnet and a superconductor from the same starting Hamiltonian, only by changing the sign and the geometry of one interaction term. These three cases are broken translational symmetry, broken time-reversal symmetry and broken gauge symmetry, and they are the structure on which the rest of the course is built, so it is useful to keep them in mind from the start. Note also that this session is the *only* place in the course where we discuss the general logic of spontaneous symmetry breaking in detail. The next session goes directly to symmetry and band structure and does not come back to it, and magnetism and superconductivity are discussed in full only much later, in sessions 6 and 5 respectively. For this reason, what we build here has to carry these concepts on its own.

1.1 Why quantum materials, and why now

Before we start with the mathematics, let us say a few words about why this subject is relevant. Every period of human technology has been named after the material that defined it. There was the stone age, the copper age and, very recently, the silicon age, which gave rise to the whole information revolution in which we live now. If we look ahead, it is plausible that the next period will be a *quantum* age, in which our technology directly uses quantum mechanical behavior at microscopic scales instead of avoiding it. Quantum materials are the physical platform for this change, since they are the systems that realize quantum matter in the lab.

At the end of this session we should be able to describe what a quantum material is and to explain why we need quantum mechanics, and not classical physics, to understand it. We should also be able to work with the basics of second quantization, and to rationalize why mean-field theory gives rise to spontaneous symmetry breaking.

Quantum materials are already the basis of technologies that we use today. Superconductors make MRI machines possible, since they generate the very large magnetic fields that nuclear magnetic resonance needs, and semiconductors are the basis of solar cells, on which a large part of our plan for renewable energy depends. In the future, almost every serious platform for quantum computing, whether it is built from superconductors, semiconductors or a combination of the two, is in the end a quantum materials problem. Let us also mention

briefly that this is connected to several of the UN Sustainable Development Goals. Quantum computing is connected to goals 8 and 9, which are economic growth, industry and innovation. Solar energy and superconducting or topological electronics with low power consumption are connected to goals 7 and 13, which are clean energy and climate action. The international and cross-disciplinary character of the field is connected to goal 17, which is partnerships. None of this is mathematics, but it is good to remember why we do the mathematics.

In the course we will see two kinds of examples, and it is useful to compare them. Semiconductors are in some sense the easy case, since their properties follow from single-particle physics, which you have likely seen already in an earlier course on materials. Superconductors are the harder case, because to understand them we have to account for how electrons interact with each other collectively, and this is the many-body formalism that we start to build in this session.

1.2 A tour of the quantum materials we will meet

Let us first give a short preview of the materials that this formalism will allow us to understand, so that the abstract tools of this session have concrete targets.

- **Insulators**, such as magnesium oxide, sodium chloride or boron nitride, where a finite band gap does not allow electrons to propagate. *Topological* insulators look the same in the bulk, but they behave in a completely different way at the edge, and we will need one full session to explain this difference.

- **Semiconductors**, such as gallium nitride, silicon and gallium arsenide, which have the same single-particle description as insulators but are lightly doped. For this reason they have free carriers, and these carriers are the ones that do the work in electronics and solar cells.

- **Magnets.** Ferromagnets, antiferromagnets and spin spirals are all *correlated* materials, in the sense that they exist only because electrons repel each other through the Coulomb interaction. A different and more exotic case is the *quantum magnet*, where there are local moments at every site but they are so strongly entangled with each other that the material has no net magnetization at all at the microscopic level, and every site forms a singlet with its neighbors. We will come back to these systems in detail later in the course.

- **Dirac and topological matter.** The prototype is graphene, which is a single honeycomb layer of carbon, where electrons propagate with a topologically protected phase that is called a Berry phase. The same family includes quantum spin Hall insulators, which have helical edge states, Chern insulators, which have chiral edge states, and chiral superconductors, which host Majorana excitations.

- **Superconductors.** These go from conventional superconductors mediated by phonons, such as aluminum, to *correlated* superconductors such as the cuprates and the pnictides, whose mechanism is still one of the biggest open problems in the field.

- **Fractional quantum matter.** Here the fundamental excitations are real fractions of an electron. The fractional quantum Hall state is the best established example, and it has been observed for decades in materials such as gallium arsenide. Fractional Chern insulators are a newer and more accessible way to obtain the same physics, and they are a promising building block for new quantum devices.

1.3 Emergent quasiparticles and what the lattice gives us

Let us now state the puzzle that motivates everything that follows. All this phenomenology is built from the same four ingredients, which are electrons, protons, neutrons and photons, and a single isolated atom shows none of it. In a single atom there is no fractionalization, there are no vortices, no magnons and no spinons. So how can a lattice of identical atoms give rise to so many different kinds of collective behavior? Before we build the formalism to answer this question, it is useful to know what we are looking for. Some of the emergent quasiparticles that we will meet are the following.

- **Phonons.** These are lattice vibrations, and they are the simplest quasiparticle of all, since they exist also in classical physics.
- **Magnons.** These are spin-1 excitations of a magnetically ordered state, which are small precessions of the local moment that carry energy and spin through the material. Note that magnons can propagate also in a Mott insulator, where charge cannot propagate, so a magnetic insulator can still carry spin and heat.
- **Spinons.** These are spin-1/2 excitations without charge that live in quantum magnets, where time-reversal symmetry is *not* broken. In some sense a spinon is only the spin part of an electron, without its charge part, and this is an example of electron fractionalization that is called spin-charge separation.
- **Dirac fermions.** These are electrons with a linear dispersion that look massless, and they do not come from interactions but from the geometry of the lattice, as in the standard example of the honeycomb lattice of graphene.
- **Vortices.** These are topological defects in the phase of a superconductor, and they are dynamical quasiparticles themselves. The same superconducting phase gives rise to flux quantization and to the Josephson effect, and these are the basis of some of our most precise measurements of the ratio between Planck's constant and the electron charge.

The summary slide of the course for this list also shows Majorana fermions, which we will meet again soon in connection with chiral superconductors, and anyons, next to these five. Anyons in particular are not discussed further in this session, neither in the slides nor in the lecture. They belong to the material on the fractional quantum Hall effect in a later session, and we mention them here only so that the name is familiar.

1.4 Setting up the formalism with the Schrödinger equation in two ways

Every quantum mechanical system is described by the Schrödinger equation, which is built from a Hamiltonian acting on a wave function, where for our purposes we can think of the Hamiltonian as a matrix. If we know the wave function at time zero and the Hamiltonian, in principle we can integrate this equation forward in time and know the state at any later time. This statement is completely general, and it holds both when the wave function describes one particle and when it describes a whole many-body system. In practice, however, the difference is very large. Single-particle problems can always be solved exactly, in every case that is relevant for us. Many-body problems in general cannot, and the difficulty grows very fast with the size of the system, as we will see in a concrete way below. Since most of the quantum matter that we are interested in is truly many-body, we need a systematic way to deal with this difficulty, even if this way is often approximate, and this is what we build in this session.

1.4.1 First quantization

When the electrons can be treated as independent of each other, we can describe the system with a single-particle wave function that lives in an ordinary Hilbert space. This is what is called *first quantization*, which is the language that you already know from a standard course on quantum mechanics. It is a very capable tool, and it is enough to describe metals, semiconductors, insulators and even topological insulators.

1.4.2 Second quantization

First quantization assumes that the number of particles is fixed, and this assumption fails for a whole class of quantum materials. It fails for any system where the electrons are not independent, and also, which is more surprising, for systems where the *number* of electrons is itself not fixed. For these systems we need *second quantization*, which is a formalism that is not built around the question of "how many electrons are present" but around how electrons are created and destroyed. This formalism is essential for superconductors, superfluids and correlated materials in general, and it is what makes phenomena such as spinon fractionalization straightforward to describe at all.

1.5 A warm-up in first quantization, and the puzzle that breaks it

Let us take the simplest single-particle system, which is a lattice of sites with one level each and one electron, that forms a finite island such as a quantum dot. Without many-body interactions this is an ordinary diagonalization problem, in the same way as a particle in a box, with a lowest energy state, a first excited state and so on. In this language, "one electron in the ground level" and "one electron in the first excited level" are two different single-particle states, and we can also take superpositions of the two. In every case, however, there is exactly one electron in the system.

Now let us say that we want to describe a quantum dot in a coherent superposition of zero electrons and two electrons, each with 50% probability. We cannot write this as an ordinary wave function, because a wave function needs a fixed number of coordinates, and it makes no sense to have zero coordinates and two coordinates "at the same time." This example is not artificial. This kind of reasoning, about a number of electrons that fluctuates, is what allows us to understand how local magnetic moments couple to each other in real oxides, which is the basis of the Goodenough-Kanamori rules that predict ferromagnetism or antiferromagnetism. It is also part of the definition of a superconductor, whose mean-field order parameter conserves only the *parity* of the electron number and not the number itself. This description of superconductors, which does not conserve the number of particles, has its own name, the Bogoliubov-de Gennes formalism. The slides introduce it here, but we will only use it in practice when we reach the session on superconductivity.

1.6 Building second quantization with creation and annihilation operators

To describe systems of this kind we introduce two operators at every site i. The first one is the annihilation operator c_i, which destroys an electron at site i, and the second one is the creation operator $c_i^\dagger$, which creates one. The Hamiltonian, and all the dynamics of the system, is written only in terms of these operators.

We define the vacuum state, which is the empty state $|0\rangle$, by

$$c_i|0\rangle = 0 \quad \text{for every site } i. \tag{1.1}$$

A hopping term between two sites, which describes an electron that can jump between sites 0 and 1, is then written as

$$H_{\text{hop}} = t\left(c_0^\dagger c_1 + c_1^\dagger c_0\right), \tag{1.2}$$

and we read it as "destroy an electron at 1 and create one at 0" plus the reverse process. If we start from the vacuum and apply creation operators, we obtain all the states of the two-site system. These are $|0\rangle$, then $c_0^\dagger|0\rangle$ with one electron at site 0, then $c_1^\dagger|0\rangle$ with one electron at site 1, and finally $c_0^\dagger c_1^\dagger|0\rangle$ with both sites occupied.

Notebook check-in: `session1.ipynb`, "The spectra of a dimer" (Core)

Build this two-site system and diagonalize it. Check that the Fock states above give the same spectrum that you would obtain from an ordinary two-level treatment in first quantization, before we add any interactions.

1.7 Enforcing fermionic statistics

Up to now nothing distinguishes fermions from bosons, and everything above applies to both in the same way. The difference is introduced through the algebra of the operators. Fermi statistics requires that the many-body wave function changes sign when we exchange two particles, and a direct consequence is that a given single-particle level can have zero fermions or one fermion, but never two. We impose this with anticommutation relations,

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \qquad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0. \tag{1.3}$$

The second relation is the important one. If we take $i = j$ it gives $(c_i^\dagger)^2 = 0$, which means that we cannot create two electrons in the same level. At the same time it imposes the antisymmetry of the wave function, so the Pauli exclusion principle is part of the algebra itself and we do not need to add it by hand.

1.8 First versus second quantization for the same problem

For a tight-binding Hamiltonian that is purely single-particle, first and second quantization need in practice the same amount of work. Let us write the first-quantized Hamiltonian with hopping matrix elements t_{ij} between orbitals i, j. The second-quantized version has the same t_{ij}, multiplied by $c_i^\dagger c_j$. To diagonalize either of them we use the same unitary transformation U, and we obtain the same eigenvalues. The only formal difference is that in first quantization the coefficients that diagonalize the Hamiltonian form a vector, while in second quantization they become operators. The physics is the same in both cases.

1.9 Where the two formalisms differ and the exponential growth of the many-body problem

The difference between the two languages appears as soon as we add real many-body interactions, and it grows very fast. Let us take only three sites, with a generic quartic interaction term, which has four operators, added to the kinetic Hamiltonian. We can no longer restrict ourselves to single-particle states, and we need the full Fock space. This contains the empty

state, three states with one electron, three states with two electrons and one state with all three sites occupied, which gives a total of $2^3 = 8$ many-body states. Figure 1.1 shows the two sides of this comparison explicitly.

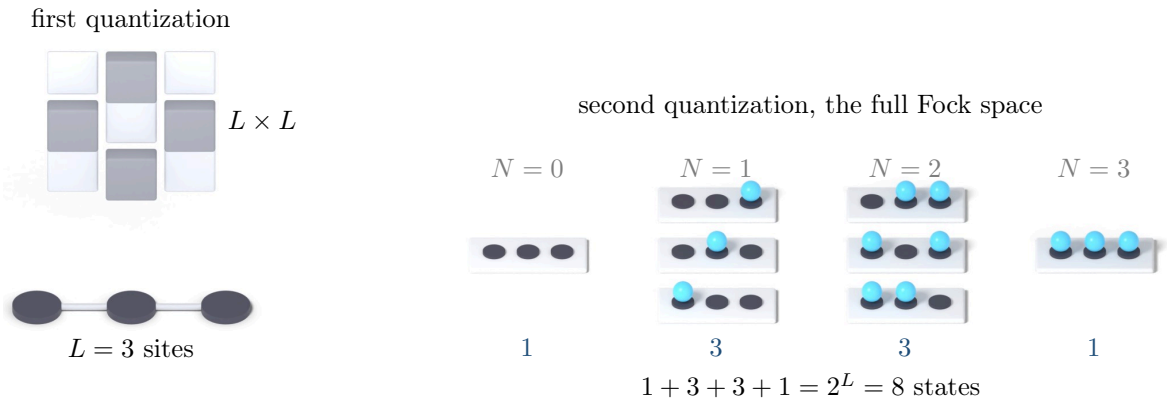


Figure 1.1: The exponential growth of the Fock space for three sites. A single-particle tight-binding problem is an $L \times L$ matrix, on the left, where the raised dark entries are the first-neighbor hoppings of the chain in front of it. The interacting problem needs the full Fock space, on the right, where every site is empty or occupied. Each plate is one many-body state with a ball on every occupied site, and the 2^L states are ordered in columns by particle number. For $L = 3$ this is 8, and for $L \approx 20$ it is already about a million, which is the limit of brute-force diagonalization.

In general, for L sites the single-particle problem is an $L \times L$ matrix, while the many-body problem is a $2^L \times 2^L$ matrix, and we need 2^L coefficients only to write the ground-state wave function. This is the reason why $L \approx 20$ is already close to the practical limit of what we can do by brute force. The exact rate of growth depends on the kind of site that we have. Spinful fermions have four states per site, which are empty, up, down and doubly occupied. Spinless fermions have two, and they are relevant when a single spin species is isolated by strong magnetism or a strong field. Pure spin sites, which we will need later in the course for quantum magnets and Kondo systems, also give an exponential growth, but with a different underlying algebra.

1.10 A first look at eigenstates in finite tight-binding islands

Before we go to interactions, let us look at what a purely single-particle Hamiltonian already gives us, because this is something that we can compare directly with experiment. Let us take a tight-binding Hamiltonian $H = \sum_{ij} t_{ij}\, c_i^\dagger c_j$ on a honeycomb lattice, cut into a finite island that is triangular, hexagonal or rectangular, and let us diagonalize it. The probability density of a given eigenvector in real space can be compared directly with what a scanning tunneling microscope measures. Qualitatively, these states look like the finite-size states of a "particle in a box". The lowest-energy state has its maximum close to the center of the island, and the higher states have more and more nodes. This is a good first check, since single-particle diagonalization, with nothing more, already gives a structure that we can recognize and that we can measure in experiments.

Notebook check-in: `session1.ipynb`, "Finite sized chains" (Core), and "Density of states of a finite system as a function of the system size" (Challenge)

Diagonalize a finite chain or island, and look at how the spatial profiles of the eigenstates and the density of states change when you make the system larger. This is the numerical version of the particle-in-a-box argument above, and it is a good place to build intuition for how we approach an infinite periodic system, which we treat in session 2, starting from the finite-size case.

We will use two software tools in all sessions. The first one is **PyQula**, which is a Python library for tight-binding and mean-field calculations, topological properties and spectral functions, and which we use through the Jupyter notebooks of the course. The second one is **Quantum Lattice**, which is a graphical interface built on PyQula that does not need any coding. Both are open source, and it is useful to have both open next to these notes.

1.11 Spontaneous symmetry breaking and why a symmetric Hamiltonian can have an asymmetric ground state

Before we discuss interactions, let us ask a more general question, since everything that we do in the rest of the session is an example of it. Physical theories usually have a lot of symmetry. Space is symmetric, time is symmetric, and so on. However, reality is not symmetric, since an object is in one place and not in another, and a system evolves forward in time and usually not backward. So why do the symmetries of the Hamiltonian so often not survive in the ground state?

Let us start with an example from daily life, which is a group of colleagues who sit at a table for a meal, with bread on both sides of each person. Each person has to choose between the bread on the left and the bread on the right. There is of course a "correct" answer from etiquette, but let us say that nobody knows it. Some people will take the left one and others the right one, and once the choice is made, everybody at the table ends up in one of two equally good configurations, which are everybody-left or everybody-right. The initial setup, with bread on both sides, was perfectly symmetric. The final result is not. This difference, a symmetric problem with a solution that is not symmetric, is spontaneous symmetry breaking on a small scale.

The same logic, in an exact form, is what happens in a classical Ising ferromagnet. The set of ground states of the Hamiltonian is symmetric, since all spins up is as good a solution as all spins down. However, if we take a *very* large system, which formally means an infinitely large one, or the thermodynamic limit, the system has to choose one of the two, all spins up or all spins down. Which one we measure is decided by fluctuations and not by the Hamiltonian, and once it is decided, the state that we observe is no longer symmetric, even though the Hamiltonian that produced it was always symmetric. The requirement of the thermodynamic limit is essential. Strictly speaking, spontaneous symmetry breaking is a statement about infinite systems, and it is a generic feature of many-body interactions and correlations, not something that appears for a few particles. Figure 1.2 puts all of this in one picture. It shows a symmetric energy landscape whose minima are not symmetric, with a barrier between them that becomes impossible to cross only in the thermodynamic limit.

This gives us a useful way to classify states of matter, according to *which* symmetry they break spontaneously.

- Broken **time-reversal symmetry** gives classical magnets. The magnetization is odd under time reversal, so a magnetic ground state is not invariant when we run the movie backward.

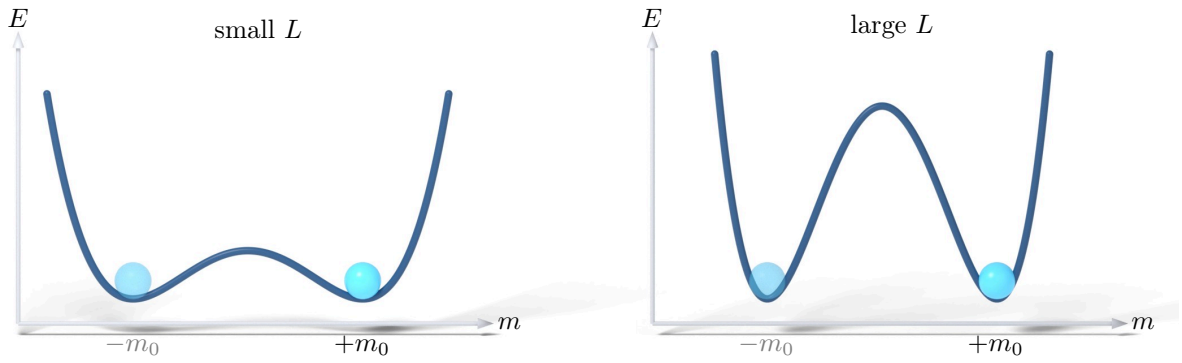


Figure 1.2: Spontaneous symmetry breaking. The energy as a function of the order parameter is symmetric under $m \to -m$, in the same way as the Hamiltonian, but its minima are not, since the system has to be at $+m_0$ or at $-m_0$. The system is the ball at $+m_0$, and the ghost ball at $-m_0$ is the minimum with the same energy that it did not choose. The choice is made by fluctuations and not by the Hamiltonian. The barrier between the minima grows with the system size, from the small system on the left to the large one on the right, which is why spontaneous symmetry breaking is strictly a statement about the thermodynamic limit.

- Broken **inversion symmetry** gives ferroelectrics. These are materials that develop an electric polarization spontaneously, which is the electric analog of a magnetization, and in this way they break the inversion symmetry of the underlying crystal.
- Broken **gauge symmetry** gives superconductors. The phase of the pairing order parameter, or equivalently the phase of the creation and annihilation operators, becomes physical once the order appears. Many observable superconducting phenomena, such as the Josephson effect, are direct consequences of fluctuations of this broken phase, and we will discuss this in detail in the session on superconductivity.

This classification by symmetry is useful, but it does not capture everything. Topological materials, such as topological insulators and fractional Chern insulators, can have *exactly* the same broken or unbroken symmetries as an ordinary trivial material and still behave in a completely different way. The reason for this is that they differ in their topological classification and not in their symmetry classification. This difference is the topic of a full session later in the course. For now, we only keep in mind that the question of "which symmetry is broken" does not give the full description of quantum matter.

Every case of spontaneous symmetry breaking also comes with its own emergent quasiparticle, which is the low-energy fluctuation of the order that has formed. The following table lists three cases.

Broken symmetry	**Realized in**	**Quasiparticle**	**Spin**	**Charge**	**Gap**
Continuous translational	Crystals	Phonon	0	0	Gapless (always)
Spin-rotational $SO(3)$	Magnets	Magnon	1	0	Gapless (gapped with anis
Gauge	Superconductors	Higgs mode	0	0	Gapped

Note that magnons and the Higgs mode both have zero charge but very different spin, and that only the phonon is gapless in every material. Magnons are gapless when the broken spin-rotational symmetry is continuous, and they become gapped when a magnetic anisotropy breaks this symmetry explicitly. The Higgs mode, which is an oscillation of the amplitude of the order parameter, is always gapped.

Since we cannot run time backward in the lab to check if a material breaks time-reversal symmetry, we measure instead quantities that are themselves odd under time reversal. Two

of them are especially useful. The first one is the **Hall conductivity**, which is a transverse current generated by a longitudinal voltage. The second one is the **magneto-optical Kerr effect**, which is a different optical reflection for left-handed and right-handed circularly polarized light. A nonzero signal in either of them is direct evidence of broken time-reversal symmetry.

With this we can finally give a precise and formal definition of a magnet, which goes beyond "the object that sticks to your fridge". A magnet is a many-body state whose wave function is not invariant under the time-reversal operator $\hat{T}$,

$$\hat{T}|\Psi\rangle = |\Psi\rangle \;\Rightarrow\; \text{not magnetic}, \qquad \hat{T}|\Psi\rangle \neq |\Psi\rangle \;\Rightarrow\; \text{magnetic}. \tag{1.4}$$

The north and south poles of a bar magnet are exchanged under time reversal, since a magnetic pole is connected to an orbital angular momentum and angular momenta change sign under $\hat{T}$. For this reason a bar magnet does not pass the invariance test in (1.4), and it is magnetic according to this definition. A plain stick that is not magnetic, on the other hand, looks the same under time reversal and passes the test.

Let us now build this general picture explicitly, three times.

1.12 The general interacting Hamiltonian, and the tool we will use on it

We can now state the problem that this whole session leads to. The Hamiltonians that we are interested in have the generic form

$$H = \sum_{ij} t_{ij}\, c_i^\dagger c_j \;+\; \sum_{ijkl} V_{ijkl}\, c_i^\dagger c_j^\dagger c_k c_l, \tag{1.5}$$

which is a free part, single-particle and exactly solvable, plus a quartic interaction. The interaction itself can come from the plain electrostatic Coulomb repulsion, or from an effective interaction mediated by some other quasiparticle, such as phonons, magnons or plasmons, which can make the total interaction attractive instead of repulsive. We will treat these interactions in a phenomenological way, as given effective models, and we will not derive them microscopically from first principles.

To solve (1.5) exactly is in general impossible, for the exponential reason of Section 1.9. **Mean-field theory** is the tool that still allows us to obtain systematic solutions from it, even if they are approximate, and it can predict a wide range of symmetry-broken states. In this session we will study three of them, which are the same three cases that we classified in Section 1.11.

1. **Broken time-reversal symmetry**, which is magnetism. If we run the movie of the system backward, it looks different from the original.

2. **Broken (lattice translational) symmetry**, which is the charge density wave. A Hamiltonian that is the same at every site at the start ends up with charge on some sites and not on others.

3. **Broken gauge symmetry**, which is superconductivity. The system develops an order parameter with a physical phase, which the symmetry of the original Hamiltonian does not allow.

Figure 1.3 shows these three branches next to each other, together with the equation from which each one is developed and the session where it appears again.

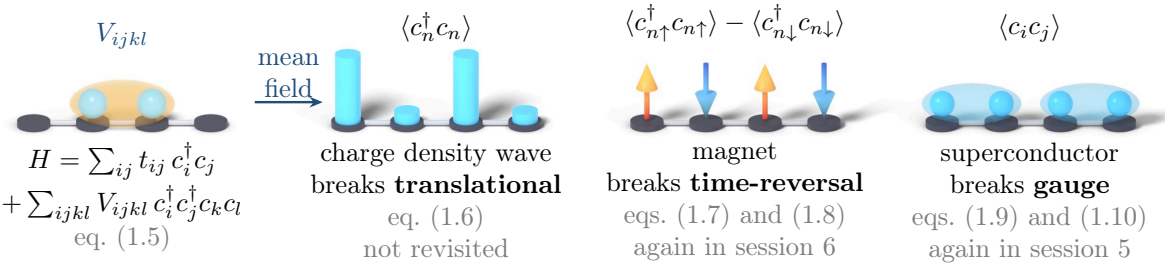


Figure 1.3: One Hamiltonian and three ordered states. The same generic interacting Hamiltonian (1.5), on the left, where the glow is the interaction between two electrons on neighboring sites, is decoupled with three different bilinear expectation values. This gives a charge density wave, a magnet or a superconductor, each breaking a different symmetry. The charge is drawn as columns of alternating height, the spin as alternating arrows, and the electrons bound in pairs as two balls inside one capsule. These three branches are the structure on which the rest of the course is organized.

Two of the three cases of this session, which are magnetism and superconductivity, will come back later in the course in their own sessions, where we treat them in much more detail. You can think of what follows as a first pass, whose goal is to set up the mean-field method itself, and not as the final discussion of either of them.

1.13 Worked example 1, the charge density wave

Let us take the simplest interacting case, which is a spinless one-dimensional chain with nearest-neighbor hopping t and a nearest-neighbor Coulomb repulsion V,

$$H = -t \sum_n \left(c_n^\dagger c_{n+1} + \text{h.c.} \right) + V \sum_n c_n^\dagger c_n \, c_{n+1}^\dagger c_{n+1}. \tag{1.6}$$

In this form, every site is equivalent because of translational symmetry. The quartic term is the difficult one, so mean-field theory factorizes it. We replace one pair of operators by its expectation value, and we keep the rest as it is. This is an *uncontrolled* approximation, and a more rigorous way exists with a path integral and a saddle-point approximation, but it is the practical method that we will use in the whole course. As a result, the interaction term becomes a single-particle onsite energy that depends on the position. Nothing forces the expectation value $\langle c_n^\dagger c_n \rangle$ to be the same at every site, so even if the bare Hamiltonian (1.6) is translationally symmetric, the mean-field solution does not have to be. This difference, a symmetric Hamiltonian with an asymmetric ground state, *is* the charge density wave, and it is the same difference that Section 1.11 described in the abstract.

More rigorously, mean-field theory is a variational search over single-particle product states, where we minimize $\langle H \rangle$ with respect to the bilinear expectation values $\langle c_i^\dagger c_j \rangle$ that define the mean-field Hamiltonian. Since the Hamiltonian depends on these expectation values and the expectation values depend on the Hamiltonian, we solve the problem self-consistently. We guess a Hamiltonian, we compute the expectation values that it gives, we build a new Hamiltonian from them, and we repeat until the process converges. Figure 1.4 shows this loop, and every worked example below and every notebook box in this chapter uses it.

Numerically, at $V = 0$ the spectrum is gapless, which is a metal. When V grows, a gap opens in the single-particle spectrum and becomes larger, and at the same time the charge imbalance between even and odd sites grows from zero toward its maximal value, where it is fully polarized. Figure 1.5 shows the two limits in real space, together with the spectrum of each one.

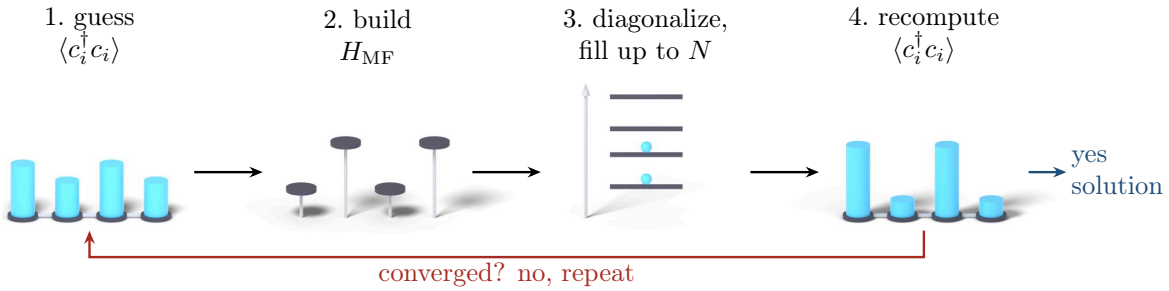


Figure 1.4: The mean-field self-consistency loop. We guess the bilinear expectation values, build the single-particle Hamiltonian that they give, and diagonalize it. Then we compute the new expectation values and repeat until they do not change anymore. The four stations are one pass of the loop, computed for the charge density wave of (1.6) on a periodic chain of four sites with $V = 3t$. The guessed charges 0.6 and 0.4, drawn as columns, raise the onsite energy of the neighboring sites, so the sites of the mean-field Hamiltonian stand at two heights. Filling its two lowest levels gives the new charges 0.82 and 0.18, which are not the guess, so the loop goes round again. The three worked examples of this session and every notebook box in this chapter use this same loop.

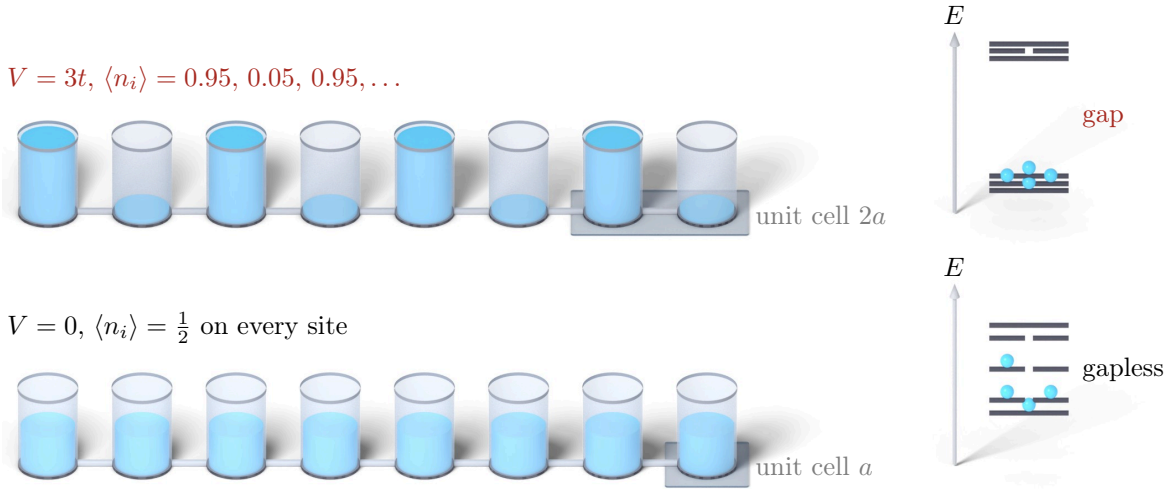


Figure 1.5: The charge density wave of the chain (1.6), solved in mean field for a periodic chain of eight sites at half filling. The height of the liquid in each cup is the charge $\langle n_i \rangle$, and next to each chain is its single-particle spectrum, with an electron on every occupied level and degenerate levels side by side. In front, at $V = 0$, every site has the same charge and the spectrum is gapless, since the Fermi level is inside a pair of degenerate levels. Behind, the nearest-neighbor repulsion drives the mean-field instability (here $V = 3t$), and the charge accumulates on alternating sites, which doubles the unit cell, drawn as the translucent slab, and opens a gap in the spectrum. The Hamiltonian is translationally symmetric in both cases, but the ground state behind is not.

Notebook check-in: `session1.ipynb`, "Symmetry breaking in the spinless fermionic chain" (Core), and "Gap opening in a spinless interacting model" (Challenge)

Run the self-consistency loop of Section 1.13 yourself. Start from an unbiased guess, iterate, and look at how the charge imbalance on each site grows with V, together with the opening of the gap in the spectrum and in the density of states.

1.14 Worked example 2, magnetism

Let us now include again the spin degree of freedom, and let us change the interaction to an onsite Hubbard term between spin-up and spin-down electrons,

$$H = -t \sum_{n,\sigma} \left(c^\dagger_{n\sigma} c_{n+1,\sigma} + \text{h.c.} \right) + U \sum_n c^\dagger_{n\uparrow} c_{n\uparrow}\, c^\dagger_{n\downarrow} c_{n\downarrow}. \tag{1.7}$$

The sign of U alone decides the physics. An attractive $U < 0$ drives superconductivity, which we discuss in Section 1.15, and a repulsive $U > 0$ drives magnetism, which is the topic of this section.

The mean-field decoupling is done in the same way as before, but now we apply it separately to $\langle c^\dagger_\uparrow c_\uparrow \rangle$ and $\langle c^\dagger_\downarrow c_\downarrow \rangle$, and we obtain a bilinear Hamiltonian with onsite energies that depend on the spin. Let us define the magnetization

$$m = \langle c^\dagger_\uparrow c_\uparrow \rangle - \langle c^\dagger_\downarrow c_\downarrow \rangle. \tag{1.8}$$

Self-consistency gives $m = 0$ automatically for $U < 0$, and a finite m for $U > 0$, which is a real magnetic solution.

Let us say precisely *which* symmetry is broken here, using the formal definition in (1.4). Time-reversal symmetry reverses the direction of time and, together with it, every angular momentum in the system, including the spin. On the fermionic operators this means that it exchanges $c^\dagger_\uparrow c_\uparrow \leftrightarrow c^\dagger_\downarrow c_\downarrow$. Since m in (1.8) changes sign under this exchange, a nonzero m means that the ground state does not pass the invariance test in (1.4), so it is a magnet by definition. More generally, this is a breaking of the *continuous* spin-rotational symmetry, and this continuous symmetry breaking is what gives rise to the gapless magnon excitations that appear in the table of Section 1.11. We will study them in detail in a later session, and in this session we only need the mean-field ground state itself.

Numerically, when we increase U a **Mott gap** opens in the spectrum, which grows linearly with U at large U and is associated with the formation of a real local moment on every site.

Notebook check-in: `session1.ipynb`, "The interacting Hubbard dimer" (Core), "Magnetic order driven by interactions in a chain" (Core), "Gap opening of the Hubbard model in a chain" (Challenge), and "Gap opening in the Hubbard model on the honeycomb lattice" (Challenge)

Start with the two-site case, the dimer, to see how the self-consistency loop converges in a small system that you can check by hand. Then go to the extended chain and finally to the honeycomb lattice, which uses the same finite-island geometries of Section 1.10 but now with interactions, and look at how the Mott gap opens and grows with U.

1.15 Worked example 3, superconductivity and broken gauge symmetry

Let us go back to (1.7) with $U < 0$. The mean-field decoupling that we used above always keeps one creation operator and one annihilation operator together, but nothing forces this

choice. We can also take an expectation value of two creation operators, or of two annihilation operators,

$$\Delta = \langle c_i c_j \rangle. \tag{1.9}$$

This anomalous term, which is called a "pairing" term, looks qualitatively different from all the terms that we wrote so far, but it is a valid mean-field term. In the self-consistent solution, Δ is zero for repulsive interactions and can become nonzero for attractive ones, and this is the superconducting order parameter. This worked example is only in the slides, and there is no notebook section for it in this session. The pairing formalism, which is the Bogoliubov-de Gennes formalism that we need to solve it numerically, is left for the session on superconductivity.

Why does this order break gauge symmetry? A gauge transformation acts on the operators as

$$c \to c\, e^{i\varphi}, \qquad c^\dagger \to c^\dagger e^{-i\varphi}. \tag{1.10}$$

Any term built from one creation operator and one annihilation operator, $c^\dagger c$, is invariant under (1.10), since the two phases cancel. However, the pairing term in (1.9) is built from *two* annihilation operators, or two creation operators, so under the same transformation it gets a phase $e^{2i\varphi}$, or $e^{-2i\varphi}$, which does not cancel. For this reason a superconductor with $\Delta \neq 0$ breaks the gauge symmetry of the Hamiltonian spontaneously, which is the "broken gauge symmetry" case in the classification of Section 1.11. This is directly connected to the puzzle of Section 1.5 about a fluctuating number of electrons. To break gauge symmetry and to lose a fixed number of particles are, physically, the same statement. A mean-field superconductor does not have a fixed number of electrons, only a fixed parity.

1.16 Closing summary

- Quantum materials realize a very large range of phenomena, such as topological order, fractionalization and correlated superconductivity, that no single atom shows on its own. The central question of this course is to understand how these phenomena emerge from electrons in a lattice.

- **Second quantization** is the language that allows us to treat in the same way systems with a fixed number of particles, which are single-particle systems, and systems with a fluctuating number of particles, which are many-body systems. It is built from creation and annihilation operators that satisfy the anticommutation relations (1.3).

- The many-body Hilbert space grows as 2^L, as we saw in Section 1.9, and for this reason interacting problems need a systematic approximation and not brute-force diagonalization.

- **Spontaneous symmetry breaking** is the general phenomenon where a symmetric Hamiltonian gives an asymmetric ground state in the thermodynamic limit. We can classify states of matter by the symmetry that they break, which is time-reversal for magnets, inversion for ferroelectrics and gauge for superconductors, and each case has its own emergent quasiparticle. A magnet has the precise and formal definition given in (1.4). Topological materials show that this classification does not cover all cases.

- **Mean-field theory** gives the practical method. We factorize the interacting Hamiltonian (1.5) by taking an expectation value of part of the quartic term, and then we solve the resulting single-particle problem self-consistently.

- The same procedure, applied to three different interaction channels, gives three different symmetry-broken states. These are a charge density wave from (1.6), with broken translational symmetry, a magnet from (1.7)-(1.8), with broken time-reversal symmetry, and a superconductor from (1.9)-(1.10), with broken gauge symmetry. Magnetism and superconductivity each have a full session later in the course, and the versions of this session are a first pass.

In the next session we leave interactions aside for some time, and we go back to the single-particle part of the Hamiltonian of this session. We will see how *symmetry*, and in particular translational symmetry, allows us to diagonalize it exactly, without any approximation, using Bloch's theorem. The general framework of symmetry breaking that we built in Section 1.11 will not be discussed again there, so it is useful to keep it from here and not to expect a summary of it later.

Session 2

Symmetries, Tight-Binding Band Structure Theory, and Effective Models

In the last session we introduced second quantization and used it in practice. Mean-field theory turned an interacting, quartic Hamiltonian into an effective single-particle one, and we saw that the same method gives a charge density wave, a magnet and a superconductor, one for each of the three symmetries that a many-body ground state can spontaneously break. We also spent some time on what this breaking means. We used the example of the bread on the table, where there is a set of solutions with the same energy and no single one of them has the full symmetry. We saw that this picture only becomes sharp in the thermodynamic limit, we discussed the ferroelectric as a fourth example next to magnets and superconductors, and we compared all this with topological order, which can distinguish two materials that have the same broken symmetries. This is a complete topic by itself, and in this session we will not repeat any of it.

In this session we ask a different question, which is about the structure of the problem. Once the interactions are included in a mean-field, single-particle Hamiltonian, how do we solve it? For a few sites we can diagonalize a matrix directly, but a real material has of order 10^{23} sites, and no computer can diagonalize a matrix of this size. The answer is again symmetry, but now it has a different role than in the last session. We do not use it to classify which order a ground state breaks, and instead we use it to simplify the diagonalization. When the ground state of a Hamiltonian *shares* the symmetry of the Hamiltonian, the symmetry is not only a classification tool, and it becomes a real shortcut. It turns an infinite-dimensional problem that we cannot solve into infinitely many finite-dimensional problems that are easy to solve. For a periodic system, this shortcut is called Bloch's theorem. At the end of this session we will have used it to derive several real band structures, including the Dirac cone of graphene, which the topology lecture that comes right after this one will take and gap open.

2.1 Motivation and recap

Let us recall the form of a many-body Hamiltonian once it is written in second-quantized, tight-binding form,

$$H = \sum_{ij} t_{ij}\, c_i^\dagger c_j,$$

where t_{ij} are the hopping matrix elements between sites i and j. In this language, to solve a quantum material means to diagonalize this Hamiltonian, which is to find the unitary

transformation that brings it into the diagonal form

$$H = \sum_n \epsilon_n \, \Psi_n^\dagger \Psi_n, \qquad \Psi_n^\dagger = \sum_i a_{n,i} \, c_i^\dagger.$$

Symmetries allow us to find this diagonal form, or at least strong qualitative and quantitative properties of it, without doing the diagonalization explicitly. This is the motivation for everything in this session. We will use symmetry not to classify order, but to simplify how we solve a quantum many-body problem.

2.2 Simplifying problems with symmetry in simple examples

We have all used this method before, also outside quantum mechanics. Let us take an infinite plane with a uniform positive charge, and let us ask in which direction the electric field points at some position away from the plane. It can point straight out, or it can be tilted by some angle. We can apply a mirror symmetry that exchanges left and right. The charge distribution does not change under this reflection, so the true field must also map to itself under the same reflection. A tilted field does not pass this test, since its mirror image is tilted in the other direction, while a field that points straight out does pass it. Since the electric field at that point is unique, symmetry alone removes every answer except the one perpendicular to the plane, and we never have to compute the integral from Coulomb's law.

The same argument also works for a less simple shape. Let us replace the infinite plane by an ellipse with a uniform charge, which is an egg-shaped charge distribution, and let us ask the same question about the field at some point on its mirror axis. Mirror symmetry again gives the answer without tilt, this time with the reflection axis of the ellipse, and again without any explicit integration. Note that this only works on the mirror axis, since at a point away from it the reflection maps the point to a different point, and symmetry alone does not fix the direction there.

Mirrors are only the simplest case. A circle stays the same under any rotation. A square stays the same only under rotations by 90°, 180°, and 270°, and a shape with less symmetry may stay the same only under a single mirror. Quantum systems can have much richer symmetries, which are also much less obvious than these, and in the following we turn this richer structure into a systematic tool.

2.3 Symmetry operators in a formal way

To make the argument from electromagnetism precise for a quantum system, we need three objects. The first is a wavefunction ψ, the second is the transformed wavefunction $\phi = S\psi$ that we obtain when we apply a symmetry operation, and the third is the operator S itself, which is a linear map from one to the other and a matrix in a finite-dimensional representation. A concrete case makes this clear. If ψ is a p_x orbital and S is a 90° rotation, the transformed state $\phi = S\psi$ is a p_y orbital, meaning that the two lobes on the x-axis are rotated into two lobes on the y-axis. Note that this p_x orbital is *not* an eigenstate of the rotation, since it is mapped to a different state.

Symmetry operators are always unitary. In physical terms, a symmetry is only a change of labels of our reference frame, for example a rotation of the axes, and any such change of labels must keep the norm of every state in the Hilbert space. In equations,

$$S^\dagger S = S S^\dagger = \mathbb{1}, \qquad S^\dagger = S^{-1}. \tag{2.1}$$

A direct consequence is that all the eigenvalues of S have unit modulus, since $\langle\psi|S^\dagger S|\psi\rangle = \langle\psi|\psi\rangle$ gives $|\lambda|^2 = 1$ for any eigenvalue λ. So a symmetry eigenvalue can always be written as a pure phase,

$$S|\psi\rangle = e^{i\varphi}|\psi\rangle, \qquad \varphi \in (-\pi, \pi]. \tag{2.2}$$

Before we include a Hamiltonian, let us look at two small examples. We take a wavefunction with real, staggered values $(+1, -1, +1, -1)$ on four sites on a ring, and we apply a 90° rotation that shifts the sites by one in a cyclic way. The transformed wavefunction is $(-1, +1, -1, +1)$, which is -1 times the original. So this state is an eigenstate of the rotation with eigenvalue -1, that is $\varphi = \pi$ in (2.2). Now we take a complex wavefunction $(+1, -i, -1, i)$ on the same ring and apply the same rotation. The transformed wavefunction is i times the original, so it is an eigenstate with eigenvalue i, that is $\varphi = \pi/2$.

The next example is less visual but more useful in practice. A two-site tight-binding Hamiltonian with a single hopping t between sites 1 and 2,

$$H = t\left(c_1^\dagger c_2 + c_2^\dagger c_1\right),$$

has two eigenstates, the bonding and antibonding combinations $c_1^\dagger + c_2^\dagger$ and $c_1^\dagger - c_2^\dagger$. Under the mirror symmetry that exchanges sites 1 and 2, the bonding state has eigenvalue $+1$ and the antibonding state has eigenvalue -1. This means that different eigenstates of the same Hamiltonian can have different eigenvalues under the same symmetry.

We now come to the central fact of this session. Let us assume that a Hamiltonian is invariant under a symmetry operation, $SHS^{-1} = SHS^\dagger = H$, which is the same as the statement

$$[H, S] = 0. \tag{2.3}$$

From linear algebra we know that two operators that commute have a common eigenbasis. So whenever (2.3) holds, every eigenstate of H can be chosen so that it is also an eigenstate of S. In the words of the lecturer, this is perhaps the most important idea of this session. The reason for this is that it is very often much easier to classify the eigenstates of a symmetry operator, whose eigenvalue is a simple phase by (2.2), than to classify the eigenstates of the Hamiltonian directly. In practice, this means that to find all the symmetries of the Hamiltonian is the first step to solve it, and the most useful one, not a side remark. Once we know that a matrix S commutes with H, to classify the eigenstates of S is the same task as to classify the eigenstates of H.

2.4 Exercises where we guess eigenstates from symmetry only

We will now see with a sequence of small examples how much (2.3) gives us. In each example we add sites, and at the end we take the infinite limit.

Three sites on a ring. We take a tight-binding Hamiltonian on three sites in a triangle, with a symmetry R that rotates $1 \to 2 \to 3 \to 1$ by 120°. Since $R^3 = \mathbb{1}$, any eigenvalue $\lambda = e^{i\varphi}$ of R must satisfy $\lambda^3 = 1$, so

$$\varphi_n = \frac{2\pi n}{3}, \qquad n = 0, 1, 2. \tag{2.4}$$

So we know a lot about the eigenstates of H before we write H explicitly. They must have the form $\alpha_1 c_1^\dagger + \alpha_2 c_2^\dagger + \alpha_3 c_3^\dagger$. Since R permutes the sites in a cyclic way and multiplies the whole state by $e^{i\varphi_n}$, the coefficients must have the same phase from one site to the next, $\alpha_j \propto e^{i\varphi_n j}$. The eigenstates of any Hamiltonian with this rotational symmetry are exactly the

three discrete Fourier modes on the ring, regardless of the values of the hopping amplitudes. Figure 2.1 shows the three modes explicitly, and it also shows how the set of allowed phases changes when the ring becomes larger.

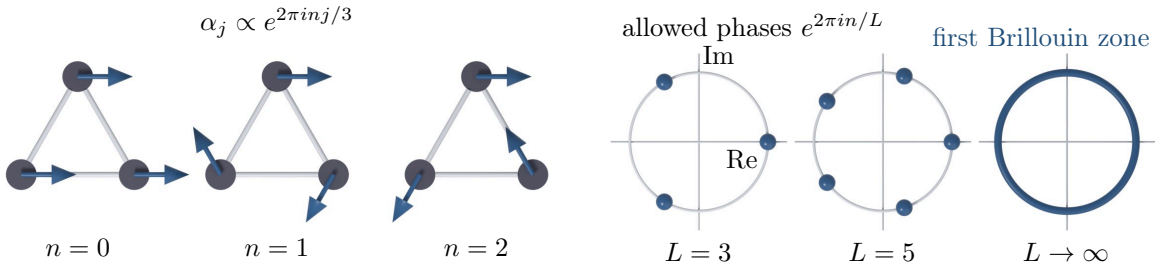


Figure 2.1: Symmetry fixes the eigenstates before we write down the Hamiltonian. On a three-site ring the rotation R can only have eigenvalues $e^{2\pi i n/3}$ by (2.4), and so the coefficients of the state must change by this same phase from one site to the next. This means that the eigenstates are the discrete Fourier modes, for any values of the hopping amplitudes. The arrow on each site is the complex coefficient $\alpha_j \propto e^{i\varphi_n j}$, which turns by $2\pi n/3$ from one site to the next. On the right, the balls are the allowed phases for $L = 3$ and $L = 5$. In the limit $L \to \infty$ they fill the circle, and this circle is the first Brillouin zone.

Five sites on a ring. The same argument on a five-site ring with nearest-neighbour hopping, $H = \sum_{\langle ij \rangle} c_i^\dagger c_j$, gives $\varphi_n = 2\pi n/5$, $n = 0, \ldots, 4$, and the eigenstates $|\Psi_n\rangle \propto \sum_{j=0}^{4} e^{i\varphi_n j}\, c_j^\dagger |0\rangle$. With this we can already answer a quantitative question without diagonalizing anything. Let us ask what the weight of site 1 is in the lowest-energy eigenstate built from the vacuum $|\Omega\rangle$, in the sense of the overlap $\left|\langle \Omega | c_1 \Psi_{\text{GS}}^\dagger | \Omega \rangle\right|^2$. Every $|\Psi_n\rangle$ is a superposition over all five sites with the same modulus on each site, so this overlap is $1/5$ for every n, for the lowest energy and for all other states. This follows from the symmetry only, and it does not depend on the specific hopping amplitudes. The same argument stays valid if the nearest-neighbour hopping is replaced by any long-range hopping, and a six-site ring with additional long-range couplings shows the same thing. As long as the rotational symmetry of the ring is present, the symmetry eigenvalues, the eigenstates and expectation values such as this one are all fixed by symmetry only, however complicated the Hamiltonian becomes.

Note that such models with few sites are not only of academic interest. With scanning tunnelling microscopy one can place single atoms with essentially arbitrary precision, and tight-binding rings with few sites like these have been built and measured in this way. There is a famous stop-motion film made only from atoms moved with STM, which is sometimes called "the smallest film ever made," where every dot on the screen is a single atom. It shows how precisely such models can now be realized in the lab.

The infinite chain. Now we let the ring grow without limit, so that it becomes an infinite one-dimensional chain with nearest-neighbour hopping. The relevant symmetry is a translation by one site, $T : c_n^\dagger \to c_{n-1}^\dagger$, which shifts a wavefunction ψ_n on the sites to ψ_{n+1}, and $[H, T] = 0$ for any Hamiltonian that only has hoppings that depend on the distance between sites and not on the absolute position. By Section 2.3, H and T have a common eigenbasis, and by (2.2) the eigenvalue of T in this common eigenbasis is a pure phase $e^{i\varphi}$. This φ is called the **Bloch phase**. The statement that H and T have a common eigenbasis, when we apply it to a system with translational symmetry, *is* Bloch's theorem, which says that if two operators commute, there is a common basis for both of them. The relation between this phase and the corresponding energy of the Hamiltonian is what we call the **dispersion relation** of a periodic system.

2.5 Bloch's theorem and the first band structure

Let us now do this fully explicitly for the one-dimensional tight-binding chain,

$$H = -t \sum_n \left(c_n^\dagger c_{n+1} + \text{h.c.} \right). \tag{2.5}$$

We take $t > 0$ in all of this session, the same sign convention as in (1.6) of the previous session.

Since $[H, T] = 0$, we look for eigenstates that are a superposition over all sites that transforms in a simple way under translation,

$$|\Psi_\varphi\rangle = \sum_n e^{in\varphi}\, c_n^\dagger |0\rangle. \tag{2.6}$$

It is cleaner to invert this relation and to define new operators that are labelled by φ instead of by site,

$$\Psi_\varphi^\dagger = \sum_n e^{in\varphi}\, c_n^\dagger, \qquad c_n^\dagger = \sum_\varphi e^{-in\varphi}\, \Psi_\varphi^\dagger, \tag{2.7}$$

which is an ordinary Fourier transform between real space and the space of Bloch phases. Putting (2.7) into (2.5) gives

$$\sum_n c_n^\dagger c_{n+1} = \sum_{n,\varphi,\varphi'} e^{-in\varphi}\, e^{i\varphi'(n+1)}\, \Psi_\varphi^\dagger \Psi_{\varphi'} = \sum_{\varphi,\varphi'} e^{i\varphi'}\, \delta_{\varphi\varphi'}\, \Psi_\varphi^\dagger \Psi_{\varphi'} = \sum_\varphi e^{i\varphi}\, \Psi_\varphi^\dagger \Psi_\varphi,$$

where the sum over n gave the orthogonality relation $\sum_n e^{in(\varphi'-\varphi)} = \delta_{\varphi\varphi'}$. The Hermitian conjugate term gives the complex conjugate phase, so

$$H = -t \sum_\varphi \left(e^{i\varphi} + e^{-i\varphi} \right) \Psi_\varphi^\dagger \Psi_\varphi = \sum_\varphi \epsilon_\varphi\, \Psi_\varphi^\dagger \Psi_\varphi, \qquad \boxed{\epsilon_\varphi = -2t \cos\varphi}. \tag{2.8}$$

This is our first band structure, and we obtained it without writing down an infinite matrix, and without diagonalizing it. We only needed the fact that the eigenstates must have a definite phase under translation, and then a single Fourier transform gave the rest.

Some features of this dispersion should be discussed explicitly, since they appear again and again in the course. Near the bottom of the band ($\varphi = 0$), $\epsilon_\varphi \approx -2t + t\varphi^2$, which is a parabola that opens upward. Near the top of the band ($\varphi = \pi$), we write $\varphi = \pi + \delta$ and obtain $\epsilon_\varphi \approx 2t - t\delta^2$, which is a parabola that opens downward. The two curvatures have *opposite* sign. Once we define the effective mass properly in Section 2.9, this means that low-energy quasiparticles near the bottom of the band behave like ordinary electrons with positive mass, while those near the top behave like holes with negative mass. Near the middle of the band, at $\varphi = \pi/2$ where ϵ_φ crosses zero, the dispersion is closer to linear than to quadratic, and this is a first sign of the Dirac-like physics that we will see in detail in Section 2.11. All of this is shown in figure 2.2, which also shows the density of states of this band.

Notebook check-in: `session2.ipynb`, "One dimensional chain" (Core)

Build the chain of (2.5), diagonalize it numerically, and check that the numerical spectrum follows $\epsilon_\varphi = -2t\cos\varphi$ from (2.8) when you scan φ across the Brillouin zone.

A natural extension, which we do in class, is to check that the same method also works beyond nearest-neighbour hopping. We add a second-neighbour hopping term $-\eta \sum_n (c_n^\dagger c_{n+2} +$ h.c.) to (2.5). This term has the same translational symmetry as before, so the eigenstates

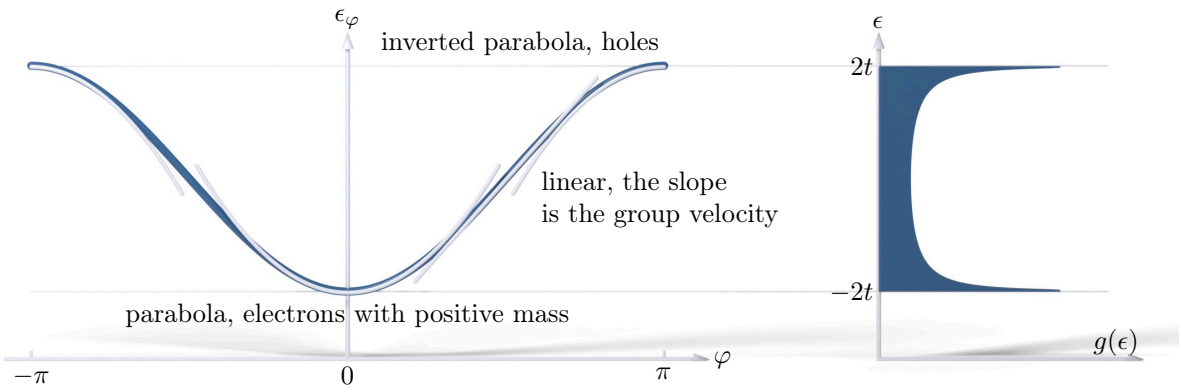


Figure 2.2: The one-dimensional band $\epsilon_\varphi = -2t\cos\varphi$ of (2.8), drawn for $t = 1$, with the grey tubes showing its local forms. Near the band bottom the dispersion is an upward parabola, and quasiparticles behave as ordinary electrons with positive mass. Near the top the curvature has the opposite sign and they behave as holes. Near midband it is locally linear, which is a first sign of the Dirac physics of Section 2.11. The slope is the group velocity (2.17) and the curvature is the inverse effective mass (2.18), and both are read from this one curve. On the right is the resulting density of states (2.16), cut at the top of the drawing, which diverges at both band edges, at the same points where the group velocity vanishes.

(2.6) do not change at all, and we only need to compute the eigenvalues again. We repeat the Fourier substitution for the new term, which couples $c_n^\dagger$ with c_{n+2} and so gives a phase $e^{\pm 2i\varphi}$ instead of $e^{\pm i\varphi}$, and we obtain

$$\epsilon_\varphi = -2t\cos\varphi - 2\eta\cos(2\varphi). \tag{2.9}$$

Notebook check-in: `session2.ipynb`, "One dimensional chain, with longer range hopping" (Core)

Add the second-neighbour hopping η to the chain from the previous box, and check that the band you compute numerically agrees with (2.9). It is also useful to check the density of states here (Section 2.9), since η can create a second local extremum in ϵ_φ, and with it an additional feature of Van Hove type in the spectrum.

We also mention `pyqula` by name here, which is a Python package that computes band structures such as this one in 1D, 2D, 3D and higher dimensions, and which is the computational tool we use in the whole course. Here we only give its name. The slides show an explicit installation guide (`pip install pyqula`) and example scripts for a 1D chain and for a 1D "ladder" with two sites per cell, together with the band plots that these scripts produce. This installation guide and the screenshots of its code are only on the slides, and they are not narrated in the transcript nor reproduced here. The notebook is the place where you run the equivalent code yourself.

2.6 The meaning of the phase, which is the quasi-momentum

Let us stop here and ask what the phase φ in (2.2) means physically, and not only formally. We go back to the Schrödinger equation in the continuum and ask how a wavefunction translated by the lattice constant a, $\psi(x+a)$, is related to $\psi(x)$. A Taylor expansion gives

$$\psi(x+a) = \psi(x) + a\,\partial_x\psi(x) + \frac{a^2}{2}\partial_x^2\psi(x) + \cdots = e^{a\partial_x}\psi(x) = e^{\frac{i}{\hbar}ap}\,\psi(x),$$

where $p = -i\hbar\partial_x$ is the momentum operator. If ψ is an eigenstate of the translation operator with eigenvalue $e^{i\varphi}$, when we compare the two expressions we find

$$\varphi = \frac{a\,p}{\hbar} = k\,a, \tag{2.10}$$

meaning that the symmetry eigenvalue is directly the crystal momentum $k = p/\hbar$ times the lattice constant. This is the reason why φ is called the *quasi-momentum*. It is directly related to the momentum operator of the original Schrödinger equation, only measured in units where the lattice constant is absorbed, and it takes values in $\varphi \in (-\pi, \pi]$, which is the first Brillouin zone.

This identification also tells us where the orthogonal basis with one orbital per site comes from, which we have used so far without saying it. The atomic orbitals of a real material, s, p, d, and so on, are not orthogonal between neighbouring sites. Before we write a clean tight-binding Hamiltonian, we first go to an orthogonal local basis with one orbital per site, which we obtain in a systematic way with what is called **Löwdin orthogonalization**. The contact session works this construction out properly, and here we only assume it, and the slides also call this basis "Wannier states". In this basis, symmetry alone already tells us that the eigenstates have the form $e^{in\varphi}c_n^\dagger|0\rangle$, which is the ansatz (2.6). This is true for a finite ring of 7 sites, for 11 sites, and for the infinite chain that we want in order to describe a solid. We can build the same intuition by working with finite rings that become larger and larger, before we take the limit $L \to \infty$.

2.7 Multi-orbital band structures

Real unit cells almost never have a single orbital. Graphene has two, one per sublattice, compounds of transition metals often have several d-orbitals per site, and so on. Let α label the orbital within a unit cell and n label the unit cell itself. We keep couplings only within a cell and to the nearest-neighbouring cell, and longer-range couplings work with the same method, only with more terms. The most general bilinear Hamiltonian built from these operators is then

$$H = \sum_{n,\alpha,\beta} t_{\alpha\beta}\, c_{n,\alpha}^\dagger c_{n,\beta} + \sum_{n,\alpha,\beta} \left(\gamma_{\alpha\beta}\, c_{n,\alpha}^\dagger c_{n+1,\beta} + \text{h.c.}\right), \tag{2.11}$$

where $t_{\alpha\beta}$ are the intracell hoppings and $\gamma_{\alpha\beta}$ are the intercell hoppings. The intracell term already contains both directions of each bond, so t is a Hermitian matrix, $t_{\alpha\beta} = t_{\beta\alpha}^*$, and the Hermitian conjugate is only added to the intercell term.

The symmetry that diagonalizes (2.11) is a slightly richer version of a pure translation. It shifts the unit-cell index $n \to n+1$ *and* applies a unitary rotation among the orbitals within each cell. The corresponding eigenstates are

$$\Psi_{\varphi,\alpha}^\dagger \sim \sum_{n,\beta} e^{in\varphi}\, U_{\alpha\beta}\, c_{n,\beta}^\dagger,$$

for some unitary U. We put this ansatz into (2.11) and repeat the Fourier argument of Section 2.5 for each orbital, and we find that the required U is exactly the matrix that diagonalizes the **Bloch Hamiltonian**

$$H(\varphi) = t + \gamma\, e^{i\varphi} + \gamma^\dagger e^{-i\varphi}, \tag{2.12}$$

which is a matrix whose size is the number of orbitals per unit cell. Each entry is labelled by the pair of orbitals (α, β) that it couples. The intracell hoppings $t_{\alpha\beta}$ enter this entry directly, with no phase, while the intercell hoppings $\gamma_{\alpha\beta}$ enter multiplied by the phase $e^{\pm i\varphi}$

that comes from the symmetry eigenvalue. This is the general method that we will use many times. First we count the orbitals per cell, which gives the size of the matrix. Then we put each hopping in the entry of its pair of orbitals, we multiply the intercell ones by $e^{\pm i\varphi}$, and we diagonalize. Note that an intracell hopping between *different* orbitals goes off the diagonal, as the example below shows.

A concrete example makes this clear, and it will be important again soon. We take a chain with two orbitals (A, B), with a strong intracell coupling of magnitude 1 between A and B, and a weaker intercell coupling η between B in one cell and A in the next cell. The Bloch Hamiltonian (2.12) is then purely off-diagonal,

$$H(\varphi) = \begin{pmatrix} 0 & 1+\eta\, e^{-i\varphi} \\ 1+\eta\, e^{i\varphi} & 0 \end{pmatrix}, \qquad \epsilon_\pm(\varphi) = \pm\left|1+\eta\, e^{-i\varphi}\right|. \tag{2.13}$$

This chain with alternating hoppings should be remembered by its name, since it is the Hamiltonian of Su-Schrieffer-Heeger type that appears again, now with a topological aspect, in the topology lecture right after this one. Figure 2.3 shows this chain as the chain of (2.8) described with a unit cell of two sites, so that its single band folds into the two bands of (2.13), and it also shows how a gap opens at the zone edge when η moves away from 1.

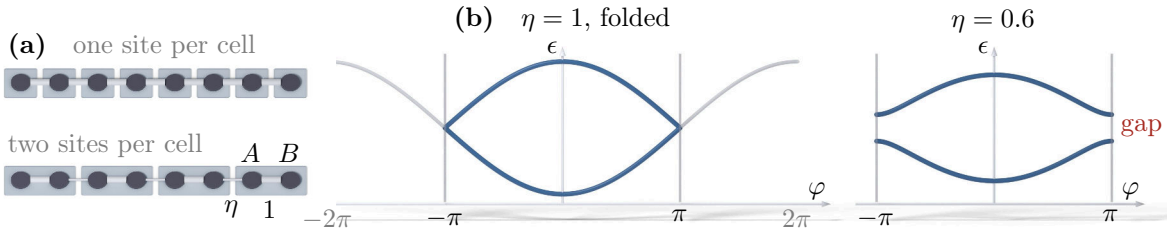


Figure 2.3: Reading a Bloch Hamiltonian from a lattice, shown as band folding. The number of orbitals per unit cell gives the size of the matrix. An intracell hopping enters the entry of its orbital pair with no phase, and an intercell hopping enters the corresponding entry multiplied by $e^{\pm i\varphi}$. (a) The same chain with one site per cell, behind, and with two sites A, B per cell, in front, where the intracell hopping 1 is the thick bond and the intercell hopping η is the thin one. Note that the intracell A-B hopping goes *off* the diagonal. For the chain with alternating hoppings (2.13) this gives the two bands $\epsilon_\pm = \pm|1+\eta e^{-i\varphi}|$. (b) For $\eta = 1$ the two descriptions are the same chain, and the band of the one-site cell, the grey tube, folds into the zone of the two-site cell, which is half as wide and ends at the translucent planes at $\varphi = \pm\pi$. The two folded branches are the two bands, which touch at the zone edge. For $\eta = 0.6$ they have a gap, and the gap closes exactly at $\eta = 1$. This is the Su-Schrieffer-Heeger chain of the topology session.

Notebook check-in: `session2.ipynb`, "One dimensional chain, band folding" (Challenge), and "One dimensional ladder, band folding" (Challenge)

The "band folding" section builds the same picture from the other direction. You take a chain with a single orbital and describe it again with a larger unit cell, which is the same as having more than one site per cell, and you see how a single band from (2.8) folds into several bands, as (2.12) predicts. The ladder section then works with a real model with two orbitals per cell, in the same spirit as (2.13), and there you can check that the numerical bands agree with $\epsilon_\pm(\varphi)$.

2.8 Higher-dimensional band structures

A two-dimensional lattice has two independent translations that commute with each other, T_x and T_y, so Bloch eigenstates have two independent phases, φ_x and φ_y, one for each direction. These phases are related to the crystal momentum along the two lattice vectors by the same Taylor-expansion argument as in (2.10), applied once for each direction. The space of allowed

pairs (φ_x, φ_y) is the **Brillouin zone**. Since $\varphi = 0$ and $\varphi = 2\pi$ give the same translation eigenvalue in both directions, the Brillouin zone is not an open square but a periodic one. When we identify the left and right edges, and the top and bottom edges, it becomes a torus, in the same way as it is a circle in the one-dimensional case of Section 2.5. This compactness of the Brillouin zone will be very important when topological invariants appear in the next session.

When we apply the Fourier method of Section 2.5 once for each spatial direction, we obtain for a square lattice with a single orbital and nearest-neighbour hopping t, with the same sign convention as in (2.5),

$$\epsilon(\varphi_x, \varphi_y) = -2t \left(\cos \varphi_x + \cos \varphi_y\right), \tag{2.14}$$

and, when we apply the same argument a third time, for a cubic lattice with a single orbital,

$$\epsilon(\varphi_x, \varphi_y, \varphi_z) = -2t \left(\cos \varphi_x + \cos \varphi_y + \cos \varphi_z\right). \tag{2.15}$$

The same method extends in a systematic way to four, five or more dimensions, since nothing in the argument depends on being in three-dimensional space. Equations (2.14) and (2.15) come from the slides and the transcript, and there is no counterpart for them in `session2.ipynb`. The two-dimensional example of the notebook goes directly to the honeycomb lattice of Section 2.11 and does not treat the plain square lattice, and there is no section on the three-dimensional cubic lattice. If you want to check (2.14) or (2.15) numerically, the same construction that we use for the honeycomb lattice in the box below can be extended directly, with a square or cubic Bravais lattice with a single orbital instead.

Before we continue, let us say explicitly that everything above applies far beyond electrons. The same Bloch-Hamiltonian method diagonalizes phonon bands, magnon bands and the bands of Bogoliubov-de Gennes quasiparticles in a superconductor. It applies in the same way to artificial lattices that are engineered and have no relation to real crystals, such as cold atoms trapped in optical lattices. We will focus on electronic bands for concreteness, but this generality is the reason why the same method appears again for magnons, and for Bogoliubov quasiparticles, later in the course. The physics that single-particle band theory gives in this way is very rich. It includes Dirac fermions with zero Dirac mass, Weyl semimetals, which are Dirac points generalized to higher dimensions, quantum spin Hall insulators, Chern insulators, crystalline topological phases and topological superconductors. All of this is built only on the Bloch Hamiltonian $H(\mathbf{k})$ that we constructed here.

2.9 Properties of an electronic dispersion

Once we have a dispersion $E(\mathbf{k})$, from (2.8), (2.13), (2.14) or from a more complicated model, several quantities that we derive from it have a direct physical meaning.

The **density of states**,

$$g(\omega) = \int_{\mathrm{BZ}} \frac{d^d k}{(2\pi)^d} \, \delta(\omega - E(\mathbf{k})), \tag{2.16}$$

counts how many states there are at energy ω. It controls how strongly a material responds to interactions. A large or divergent density of states means that even a weak interaction can drive an instability toward magnetism, a charge density wave or superconductivity, which are the three cases of the last session. A vanishing density of states means that interactions can hardly have any effect, however strong they are.

The **group velocity**,

$$v(\mathbf{k}) = \frac{1}{\hbar}\nabla_{\mathbf{k}} E(\mathbf{k}), \tag{2.17}$$

is the local slope of the dispersion. Where $v(\mathbf{k}) = 0$, electrons at that $\mathbf{k}$ do not move in practice. Where $v(\mathbf{k})$ is large, they propagate quickly through the lattice.

The **effective mass**,

$$\left(\frac{1}{m^*}\right)_{ij} = \frac{1}{\hbar^2}\frac{\partial^2 E(\mathbf{k})}{\partial k_i \, \partial k_j}, \tag{2.18}$$

is the curvature of the dispersion, and it is the mass that is measured in essentially every transport or cyclotron-resonance experiment, since the bare electron mass cannot be observed directly inside a solid. Depending on the band, m^* can go from nearly zero to very large values. For a Dirac dispersion, as we will see soon, it is formally not defined at the point where the bands touch.

Finally, the **Fermi surface** is the set of $\mathbf{k}$-points that have the same energy as the highest occupied state. Its shape controls which electronic instabilities are favoured and how a material responds to an external magnetic field, and it is one of the quantities that ARPES (Section 2.12) can map directly.

Notebook check-in: `session2.ipynb`, "Density of states of the honeycomb lattice" (Core), and "Fermi surface of the honeycomb lattice" (Challenge)

Once you have the honeycomb dispersion of Section 2.11, these two sections let you compute (2.16) and the Fermi surface directly, and see how both change when the Fermi level is tuned through the Dirac point.

2.10 Three types of dispersion, which are parabolic, Dirac and flat

Real dispersions are complicated functions of $\mathbf{k}$, but almost every band structure in this course is built locally from three limiting types.

1. **Parabolic bands**, $E \propto k^2$, which is the case similar to free electrons, describe most semiconductors and conventional metals. They appear essentially from a Taylor expansion near any extremum of a band. We already saw this explicitly near $\varphi = 0$ and $\varphi = \pi$ in (2.8), with effective masses of opposite sign at the two extrema.

2. **Dirac (linear) dispersions**, $E \propto |k|$, are realized in graphene and in several topological materials, such as quantum spin Hall insulators, Chern insulators and other symmetry-protected topological phases. They have finite topological invariants, in the form of a Berry phase, and they give rise to a phenomenology that is similar to relativistic physics, even though the electrons themselves are not relativistic at all.

3. **Flat bands**, $E(\mathbf{k}) = \text{const}$, may look like the least interesting case, but by (2.16) a flat band gives a *divergent* density of states. As a result, even weak interactions have a very large effect when a flat band is near the Fermi level. When the flatness has a nontrivial origin, which is destructive interference in real space and not a hopping set to zero, the flat band can also have a finite Chern number, which is central to the quantum Hall effect and to moiré materials.

Figure 2.4 shows the three types next to each other, together with the density of states of each one, which decides whether interactions can have an effect on them.

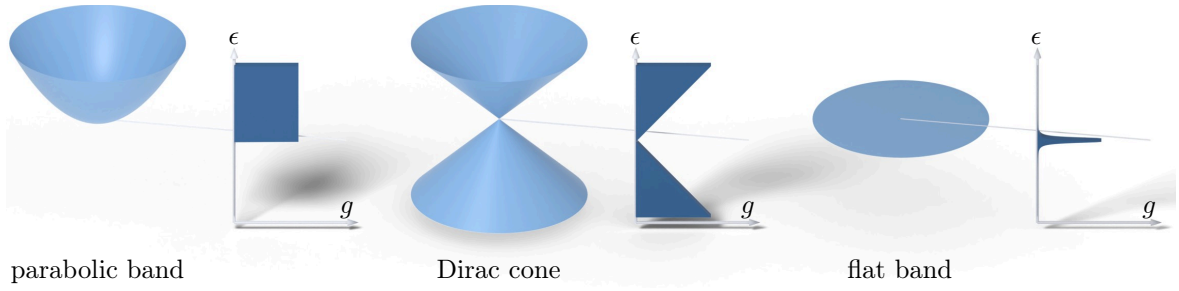


Figure 2.4: The three types of dispersion and the densities of states that they give, all in *two* dimensions. Each band is a surface over the two-dimensional momentum, with its density of states as a slab next to it on the same energy axis. A parabolic band gives a constant density of states, which is a step. A Dirac point makes it vanish linearly, so a finite interaction strength is needed before anything happens. A flat band makes it diverge, drawn here with a small broadening, so an arbitrarily weak interaction can drive the symmetry-broken states of Session 1. Almost every band structure in this course is locally one of these three.

Next we derive the Dirac case explicitly, since it is the defining feature of graphene and the direct link to the topology lecture, and then we look at two lattices that realize flat bands. The tools we use in the whole course to compute such dispersions are `pyqula`, which we named in Section 2.5, and its graphical companion, which is a second computational tool shown on the slides with the name **Quantum Lattice**. It needs no scripting, and it is not named in the lecture, which only refers to "a user interface we will use at some points".

2.11 The Dirac cone of the honeycomb lattice of graphene, and flat-band lattices

The honeycomb lattice of graphene has two orbitals per unit cell, one on each of two triangular sublattices, which are usually called A and B, and which sit on an underlying triangular Bravais lattice. Nearest-neighbour hopping only connects A with B, and never A with A or B with B, so by the method of Section 2.7 the Bloch Hamiltonian (2.12) is purely off-diagonal,

$$H(\mathbf{k}) = \begin{pmatrix} 0 & f(\mathbf{k}) \\ f^*(\mathbf{k}) & 0 \end{pmatrix}, \qquad f(\mathbf{k}) = t \sum_{j=1}^{3} e^{i\mathbf{k}\cdot\boldsymbol{\delta}_j}, \tag{2.19}$$

where $\boldsymbol{\delta}_j$, $j = 1, 2, 3$, are the three vectors that connect an A site with its nearest B neighbours. When we diagonalize (2.19) we obtain $E_\pm(\mathbf{k}) = \pm|f(\mathbf{k})|$. The function $f(\mathbf{k})$ vanishes at two inequivalent points of the Brillouin zone, which are usually called K and K', since there the three phases in the sum cancel exactly. So both bands touch at exactly zero energy at K and K'. It is useful to see this cancellation explicitly. Figure 2.5 shows the three phase factors at $\mathbf{K}$ as three unit vectors at 120° from each other, which close a triangle and so sum to zero.

We expand $f(\mathbf{k})$ to linear order around K and write $\mathbf{k} = \mathbf{K}+\mathbf{q}$ for small $\mathbf{q}$. The off-diagonal entry then becomes a linear function of $\mathbf{q}$, with one linear term for its real part and one for its imaginary part, and (2.19) reduces to the standard two-dimensional **Dirac Hamiltonian**,

$$H(\mathbf{q}) \approx \hbar v_F \left(q_x \sigma_x + q_y \sigma_y\right), \qquad v_F = \frac{3ta}{2\hbar}, \tag{2.20}$$

where σ_x, σ_y are the Pauli matrices that act on the A/B sublattice space and a is the nearest-neighbour bond length. When we diagonalize (2.20) we obtain

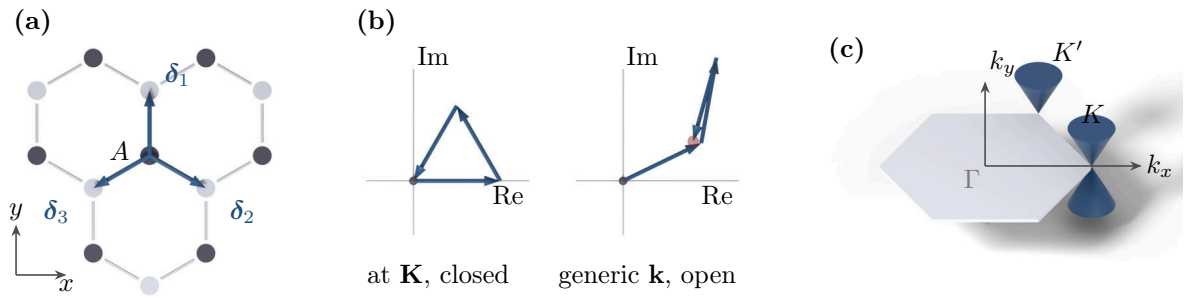


Figure 2.5: The Dirac cones of graphene. **(a)** The honeycomb lattice, with two sublattices A, B per unit cell, drawn as dark and grey discs, and the three nearest-neighbor vectors $\boldsymbol{\delta}_{1,2,3}$ from an A site. **(b)** The reason why the bands touch. At $\mathbf{k} = \mathbf{K}$ the three phase factors $e^{i\mathbf{K}\cdot\boldsymbol{\delta}_j}$ are three unit vectors at 120° from each other in the complex plane. When we place them one after the other they close a triangle, so $f(\mathbf{K}) = 0$ in (2.19) and both bands meet at zero energy. At a generic $\mathbf{k}$ they do not close, the ghost ball marks where the sum ends, and the bands are split. **(c)** The two inequivalent points K and K' of the Brillouin zone with zero energy, at two corners of the hexagonal tile, around which the dispersion is the linear cone $E_\pm = \pm\hbar v_F|\mathbf{q}|$ of (2.21).

$$E_\pm(\mathbf{q}) = \pm\hbar v_F|\mathbf{q}|, \tag{2.21}$$

which is a strictly linear dispersion with the shape of a cone around K, and the same calculation gives the same result around K'. This is what is called the **Dirac cone**. Note that the curvature of the cone along the radial direction vanishes, so (2.18) gives an infinite effective mass in this direction, and at the tip of the cone the effective mass is not defined at all. When we say that the electrons of graphene are "massless", which is the phenomenology for which graphene is famous, we refer to the Dirac mass term, which is the gap, and not to the curvature of the band. This is the low-energy physics of every material with a honeycomb lattice, and of graphene in the first place. There is much more structure at K and K', namely a Berry phase and with it a topological classification, which we leave for the topology lecture right after this one.

Notebook check-in: `session2.ipynb`, "Infinite two-dimensional graphene" (Core)

Build the two-dimensional honeycomb Bloch Hamiltonian (2.19) directly and diagonalize it across the Brillouin zone. Zoom in on K and K' and check numerically that the dispersion there is really linear, which means that it agrees with (2.21) once you are close enough to the Dirac point.

Two standard lattices realize truly flat bands only from their geometry in real space, and not from a fine-tuning of hopping parameters to zero. The **Lieb lattice** is built from a square lattice where one site out of every four is removed in a regular pattern. Its dispersion has two dispersive bands, which come from the bands of the square lattice (2.14), and in addition one exactly flat band at zero energy. Near the corner M of the Brillouin zone the two dispersive bands form a linear, Dirac-like cone, and the flat band goes through the tip of this cone. The **kagome lattice** is built instead from a triangular lattice where one site out of every four is removed. Its dispersion is closer to the one of the honeycomb lattice, with linear crossings similar to Dirac cones, and in addition it has a flat band at the edge of the spectrum. With the sign convention of (2.5) this flat band is at the top, at $\epsilon = 2t$, and it would be at the bottom for the opposite sign of t. In both cases the flat band comes from destructive interference between hopping paths in real space. This is a purely geometric effect and not an effect of interactions, and the physics of strong correlations only enters once electrons are placed into these bands, which are already flat. Figure 2.6 shows both lattices, both dispersions, and the localized state where the hopping paths cancel.

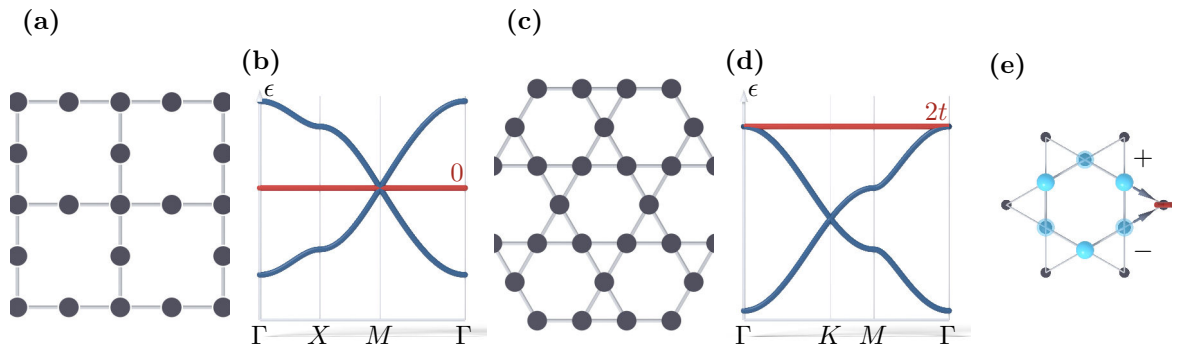


Figure 2.6: Flat bands that come only from geometry, with the bands computed from the Bloch Hamiltonian of each lattice and the flat band drawn as the red tube. **(a)** The Lieb lattice, which is a square lattice with one site in four removed, and **(b)** its bands. There are two dispersive bands and a band that is exactly flat at $\epsilon = 0$, and all three touch at M. **(c)** The kagome lattice, made of triangles that share corners, and **(d)** its bands, with a Dirac-like crossing at K and a flat band at $\epsilon = 2t$. **(e)** The reason why the flat band exists. An eigenstate can be confined to a single kagome hexagon, with amplitudes of alternating sign around it, the filled and the ghost balls. Every path by which it could hop onto a site outside arrives twice with opposite sign, the two arrows, so the two contributions cancel at the cross. Nothing is fine-tuned to zero, since the flatness is a property of the geometry of the lattice, and by (2.16) it makes the density of states diverge.

This material is only on the slides and has no counterpart in `session2.ipynb`. The notebook treats the honeycomb lattice in depth, but it has no section on the Lieb lattice or on the kagome lattice. The band-structure plots for both lattices are shown directly on the slides. The description above gives their qualitative structure, but to see the actual flat-band dispersion one needs the figure itself.

2.12 Measuring band structures with ARPES

Once we have all this formalism, a natural question is whether any of it can be observed. The energy eigenvalue $E(\mathbf{k})$ is clearly physical, but the quasi-momentum $\mathbf{k}$ itself was introduced as a symmetry eigenvalue in Section 2.6. So we ask whether we can measure it directly, and whether we can measure the *relation* between the two.

The answer is yes, with **angle-resolved photoemission spectroscopy (ARPES)**. The idea is simple in principle. We send photons of known energy and momentum onto a material, and each photon can remove an electron from the material completely, which is the photoelectric effect. Conservation of momentum and energy relates the momentum and kinetic energy of the electron, which are both measured directly outside the material, to its momentum and energy when it was still inside. So when we scan the emission angle and the kinetic energy of the emitted electrons, we map $E(\mathbf{k})$ directly. This is how the band structure and the Fermi surface of graphene are measured in practice, and the measured Dirac cone agrees well with the tight-binding derivation of Section 2.11. This material is only on the slides and has no counterpart in the notebook. The actual ARPES images of the measured band structure and Fermi surface of graphene are shown on the slides, and there is no way to reproduce a real photoemission experiment in a tight-binding notebook. The description above gives the physical principle, but to see the measured data itself one needs the figure.

2.13 Closing summary

- **Symmetry as a tool to simplify.** When the ground state shares the symmetry of the Hamiltonian, we can use a symmetry operator that commutes with it, $[H, S] = 0$ in (2.3), and find the eigenbasis of H by finding the eigenbasis of S instead. We worked this out explicitly for rings of three and five sites in Section 2.4.

- **Bloch's theorem** is this same idea applied to translational symmetry on an infinite lattice. Eigenstates have a definite phase φ under translation, which is the quasi-momentum of (2.10), and this turns one infinite-dimensional diagonalization into a continuous family of finite ones, given by the **Bloch Hamiltonian** $H(\varphi)$.

- We derived the band structure of the one-dimensional chain explicitly, $\epsilon_\varphi = -2t\cos\varphi$ in (2.8). We generalized the same method to second-neighbour hopping (2.9), to several orbitals per unit cell with the matrix Bloch Hamiltonian $H(\varphi) = t + \gamma e^{i\varphi} + \gamma^\dagger e^{-i\varphi}$ in (2.12)-(2.13), and to higher spatial dimensions, (2.14) and (2.15).

- A dispersion $E(\mathbf{k})$ directly determines the measurable properties of a material, which are the density of states (2.16), the group velocity (2.17), the effective mass (2.18) and the Fermi surface. Every real dispersion is built locally from three types, which are parabolic, Dirac and flat.

- The **Dirac cone** of the honeycomb lattice of graphene, $H(\mathbf{q}) \approx \hbar v_F(q_x\sigma_x + q_y\sigma_y)$ with $E_\pm = \pm\hbar v_F|\mathbf{q}|$ in (2.20)-(2.21), is the most important example of this session. We derived it directly from the vanishing of the off-diagonal Bloch Hamiltonian (2.19) at the points K, K'. The Lieb and kagome lattices realize flat bands with the same kind of interference in real space, and with no fine-tuning.

- All of this band-structure method generalizes directly to phonons, magnons and Bogoliubov-de Gennes quasiparticles, and not only to electrons, and it is tested against real materials in experiments with ARPES.

The main message of this session is the following. The spectra of periodic systems are computed with band-structure theory, by choosing the common eigenbasis of the symmetry operator and the Hamiltonian, and the resulting dispersion relation $E(\mathbf{k})$ alone determines many of the electronic properties of a material, before interactions are included again. The Dirac cone that we derived in Section 2.11 is not only a fact about graphene. It is the object that the next session takes up, where we ask what happens when it is gapped open, and this is where topological band structure, Chern numbers and the SSH model, which we already saw in the two-orbital chain of (2.13), enter the course in detail. The flat bands are also not the end of the topic. The kagome and Lieb constructions of this session lead directly to the flat bands of the Landau levels, which the quantum Hall session builds several sessions from now.

Session 3

Topological Band Structure Theory

In the last session we ended with the most important feature of the honeycomb lattice. Close to the corners of its Brillouin zone, the tight-binding band structure is not parabolic anymore and becomes linear, which is the Dirac cone. This is one of the three types of dispersion that we chose to organize the rest of the course, which are parabolic, Dirac and flat. We also saw that this is not only a convenient theoretical picture, since ARPES measures this linear dispersion directly in real graphene. The question for this whole session is what happens if we open a gap at this Dirac point. A Dirac cone with a gap turns a metal into an insulator, but this insulator is not always an ordinary one. Depending on *how* the gap opens, the insulator can have a hidden invariant, which is called a topological invariant. This invariant does not appear anywhere in the bulk spectrum, but it forces the material to have conducting states at its edge, and no local disorder can remove these states. To reach this statement in an exact way we need several steps. First, we need linear response theory as a tool. Then we need the Hall conductivity, which is the observable where the invariant first appears, and the Chern number, which is the invariant itself. Finally, we need the general bulk-boundary correspondence, which connects the invariant to the physics of the edge. We will build all of this with the quantum Hall state, which is the simplest example. Then we will use the same tools for a purely one-dimensional model, the SSH chain. At the end, in Section 3.10, we will come back to the Dirac cone of the honeycomb lattice, open a gap in it, and see a real Chern insulator appear without any magnetic field.

3.1 What topological materials are

Before we go into the formalism, it is useful to have a rough classification in mind, so that the abstract tools that we are going to build have concrete targets. Topological materials are systems with an excitation gap in the bulk that at the same time have protected excitations at their surface or edge. This combination of a gapped bulk and a gapless boundary will appear again and again in this session, and it is the defining feature of these materials. The point is not that the edge happens to have low-energy states. The point is that the topology of the bulk *forces* them to be there.

Within this class, we will see the following systems in this session and in the next one.

- The **quantum Hall effect**, which appears in two-dimensional electron gases and in graphene under a strong perpendicular magnetic field. It is the simplest topological state, and it is our starting point.
- The **quantum anomalous Hall effect**, which is the same physics without any external magnetic field, and which we will construct explicitly in Section 3.10.

- **Quantum spin Hall insulators**, which are in practice two copies of a quantum Hall state related by time reversal and put together, as we will see in Section 3.11.
- **Topological superconductors**, whose surface excitations are Majorana modes. They are the basis of one route to topological quantum computing, and we will come back to them in detail in session 5.
- **Topological semimetals**, including Weyl semimetals, whose low-energy quasiparticles are Weyl fermions.
- **Topological Kondo insulators**, which are the example given in the lecture this year of topology that survives in a strongly correlated many-body system, "and many others that do not fit on this slide."

This material appears only on the slides and has no counterpart in the notebook. For each entry, the slides name a specific material realization, which is graphene for the quantum Hall effect, Cr-doped $(Bi,Sb)_2Te_3$ for the quantum anomalous Hall effect, HgTe/CdTe quantum wells for the quantum spin Hall effect, Fe chains on Pb for a topological superconductor, TaAs for a Weyl semimetal, and SmB_6 for the many-body example. These were shown but not explained in detail in the lecture, and the notebook of the course does not simulate any of these specific compounds. The many-body example on the classification slide is the topological Kondo insulator, with SmB_6 as the material realization. What we will simulate, with the models that we build in this session, are minimal tight-binding versions of the first three items of the list above.

3.2 Linear response theory and how a measurement reads the wavefunctions

To see why a bulk band structure can have an invariant at all, we need a tool that connects the wavefunctions of a Hamiltonian to something that we can measure. This tool is linear response theory, and the idea behind it is very general. We perturb the system at a position x and time t, and then we measure an observable, such as the charge, the current, the magnetization or the heat, at another position x' and time t'. Linear response theory tells us that this measured response depends only on the electronic structure of the unperturbed material.

Before we specialize to the Hall conductivity, let us look at two examples. Let us put an impurity in a two-dimensional metal and look at how the electron density, or the local density of states, is modulated around it. Linear response theory shows that this modulation pattern, which is called the Friedel oscillation, directly contains the Fermi wavelength of the electron gas, and this is what scanning tunneling microscopy uses in practice. In a similar way, when we scatter neutrons on a magnetic material, the magnetic moment of the neutron couples to the local moments in the sample. The resulting scattering pattern gives both the magnetic structure and its excitations, in particular magnons. Both examples are illustrations that appear only on the slides, in the sense that the figures of the Friedel oscillation and of neutron scattering were shown on screen without a full explanation. However, the physics of both examples was explained in the lecture, and it is useful to keep it in mind as the general result of the formalism that we build next.

Formally, we start from an unperturbed Hamiltonian H_0 and add a perturbation controlled by a dimensionless parameter λ,

$$H(\lambda) = H_0 + \lambda V,$$

so that $\lambda = 0$ gives the original system and $\lambda = 1$ is the fully perturbed one. Any observable, such as the energy, the density, the magnetic moment, the current, or the wavefunctions themselves, can then be expanded in a Taylor series in λ. The derivative of an observable with respect to λ at $\lambda = 0$ is what we call the response of the system, and in the case of a magnetic perturbation it is called the susceptibility. Since every observable can be built from the wavefunctions, the most useful object to expand is the ground-state wavefunction itself. For a state $|n\rangle$ of the unperturbed Hamiltonian H_0 with energy ε_n, standard first-order perturbation theory gives

$$|n\rangle_\lambda = |n\rangle + \lambda \sum_{m \neq n} \alpha_m |m\rangle + \mathcal{O}(\lambda^2), \qquad \alpha_m = \frac{\langle m|V|n\rangle}{\varepsilon_n - \varepsilon_m}. \tag{3.1}$$

Everything that we do in the rest of this session, which is the Hall conductivity, the Berry curvature and the Chern number, comes directly from putting (3.1) into the definition of some observable.

3.3 Where perturbation theory breaks down

Before we use (3.1), we should say when it fails, because several of the most important phenomena in this course appear in this failure. The expansion (3.1) assumes that we are in the *perturbative regime*, where a small change in the input, which is the strength of the perturbation, gives a small change in the output. This is not always the case. We name three cases now, because we will see each of them in detail later in the course.

- An impurity in a metal gives rise to the **orthogonality catastrophe**. The many-body ground state after the perturbation becomes exactly orthogonal to the original ground state, even for a very small perturbation, so no Taylor expansion of finite order in λ can connect the two.
- A quantum spin, which is a local magnetic moment, coupled to a metal gives rise to the **Kondo effect**. The relevant energy scale, which is the Kondo temperature, is a non-perturbative function of the coupling constant, and no order of perturbation theory in this coupling gives it. We will study the Kondo lattice model in detail in session 8, and here we only mention it as a first sign that ordinary perturbation theory has a strict limit.
- **Superconductivity** is also non-perturbative, since the BCS gap does not appear at any finite order in the interaction strength if we start from the normal state. We will work this out in detail in session 5.

In this lecture, and for every model that we build in this session, we stay inside the regime where (3.1) holds. For now, it is enough to know that this is not always the case.

3.4 The Hall conductivity from linear response

Now we apply (3.1) to the measurement that will bring us to topology, which is the transverse conductivity, also called the Hall conductivity. We take a two-dimensional material, apply a voltage in the y direction, and ask what current flows in the x direction. In momentum space, a voltage in y enters as a perturbation proportional to $\partial H/\partial k_y$, since it shifts k_y through the Peierls substitution. The current operator in the x direction is, up to the electron charge, the group-velocity operator $\partial H/\partial k_x$. Both are the kind of matrix element that appears in (3.1).

When we put (3.1) into the definition of the Hall conductivity, we obtain an expression with two velocity operators and an energy denominator, which is the standard Kubo formula for a filled band n and empty bands m,

$$\sigma_{xy} \propto \sum_{n \text{ filled}} \sum_{m \text{ empty}} \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \, \frac{\langle n|\partial_{k_x} H|m\rangle\langle m|\partial_{k_y} H|n\rangle - (x \leftrightarrow y)}{(\varepsilon_n - \varepsilon_m)^2}. \tag{3.2}$$

At this point the expression still depends explicitly on the band energies through the denominator, so it seems that it should depend on all the details of the band structure. This is not the case, and the next section shows why.

3.5 The Berry curvature, and why the Hall conductivity is an integer

We take the derivative of the Schrödinger equation $H(k)|u_n(k)\rangle = \varepsilon_n(k)|u_n(k)\rangle$ with respect to k_μ and project it onto a different band $\langle u_m(k)|$, with $m \neq n$, which gives

$$\langle u_m|\partial_{k_\mu} H|u_n\rangle = \big(\varepsilon_n(k) - \varepsilon_m(k)\big)\, \langle u_m|\partial_{k_\mu} u_n\rangle. \tag{3.3}$$

This identity is the piece that we need, since it replaces a matrix element of $\partial_k H$ by a matrix element of the k-derivative of the wavefunction, divided by the energy difference. When we use (3.3) to rewrite the numerator and the denominator of (3.2) together, the explicit dependence on the energies cancels completely. What remains is an object built only from overlaps of wavefunctions, which is called the **Berry curvature** of band n,

$$\Omega_n(k) = -2\,\mathrm{Im}\,\langle \partial_{k_x} u_n(k)|\partial_{k_y} u_n(k)\rangle,$$

so that the Hall conductivity of a set of filled bands is

$$\sigma_{xy} = \frac{e^2}{h}\, C, \qquad C = \sum_{n \text{ filled}} \frac{1}{2\pi} \int_{\text{BZ}} \Omega_n(k)\, d^2k. \tag{3.4}$$

This is the central result of the session, and there are two important points about (3.4). First, σ_{xy} depends only on how the wavefunctions change across the Brillouin zone, and not on the energies. We can deform the band structure in any way, and as long as the wavefunctions do not change, the Hall conductivity stays the same. Second, and more surprising, for a gapped system, which is an insulator, C is always an integer, regardless of the number of bands and of the shape of the dispersion. This is a topological statement. The Brillouin zone of a two-dimensional system is a compact manifold, which is a torus since opposite edges of the zone are identified, and the integral of a curvature over a closed surface always gives an integer. The same happens for the Gaussian curvature of an ordinary closed surface in differential geometry. This integer C is the **Chern number**, and (3.4) is the first place in the course where a measurable quantity, the Hall conductivity, is fixed by a topological invariant.

3.6 Landau levels and chiral edge states in the quantum Hall state

The Chern number appears in its simplest form in the quantum Hall state itself, which is a two-dimensional electron gas in a strong perpendicular magnetic field. The resulting band structure is the flat-band type of dispersion from session 2, which is a set of Landau levels without dispersion, and each Landau level has Chern number +1. If we fill the first Landau

level, the total Chern number is $C = 1$. If we fill the first two, we get $C = 2$, and so on, so that $\sigma_{xy} = C\,e^2/h$ goes through successive integer plateaus as the chemical potential goes up through successive Landau levels. This is the integer quantum Hall effect, and in session 4 we will work out its full phenomenology, including the fractional case. In this session we only need to count the Chern numbers.

This raises a puzzle. We have an insulator, with a gap between Landau levels, but σ_{xy} is a *conductivity*, which means that some electron has to move somewhere. The answer is that any real sample has an edge, and at this edge the bulk gap closes. Gapless edge states appear, and they carry the current that (3.4) predicts. These edge states have another key property, which is that on a given edge the states propagate in only *one* direction. The top edge of a sample carries current in one direction only, and the bottom edge carries it in the opposite direction only. A mode that propagates in a single direction is called **chiral**, and chirality is what protects the quantum Hall state against disorder. An electron in a chiral edge mode cannot backscatter, since there is no state that propagates in the opposite direction at the same edge and the same energy. Without a backscattering channel there is no resistance in this channel. Figure 3.1 shows both parts of this picture, which are the closed bulk orbits that carry no current and the skipping orbits that carry all of it, and it also shows what happens when an impurity is put in their way.

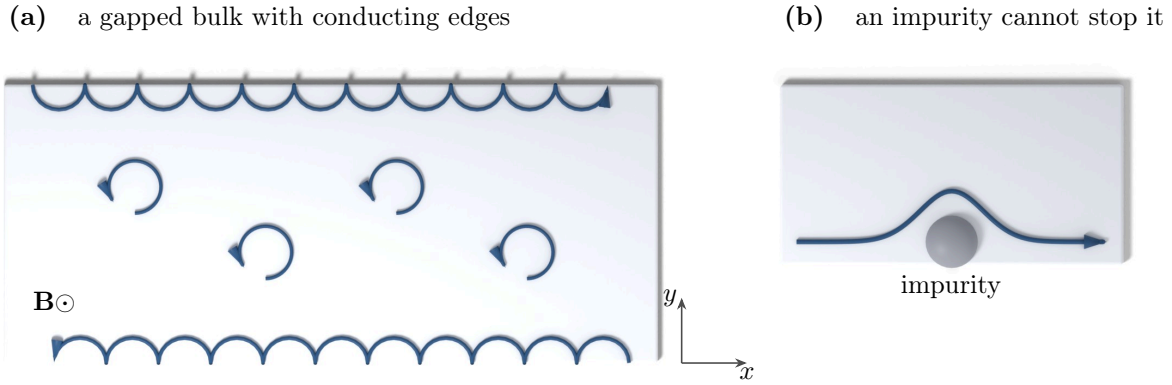


Figure 3.1: Chiral edge states, and why they cannot be backscattered, seen from above with B out of the page. (a) A quantum Hall sample. In the bulk the electrons move on closed cyclotron orbits and do not go anywhere, and the system is an insulator with a gap between Landau levels. At the boundary these orbits cannot close. They skip along the edge, and for a given direction of B they skip in one direction only, which is opposite on the two edges. Propagation in a single direction is what the word **chiral** means, and it carries the current that $\sigma_{xy} = C\,e^2/h$ predicts, even though the bulk is gapped. (b) The consequence. An impurity at the edge can deflect the mode, but there is no state that propagates in the opposite direction at the same edge and the same energy for it to scatter *into*, so the mode goes around the obstacle and continues. Without a backscattering channel there is no resistance, which is why the Hall plateaus are quantized so sharply. If we reverse the sign of B, all the arrows are reversed at the same time.

Let us look at a concrete model. We take a tight-binding square lattice with a perpendicular magnetic field and cut it into a ribbon of finite width, which has translational invariance along the ribbon but is finite across it. When we diagonalize this ribbon, we find the structure described above. There are flat bulk bands similar to Landau levels, and there are dispersive edge branches that cross the gap, with one branch confined to each edge and propagating in only one direction. If we look at the low-energy region close to zero energy, the spacing between the flat bands approaches the equal spacing of a continuum two-dimensional electron gas in a magnetic field, so the tight-binding model reproduces the standard Landau problem in the appropriate limit. This computed band structure, with the bulk flat bands in green and the two edge branches that propagate in opposite directions, is shown on the slides but was

only described qualitatively in the lecture. To reproduce it numerically is what the notebook box below asks you to do.

Notebook check-in: `session3.ipynb`, "From electrons in a ribbon to the quantum Hall state" (Core), "The quantum Hall effect in graphene" (Core), and "Localization of the edge states in the quantum Hall effect" (Challenge)

Build a square-lattice ribbon, and then a honeycomb-lattice ribbon, with a magnetic field and reproduce the figure described above, with flat Landau-level bulk bands and chiral edge branches. Check the propagation direction of the edge states at positive and at negative energy on each edge, which is the chirality argument of Section 3.6. Then look at how the edge states localize in real space as a function of energy, and how this localization changes for a weaker field or a narrower ribbon.

3.7 What topology means for a Hamiltonian, and the bulk-boundary correspondence

We have now seen one topological invariant at work. Let us now step back and say what the word topological means, in the mathematical sense from which the word is taken. Topology classifies objects by whether they can be smoothly deformed into each other without a drastic operation, such as cutting, tearing or gluing. Closed surfaces in three dimensions are classified by their number of holes. A donut and a coffee cup both have one hole and can be continuously deformed into each other, while a sphere has no hole and cannot be deformed into either of them. Closed curves embedded in three dimensions have a similar classification by knots, since curves with different knots cannot be smoothly untangled into each other.

For a Hamiltonian, the equivalent of a deformation is to change the band structure continuously while the bulk gap stays open. Two Hamiltonians related by such a deformation have the same Chern number, because C in (3.4) is quantized and so it cannot change continuously. It can only jump, and a jump requires the gap to close at some point of the deformation. So the Chern number plays the role that the number of holes plays for a surface. Hamiltonians with the same Chern number are topologically equivalent, meaning that they can be deformed into each other without closing the gap, and Hamiltonians with different Chern numbers are not.

This has a direct and strong consequence. Let us build a Hamiltonian that changes in space and is equal to H_1, with Chern number C_1, far to the left, and to H_2, with Chern number $C_2 \neq C_1$, far to the right. Since H_1 and H_2 cannot be deformed into each other without closing the gap, and the Hamiltonian in between goes continuously from one to the other, the gap *must* close somewhere in between, for any microscopic construction of the interface. An interface between two bulk phases with different Chern numbers always has a gapless state, and the existence of this gapless state depends only on C_1 and C_2, and not on any detail of the interface. This is called the **bulk-boundary correspondence**. The vacuum has $C = 0$, since it is topologically trivial, so a quantum Hall state with $C = 1$ always has a gapless edge state at every boundary with the vacuum. This is the edge physics of Section 3.6, now obtained from a completely general argument. Figure 3.2 shows the three steps of this argument in order.

The same argument explains the robustness that we used before. Let us put a local impurity at the edge of a quantum Hall sample. This only changes the microscopic details of the interface between the vacuum and the quantum Hall state. Since the topological excitation of the interface must exist for any such details, the edge state reorganizes itself around the impurity and is not scattered by it. This is a general mechanism, which is not special to the quantum Hall case, and it will appear again for every topological invariant that we see from now on.

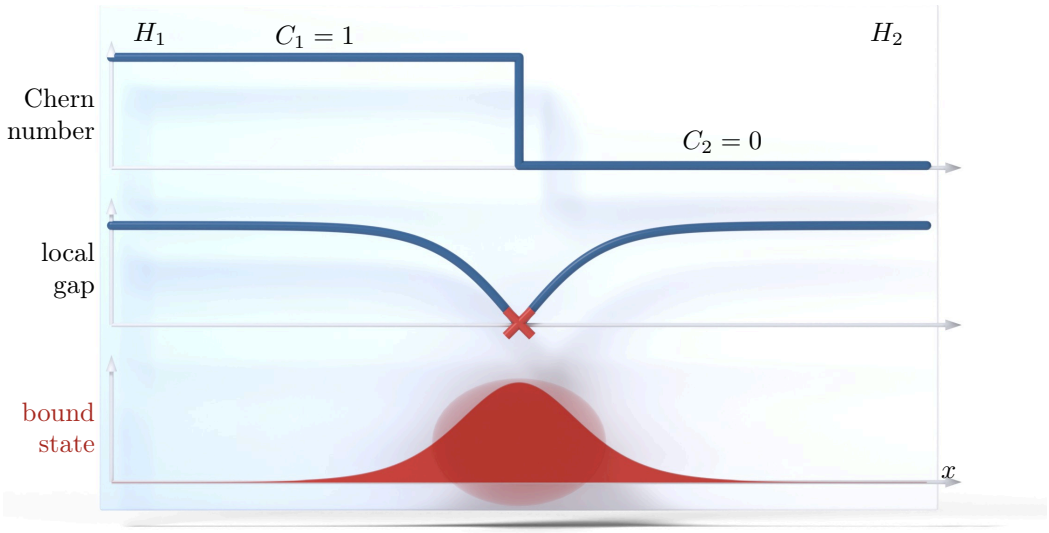


Figure 3.2: The bulk-boundary correspondence, as an argument about an integer. A Hamiltonian changes slowly in space from H_1 on the left to H_2 on the right, which is the color of the backdrop. In the top row, the Chern number is an integer, so it cannot change continuously from $C_1 = 1$ to $C_2 = 0$ and it can only jump. In the middle row, a jump is only allowed where the quantization argument itself fails, which is where the gap closes, so the local gap must go to zero somewhere in the interface region, at the cross. This is drawn schematically, in the local-gap approximation. In the bottom row, the state that appears there is a gapless state bound to the interface. This argument does not use any detail of how the interface is built, and only C_1 and C_2 enter, which is why the edge state survives rough boundaries and impurities. If we take H_2 to be the vacuum, $C_2 = 0$, we obtain the chiral edge state of figure 3.1 from a completely general argument.

3.8 Three topological material classes

With the general tools in place, let us name three classes of topological matter, before we build explicit models for two of them.

- **Chern insulators.** They have a bulk Chern number $C \neq 0$ and chiral edge states, and time-reversal symmetry is necessarily broken, since we will see in Section 3.10 that time-reversal symmetry forces $\sigma_{xy} = 0$. Without a backscattering channel there are no electrical losses at the edge, which makes them an attractive platform for dissipationless electronics.

- **Quantum spin Hall insulators.** They are two copies of a Chern insulator, one for each spin, with opposite Chern numbers, so the net Chern number vanishes and time-reversal symmetry is preserved. However, the edge states are **helical**, meaning that spin-up and spin-down propagate in opposite directions along the same edge. This is the natural platform for dissipationless spin transport, which is spintronics.

- **Topological superconductors.** They combine a bulk superconducting gap with gapless surface excitations that are their own antiparticle, which are called Majorana excitations. Since the corresponding quantum information is stored non-locally, the usual local decoherence channels do not act on it directly, and this is the basis of topological quantum computing. We will build the minimal model for this case, the Kitaev chain, in session 5, and we will see there that it is exactly the same model as the one that we construct next.

3.9 The SSH model as a minimal one-dimensional topological insulator

We now build the simplest model that shows this physics explicitly, which is the Su-Schrieffer-Heeger (SSH) chain. It looks too simple to be useful, but we should take it seriously, also because the simplest model for a topological superconductor, the Kitaev chain, is exactly this model written in terms of Majorana operators.

We take a one-dimensional lattice with two sites per unit cell, A and B, and two hopping amplitudes. The first one is an intracell hopping t, within a unit cell, and the second one is an intercell hopping fixed to 1, between neighboring unit cells. Let us first take the extreme limit in which the intracell hopping dominates, $t \gg 1$, so that the chain is fully dimerized with alternating bonds. Strong bonds of strength t couple the sites $(0,1)$, $(2,3)$, $(4,5)$, ... and the coupling between $(1,2)$, $(3,4)$, ... is negligible. Each dimer is then an isolated two-level system with bonding and antibonding eigenstates at energies $\pm t$. The spectrum of the whole chain is this pair of levels repeated once for each dimer, and there is nothing at zero energy.

Now let us take the opposite limit, $t \to 0$, in which the strong bonds are the intercell ones and the dimer pattern is shifted by one site. We leave the first site uncoupled, so that it is a dangling site, we couple $(1,2)$, $(3,4)$, ..., and we also leave the last site dangling. In the bulk this looks the same as the previous chain, since it has the same alternating dimer pattern shifted by one site, but the two ends are qualitatively different, because each end now has a site that is not coupled at all. When we diagonalize this chain, we obtain the same bonding and antibonding pairs at ± 1 in the bulk, and also **two states pinned exactly at zero energy**, one localized at each end. Figure 3.3 shows the two chains side by side together with their spectra.

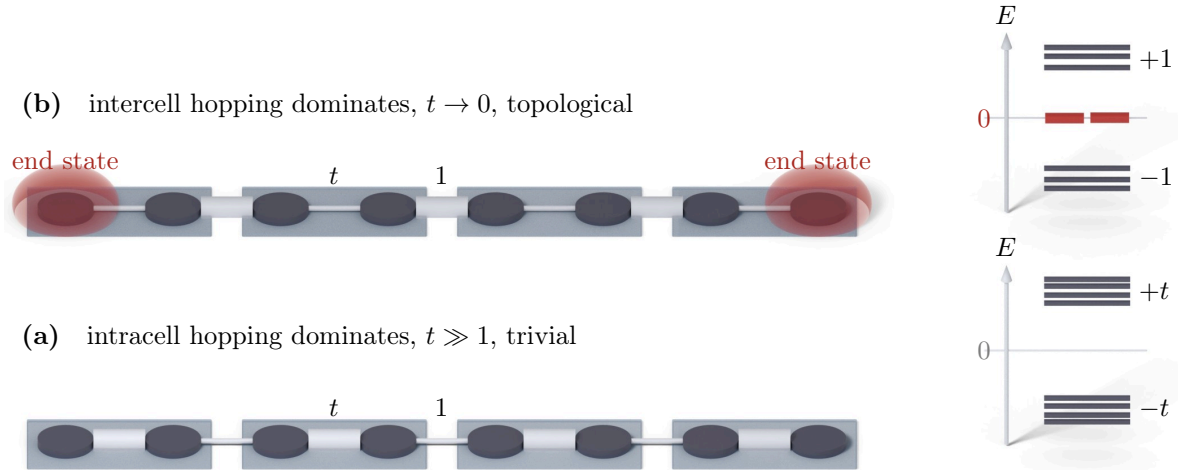


Figure 3.3: The Su-Schrieffer-Heeger chain, drawn with the same eight sites and the same unit cells in both rows, which are the translucent slabs, so that the only difference is which of the two bonds is the strong one, drawn as the thick rod. Next to each chain is its spectrum, the eigenvalues of the open chain in units of the strong hopping. (a) In front, the intracell hopping dominates, $t \gg 1$ (here $t = 4$). Every site is paired into a dimer, and the levels form two groups around $\pm t$, spread a little by the weak hopping, with a clear gap around zero, which is the trivial phase. (b) Behind, the intercell hopping dominates, $t \to 0$ (here $t = 1/4$). The dimer pattern is shifted by one site and the first and last sites, the two glowing ones, are left uncoupled. The same bulk spectrum now has two extra states at $E = 0$, the two short rods in the gap, one localized at each end, which is the topological phase. The two chains cannot be distinguished in the bulk, and the difference is only at the ends, which is what a bulk invariant has to predict. At small but finite t the two end states hybridize through the chain, and their splitting away from zero is only exponential in the chain length. For $t = 1/4$ and eight sites the two levels sit at ± 0.004, which is too small to see at this scale.

This is the minimal example of a topological phase. The fully dimerized chain of the first case is the **trivial phase**, which is gapped everywhere, including at the ends. The shifted chain of the second case is the **topological phase**, which is gapped in the bulk but has gapless modes at exactly zero energy at the edges. This construction looks very fine-tuned, and it is, since real chains have hopping between every pair of neighboring sites. The important part is what survives when we remove the fine-tuning. If we turn on a small coupling t between the sites at the two ends that were decoupled, the zero modes are *not* removed, as long as t stays below the bulk hopping strength. Intuitively, for $t \ll 1$ we weakly couple an isolated zero-energy level to a gapped bulk, and a weak coupling cannot push a level all the way across a finite gap. In a more exact way, this is the deformation argument of Section 3.7. The Hamiltonians at different small t are topologically equivalent, since they are connected without closing the bulk gap, so the invariant that protects the zero mode at $t = 0$ also protects it when the coupling is finite.

We can make this fully precise with the Bloch Hamiltonian. With intracell hopping t and intercell hopping normalized to 1, the Hamiltonian in momentum space in the (A, B) basis is the 2×2 matrix

$$H(k) = \begin{pmatrix} 0 & d(k) \\ d^*(k) & 0 \end{pmatrix} = (t + \cos k)\,\sigma_x - \sin k\,\sigma_y, \qquad d(k) = t + e^{ik}, \tag{3.5}$$

with eigenvalues $E_\pm(k) = \pm|d(k)| = \pm\sqrt{1 + t^2 + 2t\cos k}$. The gap closes exactly when $d(k) = 0$, which requires $t = 1$ and $k = \pi$. This is the topological phase transition point, and it agrees with the qualitative separation between $t < 1$ and $t > 1$ above, since at $t = 0$ the chain is in the limit where the end sites are fully dangling, which is topological, and at $t = 1$ it becomes the fully symmetric gapless chain.

The topological invariant here is the one-dimensional Berry phase that the lower-band Bloch state $|u_-(k)\rangle$ accumulates as k goes once around the Brillouin zone, which is called the **Zak phase**,

$$\gamma = i \oint_{\mathrm{BZ}} \langle u_-(k) | \partial_k u_-(k) \rangle \, dk \pmod{2\pi}. \tag{3.6}$$

Geometrically, γ/π counts how many times $d(k)$ winds around the origin as k goes from 0 to 2π. From (3.5), $d(k) = t + e^{ik}$ draws a unit circle centered at $(t, 0)$ in the complex plane. This circle encloses the origin exactly when $|t| < 1$, and it does not enclose it when $|t| > 1$. This agrees with everything above, since $\gamma = \pm\pi$ for $t < 1$, which means one winding, a topological insulator and gapless edge modes, and $\gamma = 0$ for $t > 1$, which means no winding, a trivial insulator and no edge modes. The value of the Zak phase is in exact one-to-one correspondence with the presence or absence of a zero-energy edge state, and this is the one-dimensional version of the bulk-boundary correspondence of Section 3.7. Figure 3.4 shows this winding together with the bands that it controls, and it shows that the two insulating phases cannot be distinguished from their spectra alone.

This material appears only in the lecture and on the slides, and it has no counterpart in the notebook. The notebook `session3.ipynb` does not have a dedicated section on the SSH chain, and its numerical exercises start directly from two-dimensional ribbons, which are the quantum Hall and quantum anomalous Hall constructions of Sections 3.6 and 3.10. If you want to see the zero modes and the Zak-phase winding of (3.6) numerically, the direct way is to build by hand the finite chain with N sites from Section 3.9, following the same tight-binding diagonalization method that we used for the finite islands in session 1. Then you can see how the two eigenvalues close to zero move away from zero as t is increased above 1.

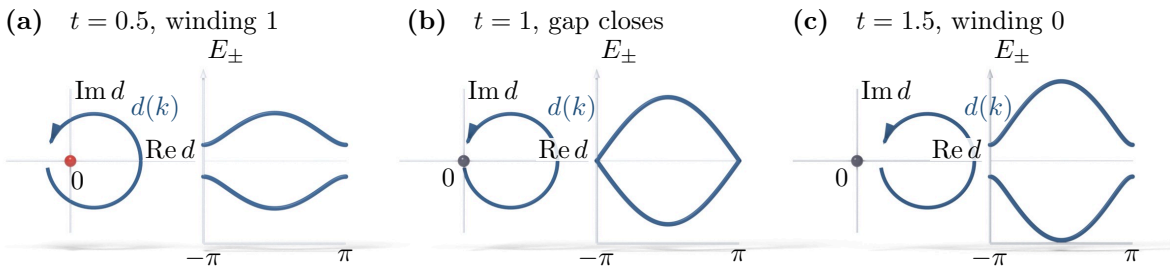


Figure 3.4: A drawing of the one-dimensional invariant. On the left of each panel is the off-diagonal element $d(k) = t + e^{ik}$ of the SSH Bloch Hamiltonian (3.5) in the complex plane, as k goes once through the Brillouin zone, and the arrow shows the direction of increasing k. It is always the unit circle, but centred at $(t, 0)$, so it encloses the origin, the ball, for $t < 1$ and does not enclose it for $t > 1$. The winding number of this curve around the origin is exactly γ/π, which is the Zak phase (3.6) in units of π. On the right of each panel are the corresponding bands $E_\pm(k) = \pm|d(k)|$, with gap $2|1 - t|$. The two halves give the same information in two ways. The only way to unwind the curve is to move it across the origin, and $d(k) = 0$ is where the gap closes, so the invariant cannot change while the system stays insulating. Note that (a) and (c) have the same gap and look like ordinary spectra. The difference between them cannot be seen in the energies and is only in the wavefunctions, which is the main point of (3.4). By the bulk-boundary correspondence, $\gamma = \pi$ is also exactly the condition for the zero-energy end modes of figure 3.3.

3.10 Chern insulators without Landau levels from a gapped Dirac cone

Now we come back to the question from the beginning of the session. A Chern insulator needs a broken time-reversal symmetry. In fact, one can show directly from (3.4) that $\sigma_{xy} \equiv 0$ whenever time-reversal symmetry holds, since time reversal changes the sign of the Berry curvature. However, nothing in the derivation of the Chern number required a magnetic field or Landau levels. So the question is if we can obtain a finite Chern number from an ordinary lattice band structure, using only terms that break time-reversal symmetry and that are entirely inside the lattice itself.

The method is to look for a lattice that has a **massive Dirac equation** in its low-energy expansion. Several two-dimensional lattices have massless Dirac points in their band structure, such as the honeycomb lattice, the kagome lattice and a square lattice with π-flux, and this is the same Dirac cone that we derived for graphene in session 2. Close to one such point, the low-energy Hamiltonian has the form

$$H(k) = k_x \, \sigma_x + \lambda \, k_y \, \sigma_y + m \, \sigma_z, \tag{3.7}$$

with $\lambda = \pm 1$ fixed by the chirality of the lattice at this Dirac point, and m a mass term that opens a gap, $E_\pm(k) = \pm\sqrt{k_x^2 + k_y^2 + m^2}$. We compute the Berry curvature of the lower band and integrate it, with the same procedure as in (3.4) but now over the small region of momentum space where the curvature is concentrated. This gives the Chern number of this single gapped Dirac cone,

$$C_{\text{Dirac}} = \frac{1}{2} \operatorname{sgn}(m) \operatorname{sgn}(\lambda). \tag{3.8}$$

The half-integer in (3.8) is not an inconsistency. A real lattice always has an even number of Dirac points related by time-reversal symmetry, and the honeycomb lattice has two, which are usually called K and K'. Only the *sum* of their contributions needs to be an integer.

This is where the choice of the mass term matters, and we will work out the two natural choices explicitly, because they give qualitatively different physics. At K and K' the chirality

λ has opposite sign, since the two points are time-reversal partners. Let us first open the gap with a staggered sublattice potential, which raises the onsite energy on one honeycomb sublattice and lowers it on the other. This mass term is an ordinary scalar potential, so it has the *same* sign m at both K and K'. From (3.8), the two Dirac points then give Chern numbers of opposite sign, $+\frac{1}{2}\,\mathrm{sgn}(m)$ and $-\frac{1}{2}\,\mathrm{sgn}(m)$, which cancel. The result is $C = 0$, which is an ordinary insulator that is topologically trivial, and this is the same kind of gap as in boron nitride. Now let us instead open the gap with a purely imaginary second-neighbor hopping, which is called the **Haldane coupling**. Physically, this term is a pattern of local orbital currents that circulate inside each hexagon, and it is odd under time reversal. This mass term changes sign between K and K' *relative to* λ, so the two Dirac points now give Chern numbers of the *same* sign, which add and do not cancel. The result is $C = \pm 1$, a real Chern insulator, built only from lattice hoppings and without any external magnetic field. This was the first model ever proposed with a finite and quantized Hall conductivity and zero net flux, and it is the mechanism to open a gap in the Dirac cone of session 2 that we were looking for. Figure 3.5 shows the two cases side by side, each on the lattice and in the spectrum of a ribbon, where only the Haldane case has edge states that cross the gap. A gap can also be opened by combining an ordinary magnetization with spin-orbit coupling, which gives the same sign pattern as the Haldane term and an equivalent quantum anomalous Hall state. Both routes have material realizations. Physics similar to the Haldane model has been realized in twisted graphene bilayers on boron nitride, and the route with magnetism and spin-orbit coupling has been realized and measured in Cr-doped $(\mathrm{Bi,Sb})_2\mathrm{Te}_3$.

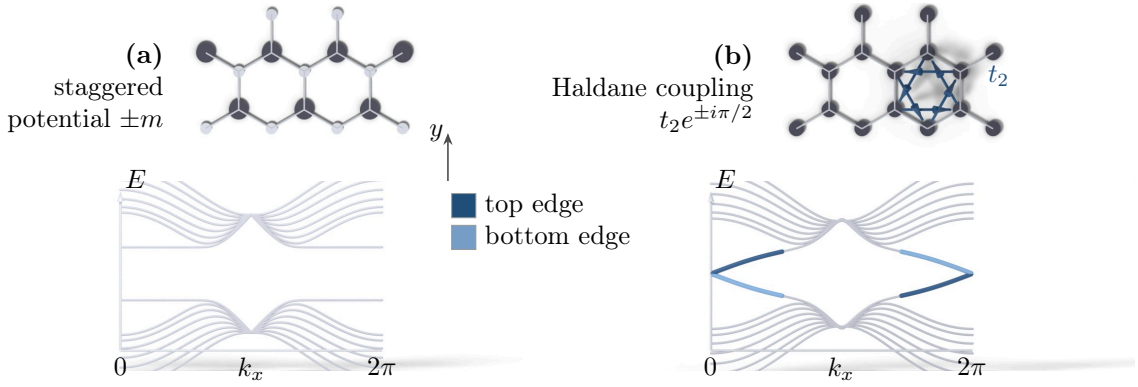


Figure 3.5: Two ways to gap the Dirac cones of the honeycomb lattice, on the lattice and in the spectrum of a zigzag ribbon. Equation (3.8) gives each gapped Dirac point a Chern number $\frac{1}{2}\,\mathrm{sgn}(m)\,\mathrm{sgn}(\lambda)$, and the two Dirac points of the honeycomb lattice are time-reversal partners with opposite chirality, $\lambda = +1$ at K and $\lambda = -1$ at K'. So the result depends only on how the mass term itself behaves under time reversal. (a) A staggered sublattice potential, $+m$ on the large dark sites and $-m$ on the small light ones, is an ordinary scalar potential, which is even under time reversal, so m has the same sign at both valleys. The two half-integers are then opposite and they cancel, which gives an ordinary insulator with the same kind of gap as boron nitride, and the ribbon spectrum below it has a gap with nothing in it. (b) The Haldane coupling, the second-neighbor hoppings drawn as arrows around one hexagon, is a pattern of circulating orbital currents, which is odd under time reversal, so m changes sign between the valleys. The two half-integers are then equal and they add, which gives $C = \pm 1$. This is a real Chern insulator with quantized σ_{xy} and chiral edge states, built only from lattice hoppings, with no external magnetic field and zero net flux, and in the ribbon spectrum one branch from each edge, the dark and the light tube, crosses the gap. The two gaps look the same in the bulk spectrum, and the difference appears only in the wavefunctions. The spectra are computed for ribbons 16 cells wide, with $m = 0.5$ and $t_2 = 0.1$ in units of the first-neighbor hopping.

Notebook check-in: `session3.ipynb`, "The quantum anomalous Hall effect" (Core), "Edge states of the quantum anomalous Hall state" (Core), "Topological states and topological invariants" (Challenge), "Topological phase transitions in the bulk" (Challenge), and "Topological phase transitions in the edge" (Core)

Build the honeycomb ribbon with the Haldane coupling and without magnetic field, and check that the same chiral edge states of Section 3.6 appear, as the argument above predicts. Then, in the exercise on the Berry curvature, integrate the curvature separately for the Haldane coupling and for a sublattice imbalance. You should find a non-zero net Chern number for the Haldane case and a vanishing one for the sublattice imbalance, which is the difference between cancellation and addition that we worked out around (3.8). If you combine both terms and follow where the bulk gap, and then the edge states, close as their relative strength changes, you obtain the numerical version of the phase transition at $t = 1$ of Section 3.9, now in two dimensions. The figures of the honeycomb Berry curvature and of the edge dispersion on the slides show this same computation, and the notebook is where you reproduce it yourself.

3.11 Quantum spin Hall insulators

A Chern insulator needs broken time-reversal symmetry. The quantum spin Hall insulator is what we obtain if we keep time-reversal symmetry and still want a topological gapped state. We take two copies of the massive Dirac construction (3.7), one for each spin, with **opposite** mass for opposite spins, $m_\uparrow = -m_\downarrow$. From (3.8), the two spin sectors then have opposite Chern numbers,

$$C_\uparrow = \pm 1, \qquad C_\downarrow = -C_\uparrow, \qquad C = C_\uparrow + C_\downarrow = 0, \qquad C_s \equiv C_\uparrow - C_\downarrow = \pm 2. \tag{3.9}$$

Here $C_\uparrow$ is the Chern number of the whole spin-up sector of the lattice. Each Dirac point gives $\frac{1}{2}$ by (3.8), and the two Dirac points add for each spin, as in the Haldane case of Section 3.10, so one spin sector has $C_\uparrow = \pm 1$.

The total Chern number C vanishes, in agreement with the unbroken time-reversal symmetry, but the **spin** Chern number C_s does not vanish, and it is C_s that controls the edge physics. At each edge, spin-up propagates in one direction and spin-down propagates in the other direction, which is a **helical** edge state and not a chiral one.

The natural mechanism that gives this opposite-sign mass is spin-orbit coupling. The relativistic spin-orbit term has the structure $\mathbf{L} \cdot \mathbf{S}$, and its z-component, $L_z S_z$, acts like an effective magnetic field that points in one direction for spin-up electrons and in the opposite direction for spin-down electrons. This is the opposite-sign mass structure that (3.9) needs, and it appears with no net magnetic field and no net breaking of time-reversal symmetry. Graphene itself, when we include its intrinsic spin-orbit coupling, which is very small, is technically a quantum spin Hall insulator through this mechanism. The effect is much larger, and accessible in experiments, in monolayer 1T′-WTe_2, where the quantum spin Hall effect survives up to 100 K.

There is a subtle point that we should state precisely. If spin were perfectly conserved, we could speak separately of the Chern number for spin-up and the Chern number for spin-down. Real materials have terms that mix spin, and spin-orbit coupling itself mixes spin to some degree, so one may worry that these terms could smoothly remove the protection of the helical edge states. They cannot, as long as time-reversal symmetry survives, and the reason is Kramers' theorem. For spin-1/2 electrons the time-reversal operator squares to -1, which forces an exact twofold degeneracy at every momentum that is its own time-reversal partner. These are the time-reversal-invariant momenta, which are Γ and X in this setup. This degeneracy exists with or without spin-mixing terms, so the crossing of the edge states at these momenta cannot be gapped by any perturbation that respects time-reversal

symmetry, but only by one that breaks it. This protection mechanism is different from, and stronger than, what the picture of two independent Chern insulators would suggest by itself. Figure 3.6 compares the chiral and helical edge dispersions and shows what a term that breaks time-reversal symmetry does to the protected crossing.

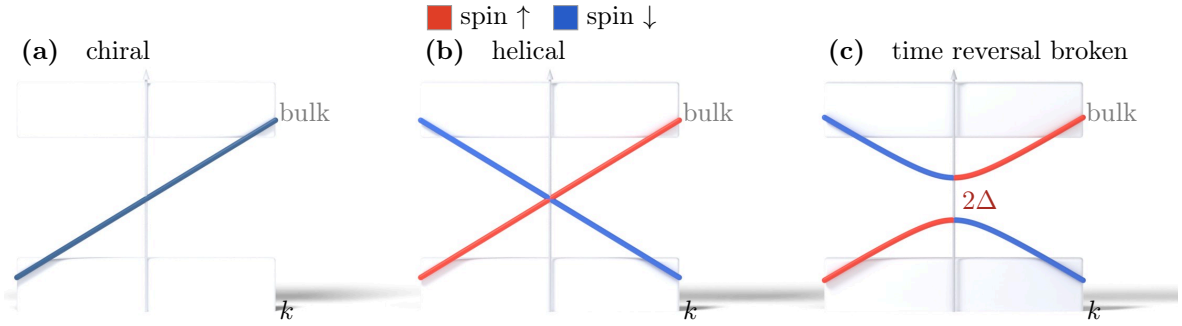


Figure 3.6: Schematic edge dispersions for the two topological classes of Section 3.8. The grey slabs are the bulk bands, and the branches that cross the bulk gap are the edge states, plotted as a function of the momentum along the edge. (a) A Chern insulator has a single branch per edge. At any energy in the gap there is one state and it moves in one direction, so a scatterer has no state to send it back into. (b) A quantum spin Hall insulator has two branches, one for each spin, drawn in a warm and a cold color, which propagate in opposite directions on the *same* edge, and this is what **helical** means. The total Chern number vanishes, but the spin Chern number does not (3.9). (c) The crossing is not protected because the two spin sectors are separately conserved. It is protected by Kramers' theorem, since the crossing is at a time-reversal-invariant momentum, where a twofold degeneracy is guaranteed for spin-1/2 electrons for any perturbation that respects time reversal. Disorder that preserves time reversal can therefore deform the branches but cannot gap them. Only a perturbation that breaks time reversal opens the gap Δ in (c), and then the protection is lost.

Disorder makes this difference concrete. Helical edge states survive disorder that preserves time-reversal symmetry, since the band structure changes but the gapless crossing at the edge remains. On the other hand, disorder that breaks time-reversal symmetry opens a real gap in the edge spectrum and removes the protection completely.

Notebook check-in: `session3.ipynb`, "The quantum spin Hall state" (Core), "The robustness of the quantum spin Hall state" (Core), "Topological invariant of the quantum spin Hall state", and "Finite size effects in topological systems" (Challenge, using exchange coupling plus Rashba spin-orbit coupling)

Build the model with two copies of a Chern insulator and identify the helical pair of edge states by their spin projection and their propagation direction, which is the pattern in (3.9). Then add disorder that preserves or breaks time-reversal symmetry, and check that the gap only opens in the second case. The note "in the lab" in this notebook section points to the measurement in 1T′-WTe_2 that we mentioned above.

3.12 Topology beyond electrons

We built all the tools of this session, which are the Chern number, the Zak phase and the bulk-boundary correspondence, around electronic band structures. However, none of the underlying mathematics requires electrons. The same invariants apply to any quantum excitation with a well-defined dispersion relation. Examples are **magnons**, which are the spin-wave excitations of an ordered magnet, and **spinons**, which are the fractionalized spin excitations of a quantum magnet. Both allow exactly the same treatment with Berry curvature and Chern number, and we will do it explicitly for magnon bands in session 6 and for spinon bands in session 8. More generally, the argument only needed a Hamiltonian, or more generally a matrix that describes

how a wave propagates, so it applies to any system with a dispersion relation. This includes photonic systems, where light propagates through a periodic medium, and even purely classical mechanical systems, such as coupled pendula, vibrating lattices and electronic circuits, which have all been shown in tabletop experiments to have topologically protected modes. In the end, topology is a statement about the internal structure of the eigenvectors as a function of momentum, and this structure exists wherever there is a wave equation, quantum or classical.

The main message of this session is the following. Topological systems have protected edge modes, and these modes are connected to a non-trivial value of a topological invariant that is computed only from the bulk. This happens in one and in two dimensions, and also in higher dimensions, and it is usually protected by a specific symmetry. For the Chern number this is translational symmetry and gauge invariance, and for the quantum spin Hall state it is time-reversal symmetry.

3.13 Closing summary

- The Hall conductivity, computed with linear response theory from the wavefunction expansion (3.1), becomes an integral of the Berry curvature over the Brillouin zone. For any gapped system it is quantized to an integer, the **Chern number**, so that $\sigma_{xy} = C\, e^2/h$ (3.4).

- The quantum Hall state realizes this with flat Landau-level bands, and each of them gives Chern number 1. The bulk gap forces the current onto **chiral** edge states, with one direction per edge, which is why they cannot backscatter and why the plateaus are quantized so sharply, as we saw in Section 3.6.

- **Topology** classifies Hamiltonians by whether their band structures can be continuously deformed into each other without closing the gap. The Chern number is the invariant of this classification, and an interface between regions with different Chern numbers must have a gapless state, which is the general **bulk-boundary correspondence** of Section 3.7.

- The **SSH model** is the minimal one-dimensional realization. It has a 2×2 Bloch Hamiltonian (3.5) whose lower-band Zak phase (3.6) is 0 in the trivial phase or π in the topological phase, depending on whether the intracell hopping is larger or smaller than the intercell hopping, and the topological phase has protected zero-energy modes at the ends of the chain.

- A lattice with a **massive Dirac equation** (3.7) gives Chern number $\frac{1}{2}\,\mathrm{sgn}(m)\,\mathrm{sgn}(\lambda)$ for each Dirac point (3.8). Whether the contributions of Dirac points related by time reversal cancel or add depends only on how the mass term transforms under time reversal. This is what separates a trivial gap, from a staggered sublattice potential, from a real **Chern insulator**, which is the Haldane model. This is the answer to the opening question of the session, which was what happens when we open a gap in the Dirac cone of the honeycomb lattice.

- If we put together two copies of (3.7) with opposite masses, one for each spin, time-reversal symmetry is preserved ($C = 0$) and the spin Chern number $C_s = \pm 2$ stays non-trivial (3.9). This is the **quantum spin Hall insulator**, with helical edge states that are protected by Kramers degeneracy and not by a broken symmetry.

- The same invariants apply beyond electrons, to magnons and spinons in sessions 6 and 8, and also beyond quantum mechanics, to any system with a dispersion relation.

In the next session we come back to the flat-band case from session 2 and study it in detail. We will see the full quantum Hall effect, Landau levels from first principles, and the fractional quantum Hall state.

Session 4

Electrons in a Magnetic Field, Quantum Hall Effect and Landau Levels

In the last session we built the general tools of topological band structure. These were the Kubo formula for the Hall conductivity, the way to write it as an integral of the Berry curvature, the Chern number as the topological invariant that comes out of it, and the bulk-boundary correspondence, which says that a nonzero Chern number gives a protected chiral edge state. We applied these tools to two concrete examples, the SSH chain and a gapped Dirac cone on the honeycomb lattice, which is the Haldane construction, and we did this on purpose without any magnetic field. This choice left one case open. In session 2 we saw that the dispersion of a Bloch Hamiltonian comes in three basic types, parabolic, Dirac and flat, and the flat-band case, which is the one where a real magnetic field gives its own topological story, was left for this session. So let us continue from the point where we stopped. In this session we apply the Chern-number tools of session 3 to the flat-band case in full. First we will see the integer quantum Hall effect and Landau levels as topological flat bands. Then we will let these flat bands become real many-body objects, and we will see the Laughlin wavefunction, fractional charge and anyonic statistics, which are the simplest example of fractionalization that we will meet in this course. At the end we will go back to the case without a field. The Haldane model gives a quantized Hall conductance with no magnetic field at all, and we will see explicitly that it and the ordinary quantum Hall state are two forms of the same topological phase.

4.1 Where the quantum Hall state shows up

The quantum Hall state is not the property of one specific material. It is a state of matter that a whole family of two-dimensional electron systems can realize, and each of these systems has its own energy scale and temperature scale.

- **Semiconductor quantum wells**, historically gallium arsenide (GaAs), were the first platform. In these systems the electrons are essentially parabolic electrons, with the ordinary quadratic dispersion, and the relevant energy scale grows linearly with the magnetic field. For the fields that we can reach in a laboratory this gives roughly one millielectronvolt, and this is the reason why quantum Hall physics in GaAs needs temperatures of the order of one kelvin.
- **Two-dimensional materials**, and graphene above all, realize the same physics but with Dirac electrons instead of parabolic ones. As we will derive explicitly below, the

energy scale then grows as the square root of the field and not linearly. The slides give the concrete numbers, which are roughly two orders of magnitude larger, around 100 millielectronvolts. This is high enough that the quantum Hall plateaus of graphene survive up to about 100 K, and in the best samples up to almost room temperature.

For all these platforms the common point is the one we already saw in the last session. An observable, the Hall conductance, is exactly quantized, and this quantization comes from topology and not from the chemistry of a specific material.

4.2 Measuring the Hall conductance

The experiment itself is simple to describe. We take a two-dimensional system, apply a voltage along one in-plane direction, which we call y, and measure the current that flows along the perpendicular in-plane direction, x. The ratio of this transverse current to the applied voltage is the Hall conductance, and this is the quantity that becomes quantized in the quantum Hall regime.

A real device for this measurement is a two-dimensional electron gas (2DEG) with several finger gates in contact with it. A voltage is applied between a top gate and a bottom gate, and the current is measured between two gates placed along x, with all the connections made through metallic leads. The geometry of the device does not add new physics beyond what we need for the concepts, so we will not discuss it further. The device figures in the slides are useful to see how a real experiment looks, but from here on the notes work with the ideal two-dimensional system.

4.3 From the Kubo formula to the Chern number, briefly

We derived this result in full in the last session, so we will not repeat fifteen lines of algebra that would look almost the same. Instead, let us recall the logic and check that it applies here without changes. The Kubo formula is linear response, so it is first order in the applied field, and its sum over states has the form of a second-order perturbation expression. It writes the transverse conductivity as a sum over all pairs of occupied and empty bands, where each term has matrix elements of $\partial H/\partial k_x$ and $\partial H/\partial k_y$ divided by energy differences. There is an identity that relates $\partial H/\partial k$ to the derivative of the eigenstates, which we obtain when we differentiate the eigenvalue equation $H(\mathbf{k})|u_{\mathbf{k}}\rangle = \varepsilon(\mathbf{k})|u_{\mathbf{k}}\rangle$ with respect to $\mathbf{k}$. With this identity every explicit energy denominator cancels, and the result depends only on the wavefunctions, through the Berry connection $\mathbf{A}(\mathbf{k}) = i\langle u_{\mathbf{k}}|\nabla_{\mathbf{k}}|u_{\mathbf{k}}\rangle$ and its curl, the Berry curvature $\Omega(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathbf{A}(\mathbf{k})$. The Brillouin zone is a closed, compact manifold, which is a torus, and when we sum over occupied bands and integrate over it we obtain an integer,

$$\sigma_{xy} = \frac{e^2}{h}\, C, \qquad C = \frac{1}{2\pi} \sum_{n\,\mathrm{occ.}} \int_{\mathrm{BZ}} \Omega_n(\mathbf{k})\, d^2k \in \mathbb{Z}, \tag{4.1}$$

which is the Chern number of the occupied bands. This is the same statement as in session 3. The Hall conductance is not only quantized, it is also a topological invariant, so it does not change under any smooth deformation of the Hamiltonian that does not close the bulk gap. What is new in this session is not this derivation. It is the specific band structure that we will use in it, which is the set of Landau levels of a magnetic field, and everything that follows when we ask how the many-body ground state of a *partially filled* Landau level looks.

4.4 Landau levels as Chern-number-one flat bands

A two-dimensional electron gas in a perpendicular magnetic field develops a set of completely flat bands, the Landau levels, with index $n = 0, 1, 2, \ldots$. In this context "flat" means without dispersion, so the energy does not depend on momentum at all, and we will derive the reason for this in Section 4.9. Even if they have no dispersion, each Landau level has a finite Chern number, which is also the smallest nonzero value, $C = 1$. If the chemical potential is between the N-th and the $(N+1)$-th Landau level, the total Chern number of the occupied bands is the sum of N contributions equal to 1,

$$C_{\text{tot}} = \sum_{n=0}^{N-1} C_n = N \quad \Longrightarrow \quad \sigma_{xy} = N \frac{e^2}{h}, \tag{4.2}$$

which is an integer that counts the number of filled Landau levels. This is the integer quantum Hall effect. It is a state built only from flat bands without dispersion, and still its Hall conductance is an exact integer multiple of e^2/h, with a sharp quantization.

4.5 Edge states and why an insulating bulk still conducts

There is an obvious puzzle. If the chemical potential is in a gap between two Landau levels, the bulk of the sample is by definition an insulator, since there is no partially filled band and there are no bulk states at the Fermi energy. So how can such a system carry a finite, quantized current?

The answer is the same as in the general bulk-boundary argument of session 3. A nonzero Chern number cannot end without consequences. At any interface where the Chern number changes, which here is the boundary between a sample with Chern number N and the vacuum, which has $C = 0$, there must be N protected, gapless and chiral edge modes. A chiral mode propagates in only one direction, and the number of edge modes at an interface is equal to the difference in Chern number across it. Every real sample is finite, so it always has an edge, and the edge states are always there, even when we do not think about them. When we apply a voltage between the top edge and the bottom edge of the sample, we create an imbalance between the edge electrons that move to the right and the ones that move to the left, and this imbalance is the quantized Hall current.

The band structure of a ribbon, which is a sample that is infinite in y and finite in x, shows this picture in a completely explicit way. Next to the flat bulk Landau levels without dispersion, we find a set of dispersive states localized at the left edge, which all propagate in one direction, and a mirror set localized at the right edge, which propagate in the opposite direction. Each propagation direction is tied to a *location*, the left edge or the right edge, so an impurity on the left edge cannot scatter a left-moving electron into a right-moving one, because the right edge is at a macroscopic distance. There is no available final state, so there is no backscattering, and the edge transport has no dissipation. Figure 4.1 shows the ribbon spectrum next to its real-space picture, together with the chiral modes that cross the chemical potential.

Notebook check-in: `session4.ipynb`, "Localization of the edge states in the quantum Hall effect in a ribbon" (Challenge), and "Localization of the edge states in the quantum Hall effect in an island" (Challenge)

Build a finite ribbon and a finite island in a magnetic field and look directly at where the edge-state wavefunctions are in real space. This confirms the separation between the left and the right edge, or along the perimeter of the island, which is the basis of the argument above for the absence of backscattering.

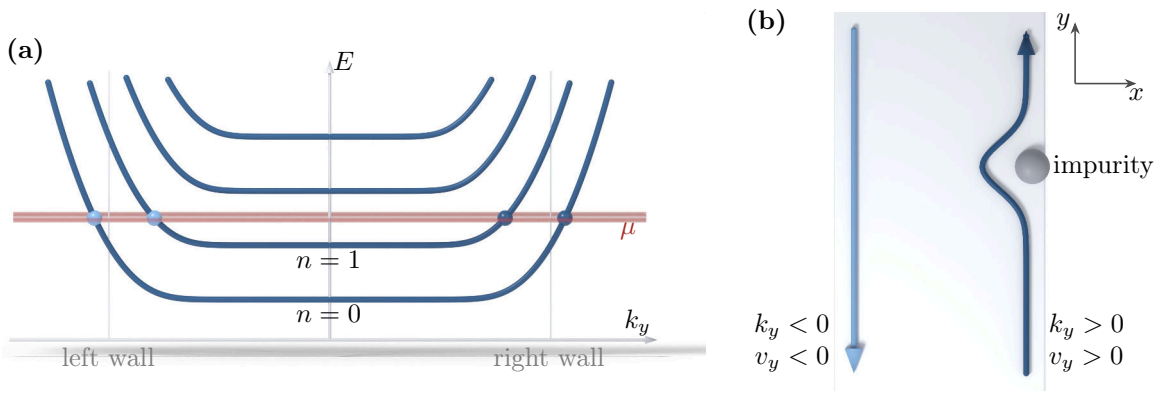


Figure 4.1: The quantum Hall ribbon. (a) The spectrum of a strip that is infinite along y and finite along x, computed for a width of $12\,\ell_B$ with hard walls. In the bulk the Landau levels (4.6) are exactly flat, so they carry no current at all, but near each boundary the guiding centre $x_0 = \hbar k_y/B$ is pushed into the wall and the level bends up strongly. These bending branches are the edge states, and each of them crosses the chemical potential, the translucent plane, exactly once per edge, at the balls, so the number of chiral modes at a given edge is equal to the number N of filled Landau levels, here $N = 2$. (b) The same statement in real space, seen from above. The branches with $k_y > 0$ have $v_y = \hbar^{-1}\partial_{k_y} E > 0$ and live on the right edge, and the branches with $k_y < 0$ move in the opposite direction and live on the left edge. To reverse the velocity of an electron we would therefore have to move it to the other side of the sample, which no local impurity can do, and this is why the edge transport has no dissipation.

4.6 Quantized Hall plateaus in experiment

When we sweep the magnetic field on a real 2DEG, we see this transition directly. At low field, before the quantum regime starts, the Hall resistance grows linearly with B, which is the classical Hall effect of ordinary electromagnetism. When the field reaches the tesla range, a staircase of sharp plateaus appears in the Hall resistance instead. The precise field scale depends on the material and follows the spacing of the Landau levels that we derive below. When the field increases further, the cyclotron energy $\hbar\omega_c$ grows, so fewer Landau levels stay below a fixed chemical potential. Each time the chemical potential goes past a Landau level, the number of filled levels N in (4.2) decreases by one, and the Hall resistance jumps to the next quantized plateau.

4.7 Coupling electrons to a magnetic field with minimal coupling and the Peierls substitution

To go beyond this phenomenology and derive the spectrum of the Landau levels, we first need to know how to include a magnetic field in a Hamiltonian. In classical physics, a magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$ enters the equations of motion through the canonical momentum,

$$\boldsymbol{\pi} = \mathbf{p} - e\mathbf{A}, \tag{4.3}$$

which replaces the ordinary momentum $\mathbf{p}$ everywhere it appears, and when we compute the equations of motion that follow from it we recover the Lorentz force law. In this session we set the magnitude of the electron charge and the speed of light to one, as the slides do without saying it, and we keep only $\hbar$ explicit where the spacing of the Landau levels needs it. The same replacement (4.3) works without changes in quantum mechanics, with $\mathbf{p} \to -i\hbar\nabla$. Minimal coupling turns the free Hamiltonian $p^2/2m$ into $\pi^2/2m$, and in the same way it changes the Dirac Hamiltonian below.

There is one subtle point that we cannot avoid, which is that $\mathbf{A}$ is never unique. Any change $\mathbf{A} \to \mathbf{A} + \nabla\chi$ with a scalar function χ gives the same physical field $\mathbf{B}$, so we always have to choose a gauge. We will need two gauges, the Landau gauge and the symmetric gauge, and we will see both in Section 4.8. A related point is a bit against intuition. A uniform out-of-plane field does not break the in-plane translational symmetry of the physical system, but *every* choice of $\mathbf{A}$ for this field breaks translational symmetry in a formal way, in one direction or in both. Which symmetry survives, and how much of it, is a choice of gauge and not a physical choice.

The minimal-coupling replacement (4.3) is the natural method for continuum models, which are Schrödinger or Dirac electrons, where we write and solve a differential equation directly. For a tight-binding model, which is written in real-space orbitals, the more natural method puts the dependence on the field into the wavefunctions and not into the Hamiltonian, with a local gauge transformation. In concrete terms, the hopping amplitude t_{ij} is by definition an overlap between the orbital wavefunctions at sites i and j. When we include the vector potential in these wavefunctions, every hopping gets multiplied by a complex phase, which is equal to the line integral of $\mathbf{A}$ between the two sites,

$$t_{ij} \;\longrightarrow\; t_{ij}\, \exp\left(i \int_{\mathbf{R}_j}^{\mathbf{R}_i} \mathbf{A} \cdot d\boldsymbol{\ell} \right), \tag{4.4}$$

and this is what is called the **Peierls substitution**. It is the tight-binding version of minimal coupling, and the two give the same physics, since at low energies the expansion of a tight-binding model with the Peierls substitution gives directly the continuum Hamiltonian with minimal coupling.

4.8 The Landau gauge and the symmetric gauge

For a uniform field $\mathbf{B} = B\hat{z}$, there are two gauges that will be important for us.

- The **Landau gauge**, $\mathbf{A} = (0, Bx, 0)$, puts all the spatial dependence into one coordinate, x. Since $\mathbf{A}$ does not depend on y, the Hamiltonian built from it still commutes with p_y, so translations along y are still a good symmetry, but it does not commute with p_x. This gauge keeps *one* translational symmetry, which is what we need to have a band structure at all, and it is the gauge that we use in every ribbon calculation in this session.
- The **symmetric gauge**, $\mathbf{A} = \frac{B}{2}(-y, x, 0)$, breaks both translational symmetries but keeps the rotational symmetry of the system instead. It is the natural gauge to write many-body wavefunctions with a good angular momentum, which is what we need when we arrive at the fractional quantum Hall state, from Section 4.13 on.

Figure 4.2 shows the difference in a concrete way, by writing both gauges as Peierls phases (4.4) on a square lattice. The phases of the single bonds are completely different in the two gauges, but the sum around any plaquette is the same flux in both. This sum is the only gauge-invariant combination, and it is the only thing that the electrons respond to.

4.9 Landau levels of Schrödinger electrons

Let us now solve for the spectrum, and let us start with the simplest case. These are free electrons with no lattice and no potential apart from the field itself, and with dispersion $p^2/2m$ before we switch on the field. With minimal coupling and the Landau gauge $\mathbf{A} = (0, Bx, 0)$ we have

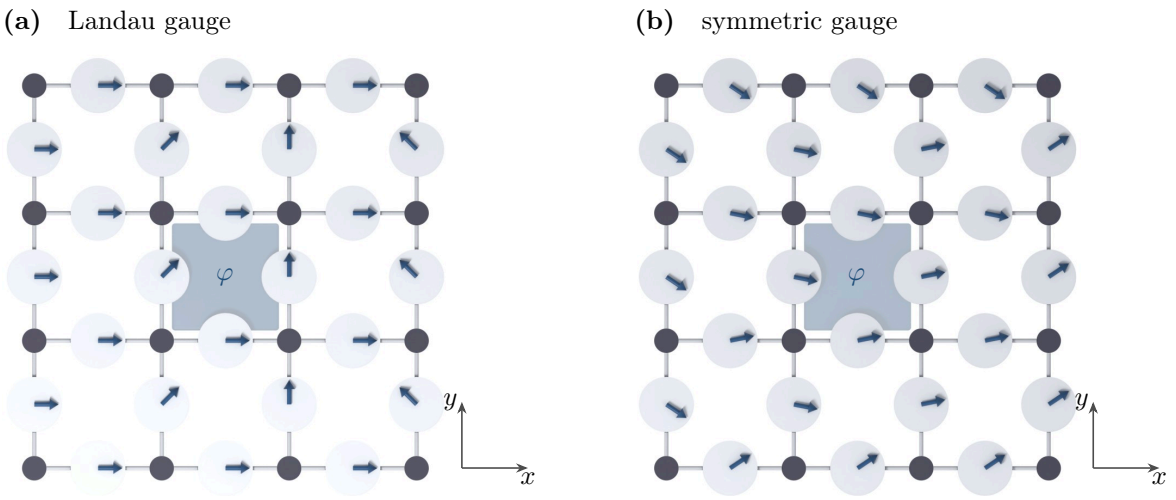


Figure 4.2: Two gauges for the same magnetic field, written as Peierls phases (4.4) on a square lattice, with a dial on every bond whose needle points at the angle of its phase, drawn for $\varphi = \pi/4$. (a) In the Landau gauge the horizontal bonds have no phase at all and the vertical bond in column x has the phase $x\varphi$. The phase pattern depends on x but not on y, so translations along y are still an exact symmetry and a band structure in k_y still exists. (b) In the symmetric gauge every bond has a phase and none of them is the same as in (a), so no translation survives, but the pattern is invariant under rotation about the centre of the plaquette. There is no physical difference between the two. When we go once around any plaquette, such as the shaded one, the phases of the single bonds add up to the same φ, which is the flux through that plaquette. The bond phases are a choice of gauge and the loop sum is not, and the electrons respond only to the loop sum.

$$H = \frac{\pi_x^2 + \pi_y^2}{2m} = \frac{p_x^2}{2m} + \frac{(p_y - Bx)^2}{2m}. \tag{4.5}$$

We take the charge of the carriers to be $+e$ in (4.3) and we set $e = 1$ in all this session, while we keep $\hbar$ explicit. This is a choice of convention, but it has consequences, since it fixes $[\pi_x, \pi_y] = +i\hbar^2/\ell_B^2$. The ladder operators of Section 4.10 and the holomorphic wavefunctions of Section 4.13 are built on this convention. For an electron, which has charge $-e$, every cyclotron orbit and every edge current below goes around in the other direction, and nothing else changes.

Since $\mathbf{A}$ does not depend on y, we have $[H, p_y] = 0$, so we can look for eigenstates of the form $\psi(x, y) = e^{ik_y y}\phi(x)$. With the replacement $p_y \to \hbar k_y$, the Hamiltonian (4.5) becomes a one-dimensional problem in x,

$$H\phi(x) = \left[\frac{p_x^2}{2m} + \frac{B^2}{2m}\Big(x - \frac{\hbar k_y}{B}\Big)^2\right]\phi(x),$$

which is exactly the Hamiltonian of a quantum harmonic oscillator centered at $x_0 = \hbar k_y/B$, with frequency $\omega_c = B/m$, the cyclotron frequency. The shift of the center is removed with a change of variable, as it is described in the lecture. When we solve it with the same ladder operators as the ordinary harmonic oscillator we obtain

$$E_n = \hbar\omega_c\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots, \tag{4.6}$$

which does not depend on k_y at all. Every value of k_y, or in other words every choice of the guiding center x_0, gives the same energy. This is the reason why the Landau levels are flat bands, and why each of them has a macroscopic degeneracy that is proportional to the area of the sample. Figure 4.3 shows this mechanism directly, with the same well placed at different

positions, and with the degeneracy counted as one orbital for each flux quantum through the sample. When we put in the real physical constants for a field typical of GaAs, we obtain the millielectronvolt scale that we gave in Section 4.1.

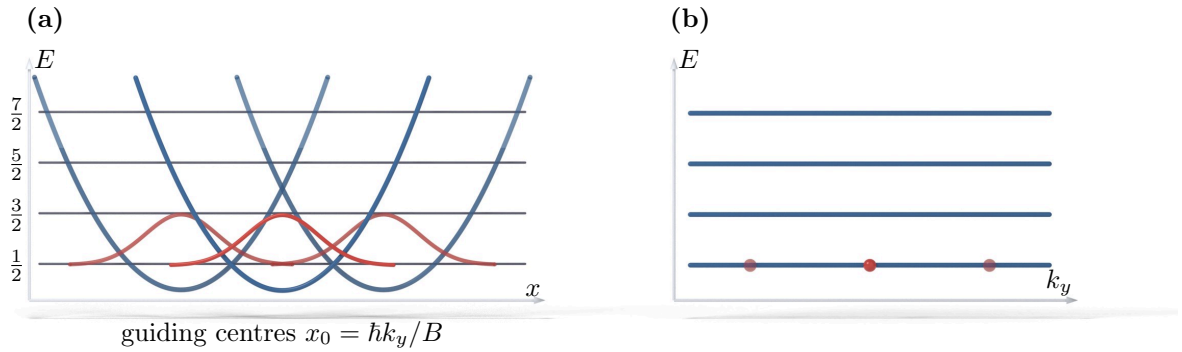


Figure 4.3: Why the Landau levels are flat. (a) In the Landau gauge each k_y reduces (4.5) to a one-dimensional harmonic oscillator in x, which is the same in every respect except that it is centred on the guiding centre $x_0 = \hbar k_y/B$. When k_y increases, the whole well moves to the right without changing its curvature. Every well therefore has the same ladder $E_n = \hbar\omega_c(n + \frac{1}{2})$, drawn here as the horizontal guides, and the ground-state wavepacket of each well, the red bump, has the same shape at a different position. (b) As a function of k_y, this is a set of bands with exactly no dispersion, and the three balls are the three values of k_y of (a). The flatness is not an artefact of the gauge. It means that the energy does not depend on the position of the electron in the sample, and this is also the origin of the very large degeneracy of the Landau levels.

4.10 Landau levels of Dirac electrons

Now we repeat the same calculation for a Dirac Hamiltonian, $H_0 = v_F(p_x\sigma_x + p_y\sigma_y)$, which is the low-energy effective description of graphene near the K point. With the same minimal coupling and the same Landau gauge, $\pi_x = p_x$, $\pi_y = p_y - Bx$, the Hamiltonian is now an *off-diagonal* 2×2 matrix,

$$H = v_F \begin{pmatrix} 0 & \pi_x - i\pi_y \\ \pi_x + i\pi_y & 0 \end{pmatrix}.$$

The matrix H is not diagonal and we cannot solve it by inspection, but its square is. We introduce the magnetic length $\ell_B = \sqrt{\hbar/B}$ and the ladder operators $a = \frac{\ell_B}{\sqrt{2}\hbar}(\pi_x + i\pi_y)$ and $a^\dagger = \frac{\ell_B}{\sqrt{2}\hbar}(\pi_x - i\pi_y)$. Since $[\pi_x, \pi_y] = i\hbar^2/\ell_B^2$, these operators satisfy the ordinary bosonic commutation relation $[a, a^\dagger] = 1$. In terms of $a, a^\dagger$ we have

$$H = \frac{\sqrt{2}\hbar v_F}{\ell_B}\begin{pmatrix} 0 & a^\dagger \\ a & 0 \end{pmatrix} \quad\Longrightarrow\quad H^2 = \frac{2\hbar^2 v_F^2}{\ell_B^2}\begin{pmatrix} a^\dagger a & 0 \\ 0 & aa^\dagger \end{pmatrix} = \frac{2\hbar^2 v_F^2}{\ell_B^2}\begin{pmatrix} N & 0 \\ 0 & N+1 \end{pmatrix},$$

where $N = a^\dagger a$ is the number operator of the harmonic oscillator. This is the statement in the transcript that H^2 has the form "$p^2 + B^2x^2 + \text{const}$, which looks like a harmonic oscillator". Since H^2 is diagonal and has the form of a harmonic oscillator, its eigenvalues are $2\hbar^2 v_F^2 n/\ell_B^2$ for $n = 0, 1, 2, \ldots$, and the energy itself is the square root,

$$E_n = \operatorname{sgn}(n)\, v_F\sqrt{2\hbar B|n|}, \qquad n = 0, \pm1, \pm2, \ldots, \tag{4.7}$$

with a real mode at exactly zero energy for $n = 0$. For this mode the lower spinor component vanishes identically, so there is no partner state to pair it with. This is the origin of

the $\sqrt{B}$ scaling in graphene. The spectrum is the *square root* of a harmonic-oscillator spectrum and not the spectrum itself, so both the dependence on the field and the level spacing change their behavior. For Schrödinger electrons the levels are equally spaced and grow linearly, while for Dirac electrons the levels are spaced as $\sqrt{n}$ and grow as $\sqrt{B}$, and this is the reason why, at the same field, the Landau levels of graphene are roughly two orders of magnitude higher in energy than the ones of a GaAs 2DEG. Figure 4.4 shows how the two ladders move with the field, linearly in one case and as a square root in the other, with the ladder at one field as a cut through each fan, including the zero mode, which has no counterpart at all in the Schrödinger case.

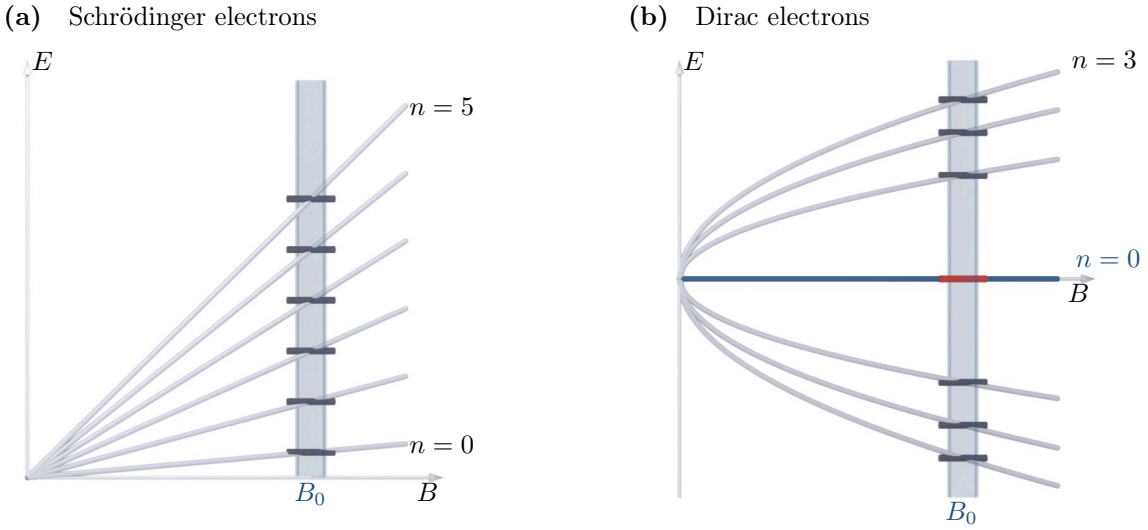


Figure 4.4: The same magnetic field with two dispersions gives two completely different ladders, drawn here as functions of the field. (a) Schrödinger electrons give the harmonic-oscillator spectrum (4.6), where every level is a constant $\hbar\omega_c = \hbar B/m$ above the one below it, so every level is a straight line through the origin and the whole ladder grows linearly in B. (b) Dirac electrons give the square root of that spectrum, equation (4.7). The levels come closer together as $\sqrt{n}$, they come in a particle-hole symmetric pair for every $|n| \geq 1$, and the overall scale grows only as $\sqrt{B}$. This is still enough to put the level spacing of graphene about two orders of magnitude above the one of a GaAs quantum well at the same field. The $n = 0$ level is the qualitatively new feature. It stays at exactly zero energy for every field strength, which is the flat line, because it is the only solution with no partner in the other spinor component. The translucent plane at the field B_0 cuts each fan, and the short rods where it cuts are the ladder of levels at that field, so the ladder at a fixed field is the vertical cut through the fan. The two vertical scales are independent of each other, since what we compare is the pattern of spacings and not the absolute energies.

Notebook check-in: `session4.ipynb`, "The dispersion of the square lattice" (Reminder), "Bands in the conventional quantum Hall state" (Reminder), "Landau levels of conventional electrons" (Core), "Magnetic field dependence of Landau levels" (Core), and "Imaging the magnetic field dependence in the Landau levels" (Core), together with "The dispersion of the honeycomb lattice" (Reminder), "Bands in the Dirac quantum Hall state" (Reminder), "Landau levels of the Dirac equation" (Core), "Magnetic field dependence of Dirac Landau levels" (Core), and "Imaging the magnetic field dependence in the Dirac Landau levels" (Core)

Build a ribbon of the square lattice, which has a parabolic band bottom, and an armchair ribbon of the honeycomb lattice, which has a Dirac cone, with the Peierls substitution (4.4) switched on, and check both spectra directly. For the square lattice you should find equally spaced flat Landau levels that grow linearly in B, as in (4.6), and for the honeycomb lattice levels with unequal spacing that grow as $\sqrt{B}$, as in (4.7). The "imaging" sections let you follow the energy of each level as a function of the field strength, which is the most direct numerical check of the linear and square-root scaling that we derived above.

4.11 Numerical check of the Landau levels and edge states on the lattice

Let us say explicitly how the continuum results of Sections 4.9 and 4.10 are connected to tight-binding lattices, since this is where the edge states of Section 4.5 first appear together with the bulk Landau levels in a single calculation. We take a ribbon of a square lattice with a finite width. The band bottom of this lattice is parabolic, so it plays the role of the Schrödinger 2DEG. We add a uniform orbital field with the Peierls substitution (4.4), and we put the chemical potential near the bottom of the band, so that the lattice model effectively describes a dilute electron gas that is almost free. The band structure that we obtain shows what the continuum calculation predicted. There is a set of flat bulk bands, the Landau levels, together with dispersive edge bands, with one branch localized at each edge of the ribbon and propagating in opposite directions, which is the chiral pair described in Section 4.5. When we repeat the same construction on an armchair ribbon of the honeycomb lattice, where the low-energy physics is a real Dirac equation at the K and K' points, we obtain the $\sqrt{B}$ Landau levels of Section 4.10 together with their own chiral edge states. The band structures of both ribbons, shown for a width of 30 in the slides, are figures that exist only in the slides, in the sense that the transcript describes them only in a qualitative way. The plots are the visual evidence, and the notebook box above asks you to build the same ribbons yourself.

4.12 Ladder operators and the many-body Landau Hamiltonian

In Sections 4.9 and 4.10 we found the structure of a harmonic oscillator by working in a specific gauge. We can show this structure in a way that does not depend on the gauge, if we define the ladder operators directly from the canonical momentum,

$$a = \frac{\ell_B}{\sqrt{2}\hbar}(\pi_x + i\pi_y), \qquad a^\dagger = \frac{\ell_B}{\sqrt{2}\hbar}(\pi_x - i\pi_y),$$

which satisfy the canonical bosonic commutation relation $[a, a^\dagger] = 1$. For this we only use $[\pi_x, \pi_y] = i\hbar^2/\ell_B^2$, which depends on the field and not on the gauge. In terms of these operators the *full* quantum Hall Hamiltonian for a Schrödinger electron becomes

$$H = \hbar\omega_c \left(a^\dagger a + \frac{1}{2} \right), \tag{4.8}$$

which is clearly a harmonic oscillator, with eigenvalues that give (4.6) directly, without any reference to a gauge. This is the clearest way to see that the statement "Landau levels behave like a harmonic oscillator" is not an artifact of the Landau gauge. It is a structural property of any electron in a uniform field.

4.13 The symmetric gauge and the lowest-Landau-level wavefunctions

To go from single-particle Landau levels to a real many-body quantum Hall wavefunction, we change to the symmetric gauge, $\mathbf{A} = \frac{B}{2}(-y, x, 0)$, and to the complex coordinate $z = x + iy$. We write $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$, and then the ladder operator of Section 4.12 in this gauge is proportional to $\partial_{\bar{z}} + z/4\ell_B^2$. The condition for the lowest Landau level (LLL), $a\psi = 0$, is therefore the first-order differential equation

$$\left(\partial_{\bar{z}} + \frac{z}{4\ell_B^2}\right)\psi(z,\bar{z}) = 0 \quad \Longrightarrow \quad \psi(z,\bar{z}) = f(z)\, e^{-|z|^2/4\ell_B^2}, \tag{4.9}$$

where $f(z)$ is an *arbitrary holomorphic function*, meaning any function built only from powers of z and never from $\bar{z}$. This is the precise form of the statement "any power z^m times a Gaussian is a zeroth-Landau-level wavefunction". Holomorphicity is not an extra assumption, since it follows directly when we solve the LLL condition in the symmetric gauge. The single-particle LLL orbitals are then

$$\psi_m(z) = z^m\, e^{-|z|^2/4\ell_B^2}, \qquad m = 0, 1, 2, \ldots, \tag{4.10}$$

and each of them has an angular momentum m, which is the eigenvalue of L_z in this gauge. All of them have the same Gaussian envelope, but the polynomial prefactor z^m moves the probability density of the orbital further from the origin when m grows. The reason is that $|\psi_m|^2$ has its maximum where $\frac{d}{d|z|}\big(|z|^{2m}e^{-|z|^2/2\ell_B^2}\big) = 0$, which is at the radius $r_m = \sqrt{2m}\,\ell_B$.

4.14 The many-body wavefunction of the filled lowest Landau level

Now we look for the many-body ground state of N electrons in which *every* LLL orbital is occupied. Two conditions fix this state completely. It must be built only from LLL orbitals (4.10), and it must be antisymmetric under the exchange of any two electrons, as fermion statistics requires. Both conditions are satisfied by

$$\Psi_1(z_1, \ldots, z_N) = \prod_{i<j}(z_i - z_j)\ \exp\!\left(-\sum_i \frac{|z_i|^2}{4\ell_B^2}\right). \tag{4.11}$$

The Vandermonde product $\prod_{i<j}(z_i - z_j)$ does two things at the same time. First, it is antisymmetric by construction, since the exchange of two coordinates $z_i \leftrightarrow z_j$ changes the sign of exactly one factor $(z_i - z_j)$, and this gives the overall minus sign that the Pauli principle requires. Second, it is exactly the Slater determinant of the single-particle orbitals $1, z, z^2, \ldots, z^{N-1}$, because of the standard identity for the Vandermonde determinant, $\det\big(z_i^{j-1}\big) = \prod_{i<j}(z_j - z_i)$. So (4.11) is the Slater determinant of the N lowest LLL orbitals, times their common Gaussian envelope, where the Gaussians of the single electrons combine into the collective exponential shown above. This is the exact many-body ground state of N electrons that fill the lowest Landau level completely, which is the integer quantum Hall state at filling $\nu = 1$.

4.15 Angular momentum, enclosed flux, and the uniform electron gas

In Section 4.13 we saw that a larger angular momentum m moves the maximum of an orbital further from the origin, to the radius $r_m = \sqrt{2m}\,\ell_B$. In other words, this orbital encloses a larger area, and so a larger magnetic flux. We will use this link between angular momentum and enclosed flux in Section 4.17 to find how much charge a quasiparticle has. For now, the important consequence is about (4.11) itself. When we sum over the full Slater determinant, we find a density profile that is essentially uniform, or flat, up to a radius set by the highest occupied angular momentum $m = N-1$, and that drops sharply to zero beyond this radius. In other words, the filled lowest Landau level is a uniform two-dimensional electron gas without any structure, and this is the property that we will now ask also from a Landau level with *fractional* filling. Figure 4.5 shows both parts of this statement, with the single orbitals as nested rings of growing radius, and the flat droplet that they give when we add them.

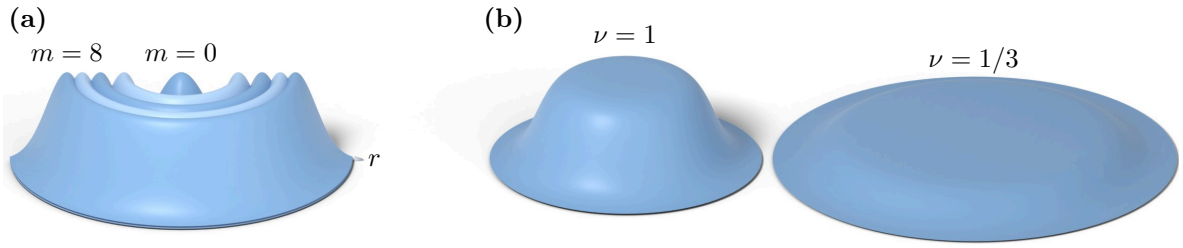


Figure 4.5: Angular momentum fixes the radius. (a) The lowest-Landau-level orbitals (4.10), $|\psi_m|^2 \propto r^{2m} e^{-r^2/2\ell_B^2}$, for $m = 0, 2, 4, 6, 8$, each renormalized here to the same peak height and cut open over half a turn. They are nested rings, and when m increases the ring moves outward, to $r_m = \sqrt{2m}\,\ell_B$, and encloses more flux. (b) When we add every orbital from $m = 0$ to $m = N-1$ we therefore fill a disc, and the rings overlap in such a way that the total density is flat inside the disc and falls off sharply at the edge. The filled lowest Landau level is a uniform electron droplet without structure, with one electron per flux quantum. The Laughlin state (4.12) at $\nu = 1/3$ is the second profile. The same N electrons now use orbitals up to $m = 3(N-1)$, so the radius of the droplet grows by $\sqrt{3}$ and the density becomes a third, while the droplet stays as uniform as before. The Laughlin construction is built to keep this uniformity, and the profiles of (b) are schematic.

4.16 Fractional filling and the Laughlin wavefunction

Let us now say that we want to occupy only a fraction of the available LLL orbitals, for example two out of every three, but in a way that keeps the electron gas uniform. If we only leave some orbitals empty, we obtain a density that is clearly not uniform and has lumps, since each empty orbital is at its own fixed radius. We look for a many-body wavefunction with three properties. First, it still lives only in the LLL, so it is built only from the coordinates z_i and never from $\bar{z}_i$. Second, it is still antisymmetric, since the particles are still electrons. Third, in a heuristic sense, it keeps the electrons as far from each other as possible, since this is what a repulsive Coulomb interaction favors when we minimize its energy.

The construction that satisfies the three conditions at the same time is very simple. We take the wavefunction of the filled LLL (4.11) and we raise the Vandermonde factor to a power,

$$\Psi_m(z_1, \ldots, z_N) = \prod_{i<j} (z_i - z_j)^m \; \exp\left(- \sum_i \frac{|z_i|^2}{4\ell_B^2} \right), \tag{4.12}$$

which is called the **Laughlin wavefunction**. The exponent m must be an odd integer, since an even power would make the wavefunction *symmetric* and not antisymmetric under

exchange, and this is not allowed for fermions. A larger exponent makes the electrons avoid each other more, since the wavefunction now vanishes to m-th order, and not only linearly, when two electrons are at the same point, and this is the correlation effect that we wanted. When we compute the electron density that follows from it, we obtain the filling factor

$$\nu = \frac{1}{m}, \tag{4.13}$$

which we can also see directly by counting. The maximum power of any single coordinate in the fully expanded polynomial (4.12) is $m(N-1)$, so the wavefunction uses the angular-momentum orbitals up to $m(N-1)$. This means that N electrons are spread over $\approx m(N-1)+1 \approx mN$ available LLL orbitals, which gives a filling fraction of $N/(mN) = 1/m$, in agreement with (4.13). So we start only from the single-particle LLL orbitals, the Pauli exclusion principle and a heuristic argument about correlations, and we arrive at the exact ground state of a strongly interacting many-body problem, which will give us fractional charge in the next section.

4.17 Quasiholes and quasielectrons

From this point to the end of Section 4.18, the material has no counterpart in `session4.ipynb`*. The part of the notebook for this session stops at the integer quantum Hall state, its Landau levels, and its edge and Chern-insulator physics, which are Sections 4.10, 4.11 and 4.19 below. The Laughlin wavefunction, the quasiholes and quasielectrons, and the Berry-phase argument for fractional charge and anyonic statistics come from the transcript and the slides, and there is no numerical exercise built around them.*

The excitations above the Laughlin ground state (4.12) also have a simple construction in these coordinates. A **quasihole** at the origin is obtained when we multiply the wavefunction by an extra overall factor z_i for every electron. A quasihole at an arbitrary position w is obtained instead when we insert a factor $(z_i - w)$ for each electron,

$$\Psi_{\rm qh}(w) = \prod_i (z_i - w)\ \Psi_m(z_1, \ldots, z_N). \tag{4.14}$$

This extra factor raises by one the effective power of every orbital. With the relation between angular momentum and radius of Section 4.15, this moves the charge density outward, away from w, and leaves a region with less charge there, which is the quasihole. When we repeat this construction m times, at m different positions, we remove in total exactly the charge of one electron, which is consistent with each quasihole having only a fraction $1/m$ of the electron charge. We will confirm this fraction directly with the Berry phase in Section 4.18. In the same language, a **quasielectron** is the inverse operation. Instead of raising the power, which we do when we multiply by $z_i - w$, we lower it, and we do this with the derivative $\partial/\partial z_i$ acting on the polynomial part instead of a multiplication. This moves charge inward instead of outward, and it gives an excitation with the opposite, negative, fractional charge.

4.18 Berry phase, fractional charge, and anyonic statistics

We now use the Berry phase, which is the same object that we used in session 3 to build the Chern number, to show directly two things. First, the quasihole excitation of Section 4.17 has a fraction of the electron charge. Second, a pair of them follow neither fermionic nor bosonic exchange statistics.

Fractional charge. We take the wavefunction with one quasihole (4.14) and move w adiabatically around a closed loop. We write $w = |w|e^{i\theta}$ and compute the Berry phase that

we obtain when θ goes from 0 to 2π. The result is an Aharonov-Bohm phase, proportional to the flux enclosed by the loop, but with an extra prefactor $1/m$ with respect to the phase that an ordinary particle with charge e would get around the same loop. The Aharonov-Bohm phase for a particle with charge q around an enclosed flux Φ is $2\pi q\Phi/\Phi_0$, where Φ_0 is the flux quantum, so the extra factor $1/m$ that we find is a direct and clear statement about the *charge* of the excitation,

$$q_{\text{qh}} = \frac{e}{m}. \tag{4.15}$$

For these excitations the electron is no longer the basic unit of charge, and the unit is a real fraction of it.

Anyonic statistics. Now we take a wavefunction with *two* quasiholes, one fixed at the origin and one at w, and we move the second one in a loop around the first, again with $w = |w|e^{i\theta}$. The Berry phase that we obtain has two parts. The first part is the same Aharonov-Bohm contribution as before, given by the charge times the enclosed flux. The second part is an additional **statistical phase**, which comes only from the fact that one quasiparticle goes around the other,

$$\theta_{\text{stat}} = \frac{2\pi}{m}. \tag{4.16}$$

To compare with the familiar cases, note that to move one quasiparticle all the way around another is *two* exchanges, one after the other, so the exchange phase is half of (4.16), which is π/m. For ordinary fermions the exchange of two particles gives a phase π, and for bosons it gives a phase 0. The exchange phase π/m is neither of the two, and it goes from one familiar case to the other as m grows, as shown in figure 4.6. These quasiparticles are what are called **anyons**, which are excitations with exchange statistics that go continuously from the fermionic to the bosonic case, and they are labeled by the same integer m that fixed the filling fraction. We started from a many-body wavefunction built only from ordinary electrons, which are real fermions, and from the correlations of this wavefunction we obtained excitations that are neither fermions nor bosons. This is the simplest example of fractionalization in a quantum many-body system that we will meet in this course.

4.19 The Haldane model and the quantum Hall effect without Landau levels

This last section goes back on purpose to a model that we already know, and it is not new material. In session 3 we already built the Haldane model, as an example of a Chern insulator obtained by opening a gap in a Dirac cone without any external field. We come back to it here because it closes the argument that we started in this session. Everything above needed a real magnetic field to give Landau levels, so we want to confirm explicitly that the quantized Hall conductance itself never needed one.

Let us recall the construction. We start from the honeycomb lattice with first-neighbor hopping only, which has a Dirac cone without a gap at K and K', and we add a second-neighbor hopping that is purely imaginary, which breaks time-reversal symmetry. In the Bloch Hamiltonian this term appears as a mass at K and K'. An ordinary staggered onsite potential also gives a mass at both valleys, but the two masses enter with opposite relative sign between the valleys. For the staggered potential the Berry curvatures of the two valleys have opposite sign and cancel, which gives $C = 0$. For the Haldane term, the imaginary second-neighbor hopping, the Berry curvatures of the two valleys have the *same* sign, so when we integrate over the Brillouin zone the two valley contributions add and do not cancel, which

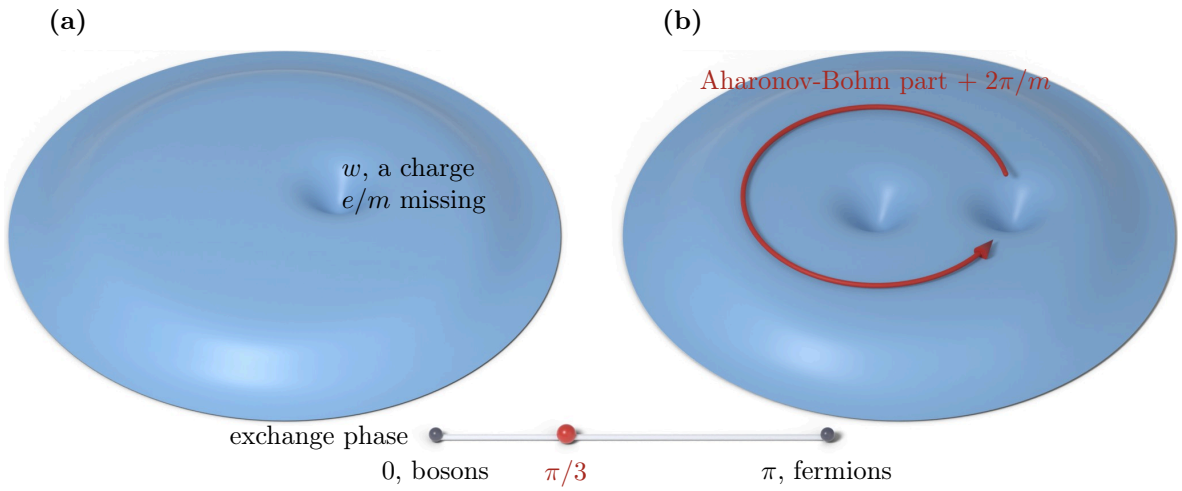


Figure 4.6: Fractional charge and fractional statistics, both obtained from a Berry phase. (a) When we multiply the Laughlin state by $\prod_i (z_i - w)$, equation (4.14), the angular momentum of every electron about w increases by one, which by figure 4.5 moves charge radially outward and leaves a region with less charge behind, the dimple. If we do this m times we remove exactly the charge of one electron, so one quasihole has the charge of an electron divided by m, which is $q = e/m$. (b) When we move one quasihole once around another, along the loop, and follow the Berry phase, it separates into two parts. The first is an Aharonov-Bohm part, proportional to the enclosed flux, and its prefactor is what fixes $q = e/m$ in (4.15). The second is a remaining part $2\pi/m$, which depends only on the fact that one quasiparticle went around the other. A full loop is two exchanges, so the exchange phase itself is π/m, drawn at the bottom for $m = 3$. For every odd $m > 1$ it is strictly between the bosonic 0 and the fermionic π, and this is what makes these excitations anyons.

gives $C = \pm 1$. The result is a net nonzero Chern number, an insulating bulk gap and a chiral edge state, all with $B = 0$.

What is really new in our version of this story is the explicit **adiabatic connection** between the Haldane model and the ordinary quantum Hall state of this session, which is driven by the field. We start in the Haldane model, with imaginary second-neighbor hopping and no field. Then we switch on a real magnetic field while we keep the Haldane coupling, and Landau levels develop from the bands that were dispersive before. Finally we switch off the Haldane coupling, and the system is left in the ordinary quantum Hall state built only from Landau levels. At every step of this path the bulk gap stays open and the chiral edge state never disappears. Since we can connect the two end points without closing the gap, the Chern insulator of the Haldane model and the quantum Hall state of the Landau levels are *topologically equivalent*. They are the same phase of matter, reached by two completely different physical routes, one that needs a magnetic field and one that does not.

Notebook check-in: `session4.ipynb`, "Topological equivalence between the Quantum Hall state and a Chern insulator" (Challenge), and "Topological domains in the quantum Hall state" (Challenge)

Build the Haldane model, switch a magnetic field on and off numerically together with the Haldane coupling, and follow the bulk gap and the Chern number along the adiabatic path described above. The "topological domains" exercise lets you study spatial regions where the two couplings compete directly, which is a practical version of the equivalence argument that we just gave.

4.20 Real materials realizing Chern and fractional Chern physics

This survey, like Sections 4.17-4.18, comes from the transcript and the slides and has no counterpart in the notebook. There is no numerical exercise in `session4.ipynb` *built around any of the specific compounds below.*

The Chern insulator at zero field is not only a theoretical idea. Chromium-doped $(Bi,Sb)_2Te_3$, which is a magnetically doped topological insulator, breaks time-reversal symmetry through real magnetism and realizes an integer quantized Hall conductance with no applied field. Twisted graphene bilayers aligned with a boron nitride substrate realize the same physics with a different mechanism, which is based on orbital currents and not on magnetic moments. On the fractional side, twisted two-dimensional materials can host topological flat bands with a mathematical structure similar to the one of an artificial pseudo-magnetic field, which comes from the elastic reconstruction of the lattice and not from a real field. Twisted bilayers of molybdenum ditelluride ($MoTe_2$) are a recent and striking example, and they show real fractional quantum Hall physics at zero magnetic field.

4.21 Closing summary

- The Hall conductivity is a topological invariant, the Chern number of the occupied bands, as we derived in more detail in session 3. Equation (4.1) is the same result, which we confirm here and do not derive again from the beginning.

- A magnetic field turns a two-dimensional electron gas into a set of flat Landau levels, each with Chern number one, so the Hall conductivity is quantized to Ne^2/h for N filled Landau levels, equation (4.2).

- An insulating bulk still conducts because a nonzero Chern number gives chiral edge states without backscattering at the boundary of the sample (Section 4.5), which we can see directly in the band structures of ribbons.

- Landau levels come from minimal coupling (4.3) or, on a lattice, from the Peierls substitution (4.4). Schrödinger electrons give equally spaced levels $E_n = \hbar\omega_c(n+\frac{1}{2})$, equation (4.6), while Dirac electrons give $E_n \propto \sqrt{nB}$, equation (4.7), and this explains the much higher temperature scale of the quantum Hall effect in graphene.

- In the symmetric gauge, the wavefunctions of the lowest Landau level are holomorphic, $\psi_m(z) \propto z^m e^{-|z|^2/4\ell_B^2}$, equation (4.10). The many-body state of the filled LLL is the Vandermonde determinant times a collective Gaussian, equation (4.11).

- When we raise the Vandermonde factor to an odd power m we obtain the Laughlin wavefunction, equation (4.12), with filling factor $\nu = 1/m$, equation (4.13). Its quasihole excitations, equation (4.14), have fractional charge e/m, equation (4.15), and follow anyonic statistics. They get a phase $2\pi/m$ for a full braid, equation (4.16), which is an exchange phase π/m strictly between the bosonic 0 and the fermionic π, and this is the simplest case of fractionalization that we will meet.

- The Haldane model shows that the quantized Hall conductance never strictly needed a magnetic field, and that it is adiabatically and topologically equivalent to the quantum Hall state driven by the field. This is a return to session 3 and not new theory.

In this session the order was driven only by a magnetic field, which is an external symmetry-breaking parameter that we switched on by hand. In the next session we will see a completely different route to order, which is the attractive interaction between the electrons themselves.

This interaction drives the system to become a superconductor without any external field, and we will see that the same classification into trivial and chiral topological states, which we built here and in session 3, comes back to sort superconducting order parameters into conventional and topological classes. It is also useful to keep in mind the form of fractionalization that we just built, since we will soon meet a second example of it, which looks quite different. Here, the correlations inside a partially filled topological flat band split the charge of the electron and made its exchange statistics anyonic. In the next session, pairing correlations in a superconductor will split the electron into two Majorana halves instead. The underlying idea is the same, which is a correlated topological state with excitations that are not the electron itself, but it is reached by two really different mechanisms.

Session 5

Superconductivity, Nambu Representation and Majorana Physics

At the end of Session 4 we kept in mind the kind of fractionalization that we had built there, and we pointed forward to a second example of it, which looks quite different. In a partially filled topological flat band, correlations split the charge of the electron and make its exchange statistics anyonic. In a superconductor, as we said there, pairing correlations instead split the electron into two Majorana halves. In this session we will work out this second case in full. As we will state carefully below, superconductors and fractional quantum Hall states are two correlated states that both host fractional excitations, but through opposite mechanisms. The first one has a mean-field description and needs an attractive interaction, and the second one has no mean-field description and needs a strong repulsive interaction. The idea behind both is the same, which is a correlated state with excitations that are not the electron itself, but the two routes to it are really different. In this session we will build all the tools for the superconducting route.

These tools also complete a topic that we left open before. In session 1 we built mean-field theory as a general tool and then applied it quickly to three symmetry-broken states, a charge density wave, a magnet and, as our third and shortest example, a superconductor. There we started from an attractive Hubbard interaction and took an expectation value of two annihilation operators instead of one creation and one annihilation operator. With this we defined the pairing order $\Delta = \langle c_i c_j \rangle$ (session 1, eq. 1.9), and we showed that this term gets a phase $e^{2i\varphi}$ under a gauge transformation, while every ordinary bilinear term stays invariant. This means that a superconductor spontaneously breaks gauge symmetry, and with it the fixed electron number. That first treatment was short on purpose. At that time we said that magnetism and superconductivity would each get "a full, much deeper treatment in their own dedicated sessions later in the course," and in this session it is the turn of superconductivity. Magnetism is moved once more to the next recording, which is session 6 in the plan of this course.

So in this session we take up this topic again and build the full set of tools around it. First, we need a new representation, the Nambu spinor, since without it we cannot even diagonalize a Hamiltonian that mixes creation and annihilation operators in the way a pairing term does. This representation gives us the Bogoliubov-de Gennes Hamiltonian, which is the central object of the whole session. We will use it to classify superconducting order by symmetry, into singlet and triplet and into s-, p- and d-wave. Then we will show that two superconductors can have the same spectrum and still have a completely different topology, and to distinguish them we will use the same Chern-number tools as in the topological-band-structure and quantum-Hall sessions. We will follow this topological difference down to its strongest consequence, which is the Kitaev chain. We will see that this model is the SSH model from the topological-band-

structure session written in terms of Majorana operators, and that its topological zero modes are their own antiparticles. At the end we will ask what these Majorana modes can be used for, which is topological quantum computing. We will also point to session 6, which does for magnetism what we are about to do for superconductivity, and this is the one point of the course where the lecture explicitly leaves a question open for the next recording. The same interacting Hamiltonian can give, at the mean-field level, either a superconductor or a magnet, and magnetism is the topic that we moved to the next recording.

5.1 Weakly versus strongly correlated matter, revisited

Before we specialize, let us state again the general setting, since it is useful to have it clearly in mind. Every interacting Hamiltonian that we consider in this course has the same general form,

$$H = \sum_{ij} t_{ij}\, c_i^\dagger c_j \;+\; \sum_{ijkl} V_{ijkl}\, c_i^\dagger c_j^\dagger c_k c_l,$$

which is a free bilinear part plus a quartic interaction, and it is exactly eq. 1.5 of session 1. When mean-field theory can turn the quartic term into a good effective bilinear Hamiltonian, we call the resulting state *weakly correlated matter*. This includes magnets, charge density waves and superconductors, which are all the states that we built in session 1. When no single-particle mean-field Hamiltonian of this kind describes the physics, we have *strongly correlated matter*, and we need exact or numerical solutions instead. The fractional quantum Hall state from the quantum-Hall session is our main example of this harder case.

Note that superconductors and the fractional quantum Hall state are at opposite ends of this classification, even though both of them in the end host fractional excitations. A superconductor has a mean-field description, comes from an *attractive* interaction, is destroyed by a magnetic field and, as we will see soon, hosts Majorana states. The fractional quantum Hall state has no mean-field description, comes from strong *repulsive* interactions, usually needs a strong magnetic field and hosts excitations with fractional charge. Roughly speaking, both Majorana excitations and fractional quantum Hall excitations are a fraction of an electron, but these fractions come from completely different mechanisms. This is one of the good examples in this course of how very different microscopic physics can lead to the same qualitative phenomenon, which is emergent fractionalization, while the mechanism behind it is not the same at all.

5.2 Superconductivity as a macroscopic quantum phenomenon

Before we write the Hamiltonian, let us see why superconductivity is one of the clearest ways that we have to observe quantum mechanics at the macroscopic scale. A superconductor is made of trillions of atoms, and still a measurement on the bulk sample can give, essentially exactly, the microscopic constants $\hbar$ and e. This works because gauge symmetry breaking gives the many-body wave function a single well-defined phase $\theta(\mathbf{r})$, and the condition that this phase is single-valued around a closed loop is a very strong constraint.

Let us take a superconducting ring threaded by magnetic flux, or in the same way a hole in a bulk superconductor, and let us take the loop deep enough inside the superconductor that the supercurrent density vanishes there because of Meissner screening. The supercurrent of Cooper pairs, which have charge $2e$ since a pair is made of two electrons, is proportional to the gauge-invariant combination $\hbar\nabla\theta - 2e\mathbf{A}$. If we set the current to zero on our loop, we obtain

$$\oint \nabla\theta \cdot d\boldsymbol{\ell} = \frac{2e}{\hbar} \oint \mathbf{A} \cdot d\boldsymbol{\ell} = \frac{2e}{\hbar}\Phi.$$

However, θ is the phase of a single-valued wave function, so after one turn around the loop it can only change by an integer multiple of 2π, which means that $\oint \nabla\theta \cdot d\boldsymbol{\ell} = 2\pi n$ for integer n. If we combine the two equations, we obtain flux quantization,

$$\Phi = n\,\Phi_0, \qquad \Phi_0 = \frac{h}{2e}. \tag{5.1}$$

Figure 5.1 shows the geometry and the argument based on the winding of the phase. The same physics is behind the precision of Josephson junctions. In the AC Josephson effect, a voltage V across the junction gives a current that oscillates with frequency $2eV/h$, which is a direct macroscopic readout of $h/2e$, and this is possible only because gauge symmetry is broken. Note the parallel with the quantum Hall state from earlier in the course. There, a macroscopic measurement also gives a ratio of the same two constants, which is the Hall conductance, quantized in units of e^2/h by the Chern number. However, the mechanism is completely different, since it is the quantization of a single-particle topological invariant and not the breaking of a continuous symmetry. In fact, if we combine the two measurements, the flux quantum and the conductance quantum of the Hall effect, we have one of the most precise ways to obtain h and e separately. This is a concrete example of the emergence that we keep coming back to in this course, where macroscopic measurements on billions of atoms give us exact microscopic constants.

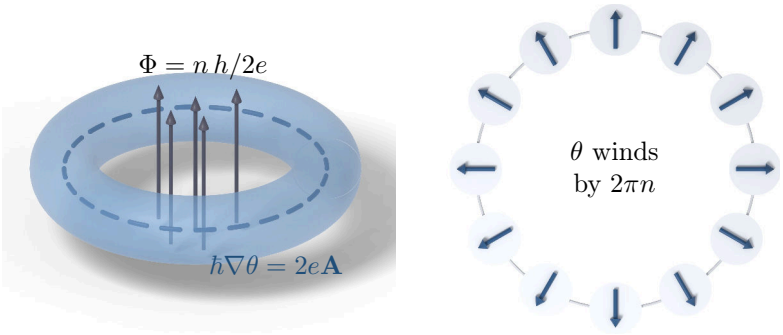


Figure 5.1: A drawing of flux quantization. On the left there is a superconducting ring. Deep in the bulk the Meissner effect has already removed the supercurrent, so on the dashed contour the gauge-invariant combination $\hbar\nabla\theta - 2e\mathbf{A}$ vanishes and the phase gradient is locked to the vector potential. On the right, the arrows are the condensate phase θ at points of this contour. When we move once around it, the phase must come back to itself, so it can only wind by an integer number of turns. Together, the two statements give $\Phi = n\,h/2e$, which means that the flux through the hole is quantized, and the 2 in the denominator is the charge $2e$ of a Cooper pair. None of this is a property of a single electron. It is a macroscopic sample that gives back a ratio of microscopic constants, only because gauge symmetry is broken.

5.3 Zero resistance and the Meissner effect in a superconductor

In the history of the field, superconductivity was discovered through its transport signature. When we cool an ordinary metal, the resistivity goes down smoothly, because there are fewer phonons from which the electrons can scatter. A superconductor instead shows a *sudden* drop of the resistance to exactly zero at a critical temperature T_c. Below T_c the resistance stays zero, and not only small. T_c is the temperature at which gauge symmetry breaking starts. This material appears only on the slides and has no counterpart in the notebook. The schematic of

the resistance as a function of temperature and the figure of the field expulsion are shown and described on the slides, but there is no notebook cell for them, since our companion notebook starts from the Hamiltonian and not from transport measurements.

The second signature is the **Meissner effect**. Above T_c a metal allows an external magnetic field inside without any problem, but below T_c the superconductor actively expels the field from its bulk. In type-II superconductors, the only field that remains inside the material is confined to quantized vortices, which are dynamical topological defects of the order parameter of the broken gauge symmetry. Both signatures, zero resistance and the expulsion of the flux, are direct consequences of the same event, which is the condensation of Cooper pairs into a single macroscopic quantum state with a rigid phase that is chosen spontaneously.

5.4 The interacting Hamiltonian and the origin of attractive interactions

As in session 1, we start from the general interacting Hamiltonian $H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l$. The interaction V_{ijkl} can come from the plain electrostatic Coulomb repulsion, which is repulsive, but it can also be an *effective* interaction mediated by another quasiparticle in the material. Examples are phonons, which are lattice vibrations, magnons, which are fluctuations of a magnetic moment, plasmons, which are collective charge fluctuations, and even more exotic mediators such as skyrmion fluctuations or valence fluctuations. For superconductivity, what matters is not the microscopic origin of V_{ijkl} but its net sign in the channel that turns out to be relevant, and superconductivity requires this channel to be effectively *attractive*.

This is the natural point to distinguish the two families of superconductors.

- **Conventional superconductors**, where the attractive channel is mediated by phonons. This is the case that we understand best. Mean-field theory, which is called BCS theory, works well at the quantitative level, and the critical temperatures of simple conventional superconductors such as aluminum or lead can be predicted from theory.
- **Unconventional superconductors**, where the attraction is mediated by anything that is not a phonon, such as antiferromagnetic or ferromagnetic magnons, plasmons, valence fluctuations, charge fluctuations and others. Here we do not have a predictive theory that is equally reliable. Even to identify *which* quasiparticle mediates the attraction in a given unconventional superconductor is often still an open research question, and it is one of the major unsolved problems that the field is working on today.

5.5 The mean-field pairing Hamiltonian in detail

Let us now build the mean-field pairing Hamiltonian with the same care that we used in session 1 for the charge-density-wave and magnetic examples, instead of only pointing at $\Delta = \langle cc \rangle$. We take the simplest interacting Hamiltonian that can give superconductivity, which is a one-dimensional tight-binding chain with an attractive onsite interaction between opposite spins,

$$H = -t \sum_{n,\sigma} \left(c_{n\sigma}^\dagger c_{n+1,\sigma} + \text{h.c.} \right) - \mu \sum_{n\sigma} c_{n\sigma}^\dagger c_{n\sigma} + U \sum_n c_{n\uparrow}^\dagger c_{n\downarrow}^\dagger c_{n\downarrow} c_{n\uparrow}, \qquad U < 0. \tag{5.2}$$

This is exactly eq. 1.7 of session 1, now with $U = -|U| < 0$. It is the attractive branch of the same Hamiltonian, and its repulsive branch gives rise to magnetism, which is the topic of session 6. This time we do the mean-field decoupling of the quartic term in the *pairing*

channel, also called the particle-particle channel, instead of the density channel. We define the order parameter

$$\Delta_n \equiv U\langle c_{n\downarrow} c_{n\uparrow}\rangle, \tag{5.3}$$

and we factorize $c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}c_{n\downarrow}c_{n\uparrow} \approx \langle c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}\rangle c_{n\downarrow}c_{n\uparrow} + c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}\langle c_{n\downarrow}c_{n\uparrow}\rangle - \langle c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}\rangle\langle c_{n\downarrow}c_{n\uparrow}\rangle$. This is the same Wick-type factorization that we used in all of session 1, now applied to a different pair of operators. The resulting mean-field Hamiltonian is

$$H_{\rm MF} = -t\sum_{n\sigma}\left(c^\dagger_{n\sigma}c_{n+1,\sigma} + \text{h.c.}\right) - \mu\sum_{n\sigma} c^\dagger_{n\sigma}c_{n\sigma} + \sum_n\left(\Delta_n\, c^\dagger_{n\uparrow}c^\dagger_{n\downarrow} + \Delta^*_n\, c_{n\downarrow}c_{n\uparrow}\right) - \sum_n \frac{|\Delta_n|^2}{U}. \tag{5.4}$$

When we solve it self-consistently, Δ_n is exactly zero for $U > 0$, the repulsive case, and it can become nonzero for $U < 0$, the attractive case. This is the superconducting order parameter, and it is the term that breaks gauge symmetry.

Notebook check-in: `session5.ipynb`, "s-wave superconductivity driven by attractive interactions" (Core), and "Dependence on the strength of the attractive interaction" (Challenge)

Build the attractive-Hubbard chain of (5.2), run the pairing self-consistency loop that gives Δ in (5.3)-(5.4), and look at how Δ grows from zero as you make U more negative. This is the same behavior as in session 1, where the charge imbalance grew with V and the magnetization grew with $U > 0$.

5.6 Why pairing does not conserve the particle number

Let us now show in an exact way the statement from session 1 that the electron number fluctuates, instead of only stating it. We take the eigenvalue equation $H_{\rm MF}|G\rangle = E|G\rangle$ for the Hamiltonian in (5.4), and we expand the ground state in the occupation-number basis, which is the Fock basis, ordered by the total particle number N, as $|G\rangle = \sum_N a_N|N\rangle$. The hopping and chemical-potential terms in (5.4) conserve N exactly, since each of them has one $c^\dagger$ and one c. The pairing terms $\Delta_n c^\dagger_{n\uparrow}c^\dagger_{n\downarrow}$ and $\Delta^*_n c_{n\downarrow}c_{n\uparrow}$ change N by $+2$ and -2 respectively. The key point is that *no term in H_{MF} ever changes N by ± 1*. So $H_{\rm MF}$ only connects Fock states whose particle numbers differ by an even integer. For $|G\rangle$ to be an eigenstate, a_N can then be nonzero only inside one fixed parity class of N, either all even or all odd. However, this says nothing about which even or odd values of N appear, and in general $a_N \neq 0$ for several values of N in that class, for example $N = 0, 2, 4, \ldots$ at the same time. The mean-field ground state is therefore *not* an eigenstate of the total number operator $\hat{N}$, even though it has a well-defined fermion parity. This is the exact meaning of the statement that a superconductor has a fluctuating electron number but a fixed parity, which session 1 pointed out in its discussion of the superposition puzzle of the quantum dot (session 1, section 1.5).

5.7 The gauge transformation in explicit form

We already wrote the gauge transformation in second quantization in session 1 (eq. 1.10), $c \to c\,e^{i\varphi}$, $c^\dagger \to c^\dagger e^{-i\varphi}$. It is the second-quantized version of the statement that the phase of a first-quantized wave function has no physical meaning. Let us now check explicitly which terms of a realistic Hamiltonian stay invariant under this transformation and which do not. Ordinary hopping $c^\dagger_i c_j$ is invariant, since the two phases cancel, as $c^\dagger_i c_j \to e^{-i\varphi}c^\dagger_i \cdot e^{i\varphi}c_j = c^\dagger_i c_j$. The same cancellation protects essentially every other bilinear term that we have seen so

far, such as magnetic terms that depend on spin, spin-orbit coupling and any density-density interaction term $c_i^\dagger c_i c_j^\dagger c_j$, because all of them are built from the same number of creation and annihilation operators, which pair up index by index.

The pairing term is the only exception. Under the same transformation,

$$\Delta\, c_{n\uparrow}^\dagger c_{n\downarrow}^\dagger \;\longrightarrow\; \Delta\, e^{-2i\varphi}\, c_{n\uparrow}^\dagger c_{n\downarrow}^\dagger, \tag{5.5}$$

it gets the phase *twice*, because both operators are creation operators and not one creation and one annihilation operator. No choice of $\varphi \neq 0 \pmod{\pi}$ leaves this term invariant, so a Hamiltonian with a nonzero Δ is not gauge invariant term by term. In an equivalent way, if we write the transformation as acting on the order parameter itself, $\Delta \to \Delta\, e^{2i\varphi}$, then any fixed phase of Δ is a choice that the ground state makes spontaneously, which is the signature of continuous symmetry breaking. This is the same logic that session 1 used to show the gauge symmetry breaking of eq. 1.9, now applied in full detail, and we have now seen why the factor is 2φ and not φ.

5.8 The Nambu spinor and the Bogoliubov-de Gennes Hamiltonian

The pairing term creates a technical problem. We cannot diagonalize (5.4) with an ordinary vector of orbital operators, because the anomalous term mixes creation and annihilation operators, instead of pairing them up in the way of every Hamiltonian that we diagonalized in sessions 1-4. The solution is to *enlarge* the operator basis, so that both the creation and the annihilation operators of the original problem become entries of a bigger spinor that behave like annihilation operators. We work in momentum space, with a single orbital and both spins, and we define the **Nambu spinor**

$$\Psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{\mathbf{k}\downarrow} \\ c_{-\mathbf{k}\downarrow}^\dagger \\ -c_{-\mathbf{k}\uparrow}^\dagger \end{pmatrix}, \tag{5.6}$$

which has an electron sector, formed by the first two components and containing ordinary annihilation operators, and a hole sector, formed by the last two components and containing creation operators at $-\mathbf{k}$, with the spins exchanged and with an extra minus sign on the last entry. This sign has a real role. It is what is needed so that, when we write H_{MF} in terms of $\Psi_{\mathbf{k}}$, the pairing term gives two identical and decoupled copies of the same 2×2 problem, instead of mixing the two spin blocks through a Pauli matrix. For a spin-singlet pairing $\Delta(\mathbf{k})$, which does not depend on momentum and is equal to Δ for the Fourier transform of the onsite case of (5.4), one can check directly, using the fermionic anticommutation relations to reorder operators where needed, that the mean-field Hamiltonian can be written exactly as

$$H_{\mathrm{MF}} = \frac{1}{2}\sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger\, H_{\mathrm{BdG}}(\mathbf{k})\, \Psi_{\mathbf{k}} + \text{const.}, \qquad H_{\mathrm{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) & 0 & \Delta(\mathbf{k}) & 0 \\ 0 & \varepsilon(\mathbf{k}) & 0 & \Delta(\mathbf{k}) \\ \Delta(\mathbf{k})^* & 0 & -\varepsilon(\mathbf{k}) & 0 \\ 0 & \Delta(\mathbf{k})^* & 0 & -\varepsilon(\mathbf{k}) \end{pmatrix}, \tag{5.7}$$

where $\varepsilon(\mathbf{k}) = -2t(\cos k_x + \cos k_y) - \mu$ is the ordinary band dispersion, which we assume to be inversion symmetric, $\varepsilon(-\mathbf{k}) = \varepsilon(\mathbf{k})$. This 4×4 matrix is called the **Bogoliubov-de Gennes (BdG) Hamiltonian**, and in a concrete sense it is the most important object to

describe the excitations of a superconductor. Note that the two diagonal entries in the lower-right part have $-\varepsilon(\mathbf{k})$ and not $+\varepsilon(\mathbf{k})$. This comes directly from writing $c^\dagger_{-\mathbf{k}\sigma}c_{-\mathbf{k}\sigma}$ in terms of $c_{-\mathbf{k}\sigma}c^\dagger_{-\mathbf{k}\sigma} = 1 - c^\dagger_{-\mathbf{k}\sigma}c_{-\mathbf{k}\sigma}$ with the anticommutation relations. This is also where the name hole comes from, since a missing electron at $-\mathbf{k}$ behaves like a particle with opposite energy.

Since the spinor (5.6) was built with this extra minus sign, the 4×4 problem in (5.7) is already block-diagonal. There is one 2×2 block built from $(c_{\mathbf{k}\uparrow}, c^\dagger_{-\mathbf{k}\downarrow})$ and an identical one built from $(c_{\mathbf{k}\downarrow}, -c^\dagger_{-\mathbf{k}\uparrow})$, and each of them has the reduced BdG matrix

$$h_{\mathrm{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) & \Delta(\mathbf{k}) \\ \Delta(\mathbf{k})^* & -\varepsilon(\mathbf{k}) \end{pmatrix}. \tag{5.8}$$

This is the structure that is shown on the slides and quoted in the lecture, a 4×4 matrix for a single orbital with two spins, with diagonal entries $\varepsilon(\mathbf{k})$, $\varepsilon(\mathbf{k})$, $-\varepsilon(\mathbf{k})$, $-\varepsilon(\mathbf{k})$ and off-diagonal entries Δ, but now we have derived it instead of only stating it. If we diagonalize (5.8), we obtain the standard BCS quasiparticle dispersion

$$E(\mathbf{k}) = \pm\sqrt{\varepsilon(\mathbf{k})^2 + |\Delta(\mathbf{k})|^2}. \tag{5.9}$$

Notebook check-in: `session5.ipynb`, "Nambu electronic structure without superconductivity" and "Nambu electronic structure with superconductivity" (both Reminder)

First build the Nambu-doubled Hamiltonian of (5.6)-(5.7) with $\Delta = 0$, and check that you get exactly two copies of the same band structure. One is the physical copy, which is the electron sector, and the other is a spurious mirror copy, which is the hole sector, with dispersion $-\varepsilon(\mathbf{k})$. Then turn on Δ and look at how the gap in (5.9) opens at every crossing between the two copies.

5.9 An example of a gap that opens independently of the dispersion

Equation (5.9) already tells us something that the slides of the lecture show with explicit band plots. The gap opens at the minimum of $E(\mathbf{k})$, which is where $\varepsilon(\mathbf{k}) = 0$, that is, exactly on the original Fermi surface, and the minimum gap there is $|\Delta(\mathbf{k})|$ evaluated on that Fermi surface. Nothing in this argument depends on the details of $\varepsilon(\mathbf{k})$, such as the range of the hopping, the lattice geometry or the position of the chemical potential. For concreteness, let us take the single-orbital square lattice with $\varepsilon(\mathbf{k}) = -2t\,(\cos k_x + \cos k_y) - \mu$, as in (2.14), and a chemical potential such that the physical red band and the spurious blue mirror band cross. When we turn on a constant Δ, a gap of size $2|\Delta|$ opens at every such crossing, for *any* choice of the chemical potential along this dispersion. The same statement holds if we do the calculation on the triangular lattice instead of the square lattice. As we move the chemical potential across the band of the triangular lattice, a uniform Δ still opens a gap everywhere the physical and the spurious bands cross. This is the general signature of *conventional* superconductivity, which is fully gapped, since a uniform $\Delta(\mathbf{k})$ that does not depend on momentum always gaps the whole Fermi surface, regardless of the microscopic details.

Notebook check-in: `session5.ipynb`, "Superconductivity, full spectral function and electron spectral function" (Core), and "The impact of superconductivity in an electronic structure" (Challenge)

Reproduce the square-lattice example above. Plot the spectral function of the electron sector with $\Delta = 0$, where there are two crossing copies, and then with $\Delta \neq 0$, where a gap of size $2|\Delta|$ opens at the crossing, as in eq. 5.9. Repeat this for the triangular-lattice dispersion and check that the gap opens regardless of where you put the chemical potential.

5.10 Metal, gapped and nodal superconductor for one Fermi surface

Not every pairing function $\Delta(\mathbf{k})$ is independent of momentum, and this gives more interesting cases. For a given electronic dispersion $\varepsilon(\mathbf{k})$ there are in general three qualitatively different outcomes.

1. **No superconductivity** ($\Delta = 0$). This is a metal, which is gapless everywhere on the Fermi surface.

2. **Conventional (s-wave) superconductivity.** Here $\Delta(\mathbf{k}) \neq 0$ everywhere on the Fermi surface, so $E(\mathbf{k})$ in (5.9) stays away from zero everywhere, and there is a full gap.

3. **Unconventional, nodal superconductivity.** Here $\Delta(\mathbf{k})$ vanishes along some lines or points of the Brillouin zone. Wherever the Fermi surface crosses one of these nodal lines, $E(\mathbf{k}) = 0$ even though $\Delta \neq 0$ nearby. The system is gapped in most of reciprocal space, but it stays gapless, which means metallic, at isolated momenta or along nodal lines.

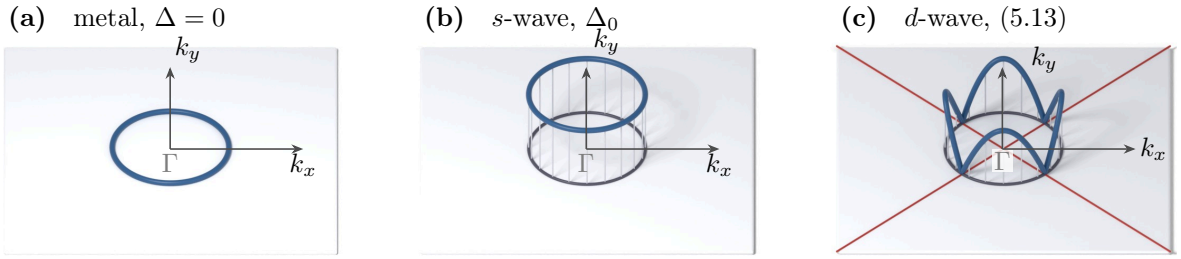


Figure 5.2: One Fermi surface with three outcomes. Each panel shows the Brillouin zone with a small Fermi pocket around Γ, and the height of the tube standing on the pocket is the minimum of the quasiparticle energy $E(\mathbf{k}) = \sqrt{\varepsilon(\mathbf{k})^2 + |\Delta(\mathbf{k})|^2}$ as we go once around it. Since $\varepsilon = 0$ on the Fermi surface, this minimum is $|\Delta|$ evaluated there, which is why the pairing function only matters *on* the Fermi surface. (a) Without pairing the whole pocket is at zero energy, which is a metal. (b) A constant Δ_0 lifts every point of it by the same amount, which gives a full gap. (c) The d-wave gap (5.13) vanishes on the zone diagonals, the red lines, and the pocket crosses them in four places, so four gapless points survive in a superconductor that is gapped everywhere else. Nodes in other parts of the zone do not appear in the spectrum, and only nodes that intersect the Fermi surface give gapless quasiparticles.

Figure 5.2 shows the three cases next to each other, in the Brillouin zone and also as the gap along the Fermi surface itself, which is the only place where the gap can be read at all. The high-temperature cuprate superconductors are the standard example of scenario 3 in real materials. They are real superconductors, but their gap function has nodes, so parts of their Fermi surface stay gapless even in the superconducting state. This division into three cases holds for any lattice geometry, and the lecture shows it explicitly for the Fermi surfaces of both the square and the triangular lattice.

Notebook check-in: `session5.ipynb`, "Unconventional singlet superconducting orders" (Challenge), and "Gapped and nodal superconducting orders" (Core)

Build a $\Delta(\mathbf{k})$ that depends on momentum and vanishes along a line in the Brillouin zone, for example the d-wave form of section 5.13 below. Then check numerically with (5.9) that the gap closes exactly where the Fermi surface crosses this nodal line, which is different from the uniformly gapped case of section 5.9.

5.11 The general pairing Hamiltonian and its symmetry classification

Up to now Δ has been a single number. In general the pairing order can depend on momentum and have a spin structure, $\Delta_{\mathbf{k},\sigma_1\sigma_2}$ with $\sigma_1, \sigma_2 \in \{\uparrow, \downarrow\}$. We can write it in a compact way as the 2×2 pairing matrix $\Delta_{\mathbf{k}} = \begin{pmatrix} \Delta_{\mathbf{k},\uparrow\uparrow} & \Delta_{\mathbf{k},\uparrow\downarrow} \\ \Delta_{\mathbf{k},\downarrow\uparrow} & \Delta_{\mathbf{k},\downarrow\downarrow} \end{pmatrix}$, and it is built from a general mean-field Hamiltonian

$$H_{\text{pair}} = -\frac{1}{2}{\sum_{\mathbf{k}}}' \sum_{\sigma_1\sigma_2} \left[\Delta_{\mathbf{k},\sigma_1\sigma_2}\, c^\dagger_{\mathbf{k}\sigma_1} c^\dagger_{-\mathbf{k}\sigma_2} + \Delta^*_{\mathbf{k},\sigma_1\sigma_2}\, c_{-\mathbf{k}\sigma_2} c_{\mathbf{k}\sigma_1} \right]. \tag{5.10}$$

The real-space and reciprocal-space pictures of Δ are related by a Fourier transform, so we can also think of this order in terms of how the pair wave function looks as a function of the distance between the two paired electrons in real space. The key step is to classify superconducting order only by the *symmetry* of $\Delta_{\sigma\sigma'}(\mathbf{k})$, because this symmetry alone controls how the superconductor responds to perturbations, such as disorder or magnetic fields, independently of the microscopic details.

Fermionic antisymmetry, which comes from the fact that the operators $c^\dagger_{\mathbf{k}\sigma} c^\dagger_{-\mathbf{k}\sigma'}$ anticommute, imposes

$$\Delta_{\sigma\sigma'}(\mathbf{k}) = -\Delta_{\sigma'\sigma}(-\mathbf{k}), \tag{5.11}$$

which is a minus sign under the *combined* operation of exchanging the spin indices and sending $\mathbf{k} \to -\mathbf{k}$. This combined statement holds for any type of pairing. The classification that has a physical meaning comes from splitting $\Delta_{\sigma\sigma'}(\mathbf{k})$ into a part that is symmetric and a part that is antisymmetric under spin exchange alone. Because of (5.11), this is the same as splitting it by its parity under $\mathbf{k} \to -\mathbf{k}$ alone,

$$\begin{aligned} &\textbf{Spin singlet (even parity):} \quad \Delta(\mathbf{k}) = \Delta(-\mathbf{k}), \qquad \Delta_{\sigma\sigma'} = -\Delta_{\sigma'\sigma} \text{ (spin-antisymmetric)}, \\ &\textbf{Spin triplet (odd parity):} \quad \Delta(\mathbf{k}) = -\Delta(-\mathbf{k}), \qquad \Delta_{\sigma\sigma'} = +\Delta_{\sigma'\sigma} \text{ (spin-symmetric)}. \end{aligned} \tag{5.12}$$

For concreteness, onsite s-wave pairing, with $\Delta(\mathbf{k}) = \Delta$ constant, which comes from a local attractive interaction as in section 5.5, is even in $\mathbf{k}$, and so it is spin-singlet. This symmetry classification has a direct physical consequence. A spin-singlet superconductor is essentially insensitive to non-magnetic disorder, which is called Anderson's theorem, since the pair wave function only depends on time-reversed partners, and non-magnetic disorder preserves them. A spin-triplet superconductor, on the other hand, is strongly suppressed even by weak non-magnetic disorder, because scattering mixes the different momentum directions between which the order parameter needs to keep coherent phase relations. This is the reason why spin-singlet s-wave superconductors are much more common in nature than spin-triplet ones. Essentially every real material has some level of non-magnetic disorder, and this disorder destroys the triplet channel and leaves the singlet channel intact.

5.12 Building a spin-triplet superconductor out of a magnet

Since spin-triplet order is so fragile, how could we ever build one on purpose? One way is a toy model that has both a ferromagnetic exchange field and an attractive interaction in the same Hamiltonian,

$$H = -t \sum_{\langle ij \rangle, \sigma} c^\dagger_{i\sigma} c_{j\sigma} \underbrace{- h \sum_i \left(c^\dagger_{i\uparrow} c_{i\uparrow} - c^\dagger_{i\downarrow} c_{i\downarrow} \right)}_{\text{exchange field}} + \underbrace{U \sum_{\langle ij \rangle} c^\dagger_{i\sigma} c^\dagger_{j\sigma'} c_{j\sigma'} c_{i\sigma}}_{\text{attractive interaction}}, \qquad U < 0.$$

If we solve it self-consistently at the mean-field level, both order parameters, the ferromagnetic exchange field and the pairing amplitude, can be nonzero at the same time. The ground state then breaks both time-reversal symmetry, since it is magnetic, *and* gauge symmetry, since it is superconducting. The pairing that survives is necessarily spin-triplet, because the exchange field explicitly splits the $\uparrow$ and $\downarrow$ Fermi surfaces, and in this way it suppresses the singlet channel, which needs them to be degenerate. Physically, s-wave singlet pairing is usually the leading instability because it can gap the *entire* Fermi surface, which gives the largest condensation energy. When the exchange field suppresses this channel, a different pairing channel becomes the leading instability instead, which here is the odd-parity, spin-triplet channel, which is effectively p-wave.

The most interesting feature of this state appears when we put it on a finite ribbon instead of an infinite plane. The bulk spectrum is gapped, as we expect for a superconductor, but the edges of the ribbon have gapless excitations. A superconductor that is fully gapped in the bulk and gapless at the edge is a signature that we already know from the topological-band-structure and quantum-Hall sessions. Indeed, these edge modes have a topological origin, and they are tied to a finite Chern number of the underlying BdG band structure, in the same way as the edge states of a Chern insulator were tied to its Chern number.

Notebook check-in: `session5.ipynb`, "The impact of a spin splitting in an electronic structure" (Reminder), "A spin triplet superconductor" (Core), "Majorana modes in a spin triplet superconductor" (Challenge), "Interaction-induced spin-triplet superconductivity" (Challenge), and "Topological modes in interaction-driven spin triplet superconductors"

Start from the plain spin-split band structure, with only an exchange field, to build intuition. Then run the self-consistency loop for the model above and check that a spin-triplet Δ emerges. Move to a ribbon geometry and find the gapless edge modes described here, and then, in the later challenge sections of the notebook, connect them to the Majorana physics of sections 5.15-5.17.

5.13 Nodal singlet order, d-wave pairing in the cuprates and the angular-momentum expansion

Inside the spin-singlet, even-parity order there is a further useful distinction, since a singlet gap can be uniformly nonzero, as in the s-wave case of section 5.9, or it can be nodal. The standard example of the second case is **d-wave** order on the square lattice,

$$\Delta_d(\mathbf{k}) = \Delta_0 \left(\cos k_x - \cos k_y \right), \tag{5.13}$$

which vanishes exactly along the diagonals of the Brillouin zone, $k_x = \pm k_y$. Together with (5.9), this means that the quasiparticle spectrum $E(\mathbf{k}) = \sqrt{\varepsilon(\mathbf{k})^2 + \Delta_d(\mathbf{k})^2}$ is gapless where the Fermi surface crosses these diagonals, which gives a nodal singlet superconductor. This is the pairing symmetry of the cuprate high-temperature superconductors, which have critical temperatures up to about 130 K at ambient pressure, above the boiling point of liquid nitrogen, and are spin-singlet, but clearly *not* s-wave.

Behind this there is a clear and general way to organize the possible cases, which is analogous to the expansion of an atomic orbital wave function in spherical harmonics. We expand the superconducting gap in an angular-momentum basis on the Fermi surface,

$$\Delta(\mathbf{k}) \sim \sum_L \Delta_L Y_L(\mathbf{k}), \tag{5.14}$$

$L = 0$: s-wave (conventional, typically phonon-driven),

$L = 1$: p-wave (spin-triplet, often ferromagnetic-magnon-driven),

$L = 2$: d-wave (spin-singlet, often antiferromagnetic-magnon-driven; cuprates).

In real and reciprocal space, s-wave order is flat and constant, with no momentum dependence at all. The $d_{x^2-y^2}$ order of (5.13) changes sign around the Fermi surface in a pattern of four lobes, $+, -, +, -$, with the same name and the same lobe structure as an atomic d orbital. There is also a third possibility, which is called *extended s-wave*. It is still $L = 0$, meaning that it has the same overall symmetry as plain s-wave under the lattice point group, but it is nodal along the *radial* direction, since it is positive near the zone center Γ, goes through a node and is negative further out. Figure 5.3 shows these sign structures over the Brillouin zone, together with the p- and f-wave cases below.

The lecture repeats the same classification into three cases, uniform s-wave, a nodal order and extended s-wave, on the triangular lattice. There the nodal case has six lobes of alternating sign around the Fermi surface, and it is an **f-wave order,** $L = 3$. The instructor confirmed this, and the "L equal to two" in the raw recording was a mistake, since a pattern of six lobes with alternating sign is the standard signature of $L = 3$ and not of $L = 2$, which is already used above for the four-lobed $d_{x^2-y^2}$ state of the square lattice. Note that this order is outside the singlet family of this section. Six lobes with alternating sign mean that the gap changes sign under a half turn, so $\Delta(-\mathbf{k}) = -\Delta(\mathbf{k})$, and by (5.12) an $L = 3$ order has odd parity and is therefore necessarily *spin-triplet*, and not singlet like the s-, extended-s- and d-wave cases above.

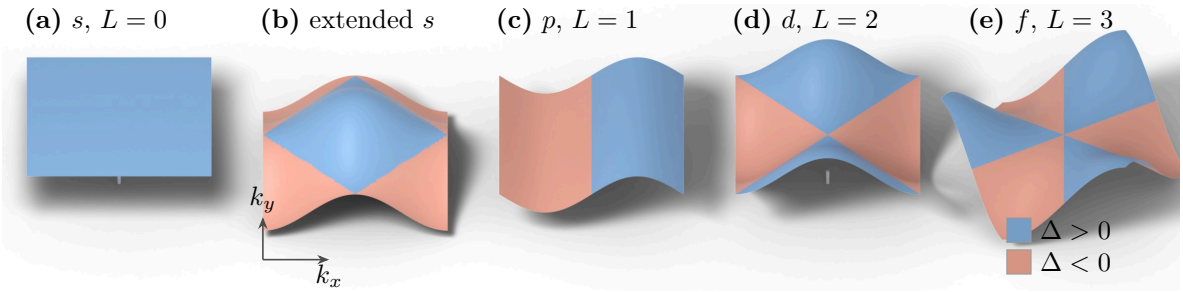


Figure 5.3: The sign structure of $\Delta(\mathbf{k})$ over the Brillouin zone, ordered by angular momentum L as in (5.14). The positive regions are raised and the negative ones are sunk, in two colors, and the nodes are where the two colors meet. A reflection of a panel through its centre sends $\mathbf{k} \to -\mathbf{k}$. Panels with even L do not change and are therefore spin singlets, and panels with odd L come back with every sign flipped and are spin triplets, which is (5.12) seen in a picture. Counting lobes is the fastest way to identify the order. Two lobes mean $L = 1$, four mean $L = 2$ and six mean $L = 3$, and any pattern whose lobes alternate an odd number of times in a half turn is necessarily odd in $\mathbf{k}$, so it is a triplet. This is why the f-wave state (e), which has six lobes on the hexagonal zone of the triangular lattice, is a spin triplet, even though it is similar to the singlet d-wave state in every other respect. Extended s-wave (b) deserves a second look, since its node is radial and not angular. It has the same even parity and the same point-group symmetry as plain s-wave, and only the position of its node ring relative to the Fermi surface decides whether it is nodal in the spectrum. The overall sign of Δ is a gauge choice, and only the sign *changes* are physical.

Notebook check-in: `session5.ipynb`, "Gapped and nodal superconducting orders" (Core)

Implement the $d_{x^2-y^2}$ gap of (5.13) on the square lattice, plot $\Delta(\mathbf{k})$ over the Brillouin zone, and check the pattern of four lobes with alternating sign and the position of its nodes relative to a given Fermi surface. The nodal $L = 3$ (f-wave) case of the triangular lattice discussed above is also in this section of the notebook, if you want to see the pattern of six lobes directly.

5.14 Same gap and different topology in trivial s-wave and chiral p-wave pairing

This is the central point of the whole session. Two superconducting gaps can have completely identical *spectral* properties, where both fully gap the Fermi surface with, for example, the same minimum gap, and still be topologically different. Let us compare plain s-wave pairing, $\Delta(\mathbf{k}) = \Delta_0$, with **chiral p-wave** pairing,

$$\Delta_{p+ip}(\mathbf{k}) = \Delta_0 \left(\sin k_x + i \sin k_y\right), \tag{5.15}$$

which is an odd-parity, spin-triplet order, in agreement with (5.12) since $\Delta_{p+ip}(-\mathbf{k}) = -\Delta_{p+ip}(\mathbf{k})$, and which winds once around the origin of the complex plane as $\mathbf{k}$ goes around the Fermi surface. Both (5.9) with Δ_0 and (5.9) with Δ_{p+ip} can fully gap the same Fermi surface, so a measurement of the bulk gap alone cannot distinguish them. However, in the same way as for the Chern-number classification of gapped bands in the topological-band-structure session, we can compute the Chern number of the occupied BdG bands for each case, and the two results are completely different. The trivial s-wave state has Chern number 0, while the chiral p-wave state has Chern number ± 2, where the sign is set by the chirality, $p_x + ip_y$ or $p_x - ip_y$.

What distinguishes them, as shown in figure 5.4, is the winding of the *phase* of $\Delta(\mathbf{k})$ when $\mathbf{k}$ goes once around the Fermi surface. This phase is constant for s-wave, and it makes one full turn for $p + ip$.

This has the consequence that we expect from the bulk-boundary correspondence. In a finite system, the s-wave superconductor stays gapped everywhere, both in the bulk and at the edge, in the same way as a trivial insulator with Chern number zero. The chiral p-wave superconductor, on the other hand, is gapped in the bulk but has gapless *chiral* edge modes. On one edge there are $|C|$ branches that all propagate in the same direction, here two, one from each of the two identical spin blocks, and the opposite edge has the same number of branches that propagate in the opposite direction, as a state with Chern number ± 2 must have. So on a ribbon we see two sets of branches that propagate in opposite directions, but they are localized on opposite edges and do not share one edge. These edge modes turn out to be fractions of an electron in a very specific and exact sense, since they are Majorana modes, and the rest of this session is about what this means.

Notebook check-in: `session5.ipynb`, "Trivial and topological superconducting order parameters, bulk spectra" (Core), and "...edge spectra" (Core)

Build the s-wave and chiral p-wave BdG Hamiltonians next to each other, and check that their bulk spectra look similar, since both are gapped with a comparable gap size. Then move to a ribbon geometry and check that only the p-wave case has gapless chiral edge states, with branches of opposite velocity on the two edges of the ribbon, which is the distinction driven by the Chern number of section 5.14.

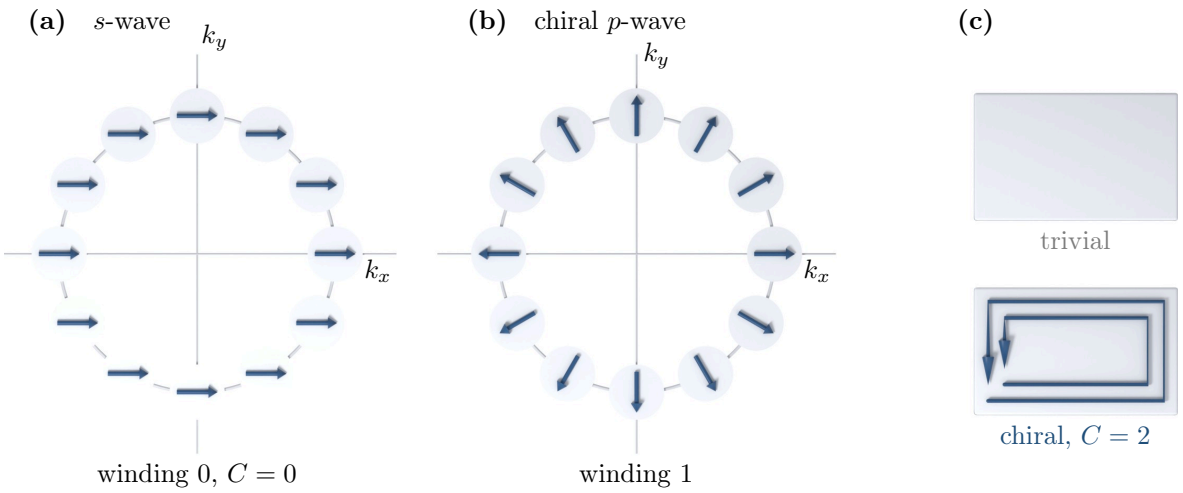


Figure 5.4: Two gaps that a spectroscopy experiment cannot distinguish, seen from above. Each dial is the phase of $\Delta(\mathbf{k})$, an arrow in the complex plane, drawn at the point of the Fermi pocket where it is evaluated. (a) For s-wave pairing this phase is the same everywhere, so as $\mathbf{k}$ goes once around the pocket the value of Δ stays at a single point of the complex plane, with winding number zero and Chern number zero. (b) For chiral p-wave pairing (5.15) the phase instead follows the direction of $\mathbf{k}$, so Δ draws a loop that encircles the origin once. The modulus is essentially the same in both cases, so both fully gap the same pocket by the same amount. Only the winding is different, and the winding is what the Chern number counts. (c) The consequence in a finite sample. The trivial state is gapped at its boundary as well as in its bulk, while the chiral state must have $|C|$ gapless branches on each edge, all going in the same direction, and in the opposite direction on the opposite edge. Here $C = \pm 2$, with one branch from each of the two identical spin blocks of (5.8).

5.15 Majorana bound states and how to split an electron into two Majoranas

A **Majorana fermion** is defined algebraically by the condition that its field operator is equal to its own Hermitian conjugate,

$$\gamma^\dagger = \gamma, \tag{5.16}$$

which means that the creation and the annihilation of the quasiparticle are the same operation, so the particle is its own antiparticle. At first this seems to have no place in an ordinary material, since an ordinary electron is clearly not its own antiparticle ($c^\dagger \neq c$), and a semiconductor, for example, never hosts anything of this kind.

However, there is a purely algebraic fact that makes it possible. *Any* ordinary fermionic operator can be written as a combination of two Hermitian Majorana operators that anticommute with each other,

$$\gamma_A = c + c^\dagger, \qquad \gamma_B = -i\left(c - c^\dagger\right), \tag{5.17}$$

so that $\gamma_A^\dagger = \gamma_A$, $\gamma_B^\dagger = \gamma_B$, $\{\gamma_A, \gamma_B\} = 0$ and $\gamma_A^2 = \gamma_B^2 = 1$, where we used $\{c, c^\dagger\} = 1$. If we invert (5.17), we obtain

$$c = \frac{1}{2}\left(\gamma_A + i\gamma_B\right), \qquad c^\dagger = \frac{1}{2}\left(\gamma_A - i\gamma_B\right). \tag{5.18}$$

At first this is only a change of variables, a mathematical trick without new physics. Every electron is *always* made of two Majorana operators that are tightly bound together into an ordinary fermion, and normally there is no way to observe γ_A or γ_B separately, only their fixed

combination c. Superconductors are special because BdG eigenstates are already mixtures of electrons and holes by construction, since the Nambu spinor of section 5.8 explicitly combines $c_{\mathbf{k}\sigma}$ with $c^\dagger_{-\mathbf{k}\sigma'}$. So a superconductor is the kind of system where, with the right pairing symmetry and the right geometry, the two Majorana components of an electron can become spatially *separated*, or deconfined, instead of staying together at one site. This deconfinement is the topic of the rest of this section.

5.16 The Kitaev chain as the minimal model with isolated Majoranas

The simplest superconductor that can host isolated Majoranas is also one of the simplest superconducting Hamiltonians that we can write, which is a one-dimensional *spinless* chain with nearest-neighbor hopping and nearest-neighbor pairing,

$$H_{\text{Kitaev}} = -t \sum_n \left(c_n^\dagger c_{n+1} + \text{h.c.} \right) - \mu \sum_n c_n^\dagger c_n + \Delta \sum_n \left(c_n c_{n+1} + c_{n+1}^\dagger c_n^\dagger \right). \tag{5.19}$$

Here the pairing has to be on a bond and not onsite. For spinless fermions, an onsite term $c_n c_n$ vanishes identically because of the Pauli principle ($c_n^2 = 0$), so the simplest allowed pairing is the nearest-neighbor term above. After a Fourier transform, the dispersion and the pairing become

$$\varepsilon(k) = -2t \cos k - \mu, \qquad \Delta(k) = 2i\Delta \sin k, \tag{5.20}$$

so the anomalous term is explicitly odd in k. As (5.12) already tells us, this means that this is necessarily a superconductor of spin-triplet type, here a spinless p-wave one, which agrees with the fact that the onsite channel is forbidden by the Pauli principle.

Now let us set $\mu = 0$ and go to the special point $\Delta = t$, and let us substitute the Majorana decomposition (5.17)-(5.18) at every site, $c_n = \frac{1}{2}(\gamma_{2n-1} + i\gamma_{2n})$. We do this substitution in both the hopping and the pairing term. The algebra is direct but a bit long, and it uses the anticommutation relations of the γ's, so we suggest that you check it yourself. The two terms combine so that every $\gamma_{2n-1}\gamma_{2n+2}$ contribution cancels exactly between hopping and pairing, and the result is very simple,

$$H_{\text{Kitaev}}\Big|_{\Delta=t,\ \mu=0} = it \sum_{n=1}^{N-1} \gamma_{2n}\, \gamma_{2n+1}. \tag{5.21}$$

Every Majorana operator $\gamma_2, \gamma_3, \gamma_4, \ldots, \gamma_{2N-1}$ appears in this sum, paired between neighboring *sites*, but the first Majorana, γ_1, and the last one, γ_{2N}, do not appear at all. They are completely absent from the Hamiltonian, so they cost zero energy and are completely decoupled from the rest of the system. This is the same structure that we saw in the SSH model of the topological-band-structure session, in its dimerized strong-coupling limit. There, at full dimerization, the bulk forms bonding and antibonding dimers, while one remaining site at each end decouples and gives a zero-energy edge mode. The Kitaev chain in (5.21) *is* this same structure, written in terms of Majorana operators instead of ordinary fermion operators, and the two decoupled end Majoranas play the role of the two decoupled end sites of the SSH model. We do not derive the SSH model again here, and we only recognize it in terms of a new set of operators. Figure 5.5 shows which Majorana is bound to which, in this phase and in the trivial one.

trivial, binding inside each site



topological, binding between sites



Figure 5.5: The Kitaev chain in terms of Majorana variables, for $N = 5$ sites (so $\gamma_{2N} = \gamma_{10}$), seen from above. Each site carries two balls, its two Majorana halves, and a translucent capsule binds two halves together. Splitting every electron into its two Majorana halves (5.17) has no physical content by itself, and what matters is which halves the Hamiltonian binds together. In the trivial phase, at the top, the binding is inside each site. The two halves of each electron stay bound to each other, as they were in the vacuum, and every mode costs energy. In the topological phase, at the bottom, the binding is between sites, γ_{2n} with γ_{2n+1}, and the pattern has one bond missing at each end. So γ_1 and γ_{2N} do not appear in (5.21), cost no energy and are at opposite ends of the sample. The ordinary fermion that they would form together has been split into two halves, which are now separated by the length of the chain, and this is what *deconfined* means here. This is the fully dimerized SSH chain of the topological-band-structure session with its two remaining end sites, with new labels, so the counting is the same and only the operators are new.

Notebook check-in: `session5.ipynb`, "Emergence of Majorana modes as a function of the size of the superconductor" (Core)

Build the Kitaev chain of (5.19), diagonalize it at the fine-tuned point $\Delta = t$, $\mu = 0$, and check directly, as the system size grows, that two states are pinned exactly to zero energy and are localized at the two ends. These are the two decoupled Majoranas γ_1 and γ_{2N} predicted by (5.21).

5.17 Topological zero modes with periodic and open boundaries

If we diagonalize (5.19) on a *periodic* chain, we obtain a fully gapped spectrum, with a superconducting gap set by Δ and a nontrivial topological invariant of the bulk. This is the same kind of $\mathbb{Z}_2$ or winding invariant that classified the two phases of the SSH model through the Zak phase in the topological-band-structure session, now computed from the BdG bands instead of an ordinary Bloch Hamiltonian. On an *open* chain, for example with $N = 100$ sites and the two ends disconnected, the bulk spectrum keeps essentially the same gap, but two extra states appear, pinned exactly at zero energy and localized at the left end and at the right end of the chain. These are the unpaired Majoranas γ_1 and γ_{2N} from (5.21), which we now see as real mid-gap states of the open chain.

These zero modes are very robust. When we move away from the fine-tuned point, with $\Delta \neq t$ and finite μ, the zero modes survive as long as we stay in the same topological phase. The standard criterion for the general Kitaev chain (5.19) with $\Delta \neq 0$ is

$$|\mu| < 2t \implies \text{topological phase (unpaired end Majoranas)}, \qquad |\mu| > 2t \implies \text{trivial phase}. \tag{5.22}$$

In the trivial phase, a large chemical potential effectively fixes each site to be either empty or occupied in the fermionic language, since the chain is spinless, and the two Majoranas that form the electron of each site combine again locally into an ordinary confined fermion, so nothing is left to deconfine. In the topological phase, on the other hand, p-wave pairing dominates over the trivial onsite physics, and the two Majoranas at each site pair with the Majoranas of the neighboring sites instead of with each other, which leaves the two end Majoranas without any partner. This is the same contrast between the dimerized strong-coupling limit and the atomic weak-coupling limit that distinguished the topological and trivial phases of the SSH model. The Kitaev chain reproduces it because, as we saw in section 5.16, it is the SSH model with new labels.

5.18 Engineering a topological superconductor from a quantum spin Hall insulator

Chiral and p-wave superconductors of the kind we need are rare in nature, so in practice they are usually *engineered* and not found. One instructive recipe uses directly the quantum spin Hall (QSH) insulator from the topological-band-structure session. We take a QSH insulator, and the concrete material used in the lecture is tungsten ditelluride (WTe_2), whose edge has helical modes, meaning that spin-up electrons propagate in one direction and spin-down electrons propagate in the other direction on the same edge. Now we place ordinary ferromagnets on two segments of this edge. A magnet breaks time-reversal symmetry locally, and this opens a gap in the helical edge modes below it. We recall from the topological-band-structure session that the protection of the QSH edge relies on time-reversal symmetry through Kramers degeneracy, so if we break this symmetry locally, the protection is lost there. What remains without a gap is a short isolated segment of helical edge between the two magnetic regions.

This isolated segment only has spin-up electrons moving right and spin-down electrons moving left, with no spin-down electrons moving right and no spin-up electrons moving left. If we count states only by their direction of propagation, this segment is *effectively spinless*, which is the ingredient that the Kitaev-chain construction of section 5.16 needs. If we then couple this segment by proximity to an ordinary s-wave superconductor, we reproduce on this effectively spinless one-dimensional system the physics of (5.19), which is a one-dimensional p-wave superconductor with Majorana zero modes pinned at the two ends of the isolated segment. Figure 5.6 shows the whole setup.

Notebook check-in: `session5.ipynb`, "Artificial topological superconductivity" (Challenge), "Edge states of an artificial topological superconductor" (Challenge), and "Topological phase transition as a function of the exchange field"

Build the construction described above, with the QSH edge, the magnets and superconductivity, and check that zero modes localized at the ends emerge as you change the strength of the exchange field. Look at how the system crosses the boundary between the topological and the trivial phase, of the type of (5.22), as the magnetic gap changes.

5.19 Topological quantum computing with Majoranas

Majorana modes get so much attention, beyond their own unusual properties, because of their proposed use in fault-tolerant quantum computing. Majoranas are **non-Abelian anyons**. Exchanging, or braiding, one Majorana around another does not only multiply the many-body ground state by a phase. It acts as a unitary matrix on a degenerate ground-state manifold, so the *order* in which different braiding operations are done matters, which means that the

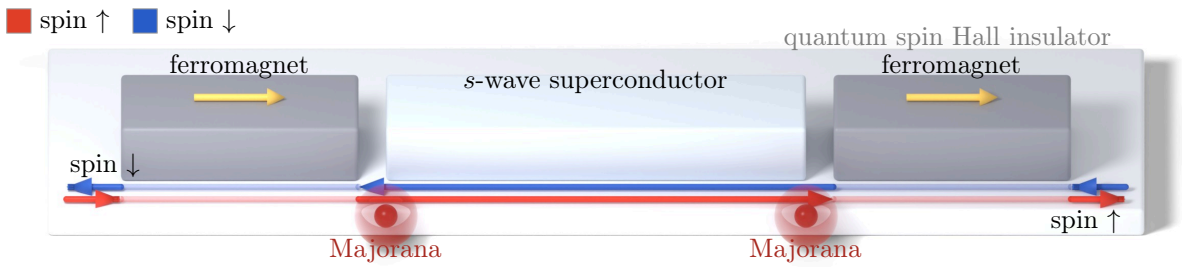


Figure 5.6: How to build the Kitaev chain from parts that exist. The edge of a quantum spin Hall insulator is helical, which means that on the same edge spin-up electrons move in one direction and spin-down electrons in the other, drawn in a warm and a cold color. A ferromagnet placed on a part of this edge breaks time-reversal symmetry locally and removes the Kramers protection there, which opens a gap in the edge below it, where the channels are drawn faded. With two ferromagnets, one short helical segment survives between them. This segment has only one direction of propagation for each spin, so if we count channels it is a *spinless* one-dimensional wire. This is the ingredient that (5.19) needs, and nature does not give it to us directly. Coupling it by proximity to an ordinary s-wave superconductor gives the pairing, and the result is a one-dimensional p-wave superconductor with a Majorana zero mode at each end of the segment, which means at the two ferromagnet-superconductor interfaces. Note what was *not* needed. No unconventional superconductor appears in the recipe, only an s-wave one, a magnet and a topological insulator.

algebra does not commute, and this is what is called non-Abelian. This follows directly from the structure of the BdG Hamiltonian when more than two Majoranas are present.

For concreteness, four Majorana operators $\gamma_1, \gamma_2, \gamma_3, \gamma_4$ can be combined again, with pairings of the type of (5.18), into two ordinary fermionic operators, which we can call artificial fermions, so a system with four Majoranas encodes two ordinary fermionic modes. The resulting many-body ground-state manifold, with the states empty, one mode filled, the other mode filled and both filled, is four-fold degenerate. All four states cost zero energy, by the same argument that made γ_1 and γ_{2N} zero-energy modes in the Kitaev chain, and this degenerate manifold is the proposed computational basis for a topological qubit.

Braiding two Majoranas, for example γ_1 around γ_2, acts on the ground-state manifold as a unitary

$$U_{12} = \exp\left(\frac{\pi}{4}\,\gamma_1\gamma_2\right), \tag{5.23}$$

which multiplies states by a relative phase built from $e^{i\pi/4}$ times the Majorana product. The exact form is less important than what it implies. Two exchanges that share a Majorana do not commute. Braiding the pair $(1,2)$ and then the pair $(2,3)$ gives a different final state than braiding $(2,3)$ first and $(1,2)$ second, since $\gamma_1\gamma_2$ and $\gamma_2\gamma_3$ anticommute. Exchanges of disjoint pairs, such as $(1,2)$ and $(3,4)$, commute instead. This braid algebra that does not commute is what allows sequences of purely topological and geometric moves, which physically means moving Majoranas around one another, to implement non-trivial quantum logic gates. Since the resulting qubit is stored non-locally, divided between spatially separated Majoranas instead of being in one local degree of freedom, it is intrinsically resistant to local decoherence and to weak local disorder, and this robustness is the main reason for the interest in topological quantum computing. This material on braiding and quantum computing appears only in the slides and the transcript and has no counterpart in `session5.ipynb`. None of the sections of the notebook implement braiding dynamics or a qubit manifold, which is reasonable, since this is far from a static tight-binding diagonalization and it is a separate dynamical calculation.

5.20 Closing summary

- **Superconductivity is a mean-field, attractive-interaction phenomenon.** Starting from an attractive Hubbard chain (5.2), pairing mean-field theory gives the order parameter $\Delta_n = U\langle c_{n\downarrow} c_{n\uparrow}\rangle$ (5.3) and the mean-field Hamiltonian (5.4).

- **Pairing breaks particle-number conservation down to parity** (section 5.6), and this is the same as the statement that gauge symmetry is spontaneously broken. The pairing term gets a phase $e^{-2i\varphi}$ under the gauge transformation (5.5), unlike ordinary bilinear terms.

- **Flux quantization**, $\Phi = n\,h/2e$ (5.1), is a direct and measurable macroscopic consequence of this broken gauge symmetry, at the same level as the conductance quantum of the quantum Hall effect from earlier sessions.

- **The Nambu spinor (5.6) and the Bogoliubov-de Gennes Hamiltonian (5.7)-(5.8)** are the tools that allow us to diagonalize any mean-field superconducting Hamiltonian. They give the quasiparticle dispersion $E(\mathbf{k}) = \pm\sqrt{\varepsilon(\mathbf{k})^2 + |\Delta(\mathbf{k})|^2}$ (5.9), and a gap opens at the original Fermi surface regardless of the details of the microscopic dispersion.

- **The symmetry of $\Delta(\mathbf{k})$ classifies the superconductor.** It can be singlet, even in $\mathbf{k}$, or triplet, odd in $\mathbf{k}$, as in (5.11)-(5.12), with a further classification by angular momentum ($L = 0, 1, 2, \dots$) (5.14) that gives s-, p- and d-wave order. The last one is realized in the cuprate high-T_c superconductors through the nodal $d_{x^2-y^2}$ gap (5.13).

- **Gaps with the same spectrum can have different topology.** Trivial s-wave ($C = 0$) and chiral p-wave ($C = \pm 2$, eq. 5.15) pairing both fully gap the same Fermi surface, but only the topological one has gapless edge modes, which is the bulk-boundary logic of the Chern number from the topological-band-structure and quantum-Hall sessions.

- **The Kitaev chain (5.19) is the SSH model in Majorana language.** At the fine-tuned point it reduces to $H = it\sum_n \gamma_{2n}\gamma_{2n+1}$ (5.21), which leaves two unpaired end Majoranas with zero energy. These survive away from the fine-tuned point in the whole topological phase $|\mu| < 2t$ (5.22).

- **Majoranas are non-Abelian anyons.** Braiding them acts as a unitary (5.23) on a topologically degenerate ground-state manifold, and exchanges that share a Majorana do not commute, which is the basis of the proposed schemes for topological quantum computing. This last part is material from the slides and the transcript, with no counterpart in the notebook.

In this session we followed the topic of superconductivity from session 1, from the bare pairing order parameter to Nambu spinors, Bogoliubov-de Gennes Hamiltonians, the singlet and triplet classification and finally Majorana zero modes in the Kitaev chain. The division that we made in session 1 had a second branch that we have not considered since its first short treatment, which is magnetism, where time-reversal symmetry is broken instead of gauge symmetry. This is where we go next. Session 6 comes back to the same Hubbard Hamiltonian that opened section 5.5 of this session, now on its repulsive ($U > 0$) side, and gives magnetism the same depth of treatment that we have now given to superconductivity.

Session 6

Classical Magnetism and Magnons

The last session ended with a sentence that we repeat here, because it is also the sentence that starts this lecture. The same Hubbard-like Hamiltonian can give, at the mean-field level, either a superconductor or a magnet, and magnetism was left for the next recording. This session is that next recording. We already did a first mean-field treatment of magnetism in session 1, with the Hubbard chain, the magnetization m and the Mott gap. That treatment was only an introduction on purpose. It was one change of sign away from the superconducting case, and we only went far enough to show that magnetism *is* a symmetry-broken mean-field state. In this session we go much further. First, we will solve the magnetism of the Hubbard model numerically, and we will see that ferromagnetism and antiferromagnetism compete as two different self-consistent solutions of the same Hamiltonian. Then we will find the exact value of the interaction above which magnetism appears. This threshold is in the same regime where mean-field theory stops being reliable, so after that we will leave mean field and derive from first principles the effective spin Hamiltonian of the strongly correlated limit, which is the Heisenberg model. After this we go one step further and ask about the *excitations* on top of a magnetically ordered state. This leads us to magnons, which are the magnetic analogue of a phonon, and we will construct them exactly with the Holstein-Primakoff transformation. Once magnons have a real band structure, we will see that the topological methods of sessions 3 and 4, namely Chern numbers, Berry curvature and chiral edge states, also apply to them. This gives us topological magnon bands, which is a third application of a tool that we have now used twice on electrons and once on magnons.

Before we start, we comment on how the session is organized, because this explains how it ends. This recording is one self-contained lecture on classical magnetism and magnons only. It does not include a second recording on quantum magnetism and the Kondo model, even if the preview of spinons below may give that impression. That material belongs to its own session later in the course. We should also say clearly that this recording stops when it reaches the end of the "topological magnons" material. There is no summary slide narrated at the end, and the lecture ends when the planned material ends. For this reason these notes also end with topological magnons, and we do not add a final sentence that the lecture itself does not say.

6.1 Two classes of magnets and their excitations

Let us start with a map of where we are going, since this lecture first previews material from two later sessions and only then moves to its own topic. There are two big classes of magnetic systems.

The first class contains the magnets that break time-reversal symmetry. These materials have a net magnetization on every site, or a staggered one, and a finite spin angular momentum

is what makes a system look different when we play its "movie" backwards, which is the definition of broken time-reversal symmetry. Ferromagnets, antiferromagnets and noncollinear spin spirals are in this class. Iron and cobalt are the two materials that the lecture names, several times, as concrete examples of this family. In this session we spend all our time on this class.

The second class contains the magnets that do *not* break time-reversal symmetry, at least not classically. These are quantum magnets, and quantum spin liquids are among them. They have a local moment on every site, but the spins are in a quantum superposition and do not form a static classical arrangement, so time-reversal symmetry does not need to be broken overall. A related case, which is also mentioned only briefly in this lecture, is the heavy-fermion Kondo system. It is a combination of local moments in a quantum superposition with a sea of itinerant metallic electrons. The "Classical and quantum magnetism" slide shows the three families with photographs of materials, which are iron for the ferromagnet and antiferromagnet family, herbertsmithite for quantum spin liquids and samarium hexaboride SmB_6 for heavy-fermion insulators. None of these three compound names is said in this year's narration. They are only labels on the slide, which are good to know but which the lecture does not say out loud.

Each class has its own emergent excitation. Magnets that break time-reversal symmetry have **magnons**, which are spin-1 and charge-neutral fluctuations of the order parameter, and they are the direct magnetic analogue of a phonon. Magnons have no charge, so electric fields do not affect them at all, and they can propagate in materials that are electronic insulators. A ferromagnetic or antiferromagnetic Mott insulator does not let any current through, but a magnon can still carry spin and heat across it. Quantum magnets have instead **spinons**, which are spin-1/2 and charge-neutral excitations. They are sometimes described informally as "half an electron," since the spin and the charge degrees of freedom of the electron have in practice separated. In the rest of this session we will construct magnons in detail. Spinons and the Kondo lattice model only appear here as a name and a spin quantum number. Their full construction is the material of a later session, session 8, and this lecture does not come back to them.

6.2 A recap of symmetry breaking and the mean-field philosophy

We have already used this idea twice in this course, first in a small form in session 1 and then in full detail for superconductivity in session 5. Let us state it once more, in a short way, before we apply it to magnetism. Many-body interactions can spontaneously break a symmetry that the Hamiltonian itself has. Three cases appear in the whole course. Magnets break time-reversal symmetry, and they also break spin-rotation symmetry, since a magnetic moment breaks both. Charge density waves break the translational symmetry of the lattice. Superconductors break gauge symmetry. All three emerge from the same generic starting point, which is a kinetic term that is single-particle and exactly solvable, plus a quartic many-body interaction. All three are also obtained in the same way. Mean-field theory factorizes the quartic term by replacing part of it with its expectation value, which turns a many-body problem that we cannot solve into a single-particle problem that we can solve, and we solve it self-consistently.

Where do the interactions themselves come from? There are two sources. The electrostatic, or Coulomb, repulsion between electrons is the classical contribution, and it is always present. Interactions can also be mediated by other quasiparticles, such as phonons, magnons or plasmons, and these can change the *net* effective interaction from repulsive to attractive. Magnetism needs a net interaction that is repulsive. The bare Coulomb repulsion is already

repulsive, so this case is easy to realize, which is one of the reasons why magnetism is so common in nature.

6.3 The Hubbard model and mean-field magnetism, revisited

The minimal Hamiltonian for all of this is the one-dimensional Hubbard chain that we first saw in session 1,

$$H = -t \sum_{n,\sigma} \left(c^\dagger_{n\sigma} c_{n+1,\sigma} + \text{h.c.} \right) + U \sum_n c^\dagger_{n\uparrow} c_{n\uparrow}\, c^\dagger_{n\downarrow} c_{n\downarrow}, \tag{6.1}$$

which is the hopping t plus a purely local onsite density-density interaction between opposite spins, called the Hubbard term. A negative U, which is attractive, gives rise to superconductivity, which was the topic of session 5. A positive U, which is repulsive, gives rise to magnetism, which is our topic in this session.

The mean-field decoupling is done in the same way as before. We replace $c^\dagger_{n\uparrow} c_{n\uparrow}$ and $c^\dagger_{n\downarrow} c_{n\downarrow}$ by their expectation values, and we obtain a bilinear Hamiltonian with onsite energies that depend on the spin. Nothing forces these two expectation values to be equal, and the difference between them is the magnetization,

$$m = \langle c^\dagger_\uparrow c_\uparrow \rangle - \langle c^\dagger_\downarrow c_\downarrow \rangle, \tag{6.2}$$

which plays the role of an exchange field. In a schematic form, the resulting mean-field Hamiltonian is $H_{\text{MF}} \sim -\frac{U}{2}\, m\, \boldsymbol{\sigma} \cdot c$, where $\boldsymbol{\sigma}$ is the vector of Pauli matrices. Self-consistency gives $m = 0$ for $U < 0$ and a magnetic solution with $m \neq 0$ for $U > 0$, as in session 1.

There is a more careful point here, which the slides show but the spoken explanation only mentions. Nothing in this construction chooses the z axis. For a single site, the Hubbard interaction can be rewritten exactly in terms of the local spin operator $\mathbf{S} = \frac{1}{2} c^\dagger_\alpha \boldsymbol{\sigma}_{\alpha\beta} c_\beta$ and the total occupation $n = n_\uparrow + n_\downarrow$, as $U\, n_\uparrow n_\downarrow = \frac{U}{2} n - \frac{2U}{3} \mathbf{S}^2$, where we used that a single site has $\mathbf{S}^2 = \frac{3}{4}(n - 2 n_\uparrow n_\downarrow)$, since only a singly occupied site has a spin. Mean-field theory then factorizes the quadratic operator $\mathbf{S}^2$ in the same way as any product of two operators, $\mathbf{S}^2 \approx 2\langle \mathbf{S} \rangle \cdot \mathbf{S} - \langle \mathbf{S} \rangle^2$, and the resulting exchange field points along $\mathbf{m} \equiv \langle \mathbf{S} \rangle$, in *any* direction where this expectation value is nonzero. If we choose $\langle c^\dagger_\uparrow c_\uparrow \rangle \neq \langle c^\dagger_\downarrow c_\downarrow \rangle$, as in (6.2), we select $m_z \neq 0$. If we choose instead $\langle c^\dagger_\uparrow c_\downarrow \rangle \neq 0$, we select m_x or m_y. For this reason mean-field magnetism is not restricted to the z axis. It gives magnets with a magnetization along any direction in space, and this includes noncollinear arrangements where $\mathbf{m}$ points in different directions on different sites. This is the "non-collinear mean-field Hamiltonian" figure on the slides, which the spoken lecture only names briefly.

Mean field can only be trusted when $U \ll t$. We will see in Section 6.6 that magnetism usually needs the opposite regime, and this is the reason why the strongly localized limit, from Section 6.7 on, is so important.

6.4 Exchange splitting and antiferromagnetic order in numerical mean-field theory

Let us now see the method of Section 6.3 in a real calculation. First we take the one-dimensional chain (6.1) with a unit cell of one site at 20% filling, and we solve it self-consistently. The resulting band structure shows an exchange splitting. The spin-up band, which is occupied up to the chemical potential, is at lower energy than the spin-down band, which is empty at this filling. The splitting between them is called the **exchange splitting**,

and it grows with U. Only spin-up states are occupied, so this is a fully polarized ferromagnetic solution with $m \neq 0$.

Now we take a unit cell with two sites, and we work at half filling. At $U = 0$ there are two bands, one per site in the cell, and they cross at one point, so the system is metallic. When we turn on $U > 0$ a mean-field gap opens, and in real space the two sublattices get opposite spin polarization, with one site mostly up and its neighbor mostly down. The *net* magnetization is zero, but the *staggered* magnetization is not, and this is antiferromagnetic order. As always when a mean-field gap opens, it lowers the total electronic energy with respect to the state without a gap, because the states that the gap pushes down in energy are the occupied ones.

Notebook check-in: `session6.ipynb`, "(Reminder) Bandstructure of a tight binding chain with two sites" and "(Exercise) Magnetic order driven by interactions in a chain"

Start from the band structure of the two-site chain without interactions, which is the metallic case with a crossing. Then turn on U and see how the mean-field gap opens and how the spin pattern in real space becomes antiferromagnetic, which reproduces the exchange splitting of (6.2) directly. The ferromagnetic case with one site per cell at 20% filling, from the same section, is the companion calculation.

6.5 Competition between the ferromagnet and the antiferromagnet

The Hubbard chain (6.1) has *both* a uniform self-consistent solution, with one site per cell and without the gap of the doubled cell, which is the paramagnet below $U \approx 4t$ and a fully polarized ferromagnet above it, where the spin-up band is full and the spin-down band is empty, and an antiferromagnetic one, with two sites per cell, insulating and gapped, at the same filling and the same U. Both are real fixed points of the self-consistency loop. Only one of them can be the true ground state, and we decide which one by comparing total energies. Here the gapped antiferromagnetic solution always has the lower energy. Opening a gap pushes the occupied single-particle energies down, so the insulating configuration has a lower total energy than the gapless ferromagnetic one for every $U > 0$ that we can reach numerically in this half-filled chain.

Let us follow how large this energy difference is as a function of U. At $U = 0$ the two configurations are the same, so the energy difference is zero. When U grows from zero the gap opens and the energy difference grows with it. However, when U keeps growing, the energy difference starts to *decrease* again, which is not what one would expect. In the limit $U \to \infty$, which is the same as the limit where the kinetic energy can be neglected, the two configurations become degenerate. Numerically, this energy difference follows a $1/U$ form. In Section 6.8 we will derive this exact $1/U$ scaling analytically from a completely independent calculation, which is a real cross-check between the mean-field numerics and a many-body perturbative treatment.

6.6 The Stoner threshold as a criterion for magnetism

A natural question is whether *any* repulsive $U > 0$ gives magnetism, or if there is a threshold. We write the mean-field bands as $E_\sigma(\mathbf{k}) = E(\mathbf{k}) - \sigma \frac{U}{2} m$ for $\sigma = \pm 1$, meaning spin up and down, where m is the self-consistent magnetization of (6.2). At zero temperature, the self-consistency condition is

$$m = \sum_{\mathbf{k}} \Big[n_F\big(E(\mathbf{k}) - \tfrac{U}{2}m\big) - n_F\big(E(\mathbf{k}) + \tfrac{U}{2}m\big) \Big].$$

If we expand the right-hand side for small m we obtain $m \approx U\, m\, D(E_F)$, where $D(E_F)$ is the density of states at the Fermi level. The trivial solution $m = 0$ always exists, but a *nontrivial* solution appears when the linear coefficient reaches one, which gives the condition

$$U\, D(E_F) \geq 1. \tag{6.3}$$

This condition is called the Stoner criterion. Below the threshold only $m = 0$ is stable, and the system stays nonmagnetic for any perturbation. Above the threshold the nonmagnetic solution becomes unstable, and a finite m appears spontaneously. This is the reason why some metals with a sizable Coulomb repulsion, such as aluminum or lead, stay nonmagnetic, since their density of states at the Fermi level is too small for the product in (6.3) to reach one. Iron and cobalt, on the other hand, have a large density of states from their narrow d bands together with a sizable effective U, so they are clearly above the threshold. In the numerics we see what (6.3) predicts. The self-consistent magnetization of the chain stays at zero below a critical U, and it grows continuously once U is above this threshold. Figure 6.1 shows both parts of this statement together. It shows why a nontrivial solution of the self-consistency condition only exists above the threshold, and how the resulting magnetization curve looks.

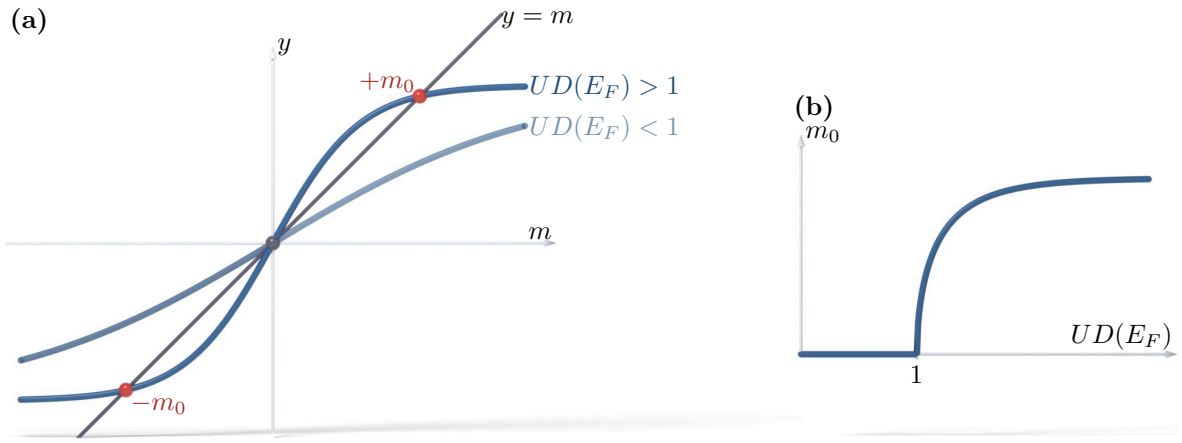


Figure 6.1: A graphical reading of the Stoner criterion (6.3). (a) The self-consistency condition $m = f(m)$, where f is the right-hand side of the zero-temperature gap equation, drawn here with the shape tanh for two values of the slope. It is an odd function that saturates once the bands are fully polarized, and its slope at the origin is exactly $UD(E_F)$. If this slope is below one, the only crossing with $y = m$ is the trivial solution $m = 0$. Once the slope is above one, a pair of nontrivial solutions $\pm m_0$ appears and the nonmagnetic state becomes unstable. (b) The resulting order parameter, which stays at zero until $UD(E_F)$ reaches one and then grows continuously. For this reason, whether a metal becomes magnetic is not decided by U alone but by the *product* of U with the density of states at the Fermi level. This is why aluminium and lead stay nonmagnetic even with a sizeable Coulomb repulsion, while the narrow d bands of iron and cobalt are clearly above the threshold.

Notebook check-in: `session6.ipynb`, "(Exercise) Intermediate coupling regime and Stoner instability"

Reproduce the threshold behavior of (6.3) directly. Compute the self-consistent magnetization as a function of U, and find the critical interaction at which m first becomes nonzero.

Notebook check-in: `session6.ipynb`, "(Exercise) Magnetism in flat bands"

In principle, a flat band has a divergent density of states at E_F, so the Stoner product $UD(E_F)$ in (6.3) is above the threshold for practically any $U > 0$. This exercise is not narrated in the transcript, but it is the natural numerical extension of Section 6.6, since a flat-band system is the case where it is easiest for magnetism to appear.

One important consequence ends this section and starts the next one. From (6.3), magnetism in general needs $UD(E_F) \gtrsim 1$, and for a reasonable bandwidth this usually means that U has to be comparable to t or larger, and not smaller. However, mean-field theory itself is only reliable for $U \ll t$, as we saw in Section 6.3. So the regime where magnetism appears is, in general, the same regime where the method that we have been using stops working. We need a different approach for $U \gg t$, and we construct it in the rest of this session.

6.7 The strongly localized limit, where mean field breaks down

Let us look again at the half-filled chain of Section 6.4, but now we follow the magnetization per site as U grows from zero to very large values. At $U = 0$ there is no interaction and $m = 0$. When U increases, m grows, and for $U \to \infty$ it saturates at exactly one Bohr magneton per site. This is a full localized spin-1/2 moment on every site, and the moments point in alternating directions on the two sublattices, which is the "Ising-like" antiferromagnetic state to which the numerics converge.

This saturation tells us something physical, which is that in the large-U limit the electrons stop moving completely. If an electron hops onto a neighboring site that is already occupied, this costs an energy U. When $U \gg t$ this cost is too large, so each site keeps its one localized electron, and only its spin can still fluctuate. For this reason, in this limit perturbation theory should *not* be organized around the kinetic term, as mean field does implicitly, but around the localized state. We take the interacting and localized configuration as the ground state at zeroth order, and we treat the hopping as a small perturbation on top of it.

6.8 From the Hubbard dimer to the Heisenberg model

Let us make this precise in the smallest possible example. We take two sites, 0 and 1, with a hopping t between them, a local repulsion U on each of them, and exactly two electrons in total, which is half filling. The Fock space at this filling has $\binom{4}{2} = 6$ states. Four of them have one electron on each site, where the electron of each site can be up or down independently. The other two have both electrons on the same site, with one electron up and one down because of Pauli exclusion.

Let us group these states by energy in the atomic limit ($t = 0$). The four configurations with single occupation, $|\uparrow,\uparrow\rangle$, $|\downarrow,\downarrow\rangle$, $|\uparrow,\downarrow\rangle$ and $|\downarrow,\uparrow\rangle$, have no interaction energy. We call them the **low-energy manifold**, at energy 0. The two configurations with double occupation, $|\uparrow\downarrow,0\rangle$ and $|0,\uparrow\downarrow\rangle$, each have the full charging energy U, and they form the **high-energy manifold**. In the low-energy manifold every state has exactly one electron per site, so the charge degree of freedom is completely frozen and only the spin on each site can still fluctuate. For this reason the natural low-energy effective Hamiltonian is a Hamiltonian for the spin only, and rotational invariance fixes its form up to an overall coupling, $H_{\text{eff}} = J\,\mathbf{S}_0 \cdot \mathbf{S}_1$, which is the two-site Heisenberg Hamiltonian. What remains is to obtain J from the microscopic t and U.

The hopping, which is the perturbation, conserves the total spin, so for the *parallel*-spin states it cannot connect the low-energy manifold to the high-energy one at all. If we apply H_t to $|\uparrow,\uparrow\rangle$, we would need to move a spin-up electron onto a site that already has one, and Pauli exclusion forbids this completely. The three parallel or triplet-like states, $|\uparrow,\uparrow\rangle$, $|\downarrow,\downarrow\rangle$

and the symmetric combination $\frac{1}{\sqrt{2}}(|\uparrow,\downarrow\rangle + |\downarrow,\uparrow\rangle)$, are therefore exact eigenstates of the *full* Hamiltonian at energy exactly $E_T = 0$, since the hopping does not act on them at any order. The antiparallel or singlet combination $|S\rangle = \frac{1}{\sqrt{2}}(|\uparrow,\downarrow\rangle - |\downarrow,\uparrow\rangle)$ is different. The hopping *can* connect it to the symmetric combination with double occupation, $|D\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow,0\rangle + |0,\uparrow\downarrow\rangle)$, with matrix element $\langle D|H_t|S\rangle = -2t$. In the two-dimensional subspace $\{|S\rangle, |D\rangle\}$, the Hamiltonian is exactly

$$H = \begin{pmatrix} 0 & -2t \\ -2t & U \end{pmatrix},$$

with exact eigenvalues $E_\pm = \frac{U}{2} \pm \sqrt{(U/2)^2 + 4t^2}$. The energy of the singlet ground state is the lower root, $E_S = \frac{U}{2} - \sqrt{(U/2)^2 + 4t^2}$, and if we expand it for $U \gg t$ we obtain

$$E_S \approx -\frac{4t^2}{U} + O(t^4/U^3), \qquad E_T = 0 \text{ exactly.}$$

We compare this with the exact spectrum of the Heisenberg dimer, $E_{\text{singlet}} = -\frac{3}{4}J$ and $E_{\text{triplet}} = +\frac{1}{4}J$, where the singlet-triplet gap is $\Delta E = E_T - E_S = J$, and we obtain

$$J = \frac{4t^2}{U}. \tag{6.4}$$

We have $J > 0$, which is antiferromagnetic, and the singlet is below the triplet. This confirms, with the correct numerical prefactor, the second-order perturbation theory that the lecture sketches, where the unperturbed Hamiltonian is the Hubbard term, the perturbation is the hopping, the matrix element is t and the energy denominator is U. It also reproduces exactly the $1/U$ scaling that we found numerically for the energy difference between the ferromagnet and the antiferromagnet in Section 6.5. That mean-field energy difference per bond *is* one half of the exchange coupling (6.4), so the two completely different calculations, the mean-field numerics and the many-body perturbation theory, agree. Figure 6.2 shows the whole argument. It shows the Pauli blockade that protects the triplet, the virtual process that lowers the energy of the singlet, and the exact spectrum, whose splitting is J.

Notebook check-in: `session6.ipynb`, "(Reminder) The spectra of a dimer," "(Exercise) The interacting Hubbard dimer," "Gap opening in the Hubbard model," and "(Exercise) The strongly interacting limit"

Diagonalize the Hubbard dimer with six states directly, and check that the singlet-triplet splitting agrees with (6.4). Then take large values of U/t and check that the gap and the low-energy spectrum converge to the prediction of the Heisenberg model. This compares the exact result of this section with a direct numerical diagonalization.

6.9 The Heisenberg model on a general lattice

The argument of Section 6.8 did not use the fact that we had only two sites. For a generic lattice with hopping t_{ij} between sites i and j and a local interaction U, the same second-order argument applies to every connected pair independently in the limit $U \gg t_{ij}$, and it gives the generalized Heisenberg Hamiltonian

$$H_{\text{Heis}} = \sum_{\langle ij \rangle} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j, \qquad J_{ij} = \frac{4t_{ij}^2}{U}. \tag{6.5}$$

Since $J_{ij} \propto t_{ij}^2$, this superexchange coupling, derived directly from a single-orbital Hubbard model, is always positive and antiferromagnetic, regardless of the sign or the geometry of t_{ij}.

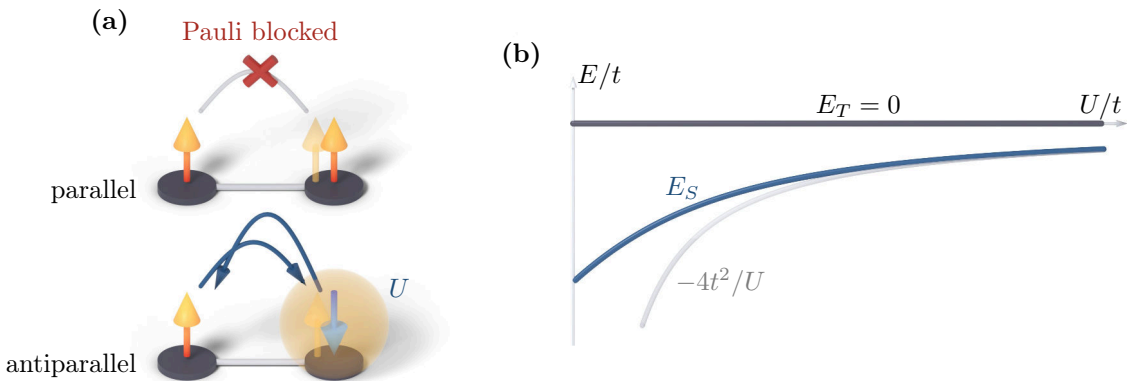


Figure 6.2: Superexchange in the two-site Hubbard model, and the reason why it is always antiferromagnetic. (a) With parallel spins, the only possible hop would put two ↑ electrons on one site, and Pauli exclusion forbids this completely. As a result the triplet states cannot lower their energy at any order, and stay at exactly $E_T = 0$. With antiparallel spins the hop is allowed, so the singlet goes through a virtual double occupation of energy U, the glow, and comes back, and gains a second-order energy $-4t^2/U$. Here the magnetic order is a purely kinetic effect, since antiparallel spins have lower energy because only they can move. (b) The exact singlet energy $E_S = \frac{U}{2} - \sqrt{(U/2)^2 + 4t^2}$ as a function of U/t, with $E_T = 0$ exact for every U. The singlet-triplet splitting *is* the Heisenberg coupling J of (6.4), and it decays as $4t^2/U$, the grey curve. This is the same $1/U$ law that we found numerically for the mean-field energy difference between the ferromagnet and the antiferromagnet in Section 6.5.

More general microscopic models, which are not Hubbard models or have several orbitals, can give couplings of both signs, and they do. In Section 6.11 we will see explicitly how a ferromagnetic coupling appears when a second, intermediate orbital takes part. In all of this session we use the following sign convention. A coupling $J_{ij} > 0$ favors antiparallel neighboring spins, which is antiferromagnetic, and $J_{ij} < 0$ favors parallel neighboring spins, which is ferromagnetic. If we treat the spins in (6.5) as classical vectors, we already obtain both classical ground states on a bond between two sites, which are antiparallel for $J > 0$ and parallel for $J < 0$.

6.10 Bipartite order versus geometric frustration

On a **bipartite** lattice, such as the square or the honeycomb lattice, the sites split into two sublattices A and B so that every bond connects an A site with a B site. On such a lattice the antiferromagnetic Heisenberg model (6.5) has an obvious classical ground state. All spins on A point in one direction and all spins on B point in the opposite direction, and in this way every bond has antiparallel neighbors at the same time, which is what it prefers. This is the collinear antiferromagnetic order with two sublattices that we already saw at the mean-field level in Section 6.4, and (6.5) explains *why* it appears. In the large-U limit, the mean-field Hubbard model on a bipartite lattice reduces exactly to this Heisenberg model.

Frustrated lattices are different. On a **non-bipartite** lattice, of which the kagome and the triangular lattices are the standard examples, it is not possible to color the sites with two colors so that every bond connects opposite colors. For instance, a triangular plaquette has three sites that are all neighbors of each other, and no choice of up and down spins makes all three bonds antiparallel at the same time. The antiferromagnetic tendency of $J_{ij} > 0$ is still present on every single bond, but it cannot be satisfied everywhere at the same time, and this is what is called **geometric frustration**. The best compromise for the system is usually a noncollinear arrangement, and the standard case is spins at 120° from each other on a triangular plaquette. This arrangement maximizes the total antiferromagnetic character bond by bond, without fully satisfying any single bond. This noncollinear character also appears

directly in the mean-field solution of the corresponding electronic Hamiltonian, and not only in the spin model. Figure 6.3 shows next to each other the bipartite case, the frustrated collinear attempt and the 120° compromise, together with the classical bond energies that decide between the last two.

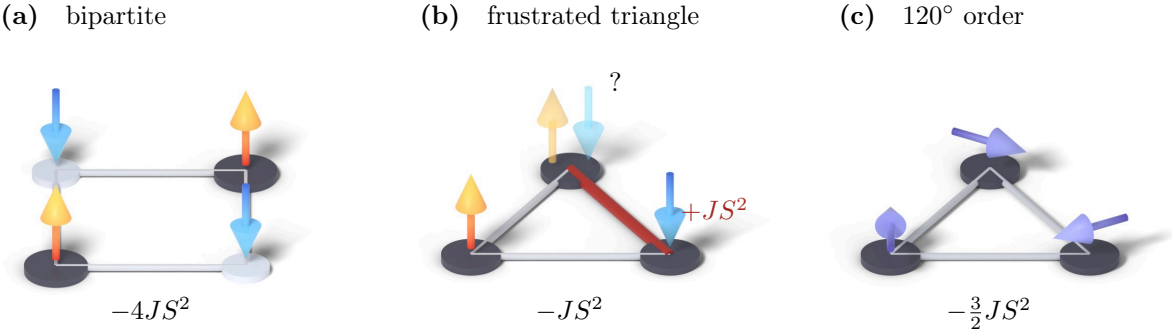


Figure 6.3: How the geometry decides if an antiferromagnet can order. (a) On a bipartite lattice the sites can be coloured with two colours, here dark and light, every bond joins the two colours, and the Néel state of Section 6.10 satisfies all bonds at the same time. (b) On a triangle it is not possible to use two colours. If we fix two spins antiparallel, the third one is parallel to one of its neighbours for any choice, the two ghost arrows, and one bond stays frustrated. For classical spins, this collinear compromise costs $+JS^2$ on the frustrated bond, and the total is $-JS^2$ per triangle. (c) The energy is lower if no single bond is fully satisfied. Three spins at 120° in the plane give $\mathbf{S}_i \cdot \mathbf{S}_j = -S^2/2$ on *every* bond and a total of $-\frac{3}{2}JS^2$, which is lower than (b) by half of its value. This is the origin of the noncollinear order to which the mean-field solution of the corresponding electronic Hamiltonian also converges.

Notebook check-in: `session6.ipynb`, "(Exercise) Magnetic instabilities of the square lattice" and "(Exercise) Non-collinear magnetic order from geometric frustration"

Compare at the mean-field level a bipartite lattice, the square lattice, with a frustrated one, the triangular or kagome lattice. The square lattice should converge to the collinear order with two sublattices of Section 6.10, and the frustrated lattice should converge to a noncollinear compromise. Both results can be traced back to the sign convention of (6.5) on each bond.

6.11 Superexchange as the origin of ferromagnetic coupling

The direct t^2/U mechanism of Section 6.9 is always antiferromagnetic, but many real magnetic materials are ferromagnets. So a ferromagnetic coupling needs a different microscopic mechanism, and it usually requires an intermediate atom, called a ligand, between the two magnetic ions, instead of a single direct hop. Let us take two transition-metal ions, for example cobalt, iron or nickel, connected by a ligand such as oxygen. As before, electrons can hop directly between the two metal sites, and this channel is antiferromagnetic as in Section 6.9. However, there is a second virtual process. An electron hops from the ligand onto one metal ion, and another electron hops from the ligand onto the other one. The two electrons are now on the metal ions, and this leaves a virtual state with two holes in the p shell of the ligand, which was full before. Depending on which d orbitals take part and on the metal-ligand-metal bond angle, the energy correction from this virtual channel with two holes on the ligand can favor the ferromagnetic configuration over the antiferromagnetic one, and in this way it changes the net sign of the effective coupling. The systematic version of this argument is given by the **Goodenough-Kanamori rules**, which predict the sign of the exchange coupling from the orbital filling and the bond geometry. These rules can also be derived systematically from self-consistent multi-orbital calculations, for instance with density functional theory. This ma-

terial comes from the slides and the transcript, and there is no dedicated notebook exercise for it in this session, since the exercises of the notebook stay within the single-orbital Hubbard model.

6.12 Anisotropic exchange with Dzyaloshinskii-Moriya, Ising and Kitaev terms

In (6.5) we assumed that the coupling is an isotropic scalar product, $\mathbf{S}_i \cdot \mathbf{S}_j$, which is what the rotational spin symmetry requires when there is no spin-orbit coupling. Once spin-orbit coupling is included in the underlying electronic Hamiltonian, other spin couplings with lower symmetry become allowed. Three of them appear often in real materials, and we list them now.

Antisymmetric, or Dzyaloshinskii-Moriya, exchange,

$$H_{\mathrm{DM}} = \sum_{\langle ij \rangle} \mathbf{D}_{ij} \cdot \left(\mathbf{S}_i \times \mathbf{S}_j \right),$$

which is a cross product and not a dot product. It requires broken mirror symmetry, which is the reason why it is strong at surfaces and interfaces, where the substrate itself breaks this mirror symmetry. It favors neighboring spins rotated by roughly 90° with respect to each other, so it favors noncollinear order, which is often a spiral.

Anisotropic, or Ising-like, exchange, which is a coupling of a single component such as $\Lambda\, S_i^z S_j^z$. It favors an easy axis, or an easy plane for the opposite sign, which means that the system magnetizes more easily along one direction than along another.

Kitaev interaction, which is a coupling that depends on the bond, on a lattice of honeycomb type. It has an $S_i^x S_j^x$ term on one bond direction, $S_i^y S_j^y$ on a second one and $S_i^z S_j^z$ on a third one,

$$H_{\mathrm{Kitaev}} = \sum_{\langle ij \rangle \in x\text{-bonds}} K\, S_i^x S_j^x + \sum_{\langle ij \rangle \in y\text{-bonds}} K\, S_i^y S_j^y + \sum_{\langle ij \rangle \in z\text{-bonds}} K\, S_i^z S_j^z .$$

Even on a bipartite honeycomb lattice, where the geometry alone would not give any frustration, this dependence on the bond is frustrating by itself, since no single spin orientation can satisfy the three types of bonds at the same time (figure 6.4). This interaction is what makes the Kitaev model a paradigm for quantum spin liquids, and this is why we point it out here explicitly, as an important preparation for the material on frustrated magnets in sessions 7 and 8.

Notebook check-in: `session6.ipynb`, "(Exercise) Non-collinear magnetism from spin-orbit coupling"

Add a term of Dzyaloshinskii-Moriya type to the mean-field Hamiltonian, and see how the ground state tilts away from collinear order. This is the electronic counterpart of the statement above for the spin model.

6.13 Magnons as fluctuations of the order parameter

Until now we have only asked about ground states. Let us now ask about excitations. We take the Heisenberg model (6.5) with $J < 0$, which is ferromagnetic, and we promote the spins to real quantum operators that satisfy the angular-momentum algebra $[S_i^a, S_j^b] = i\hbar\, \delta_{ij} \epsilon^{abc} S_i^c$. From here on we set $\hbar = 1$, which is the standard choice for this kind of calculation, and we ask for the low-lying excited states. In contrast with the interacting electron problem, the

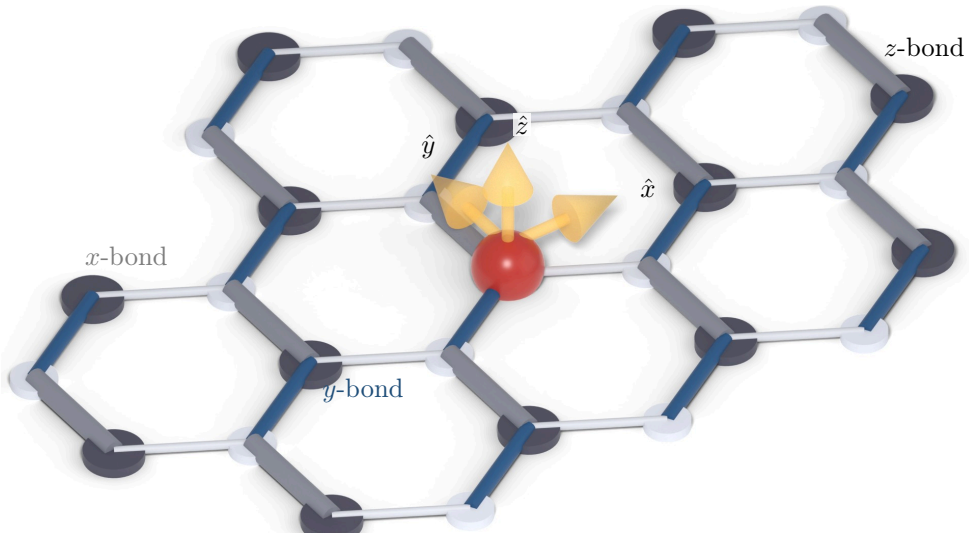


Figure 6.4: The Kitaev interaction and the second kind of frustration. The honeycomb lattice is bipartite, and its sites can be coloured with two colours, dark and light, as in figure 6.3(a). For this reason an ordinary Heisenberg antiferromagnet on this lattice is not frustrated at all. The Kitaev coupling gives frustration anyway, because each of the three bond directions, drawn in three styles, couples only *one* spin component. The x-bond only depends on S^x, so it favors both of its spins along $\pm\hat{x}$, the y-bond favors $\pm\hat{y}$, and the z-bond favors $\pm\hat{z}$. Every site belongs to one bond of each type, and no single orientation can satisfy three mutually orthogonal conditions at the same time, which the three ghost arrows on the highlighted site show. Here the frustration comes from the interaction and not from the geometry, and this is what makes the Kitaev model a paradigm for quantum spin liquids in sessions 7 and 8.

exact ground state of the ferromagnetic Heisenberg model is known immediately. It is the fully polarized state with all spins aligned, since for $J < 0$ no other configuration can lower $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle$ further. A **magnon** is a small fluctuation of the magnetization that propagates on top of this fully polarized state. Qualitatively, it is a single spin tilted slightly out of alignment, delocalized coherently over the whole lattice as a wave. The question is how to obtain these excitations systematically from a many-body spin Hamiltonian. The answer is a change of variables that turns the spin problem into a bosonic tight-binding problem.

6.14 The Holstein-Primakoff transformation

There is an exact mapping between spin operators and bosonic creation and annihilation operators $a_i^\dagger, a_i$ with $[a_i, a_j^\dagger] = \delta_{ij}$, which is called the **Holstein-Primakoff transformation**. For a spin of magnitude S on site i, it reads

$$S_i^z = S - a_i^\dagger a_i, \qquad S_i^+ = \sqrt{2S - a_i^\dagger a_i}\; a_i, \qquad S_i^- = a_i^\dagger \sqrt{2S - a_i^\dagger a_i}. \tag{6.6}$$

This is an *exact* rewriting. If we insert (6.6) and use the bosonic commutation relations, we obtain exactly the full $SU(2)$ spin algebra $[S_i^z, S_i^\pm] = \pm S_i^\pm$, $[S_i^+, S_i^-] = 2S_i^z$, without any approximation so far. The physical picture is direct. The operator $a_i^\dagger a_i$ counts how many units of spin have been removed from the fully polarized value S on site i, which means that it counts the **magnons** on that site, and $S_i^z = S - a_i^\dagger a_i$ states exactly this. The square-root operators are what make (6.6) exact even far from full polarization. They are also the reason why the Hamiltonian, written in this form, looks more complicated and not simpler.

The simplification comes from linearizing. When the expectation value of $a_i^\dagger a_i$ stays much smaller than S, which means when we stay close to the fully polarized ground state, we can

expand the square roots and keep only the leading term. This is the regime that we want, since we are looking for low-lying excitations above this ground state, and the result is

$$S_i^+ \approx \sqrt{2S}\, a_i, \qquad S_i^- \approx \sqrt{2S}\, a_i^\dagger, \qquad S_i^z = S - a_i^\dagger a_i. \tag{6.7}$$

This linearized form is the one that is used in practice to compute magnon spectra. The counting that follows from it is the easiest way to remember the whole transformation, so we write it out. The operator S^+ *raises* S^z, meaning that it increases the local magnetization back toward saturation, and it *destroys* a magnon. The operator S^- lowers the magnetization, and it *creates* a magnon. The total magnetization of the system is the fully saturated value minus the total number of magnons, $\sum_i \langle S_i^z \rangle = NS - \sum_i \langle a_i^\dagger a_i \rangle$. Each magnon has exactly one unit of spin, so magnons are $S = 1$ excitations, in agreement with what we said about them in Section 6.1. They exist only because the ground state on which they live has already broken spin-rotation symmetry spontaneously. Figure 6.5 shows the three parts, which are the fully polarized ground state, the delocalized precession wave that carries a single magnon, and this rule for raising and lowering.

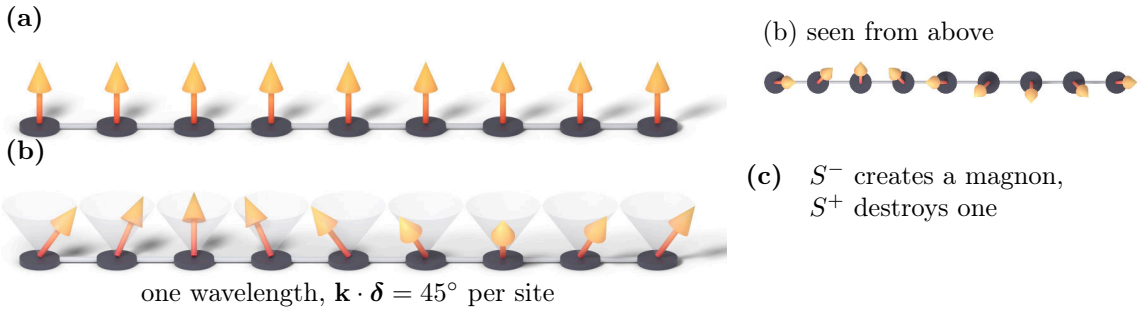


Figure 6.5: The states that (6.6) and (6.7) describe. (a) The ferromagnetic ground state, where every spin has its maximal value $S^z = S$, and so there are no magnons. (b) A single magnon is *not* one spin flipped on one site. Instead, every spin is tilted slightly out of alignment and precesses on its cone, and the precession phase advances by $\mathbf{k}\cdot\boldsymbol{\delta}$ from one site to the next. This is seen most clearly in the view from above, where the transverse component turns at a constant rate and comes back to itself after one wavelength. The missing unit of spin is therefore shared coherently by the whole lattice, and $\sum_i \langle S_i^z \rangle = NS - 1$. Since the tilting angle is small, we have $\langle a_i^\dagger a_i \rangle \ll S$ everywhere, which is the condition under which we can expand the square roots in (6.6). In the drawing the tilting angle is much larger than in reality. (c) The counting rule that makes the transformation easy to remember. The boson number counts the spin removed from saturation, so the *lowering* operator S^- creates a magnon and the *raising* operator S^+ destroys one.

We insert (6.7) into the ferromagnetic Heisenberg Hamiltonian, $H = J\sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j = J\sum_{\langle ij \rangle} \left[S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+) \right]$, and we keep the terms up to quadratic order in the bosonic operators,

$$S_i^z S_j^z \approx S^2 - S\big(a_i^\dagger a_i + a_j^\dagger a_j\big), \qquad \tfrac{1}{2}\big(S_i^+ S_j^- + S_i^- S_j^+\big) \approx S\big(a_i^\dagger a_j + a_j^\dagger a_i\big),$$

and in this way the spin Hamiltonian becomes a bosonic tight-binding Hamiltonian for magnons, and its hopping amplitude is set directly by the exchange coupling J, as we said before. The magnon "hops" between sites with an amplitude proportional to JS, in direct analogy with the electronic tight-binding Hamiltonians that we have diagonalized during the whole course.

6.15 Magnon dispersions in ferromagnets and antiferromagnets

We Fourier transform the bosonic Hamiltonian of Section 6.14 on a Bravais lattice with coordination number z and first-neighbor structure factor $\gamma(\mathbf{k}) = \frac{1}{z}\sum_{\boldsymbol{\delta}} e^{i\mathbf{k}\cdot\boldsymbol{\delta}}$. For the **ferromagnet**, with $J < 0$ and $|J| = -J$, this gives a diagonal magnon Hamiltonian with dispersion

$$E_{\mathrm{FM}}(\mathbf{k}) = |J|\, S\, z\big(1 - \gamma(\mathbf{k})\big). \tag{6.8}$$

This dispersion is zero at $\mathbf{k} = 0$, as it has to be. A uniform rotation of all the spins costs no energy, and this is the Goldstone mode of the broken spin-rotation symmetry. If we expand $\gamma(\mathbf{k})$ for small $\mathbf{k}$ we obtain $E_{\mathrm{FM}}(\mathbf{k}) \propto k^2$, which is a **quadratic** magnon dispersion, and this is the "gapless, quadratic" ferromagnetic magnon that the lecture points out.

The **antiferromagnetic** case needs one more step, because the ground state is not "all spins up." On a bipartite lattice, we apply the Holstein-Primakoff transformation (6.7) separately on the two sublattices, and on the B sublattice the roles of S^+ and S^- are exchanged, since these spins point in the opposite direction. We have $S_i^z = S - a_i^\dagger a_i$ on A, but $S_j^z = -S + b_j^\dagger b_j$ on B. If we insert this into $\mathbf{S}_i \cdot \mathbf{S}_j$ for a bond between the two sublattices, we obtain not only density terms $a^\dagger a$ and $b^\dagger b$, but also *anomalous* terms $a_i b_j$ and $a_i^\dagger b_j^\dagger$, which do not conserve the particle number. This tells us that the magnon vacuum on top of the Néel state is not the bosonic vacuum, and that to diagonalize the Hamiltonian we need a Bogoliubov transformation and not only a Fourier transform. If we do this, we obtain the antiferromagnetic magnon dispersion

$$E_{\mathrm{AFM}}(\mathbf{k}) = J\, S\, z\, \sqrt{1 - \gamma(\mathbf{k})^2}. \tag{6.9}$$

Close to $\mathbf{k} = 0$ we have $1 - \gamma(\mathbf{k})^2 \propto k^2$, so $E_{\mathrm{AFM}}(\mathbf{k}) \propto |k|$. This is a **linear** dispersion similar to a Dirac cone, which is gapless at the same point but linear and not quadratic. This is the rule "quadratic in ferromagnets, linear in antiferromagnets" from the lecture, which we have now derived and not only stated. It is also one more example, together with the Dirac cones of session 2 and 3, of a linear dispersion that comes from the structure of a symmetry and is not put in by hand. Figure 6.6 shows (6.8) and (6.9) together, with a magnified view of the zone center, so that the parabola and the cone can be distinguished.

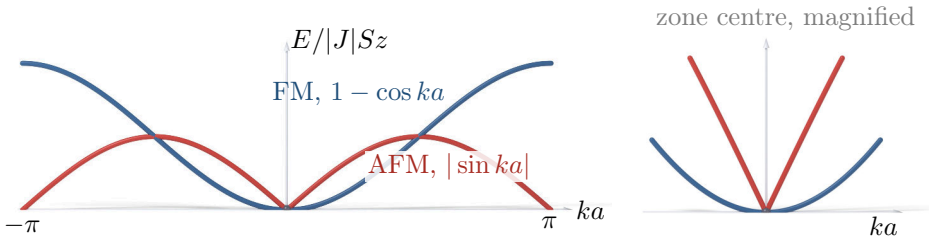


Figure 6.6: Magnon dispersions (6.8) and (6.9) for a chain ($z = 2$, $\gamma(\mathbf{k}) = \cos ka$), both in units of $|J|Sz$. Both are zero at $\mathbf{k} = 0$, since a uniform rotation of all the spins costs no energy. So the magnon is the Goldstone mode of the broken spin-rotation symmetry, and no value of the exchange can open a gap in it. The difference is *how* the two dispersions go to zero. The factor $1-\gamma(\mathbf{k})$ of the ferromagnet is quadratic in k. The factor $\sqrt{1-\gamma(\mathbf{k})^2}$ of the antiferromagnet is the square root of the same quantity, so it is linear and gives a Dirac-like cone instead of a parabola. The right panel magnifies the zone centre, where the difference is clear. This extra square root comes from the Bogoliubov transformation that the Néel state requires, since the anomalous $a_i b_j$ terms are what change a parabola into a cone.

6.16 Topological magnons

Once magnons have a real band structure, which is a dispersion $E(\mathbf{k})$ over the Brillouin zone, we can ask about *magnon* bands the same question that we have already asked twice about *electronic* bands. Can this band structure have a nontrivial topological invariant? To answer it, we use again the Chern number of sessions 3 and 4, where the Berry curvature $\Omega(\mathbf{k})$ of a Bloch band is integrated over the Brillouin zone and gives an integer,

$$C = \frac{1}{2\pi} \int_{\mathrm{BZ}} \Omega(\mathbf{k})\, d^2k, \tag{6.10}$$

and this is the third time that we apply it, now to the bosonic Bogoliubov Hamiltonian of Section 6.15 instead of an electronic tight-binding Hamiltonian. We add to the plain Heisenberg exchange a spin-orbit coupling of the Dzyaloshinskii-Moriya type of Section 6.12. This term plays the same role as the complex second-neighbor hopping of the Haldane model for electrons in session 4. It introduces complex phases in the magnon hopping amplitudes, which can open a gap in the magnon band structure at finite energy and give the bands on both sides of this gap a nonzero Chern number (6.10). When we put such a topological magnon Hamiltonian on a ribbon, the bulk-boundary correspondence guarantees chiral magnon edge modes inside the gap. These modes propagate in only one direction along each edge, and they are the magnon analogue of the chiral electronic edge states of a Chern insulator. They carry spin and energy and not charge, but they are topologically protected in the same sense as their electronic counterparts. It is the same tool that we used for electrons in sessions 3 and 4, now applied to magnon bands.

Notebook check-in: `session6.ipynb`, "(Exercise) Topological magnons in 2D magnets"

Construct a magnon Hamiltonian with a spin-orbit term of Dzyaloshinskii-Moriya type on top of the Heisenberg exchange, compute its Chern number with (6.10), and look for chiral magnon edge modes on a ribbon. This reproduces directly, for magnons, the argument of the bulk-boundary correspondence above.

Before we stop, a short comment on the scope of these notes. The companion notebook goes further than the lecture of this session in a few places. The most visible ones are its exercises on electronic and magnetic symmetry breaking in graphene multilayers, and its last sections on many-body spin response functions. These are real and useful extensions of the material of this session. However, neither the transcript nor the slides of this session narrate them, so we have not included the corresponding derivations in these notes. They should be taken as independent work with the notebook, and not as material that these notes derive.

6.17 Closing summary

So far in this session we have done the following.

- We constructed again the magnetism of the Hubbard model numerically and in full detail. This includes the direct competition between the ferromagnetic and the antiferromagnetic self-consistent solutions, and the Stoner threshold (6.3) that decides whether magnetism appears at all.
- We derived the Heisenberg model exactly from the strongly correlated limit of the Hubbard model, and we obtained the exact coefficient $J = 4t^2/U$ (6.4) and its generalization to a lattice (6.5). We also saw how frustration and superexchange modify this simple picture.

- We constructed the magnon spectrum from first principles with the Holstein-Primakoff transformation (6.6)–(6.7). We obtained the quadratic ferromagnetic dispersion (6.8) and the linear antiferromagnetic dispersion (6.9), and then we applied the topological methods of sessions 3 and 4 a third time to show that magnon bands can also be topological.

This is the point where the live lecture ends, right after topological magnons and without a narrated summary slide. This is a real feature of this year's recording and not something missing in these notes, so we also end here, and we do not add a summary sentence that the lecture does not say. Every magnet that we have constructed in this session, from the mean-field exchange field to the topological magnon edge states, is a state that breaks time-reversal symmetry. It is a net, staggered or noncollinear spin arrangement that looks different when we run time backwards than when we run it forwards.

Two topics are left open on purpose, and both were pointed out along the way without being resolved here. The Kitaev interaction of Section 6.12, which the lecture points out explicitly as "important for quantum spin liquids," comes back from two different directions in the next sessions. Session 7 treats the frustrated Hamiltonians of kagome type, to which it belongs, as the paradigm case where mean-field theory fails. Session 8 comes back to the Kitaev model itself as a way to obtain an exactly solvable quantum spin liquid. The spinon was previewed in Section 6.1 only by name and by its spin-1/2 and charge-neutral quantum numbers. It will get the same full treatment that magnons got in this session, but only in session 8, when we move from magnets that break time-reversal symmetry to quantum magnets that do not.

Session 7

Quantum Many-Body Entanglement and Tensor Networks

In session 6 we studied the magnetism of the Hubbard model of session 1 in full numerical detail. We derived the Heisenberg model as the strong-coupling limit, we introduced Holstein-Primakoff magnons, and we used the topological methods of sessions 3 and 4 for a third time, now for magnon bands. In that session we also mentioned the Kitaev interaction as an important ingredient for quantum spin liquids, and we mentioned materials such as herbertsmithite, where the local moments sit on a kagome lattice and do not order even at zero temperature. That session ended with "topological magnons," without a final summary. This was not a transition into the present session, since that session had reached the end of its material, but the open topic is the one we continue with now. In contrast with a ferromagnet or an antiferromagnet, a quantum spin liquid does not break time-reversal symmetry. Its spins keep fluctuating locally for any low temperature, and the ground state is one large entangled object and not a simple pattern of arrows. All the tools that we have built in this course so far fail for this kind of Hamiltonian. Mean-field theory, which we have used since session 1, looks for an order parameter that breaks a symmetry, such as a nonzero magnetization, a nonzero charge imbalance or a nonzero pairing amplitude. By definition, a quantum spin liquid has none of these, so the self-consistent calculation has nothing to converge to and gives back zero, which tells us nothing about the real ground state. The same problem appears in a second family of materials, the fractional Chern insulators, where the interactions are again too strong to be described by any effective single-particle model.

In this session we will build a tool that works directly in this regime and does not assume that a symmetry is broken. This tool is tensor networks, and in particular matrix product states (MPS). This method is well developed, and today it gives the best quantitative evidence for spin liquid physics on the triangular and kagome lattices, which are the Hamiltonians mentioned in session 6. To get there, we first point out something that was present in the last sessions without being said explicitly, which is that fermionic and spin Hamiltonians describe the same physics in two languages, and the link between them is the Jordan-Wigner transformation. Then we will ask a question which is not "how do we diagonalize a huge matrix" but "how do we *store* a many-body wave function in the first place." To answer this question we will use entanglement and a quantity called the entanglement entropy, and this is the main part of this session.

7.1 Two regimes of many-body physics

Let us go back to the general interacting Hamiltonian that we have used in the whole course,

$$H = \sum_{ij} t_{ij}\, c_i^\dagger c_j \;+\; \sum_{ijkl} V_{ijkl}\, c_i^\dagger c_j c_k^\dagger c_l. \tag{7.1}$$

The kinetic term is free and can be solved exactly, and the interaction term in general cannot. In all sessions so far we treated (7.1) in the same way. We decoupled the quartic term with mean field, we replaced it with an effective bilinear Hamiltonian, and we looked at which symmetry is broken by the resulting order parameter. This strategy defines what we can call **weakly correlated matter**. In this sense, magnets, superconductors and charge density waves are all single-particle physics, once we find the right symmetry-broken background.

In this session we are in the opposite regime, which is **strongly correlated matter**. Here the interactions are so strong that no effective single-particle Hamiltonian gives the ground state, for any choice of the decoupling. Quantum spin liquids and fractional Chern insulators are our two main examples, and more generally, the same problem appears when we want an accurate phase diagram of a truly interacting model. In both cases we cannot solve the problem with an ansatz for the broken symmetry. Instead, we need something close to an exact numerical solution, or at least a controlled approximation that does not assume symmetry breaking from the start. This is what tensor networks give us.

7.2 Fermions, spins and the Jordan-Wigner transformation

Before we build the tool, let us point out something that was present in the last sessions without being said explicitly. We have used two families of many-body Hamiltonians that look different. The first family is the fermionic Hamiltonians of the form (7.1), with a general two-body interaction V_{ijkl}, where density-density interactions are only a special case, which we used in sessions 1 to 5. The second family is the spin Hamiltonians, mainly the Heisenberg model $H = \sum_{ij} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j$, which we used for the magnons of session 6 and which we use again in this session for quantum spin liquids. These are not two different topics. They are the same physics written in two different languages, and in this section we make this translation precise.

Let us first look at a single site. A spinless fermionic site has a local Hilbert space of dimension two, empty or full, and a spin-1/2 site also has a local Hilbert space of dimension two, down or up. The first guess is $c_i^\dagger \leftrightarrow S_i^+$ and $c_i \leftrightarrow S_i^-$, and on a single site this guess is exactly right. However, it fails when we have more than one site, since fermion operators on different sites anticommute, while spin operators on different sites commute. As a result, the naive identification gives the wrong relative signs between sites. To fix this, we add to each site an extra phase that depends on the occupations. This phase is a "string" over all the sites to the left, chosen so that the exchange of two fermions gives the minus sign that the spin operators do not have by themselves. This mapping is called the **Jordan-Wigner transformation**, and it reads

$$S_i^+ = c_i^\dagger K_i, \qquad S_i^- = c_i K_i, \qquad S_i^z = n_i - \tfrac{1}{2}, \qquad K_i = \exp\left(i\pi \sum_{j<i} n_j \right), \tag{7.2}$$

where $n_j = c_j^\dagger c_j$ is the fermion number operator on site j. The string operator K_i is built only from the occupation numbers to the left of site i. For this reason it commutes with all local operators on sites $j \geq i$, and it counts, mod 2, how many fermions are to the left. We can check directly that (7.2), built in this way, gives the full $SU(2)$ commutation algebra of spin-1/2 operators on every site and between every pair of sites. For this check we only need the fermionic anticommutation relations for $c_i, c_i^\dagger$ and the string. In the lecture this is summarized as *spin equals fermion times string*, and this statement is correct. Figure 7.1 shows what the

string does. It counts the fermions to the left of site i and gives their parity, so a change of a single occupation anywhere in that region changes the sign of S_i^+.

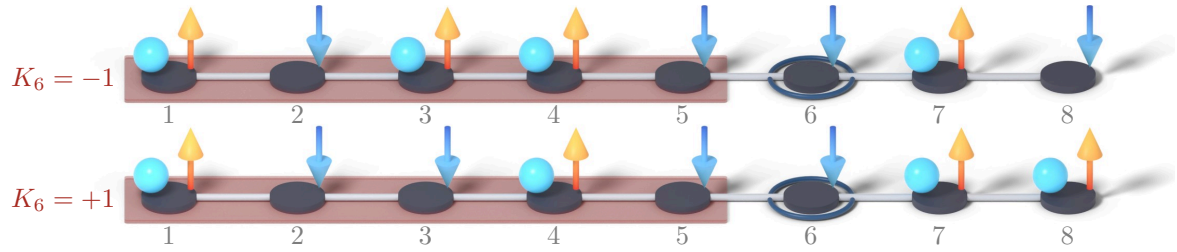


Figure 7.1: Spin equals fermion times string. On a single site the mapping is trivial, since an empty site is a down spin and an occupied site is an up spin, $S_i^z = n_i - \frac{1}{2}$, as the ball and the arrow on each site show. The naive identification gives the wrong *relative* signs between sites, because fermion operators on different sites anticommute while spin operators on different sites commute. The string K_i of (7.2) corrects exactly this, since it gives a factor -1 for every fermion to the left of site i, which is the shaded strip. The two chains in the figure differ only in the occupations to the left of the marked site $i = 6$, with three fermions in the chain behind and two in the chain in front, and this parity alone changes the sign of $S_6^+ = c_6^\dagger K_6$. The string is nonlocal, and it carries the fermionic exchange sign that spins do not have by themselves.

This is the one explicit link in the course between the two languages that we have used. The fermionic operators of sessions 1 to 5 and the spin operators of session 6 and, as we will see, of session 8 are related by (7.2), and they are not only analogous to each other. It also explains a practical choice that we make for the rest of this session. Since interacting fermionic Hamiltonians and interacting spin Hamiltonians are mathematically equivalent, we can work only in the spin representation from now on, and we will do so. The reason is that spin operators on different sites commute, so there is no fermionic sign to keep track of when we cut the chain, and this makes the counting of entanglement below clean.

7.3 Storing the wave function as the main bottleneck

Now that we use the language of spins, let us ask why a general interacting many-body Hamiltonian is hard to solve. A first answer would be "because the diagonalization of a huge matrix is slow," but the main problem is more basic, and it is the *storage* of the wave function.

Let us take the simplest nontrivial case, which is the antiferromagnetic Heisenberg dimer, $H = \mathbf{S}_0 \cdot \mathbf{S}_1$ on two spin-1/2 sites. Its ground state is the unique singlet,

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \right), \tag{7.3}$$

which is a state that does not break any symmetry, since the expectation value of every single-site spin vanishes. As we will show below, it is also maximally entangled. The question of the whole session is how this kind of state generalizes to a macroscopic system with many sites.

Let us write a general spin-1/2 wave function in the natural product basis, with one up or down label s_i per site.

$$|\Psi\rangle = \sum_{s_1,\ldots,s_L} C(s_1,\ldots,s_L)\, |s_1,\ldots,s_L\rangle, \qquad s_i \in \{\uparrow,\downarrow\}. \tag{7.4}$$

For $L = 2$ there are four basis states, up-up, up-down, down-up and down-down, so there are four coefficients C, and the Hamiltonian is a 4×4 matrix. If we add one site, $L = 3$, the number doubles to eight. If we add one more site, $L = 4$, it doubles again to sixteen.

In general, the dimension of the Hilbert space is $d = 2^L$, and this is also the number of coefficients $C(s_1, \ldots, s_L)$ that we need to write the ground state exactly. This exponential growth is the main limitation. Even if we never build the full $2^L \times 2^L$ Hamiltonian matrix, and we only compute the matrix elements when we need them, we still have to store the 2^L coefficients of the ground state once we have found it. This storage cost, and not the cost of the diagonalization, is the real obstruction, and it is the most important point of the whole session. Tensor networks are one answer to this storage problem. In the lecture, **neural network quantum states** are also mentioned briefly as a second answer to the same problem, which comes from machine learning, but this second tool is not developed in this session. In session 10 we will come back to neural network quantum states in detail, from the point of view of machine learning. The material of this session is only about the first answer, tensor networks, and in particular matrix product states.

Notebook check-in: `session7.ipynb`, "Energy of the Heisenberg model as a function of the system size" (Core)

In this exercise you compute the ground-state energy per site of the Heisenberg chain with exact diagonalization, for growing system sizes. For this you use the `dmrgpy` library, which is new in this session, together with `pyqula`. You will see how fast exact diagonalization stops being possible, and this limit is the 2^L scaling of (7.4) in a concrete calculation.

7.4 The matrix product state ansatz

The idea that solves this problem is very simple. Instead of storing $C(s_1, \ldots, s_L)$ as one large array of 2^L numbers, we write it as a product of matrices, with one matrix for each possible value of each local spin.

$$C(s_1, s_2, \ldots, s_L) \;\approx\; M_1^{s_1} M_2^{s_2} \cdots M_L^{s_L}, \tag{7.5}$$

Here each $M_i^{s_i}$ is an $M \times M$ matrix, except at the two ends of a finite open chain where it is a vector, and M is the **bond dimension**, which is a free parameter that we choose. For a spin-1/2 chain there are two matrices per site, one for $s_i = \uparrow$ and one for $s_i = \downarrow$. Therefore the total number of parameters of the state is proportional to $L \times M^2$, which is linear in the system size, in strong contrast with the 2^L of the exact representation (7.4). This is called the **matrix product state** (MPS) ansatz. At first sight, it is only a choice, and one that looks rather arbitrary. In Sections 7.5 to 7.8 we will answer two questions about it. First, why it is a *good* choice. Second, what exactly it captures, and what it does not capture, about the physical state.

We should know from the start that this is not a toy model. Today matrix product states give the most accurate results for two of the central hard problems in quantum materials. The first one is the doped two-dimensional Hubbard model with second-neighbor hopping, which is related to unconventional superconductivity. The second one is the two-dimensional frustrated Heisenberg model on the triangular and kagome lattices, and this is where the strongest numerical evidence for real quantum spin liquid physics comes from, which is the herbertsmithite example of session 6. Both examples are shown as labeled figure panels on slide 15 of the deck, with the titles "Solving the 2D Hubbard model at finite doping" and "Solving the 2D Heisenberg model in frustrated lattices." This shows that MPS are a working tool for the strongly correlated regime, and not only a formal construction.

7.5 Measuring entanglement with the entanglement entropy

To see why (7.5) works, we need a quantitative measure of how entangled a many-body state is. Let us split the system into two complementary parts, subsystem A and subsystem B. A state is **not entangled**, which is also called a product state, if it can be written as $|\Psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$. Two examples are the state $|\uparrow\downarrow\rangle$ and a product of two single-site states that are both polarized along x. The singlet (7.3) is the opposite case. It cannot be written as a product of this form, and it is our main example of an **entangled** state.

To turn the question "entangled or not" into a number, we define the full density matrix $\rho = |\Psi\rangle\langle\Psi|$, and we trace out subsystem B. This gives the **reduced density matrix** of A, $\rho_A = \mathrm{Tr}_B\, \rho$. The **entanglement entropy** is then the von Neumann entropy of ρ_A,

$$S(\rho_A) = -\,\mathrm{Tr}(\rho_A \log \rho_A)\,. \tag{7.6}$$

If $|\Psi\rangle$ is a product state, $\rho_A = |\psi_A\rangle\langle\psi_A|$ is a projector of rank one, so its only nonzero eigenvalue is 1, and we directly get $S = 0$. This means that there is no entanglement and the entropy is zero, which is what we expect.

Let us now check (7.6) on the singlet (7.3). We write $\rho = |\Psi\rangle\langle\Psi|$ in the product basis and trace over site 1, which is our subsystem B, and we obtain

$$\rho_A = \tfrac{1}{2}\,|\uparrow\rangle\langle\uparrow| + \tfrac{1}{2}\,|\downarrow\rangle\langle\downarrow|,$$

which is a maximally mixed single-site density matrix, with two eigenvalues both equal to $1/2$. Then

$$S = -\left(\tfrac{1}{2}\log\tfrac{1}{2} + \tfrac{1}{2}\log\tfrac{1}{2}\right) = \log 2. \tag{7.7}$$

This is the maximum entanglement entropy that a single spin-$1/2$ can have, since a local Hilbert space of dimension d can have at most $\log d$. This is the meaning of the statement that the singlet is "maximally entangled," since it uses all the capacity of the local Hilbert space for entanglement.

7.6 Entanglement entropy of a solvable dimerized chain

There are two exactly solvable models that show how (7.6)-(7.7) scale with the system size. They are the key to the whole session, so let us go through them carefully.

Let us take a one-dimensional chain of spin-$1/2$'s, but we couple the spins only in separate pairs. There is an antiferromagnetic bond between sites 0 and 1, another one between 2 and 3, another one between 4 and 5, and so on, and there is *no* coupling between different pairs,

$$H = \sum_n J\, \mathbf{S}_{2n} \cdot \mathbf{S}_{2n+1}.$$

This Hamiltonian is easy to solve exactly, because it is a sum of independent Heisenberg dimers. Its ground state is therefore an exact tensor product of singlets, with one singlet per dimer bond,

$$|\Psi\rangle = |\mathrm{singlet}\rangle_{01} \otimes |\mathrm{singlet}\rangle_{23} \otimes |\mathrm{singlet}\rangle_{45} \otimes \cdots\,. \tag{7.8}$$

Physically, this is a valence-bond product state, also called a dimer product state, which is the simplest example of a gapped state with short-range entanglement, and not a spin liquid. The ground state does not break any symmetry of this Hamiltonian, and it is separated from the lowest triplet excitation by a large gap, which is the singlet-triplet splitting of a single dimer. For this reason it is as far from criticality as a spin model can be.

Now we divide the chain into two parts and ask for the entanglement entropy. Since (7.8) is a tensor product over separate bonds, the reduced density matrix ρ_A factorizes bond by bond. A dimer that is fully inside A, or fully inside B, gives a pure state in ρ_A, and therefore it gives exactly zero to S, with the same argument as for the product state in Section 7.5. The only bonds that contribute are the ones that are cut by the boundary between the two parts. Each cut bond gives the single-dimer result (7.7), $\log 2$, since tracing out one half of a singlet always leaves the same maximally mixed single-site density matrix, regardless of what happens in the rest of the chain. So if the boundary cuts n_{cut} bonds, $S = n_{\text{cut}} \log 2$.

Let us take a small subsystem A whose two ends cut through two dimers, one on each side. Then two singlet bonds cross the boundary, so $S = 2 \log 2$. Let us now take a much larger subsystem A, with a width of many dimers, whose ends again cut through one dimer each. The boundary still crosses only two singlet bonds, the ones at its two edges, so we get again $S = 2 \log 2$, with no change. **The entanglement entropy of this state does not grow with the size of the subsystem**. It is fully determined by the two points where the one-dimensional boundary is, for any distance between these two points.

7.7 Two dimensions and the area law

Let us now do the same construction in one more dimension. We take a two-dimensional lattice and we group the sites into separate clusters of four sites, called tetramers, instead of pairs. The sites inside a cluster are coupled, but different clusters are not coupled, so the ground state is again an exact tensor product, now of tetramer ground states instead of dimer singlets. The same factorization argument applies. A cluster fully inside A or fully inside B gives nothing, and only the clusters that cross the boundary between the two parts contribute to S.

The only difference is geometric. Let us compare a small region A, whose boundary cuts eight bonds, with a much larger region A', whose longer boundary cuts many more bonds. Since the entropy again gets one contribution per cut bond, S is now set by the *length of the boundary*. In two dimensions this length grows when the subsystem grows. This is different from the 1D case of Section 7.6, where the boundary was always two points for any size of the subsystem. This is the central contrast. **In one dimension the entanglement entropy of a gapped ground state saturates**, and it does not depend on the size of the subsystem. **In two or more dimensions it grows with the size of the boundary**, $S_A \sim L_A \sim \partial A$ and $S_{A'} \sim L_{A'} \sim \partial A'$, so $S_{A'} > S_A$ for the larger region. Figure 7.2 shows the two cases next to each other, with the cut bonds marked, so that we can count them by eye in each case.

This behavior is called the **area law**. For the ground state of a gapped and local Hamiltonian, the entanglement entropy of a subsystem scales with the size of the boundary between the subsystem and the rest of the system, which is a fixed number of points in 1D, a length in 2D and an area in 3D,

$$S_A \;\sim\; \alpha\, |\partial A|, \tag{7.9}$$

and not with the *volume* of A. This is a strong restriction. A random state taken uniformly from the full Hilbert space of dimension 2^L would usually have S_A close to its maximum value, which is $|A| \log 2$ for a spin-1/2 subsystem of size $|A|$, and this is called a volume law. In this sense, gapped ground states are very far from typical states. They live only in the very small part of the Hilbert space that satisfies the area law. The dimer and tetramer models above are exactly in this part, and realistic gapped Hamiltonians are approximately in it.

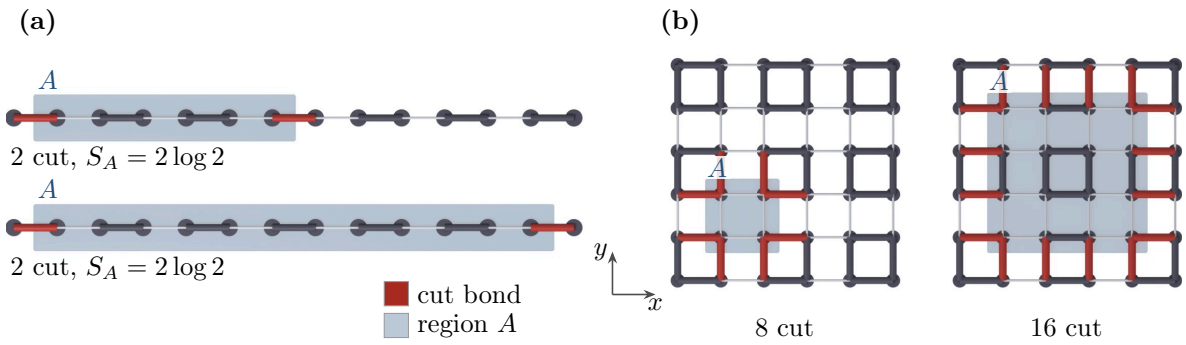


Figure 7.2: Why the entanglement entropy counts the boundary and not the volume, seen from above. In both panels the ground state is an exact product of the ground states of decoupled clusters, so a cluster that is fully inside A or fully inside B is in a pure state and gives nothing. Only the clusters that are cut by the division contribute, and the cut bonds are the red ones. (a) The dimerized chain. Each cut singlet gives exactly the single-dimer result $\log 2$ of (7.7), and the boundary of a block is two points for any length of the block. Each end of A cuts through one dimer, so making A larger, from two interior dimers to five, leaves two cut singlets and $S_A = 2\log 2$ unchanged. The entropy saturates. (b) The same construction with clusters of four sites on a square lattice. Each cluster that crosses the boundary still gives a bounded amount, at most $\log 2$ per cut bond, but now the boundary is a curve whose length grows with the subsystem. When we double the linear size of A, the number of cut bonds doubles, from eight to sixteen, and S_A grows in proportion. This is the area law (7.9), $S_A \sim \alpha\,|\partial A|$, and the contrast between the two panels is the reason why matrix product states are a one-dimensional tool.

7.8 Why MPS work in 1D, and where they stop working

The area law is the missing piece that explains why the ansatz (7.5) is a good one. Let us take an MPS at a given bond, and use it to build the Schmidt decomposition of the state for a division of the system at that bond.

$$|\Psi\rangle = \sum_{\alpha=1}^{M} \lambda_\alpha\, |\alpha\rangle_A \otimes |\alpha\rangle_B,$$

This sum has at most M terms, because the internal index, also called virtual index, that runs through that bond only takes M values. Therefore the reduced density matrix $\rho_A = \sum_\alpha \lambda_\alpha^2 |\alpha\rangle_A\langle\alpha|_A$ has rank at most M. Among all density matrices with rank $\leq M$, the entropy (7.6) is maximal when all the eigenvalues are equal, $\lambda_\alpha^2 = 1/M$, which gives

$$S \;\leq\; \log M. \tag{7.10}$$

In other words, an MPS with bond dimension M can have an entanglement entropy of at most $\log M$ across any bond, which grows only logarithmically with M. Let us compare this with (7.9). A 1D gapped ground state has an entanglement entropy that *saturates* at a finite value, which does not depend on the size of the subsystem, so a moderate and fixed bond dimension is already enough to describe it almost exactly. This is the precise and quantitative reason why matrix product states are such a good ansatz in one dimension, since the area law limits the same quantity that the bond dimension limits. Geometrically, we can think of the full many-body Hilbert space as a large set, and MPS describe only a very small part of it, which is the part of area-law states. This part is reached with a bond dimension that stays moderate ($M \sim 10$, 100) as long as we stay in the area-law regime. To reach a general state with a volume law, the bond dimension would have to become very large, $M \sim 2^{L/2}$, which is exponential in L. Figure 7.3 shows this comparison directly. It shows the upper limit $\log M$

given by (7.10) and the entropy that a gapped, a critical and a general state need when the subsystem grows.

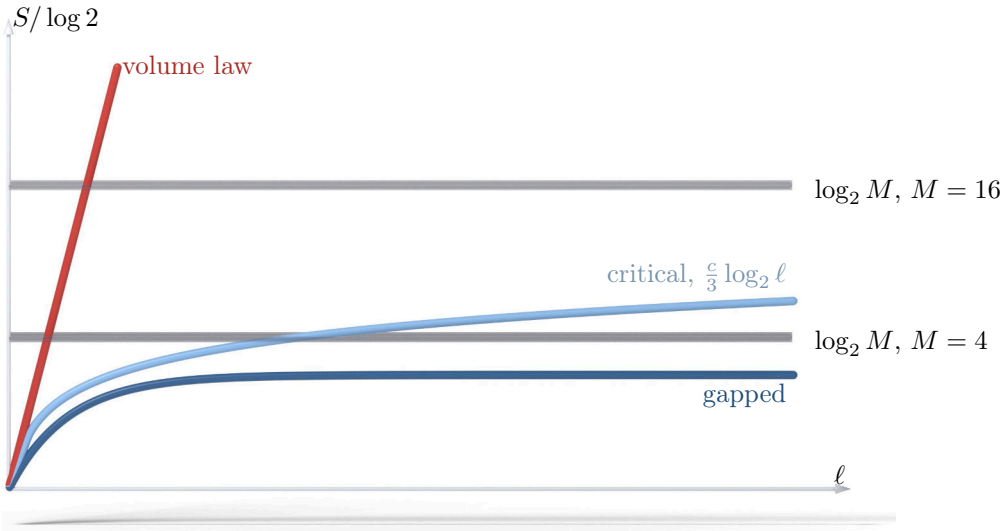


Figure 7.3: The bond dimension is an upper limit on the entanglement, and this determines when matrix product states work. An MPS with bond dimension M has at most M Schmidt terms across any bond, so by (7.10) it can have at most $S = \log M$ there, which is shown by the two horizontal bars, for $M = 4$ and $M = 16$. A gapped one-dimensional ground state follows the area law of Section 7.6, and its entropy saturates at a value set by the correlation length and not by the subsystem size, so it stays below a fixed and moderate limit for any chain length. A critical chain violates the area law logarithmically, (7.12), so it goes above any fixed limit at some point, here the lower bar, and M must be increased as the system grows, slowly but without bound. A general state of the Hilbert space of dimension 2^L follows a volume law, $S = \ell \log 2$, and it goes above any usable M after a few sites. This is why time evolution over long times, highly excited states and two-dimensional systems are all out of reach. The entropy is drawn in bits, $S/\log 2$, with central charge $c = 1$, and the curves are schematic and not fits to a specific model.

The same argument tells us directly where MPS must fail, which is whenever the true entanglement entropy is larger than what a practical bond dimension can give through (7.10). In practice this happens in four cases. The first case is dynamics over long times, since after a quench the entanglement grows *linearly* with time and not logarithmically. The second case is systems in more than one dimension, where the length of the boundary in the area law grows with the system size, as we saw in Section 7.7, so M has to grow too, often exponentially in the linear size of the subsystem. The third case is highly excited states, since the area law is a statement about ground states and not about general eigenstates. The fourth case is, more generally, states far from equilibrium. However, when the physics stays close to a gapped 1D ground state, MPS can be used for more than the ground-state problem. With a higher cost, they can be used for time evolution, excited states and open quantum systems, routinely in 1D and in some cases in 2D.

Notebook check-in: `session7.ipynb`, "Near linear scaling with system size with matrix product states" (Core) and "The role of the bond dimension in tensor networks" (Core)

In the first exercise you check directly that the cost of the DMRG ground-state calculation in `dmrgpy` grows linearly with the system size, which is the $L \times M^2$ scaling of (7.5) and not the 2^L scaling of (7.4). In the second exercise you change the bond dimension and see how the fluctuation of the ground-state energy becomes smaller and then saturates. The point where it saturates is where M is large enough to satisfy the bound (7.10) for that state.

Notebook check-in: `session7.ipynb`, "Entanglement entropy of the ground state" (Core), "Size dependence of the entanglement entropy" (Challenge), and "Bond dimension dependence of the entanglement entropy" (Challenge)

With these three exercises you can reproduce Sections 7.5 to 7.8 numerically. First, you compute the entanglement entropy at each bond for the MPS ground state of a Heisenberg chain. Then you check that it grows very little with the system size, which is the 1D area-law saturation of Section 7.6, but this time in a gapless chain and not in a gapped one. Finally, you check that it saturates once the bond dimension is large enough to satisfy (7.10). The quiz of the first exercise also asks you to replace the model with an Ising model and to see how the entropy changes. This is a quick way to see how the area law responds to a really different ground state, which in this case is classical and has zero entanglement.

7.9 Tensor diagrams as notation for networks and operators

Matrix product states come with a convenient graphical notation, which makes it much easier to think about both the ansatz and the algorithms built on it. A scalar, which is a tensor of rank 0, is drawn as a point. A vector, of rank 1, is a point with one free leg. A matrix, of rank 2, is a point with two legs. A tensor of rank 3 is a point with three legs, and so on for any rank. Each leg represents one index of the tensor, and when we join two legs of two tensors, this means that we sum over the shared index, which is called a contraction.

An MPS is then a horizontal chain of such points, with one point per site. Each point has one *physical* leg pointing down, which is the index s_i, the local spin value. It also has one or two *virtual* legs that connect it to its neighbors in the chain, and these carry the bond-dimension index that is summed over. A **matrix product operator** (MPO) represents an operator, in particular the Hamiltonian, in the same way. Now each site has *two* physical legs instead of one, which correspond to the two indices σ_i, σ_i' of the operator, and the same virtual legs that link neighboring sites. To apply an MPO to an MPS, for example to compute $H|\Psi\rangle$, we join the two diagrams along the physical legs and contract them. This means that we multiply every pair of matrices that share an index and sum over that index. If we do this in a naive way, the bond dimensions of the two objects are multiplied, which would remove the advantage of the method. In practice, we do a singular value decomposition of the result and keep only the largest singular values, so we truncate back to a bond dimension that we can handle, and we remove the least important part of the Hilbert space that is represented. This truncation step is the practical core of almost every MPS algorithm, including the ground-state search of the next section. Figure 7.4 shows together the notation, the ansatz (7.5) drawn in this notation, and this step of contraction and truncation.

7.10 The variational principle and DMRG for the ground state

An ansatz is only useful if we can optimize it. We take the MPS matrices $\{M_i^{s_i}\}$ as variational parameters and minimize the Rayleigh quotient

$$E[\Psi] = \frac{\langle\Psi|H|\Psi\rangle}{\langle\Psi|\Psi\rangle}, \tag{7.11}$$

or, in an equivalent way, we introduce a Lagrange multiplier λ for the normalization and minimize $\langle\Psi|H|\Psi\rangle - \lambda\,(\langle\Psi|\Psi\rangle - 1)$ with respect to every matrix in the network. In the diagrams, the derivative with respect to one tensor in the network, for example the matrix at site k, corresponds to "erasing" that tensor from the diagram. The reason is that each tensor appears once in the bra layer and once in the ket layer, so with the other tensors fixed the energy is a quadratic form in that tensor, and it is linear in the bra copy alone. When we

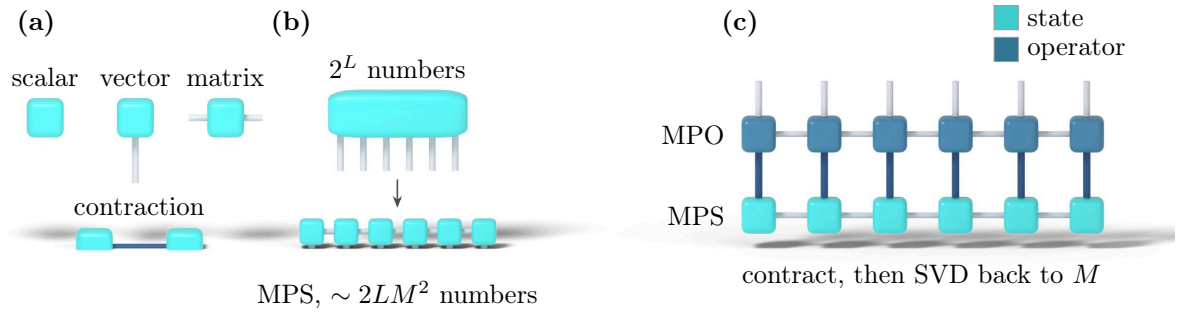


Figure 7.4: The graphical notation, and the MPS ansatz drawn in it. (a) A tensor is a block and each of its indices is a leg, so a scalar has no leg, a vector has one leg, a matrix has two, and so on. Joining two legs means summing over the shared index. (b) The exact coefficient tensor of (7.4) is a single object with one leg per site, and it contains 2^L independent numbers. The matrix product state (7.5) replaces it with a chain of small tensors. Each of them has one physical leg pointing down and virtual legs of dimension M that link it to its neighbours, for a total of $\sim 2LM^2$ numbers, so the exponential growth is replaced by a chain that grows linearly. (c) An operator, in particular the Hamiltonian, becomes a matrix product operator, the darker row, with two physical legs per site. To apply it to a state we join the diagrams along the shared physical legs and contract them, which multiplies the two bond dimensions. A singular value decomposition that keeps only the largest singular values then truncates the result back to a usable M. This truncation is the core of every MPS algorithm, including DMRG.

erase the bra tensor of site k from both the numerator and the denominator diagrams, what is left is a generalized eigenvalue problem for that single tensor, which becomes an ordinary one when the other tensors are chosen orthonormal. It involves an effective, reduced Hamiltonian that acts on the indices of site k, and its ground eigenvector is the locally optimal replacement for $M_k^{s_k}$. Figure 7.5 shows this erasure and the sweep that follows from it.

This suggests an iterative algorithm. We fix every tensor except the one of site 1, solve the local eigenvalue problem there, and update $M_1^{s_1}$ with the resulting eigenvector. Then we move to site 2, fix everything else, solve the local eigenvalue problem there, and update. We continue sweeping through the chain, site by site, until the energy stops decreasing. This sweeping procedure is the **density matrix renormalization group (DMRG)** algorithm. Its efficiency comes from the property of (7.11) that we used to build it. At every step we optimize only one matrix at a time, which is a small linear-algebra problem that we can solve, instead of searching for the ground state in the full exponentially large space at once.

Notebook check-in: `session7.ipynb`, "Low energy excitations of a Heisenberg model" (Core)

In this exercise you look at the matrix elements between the DMRG ground state and the first excited state of a Heisenberg chain, meaning what the sweeping procedure of Section 7.10 finds one level above the state that directly minimizes (7.11).

7.11 MERA and PEPS beyond MPS

Matrix product states are the simplest member of a much larger family, which is called **tensor networks**. Nothing requires the tensors to sit in a 1D chain, and the same coefficient tensor C of (7.4) can also be drawn as a 2D grid or as a general graph of connections. A richer connectivity makes the optimization harder, and MPS are the tensor network that is easiest to optimize, but in exchange it can represent a larger class of states. Let us mention two of these architectures.

The **multi-scale entanglement renormalization ansatz (MERA)** organizes its ten-

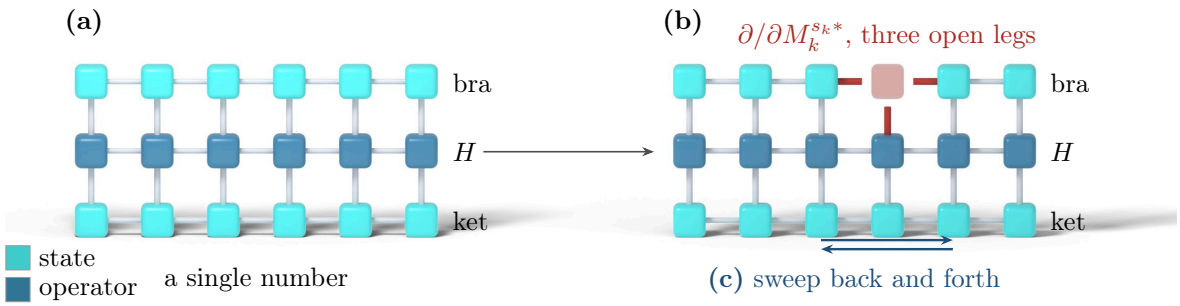


Figure 7.5: How DMRG turns an exponentially large minimization into a sequence of small ones. (a) The numerator of the Rayleigh quotient (7.11) is a single closed network, made of the ket MPS, the Hamiltonian as an MPO and the bra MPS, all contracted together. It has no free legs, so it is a number. (b) Every tensor enters that network once in the bra layer and once in the ket layer, so the energy is a quadratic form in each tensor and linear in its bra copy, and the derivative with respect to the bra tensor at site k removes it, which is the ghost. What is left has exactly three open legs, drawn in red, the two virtual bonds and the physical index of site k, which are the indices of the tensor $M_k^{s_k}$ that we want to find. Everything to the left and right of k has been reduced to an environment, and the stationarity condition is a generalized eigenvalue problem, which is an ordinary one in an orthonormal gauge, with the size of a single tensor. (c) We solve it, move one site along, solve again, and sweep back and forth until the energy stops decreasing. This is the density matrix renormalization group, and at no point do we build an object of size 2^L.

sors in a hierarchy of length scales instead of along a single chain. This extra structure allows it to describe more entanglement than an MPS with a comparable number of parameters. This is important because the area law (7.9) is strictly a statement about *gapped* systems. At a critical point, which is a continuous quantum phase transition usually described by a conformal field theory in 1+1 dimensions, the entanglement entropy of the ground state does not saturate anymore. Instead, it grows logarithmically with the size of the subsystem,

$$S(\ell) \;=\; \frac{c}{3}\log \ell \;+\; \text{const.}, \tag{7.12}$$

where c is the central charge of the CFT. This is a weak but real violation of the strict area law (7.9), and the extra hierarchical structure of MERA is built to describe this kind of weak violation. For this reason MERA is the tensor network that is used at criticality.

Projected entangled pair states (PEPS) are the direct 2D generalization of the MPS idea. Instead of tensors coupled in a line, the tensors sit on the nodes of a real 2D lattice, and they are coupled to all their lattice neighbors. This is a natural and promising architecture for 2D quantum many-body problems, but it has a real cost. For PEPS there is no analogue of the simple local eigenvalue problem for a single tensor that we have in DMRG, because the contraction of a general 2D tensor network is itself a hard computational problem. This is different from the 1D contraction, which is what made the local eigenvalue problem of Section 7.10 possible to solve. The optimization of PEPS is still an open and harder problem. Figure 7.6 shows both architectures, where the extra structure of each one comes from and what it costs.

This is also where the lecture of this year ends. In contrast with some previous editions of this course, the recording of this year stops after MERA and PEPS, and it does not continue with neural network quantum states. The single mention in Section 7.3 is all that we say about this topic in this session. The tensor-network methods built in Sections 7.4 to 7.11 are still important, since in session 10 we will use the same MPS, MERA and PEPS architectures directly as machine-learning classifiers, and we will come back to neural network quantum states in detail, from the point of view of machine learning.

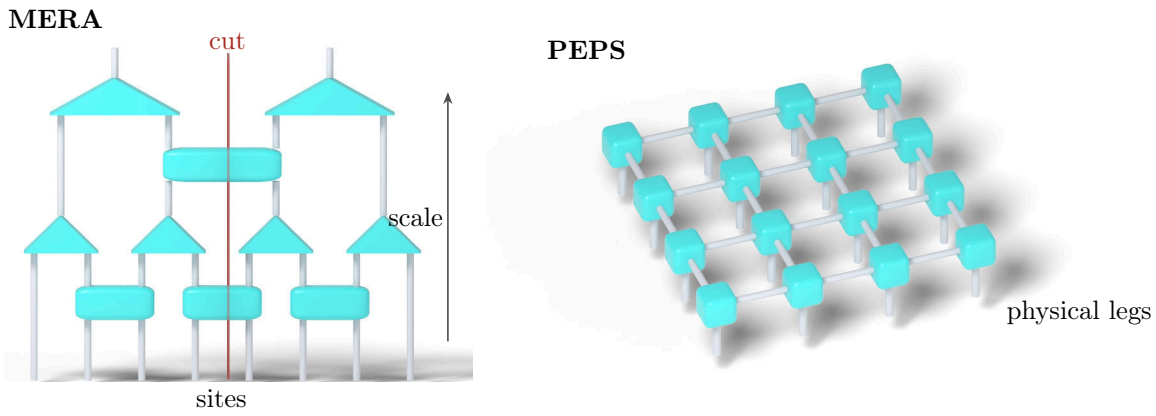


Figure 7.6: The two tensor networks beyond the chain, and what their geometry gives. Left, the multi-scale entanglement renormalization ansatz. Its tensors are stacked in layers. Each layer first removes the short-range entanglement between neighbouring blocks with the disentanglers, the boxes, and then it combines two sites into one with the triangular isometries, so the network describes the state at longer and longer length scales and not at successive positions along a chain. A division of the system, the red plane, now cuts one bond in every layer instead of one bond in total, and this is how MERA can describe the logarithmically growing entropy (7.12) of a critical chain, which an MPS with fixed bond dimension cannot reach. Right, projected entangled pair states, the direct 2D generalization of (7.5), with one tensor per lattice site that has a physical leg and four virtual bonds. The figure also shows the difficulty. There is no way to sweep along such a network one tensor at a time with the rest reduced to a simple environment, and the exact contraction of the network is itself a hard problem, which is why PEPS have no simple analogue of DMRG.

Notebook check-in: `session7.ipynb`, "The entanglement entropy as a signal for phase transitions" (Challenge)

In this exercise you follow the entanglement entropy of the ground state of the transverse-field Ising chain while you change the transverse field across its quantum critical point, using the MPS methods of `dmrgpy`. Away from criticality you should see the area-law saturation of Section 7.8. Close to the transition, you should see that the entropy does not saturate, as predicted by (7.12). The quiz also asks which regime is the hardest for MPS, and this question asks you to connect back to the reason why MERA exists.

7.12 Closing summary

- Strongly correlated matter, including quantum spin liquids and fractional Chern insulators, is the regime where the mean-field methods of sessions 1 to 6 have no order parameter to converge to, as we saw with (7.1) and in Section 7.1.
- The **Jordan-Wigner transformation** (7.2) is an exact mapping between fermionic and spin-1/2 Hamiltonians, summarized as "spin equals fermion times string." It is the link between the fermionic language of sessions 1-5 and the spin language of sessions 6 and 8, and it is the reason why we can work in the spin representation for the rest of the session.
- The real obstruction to solving many-body Hamiltonians is **storage**, since a system of L spins needs 2^L coefficients (7.4) to describe its wave function exactly.
- The **matrix product state** ansatz (7.5) replaces these 2^L coefficients with $O(LM^2)$ parameters, so we give up exactness in exchange for an approximation that we control with the bond dimension M.

- The **entanglement entropy** (7.6), which we computed for the singlet in (7.7), measures how entangled a state is. Exactly solvable dimer and tetramer models show that gapped ground states follow an **area law** (7.9), with an entropy set by the length of the boundary and not by the volume of the subsystem.

- The entanglement of an MPS is itself bounded, $S \leq \log M$ (7.10). This is the reason why MPS work so well in 1D, where the area law saturates, and why they need an exponentially large M in higher dimensions or away from the ground state.

- Ground states are found variationally, by minimizing (7.11) one tensor at a time, and this sweeping procedure is the **DMRG** algorithm. **MERA** and **PEPS** extend the idea of tensor networks to criticality, through the logarithmic correction to the entropy (7.12), and to real 2D lattices, respectively. MERA can still be optimized tensor by tensor, while for PEPS the contraction is the hard step, so the simple local optimization of DMRG is lost.

- Neural network quantum states were mentioned once, together with tensor networks, as a second answer to the same storage problem, but they were not developed in this session. In session 10 we will study them in detail from the point of view of machine learning, and we will also come back to the MPS, MERA and PEPS methods of this session as architectures for machine-learning classifiers.

Session 8

Quantum Magnetism, Spinons and Kondo Physics

In Session 6 we only studied magnets that break time-reversal symmetry, which are ferromagnets and antiferromagnets. In the last part of that session we used the Holstein-Primakoff transformation to write these ordered states in terms of bosons, the magnons, and we went as far as a topological magnon band structure with its own chiral edge modes. That session stopped suddenly, at "topological magnons," without a final recap. It also left one open point. There was a short remark about Kitaev-type interactions, which were called "important for quantum spin liquids" at that time, but we did not continue with them. In this session we come back to that point, as a lecture of its own, with its own introduction and its own plan. First, we will see minimal models of quantum magnetism. Then we will see the excitations that quantum spin liquids (QSLs) have, which are spinons and gauge excitations. At the end we will study Kondo systems and Kondo lattice models, which are a very different physical system that can be solved with the same method. The idea that connects the whole session is the same idea that we have used in this course since Session 1. We take a many-body Hamiltonian that mean-field theory cannot treat, we write it with the right auxiliary particles, and we do mean-field theory on *that* problem.

In this session we use $\hbar = 1$ for the spin operators, which is the same convention that we already used without saying it in Session 6.

8.1 Two classes of magnets and where we find them

In the language of this lecture, every magnetic material belongs to one of two classes. **Classical magnets** break time-reversal symmetry, and examples are conventional ferromagnets, antiferromagnets and spin spirals. For the electrons, these magnets can be described with mean-field theory, in the same way as we did in Sessions 1 and 6. Most magnetic materials in nature are of this type, for example iron, copper oxide, or any material with a well-defined Néel temperature.

Quantum magnets are the opposite case. They cannot be described with mean-field theory in terms of electrons, and their ground states have long-range entanglement. This entanglement is not a small detail, since it gives rise to excitations that do not exist in a symmetry-broken magnet. In this session we will study two physical realizations of quantum magnetism, which are quite different from each other.

- **Quantum spin liquids (QSLs)**, which are the first class that we build in detail. The best-known candidate material is herbertsmithite, a compound where the magnetic ions form a kagome lattice of spin-1/2 sites.

- **Heavy-fermion Kondo insulators**, which are often rare-earth compounds, and the standard example is samarium hexaboride (SmB_6). We build the theory for this class in the second half of the session.

8.2 Two emergent excitations, magnons and spinons

The strong many-body interactions between electrons are the origin of the emergent collective excitations of magnetic materials. In a symmetry-broken magnet, this collective excitation is the **magnon**, which has spin 1 and no charge. The reason is that the strong repulsion between electrons freezes the charge degree of freedom completely, so only the spin can fluctuate. In a quantum magnet, the excitation is of a different kind. It is the **spinon**, which has spin 1/2 and also no charge. It is useful to think of a spinon as the spin half of an electron. If we take this picture one step further, we obtain the standard language of fractionalization for a QSL, where the electron splits into a **chargon**, which has the charge, and a **spinon**, which has the spin.

There is a simple argument with pictures that shows why the spin of the two excitations differs by exactly a factor of two. Let us take a ferromagnet, where the ground state has all spins up, $|\uparrow\uparrow\uparrow\cdots\rangle$. If we flip one spin to build an excited state, the total spin of the excited state differs from the one of the ground state by $\Delta S = 1$, and this is a local magnon. Let us now take a quantum spin liquid, where the ground state is a collection of singlets everywhere and no spin points in a specific direction. If we build an excited state by leaving one spin without a partner, instead of putting it in a singlet, the difference in total spin is $\Delta S = 1/2$, and in this picture this is a spinon. This picture is too simple on purpose, since real spinons are more subtle than "an unpaired spin," but it is the right first intuition for why symmetry-broken magnets and quantum spin liquids have fundamentally different excitations. Figure 8.1 shows the two counting arguments next to each other.

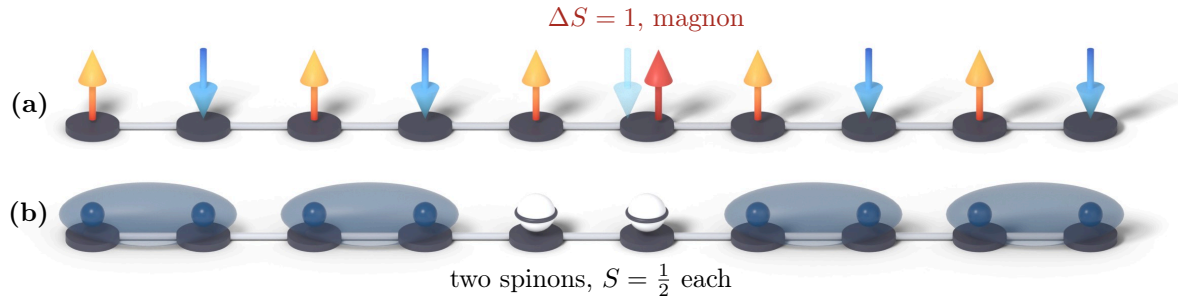


Figure 8.1: Why the two kinds of magnet have excitations whose spin differs by exactly a factor of two. (a) In a symmetry-broken magnet every site has a definite spin, shown with arrows. Flipping one costs $\Delta S = 1$, and this excitation is the magnon of session 6. The single flipped spin drawn here, next to the ghost of its old direction, only does the counting of ΔS, since the real magnon is the delocalized precession of figure 6.5 and not one flipped site. (b) In a quantum magnet no site has a definite spin at all. The sites are paired into singlets (8.3), drawn as capsules, so $\langle \mathbf{S}_i \rangle = 0$ everywhere and $S_{\text{tot}} = 0$. When we break one pair, we leave *unpaired* sites, drawn as open balls, and each one of them has $S = 1/2$. This is a spinon, which is half a magnon. For this reason spinons are always created in pairs, and figure 8.3 shows whether the pair stays bound or separates.

8.3 Two minimal dimers, a classical order parameter and a quantum singlet

The simplest way to start with quantum magnetism is to look first at the smallest possible magnet, which has two spins, and to compare two different Hamiltonians for it.

The first one is the **Ising dimer**,

$$\mathcal{H}_{\text{Ising}} = S_0^z S_1^z, \tag{8.1}$$

which couples only the z-components of two spin-1/2's. This Hamiltonian has two degenerate ground states, which are related by time reversal.

$$|GS_1\rangle = |\uparrow\downarrow\rangle, \qquad |GS_2\rangle = |\downarrow\uparrow\rangle.$$

Both are eigenstates with the same energy, but each one of them *alone* breaks time-reversal symmetry, since time reversal flips every spin and so it turns $|GS_1\rangle$ into $|GS_2\rangle$ and the other way around. Only the degenerate pair of the two states together respects the symmetry. A macroscopic symmetry-broken antiferromagnet, of the same kind as the one we built in Session 6, is the version of this same problem with an infinite number of sites, and the same argument also works for ferromagnets and spin spirals.

Let us now replace the Ising coupling by the full **quantum Heisenberg dimer**, which has rotational symmetry,

$$\mathcal{H}_{\text{Heis}} = \mathbf{S}_0 \cdot \mathbf{S}_1. \tag{8.2}$$

This is the model that we obtain from a half-filled Hubbard model with strong interactions in the large-U limit, as we derived in Session 6. When we solve it with $\mathbf{S}_0, \mathbf{S}_1$ as real spin-1/2 operators, the two spin-1/2's combine into a singlet ($S = 0$) and a triplet ($S = 1$). The value is $\mathbf{S}_0 \cdot \mathbf{S}_1 = -3/4$ for the singlet and $+1/4$ for the triplet, so the unique ground state is the **singlet**,

$$|GS\rangle = \frac{1}{\sqrt{2}}\Big(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\Big), \qquad \langle \mathbf{S}_i \rangle = 0. \tag{8.3}$$

This ground state is different from the Ising case in three ways, and these three points are important for the rest of the session. First, it is the *unique* ground state, so there is no degeneracy and no spontaneous choice between states. Second, it is *time-reversal symmetric*, since its net magnetization is exactly zero and time reversal maps $|GS\rangle$ to itself. Third, it is *maximally entangled*, since the two terms in (8.3) have the same weight, which is the largest entanglement that two qubits can have. Can we have a macroscopic version of this ground state? This question is the origin of quantum spin liquids.

8.4 Frustration as the route to a quantum spin liquid

Before we build a macroscopic version of (8.3), we need to know when the ground state of a Heisenberg model stays time-reversal symmetric in this way, instead of ordering spontaneously as the magnets of Session 1 and Session 6 did. The general rule is **frustration**. The standard example is a triangular plaquette with antiferromagnetic coupling on all three bonds. Any two spins can point in opposite directions, but then the third spin is parallel to one of the two, so one of the two remaining bonds is not satisfied, and no spin configuration can satisfy all three bonds at the same time. This geometric frustration gives many classical ground states that are degenerate, which means that they have the same energy and different spin

configurations. When we add quantum fluctuations, they mix these degenerate configurations into one quantum superposition, which in general does not break time-reversal symmetry.

This is the reason why the **triangular** and **kagome** lattices, which are both built only from triangles, are the standard place to look for quantum spin liquid physics. An antiferromagnetic Heisenberg model on either lattice is a real candidate for a QSL ground state, in the right range of couplings. This is also the reason why herbertsmithite, a compound with spin-1/2 sites on a kagome lattice, is the best-known real candidate material. Figure 8.2 shows the geometry behind the whole argument, which is a frustrated triangle with its degenerate configurations and the kagome lattice that is built only from such triangles, while the plaquette without frustration and the single frustrated triangle are the ones of figure 6.3.

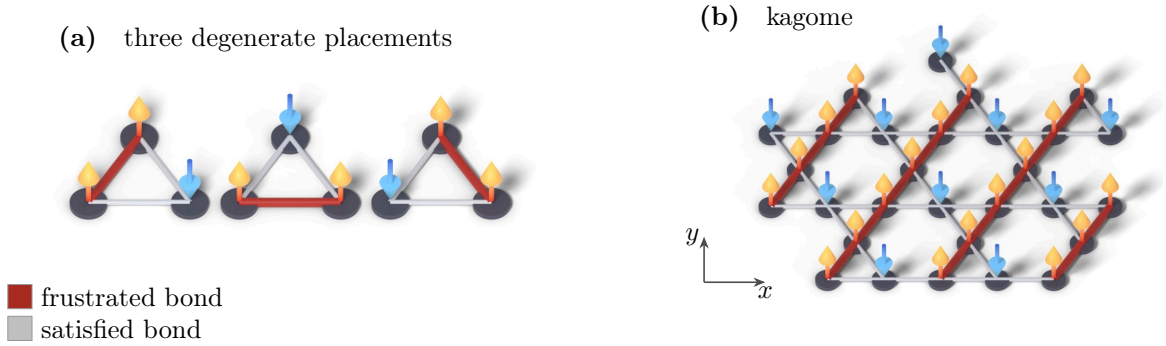


Figure 8.2: Frustration, with the spins drawn as Ising (collinear) variables, which is how the argument in the text is written. If the spins can rotate freely, the triangle goes instead into the 120° noncollinear state of figure 6.3, and the degeneracy that we discuss here is the Ising one. The plaquette without frustration and the single frustrated triangle are panels (a) and (b) of figure 6.3. On a bipartite plaquette every antiferromagnetic bond can be satisfied at the same time, and the ground state is the Néel state, with two configurations related by a global flip, each of which breaks time-reversal symmetry. On a triangle this is not possible, since once two spins are antiparallel the third bond cannot be satisfied for both directions of the last spin. (a) The frustrated bond, in red, can be on any of the three edges, so the classical ground state is degenerate and not unique. Quantum fluctuations mix the degenerate configurations into one superposition with $\langle \mathbf{S}_i \rangle = \mathbf{0}$, which does *not* break time-reversal symmetry. (b) The kagome lattice of herbertsmithite is built only from corner-sharing triangles, so all of them are frustrated at the same time, here with one frustrated bond on every triangle, and the classical degeneracy becomes macroscopic. This is why triangular and kagome lattices are the standard place to look for quantum spin liquids.

8.5 The dimerized Heisenberg model, a first step before a spin liquid

Before we study a frustrated Hamiltonian, it is useful to build the simplest possible *macroscopic* quantum magnet and to check explicitly that its ground state does what (8.3) told us. Let us take a **fully dimerized Heisenberg model**, which is made of dimers that are completely decoupled from each other, placed on a chain or on a two-dimensional lattice,

$$H = \sum_n J\, \mathbf{S}_{2n} \cdot \mathbf{S}_{2n+1}, \qquad J > 0. \tag{8.4}$$

Since every dimer is decoupled from its neighbors, we can write the many-body ground state exactly, as a product of singlets over the pairs of the dimers,

$$|\Psi\rangle \sim \bigotimes_n \Big(|\uparrow_{2n}\downarrow_{2n+1}\rangle - |\downarrow_{2n}\uparrow_{2n+1}\rangle \Big), \qquad \langle \mathbf{S}_i \rangle = 0 \text{ on every site.} \tag{8.5}$$

In one sense this is a trivial Hamiltonian, a collection of independent zero-dimensional problems. But it already shows something important. A macroscopic Heisenberg Hamiltonian *can* have a ground state that does not break time-reversal symmetry, and this state is the direct generalization of the two-site singlet (8.3) to an extensive system. If we add small couplings between the dimers, the state changes smoothly and it still does not break time-reversal symmetry.

There is, however, an important difference to make here, because this state is *not yet* a quantum spin liquid. Its fundamental excitations still have spin 1. If we turn one singlet into its triplet partner, we obtain a **triplon**, which is a local transition from singlet to triplet, and it has the same structure as a magnon, with only a different name. A state that is built in this way, from a fixed and static pattern of singlets, is called a **valence bond solid (VBS)**. To obtain a real quantum spin liquid, with spin-1/2 excitations instead of spin-1 triplons, we need more than a static dimer covering. We need the singlets to fluctuate and to resonate over the whole lattice, and this is the frustrated case that we study in the next section. Figure 8.3 shows this difference explicitly.

8.6 The resonating valence bond wavefunction as the QSL ground state

For a frustrated Heisenberg Hamiltonian $\mathcal{H} = \sum_{ij} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j$, we do not expect the ground state to be one fixed dimer covering as in (8.5). Instead, we expect a **linear combination of singlets over every possible pairing of sites**, where each pairing has its own amplitude, which is a complicated many-body wavefunction. We could try to treat this directly, for example by writing it as a tensor network as in Session 7 for the general strongly correlated case. There is, however, a way that is easier to understand physically, which is a variational ansatz based on an analogy with superconductivity.

The idea is that the singlets in a QSL are similar to the Cooper pairs in an s-wave superconductor, since both are singlet correlations between pairs. A BCS-type wavefunction is easy to write at the mean-field level in the right basis, which is the Nambu space as in Session 5. So we start from a BCS-paired state built from the same auxiliary fermions $f_{i\sigma}$ that we will justify in detail in Section 8.8,

$$|\Psi_{\mathrm{BCS}}\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}}\, f^\dagger_{\mathbf{k}\uparrow} f^\dagger_{-\mathbf{k}\downarrow} \right) |0\rangle, \tag{8.6}$$

where $u_{\mathbf{k}}, v_{\mathbf{k}}$ are variational parameters, as in the BCS state of Session 5. By itself, (8.6) is a superposition of states with every possible number of fermions per site, which can be empty, singly occupied or doubly occupied. This is not a physical spin state at all. A QSL has strong local interactions, so we need single occupation on every site, and no charge fluctuations between sites are allowed. We impose this condition with a **Gutzwiller projector** onto the singly-occupied subspace,

$$|\Psi_{\mathrm{RVB}}\rangle = P_G\, |\Psi_{\mathrm{BCS}}\rangle, \qquad P_G = \prod_i n_i(2 - n_i), \qquad n_i = \sum_\sigma f^\dagger_{i\sigma} f_{i\sigma}, \tag{8.7}$$

which gives 1 for states with exactly one fermion per site and 0 for all other states. What remains is a real many-body spin wavefunction. It is a superposition over the different ways to pair sites into singlets, in the same spirit as the singlet of a single dimer (8.3), but now over a whole frustrated lattice. The name of this construction describes what it is, and it is called the **resonating valence bond (RVB)** variational wavefunction. It turns out to be a good description of the physics of quantum spin liquids. Figure 8.3 shows what makes this

resonating state, and the excitations that it has, different from the frozen dimer covering of Section 8.5.

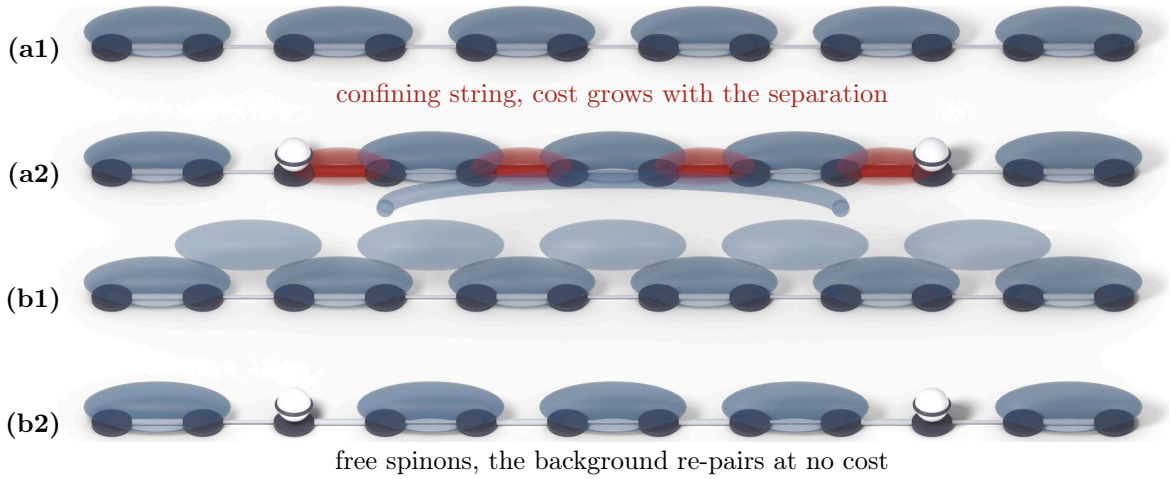


Figure 8.3: The difference between a valence bond solid and a quantum spin liquid is not whether the ground state is built from singlets, since both are, but whether the singlet pattern is frozen. (a1) The fully dimerized model (8.4) has one fixed covering, (8.5). (a2) When we separate the two unpaired spins of figure 8.1b, every strong bond between them loses its singlet, drawn in red, so the energy cost grows with the separation. The pair is confined, and only the bound object propagates, which is the spin-1 triplon. (b1) The RVB state (8.7) is a superposition over *all* pairings, near and far, drawn as coverings stacked on top of each other. (b2) Now the background pairs again behind the excitations with no cost, the confining string does not form, and the two halves separate as independent spin-1/2 spinons. Frustration is the reason why the superposition in (b1), and not a single covering, is the ground state.

8.7 Why the Holstein-Primakoff magnon approach fails for QSL excitations

It is hard to write the exact QSL ground state, so the natural next step, as in Session 6, is to ask about the excitations *on top of* the ground state, and not about the ground state itself. Up to now, the only tool that we have for this is the **Holstein-Primakoff transformation**, which writes

$$S_i^z = S - a_i^\dagger a_i, \qquad S_i^+ \approx \sqrt{2S}\, a_i, \qquad S_i^- \approx \sqrt{2S}\, a_i^\dagger, \tag{8.8}$$

where we linearized under the assumption that the number of bosonic magnons is small, $\langle a_i^\dagger a_i \rangle \ll S$. This is a very good approximation close to a fully or nearly polarized magnet, which is the regime where we used it in Session 6.

A QSL is in the opposite limit. Its defining property, $\langle \mathbf{S}_i \rangle = 0$ on every site, gives through (8.8)

$$\langle a_i^\dagger a_i \rangle = S, \tag{8.9}$$

which is the regime where the condition of a small boson density, the basis of the linearized Holstein-Primakoff mapping, fails completely. For a QSL there is no ordered reference state that it is "close to" and around which we can expand. As a result, the magnon method that worked for the ordered magnets of Session 6 is not the right tool here, and we need a different approximation for the excitations of a quantum spin liquid.

Notebook check-in: `session8.ipynb`, "Magnetization of a large Heisenberg model" (Core)

Diagonalize a quantum Heisenberg chain of moderate size and check numerically that the local magnetization stays zero on every site. This also holds with a small external field, as long as the field is weaker than the finite-size gap to the neighboring sector of total spin. This is how the defining property behind (8.5) and (8.9) appears in a numerical calculation.

8.8 The parton construction with Abrikosov fermions and the single-occupancy constraint

The construction that allows us to treat QSL excitations does not write each spin as a bosonic fluctuation around an ordered reference state. Instead, it writes each spin with an auxiliary **fermion** field $f_{i\sigma}$ ($\sigma = \uparrow, \downarrow$) on every site. This is a particle with spin 1/2 and no electric charge, which is the spinon of Section 8.2, now written in a formal way. These particles are called **Abrikosov fermions**, and the operator identity that defines the parton transformation is

$$\mathbf{S}_i = \frac{1}{2} \sum_{\alpha\beta} f^\dagger_{i\alpha}\, \boldsymbol{\sigma}_{\alpha\beta}\, f_{i\beta}, \tag{8.10}$$

where $\boldsymbol{\sigma}$ is the vector of Pauli matrices. One can check directly, from the anticommutation relations of the fermions, that (8.10) gives the correct $SU(2)$ spin algebra $[S^a_i, S^b_i] = i\varepsilon^{abc} S^c_i$ on the full Fock space of $f_{i\uparrow}, f_{i\downarrow}$, which has four states, namely empty, singly occupied with spin up, singly occupied with spin down and doubly occupied. No extra input is needed for this. The problem is that this transformation makes the local Hilbert space *artificially larger*, since a physical spin-1/2 has only two states, while $f_{i\uparrow}, f_{i\downarrow}$ give four. We remove the two extra unphysical states, which are the empty and the doubly occupied ones and which (8.10) maps to $\mathbf{S}_i = 0$ in any case. We do this by imposing by hand the **single-occupancy constraint**

$$\sum_\sigma f^\dagger_{i\sigma} f_{i\sigma} = 1 \qquad \text{at every site } i. \tag{8.11}$$

This is the fermionic analogue of the Gutzwiller projection (8.7) that we applied before to the many-body wavefunction. To project the many-body state onto the singly-occupied subspace and to fix the fermion filling to one per site are the same statement, the first one seen from the side of the wavefunction and the second one from the side of the operators. Figure 8.4a shows which two states the constraint has to remove, and why they were never spin states in the first place.

When (8.11) holds, (8.10) is an *exact* way to write a spin-1/2 operator in the language of fermions, and we can put it directly into the Heisenberg Hamiltonian $\mathcal{H} = \sum_{\langle ij\rangle} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j$. With the Fierz identity for Pauli matrices $\sum_a \sigma^a_{\alpha\beta}\sigma^a_{\gamma\delta} = 2\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta}$ and the constraint (8.11) on both sites, we obtain the simple identity

$$\mathbf{S}_i \cdot \mathbf{S}_j = \frac{1}{4} - \frac{1}{2}\, \chi^\dagger_{ij}\chi_{ij}, \qquad \chi_{ij} \equiv \sum_\sigma f^\dagger_{i\sigma} f_{j\sigma}, \tag{8.12}$$

where χ_{ij} is the spinon bond operator, or hopping operator, between sites i and j. When we insert (8.12), the original Heisenberg Hamiltonian, which is bilinear in the spins, becomes a Hamiltonian that is *quartic* in the auxiliary fermions,

$$\mathcal{H} = \sum_{\langle ij\rangle} J_{ij} \left(\frac{1}{4} - \frac{1}{2}\, \chi^\dagger_{ij}\chi_{ij} \right) = \text{const} - \frac{1}{2} \sum_{\langle ij\rangle} J_{ij}\, \chi^\dagger_{ij}\chi_{ij}. \tag{8.13}$$

This is the same problem that we had in Sessions 1 and 6 with the Hubbard interaction, which is a quartic term that we cannot diagonalize directly. We treat it in the same way, with a **mean-field decoupling**,

$$\chi_{ij}^\dagger \chi_{ij} \;\approx\; \langle \chi_{ij}^\dagger \rangle \, \chi_{ij} + \chi_{ij}^\dagger \, \langle \chi_{ij} \rangle - |\langle \chi_{ij} \rangle|^2 . \tag{8.14}$$

8.9 The mean-field spinon Hamiltonian and its dispersion classes

We put the couplings and the expectation value of (8.14) together into one mean-field parameter χ_{ij} per bond, and if needed we impose the constraint (8.11) on average with a Lagrange multiplier. In this way, the mean-field decoupling of Section 8.8 turns the original spin problem into a real Hamiltonian for the auxiliary fermions, which has the form of a tight-binding model and which we can diagonalize,

$$\mathcal{H}_{\text{MF}} = \sum_{ij,s} \chi_{ij} \, f_{i,s}^\dagger f_{j,s} . \tag{8.15}$$

This mean-field step is different from all the other mean-field decouplings of this course up to now, since it does not need to break time-reversal symmetry. If we take the same χ_{ij} for spin up and spin down, we keep $\langle \mathbf{S}_i \rangle = 0$ exactly, so the ground state that we obtain is time-reversal symmetric by construction, as (8.5) and (8.9) require. Equation (8.15) is a tight-binding model for the auxiliary fermions, with hoppings χ_{ij} that are fixed self-consistently by the geometry of the lattice and by the exchange couplings J_{ij}. Its excitations are the **spinons**, which are emergent fermionic quasiparticles with spin 1/2 and no charge. We obtain them with the same procedure that we used for electrons in the whole course, which is to take a quartic interaction, decouple it at the mean-field level and diagonalize a bilinear Hamiltonian, and the only difference is that now we apply it to the spin sector. Figure 8.4b shows the resulting picture, with spinons that hop on the lattice with amplitude χ_{ij}, at half filling because of (8.11).

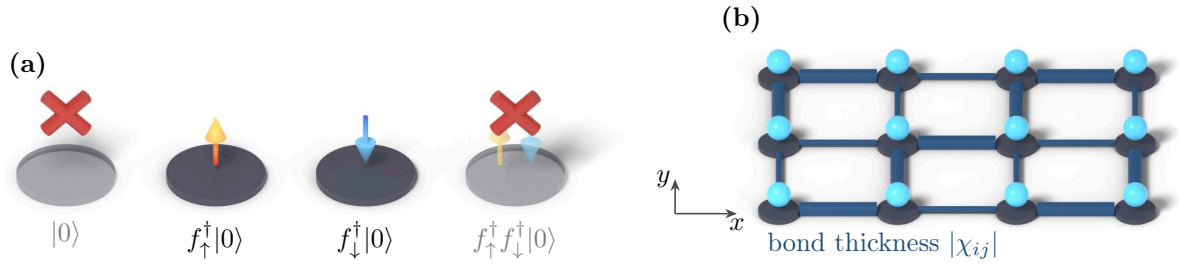


Figure 8.4: What the Abrikosov-fermion rewriting (8.10) costs, and what it gives us. (a) The four fermionic states of a site. The two singly-occupied states give the physical spin-1/2 doublet exactly. The empty and doubly-occupied states, drawn as ghosts with a cross, are extra unphysical states, which (8.10) maps to $\mathbf{S}_i = \mathbf{0}$ and which the constraint (8.11) removes by hand, and this constraint is the operator version of the Gutzwiller projector (8.7). (b) What remains after the mean-field decoupling (8.14), which is an ordinary tight-binding problem (8.15) for fermions with spin-1/2 and no charge, which hop with the self-consistent bond amplitude χ_{ij}, drawn as the thickness of the bond, here with two values for illustration. Since there is one fermion per site, the resulting spinon model must be studied at half filling. Taking the same χ_{ij} for both spin species keeps $\langle \mathbf{S}_i \rangle = \mathbf{0}$, so this mean field, unlike the magnetic mean fields of sessions 1 and 6, does not break time-reversal symmetry.

When we diagonalize (8.15) on a given lattice, we obtain a real spinon band structure. Depending on the mean-field parameters that we obtain, there are three different possible outcomes.

- **Gapless spinons**, with excitations at energies as close to zero as we want.
- **Gapped spinons**, where it costs a finite energy to create a spinon.
- **Dirac spinons**, with a linear relation between energy and momentum at isolated points in the Brillouin zone.

Notebook check-in: `session8.ipynb`, "Spinon Fermi surface in gapless quantum spin liquids" (Core)

Build the spinon tight-binding Hamiltonian (8.15) on the triangular lattice and look at how its Fermi surface changes when the ratio of hopping amplitudes t_2/t_1 changes. In the quiz you repeat this on the square and kagome lattices, and you explain why the effective spinon model must be at half filling. The reason is the single-occupancy constraint (8.11) at the mean-field level.

8.10 Emergent gauge symmetry and vison excitations

There is one more essential part of the QSL picture, and it comes directly from the fact that we introduced fermions to describe a spin. Let us take a local phase rotation of the auxiliary fermions that depends on the site,

$$f_{i\sigma} \;\to\; e^{i\theta_i} f_{i\sigma}, \qquad f^\dagger_{i\sigma} \to e^{-i\theta_i} f^\dagger_{i\sigma}. \tag{8.16}$$

Both the physical spin operator (8.10) and the single-occupancy constraint (8.11) do not change at all under (8.16). The phase $e^{i\theta_i}$ cancels between $f^\dagger_{i\alpha}$ and $f_{i\beta}$ in every term, so the Hamiltonian and the physical spin operators are exactly invariant. The mean-field bond variable χ_{ij}, however, is *not* invariant, since it gets a relative phase, $\chi_{ij} \to e^{i(\theta_i-\theta_j)}\chi_{ij}$. This is the structure of a lattice gauge theory. The variable χ_{ij} behaves as a link variable, and (8.16) is a local transformation, which is called a gauge transformation. The physical spin theory is invariant under it, while the mean-field saddle point of Section 8.9 is not. Any two choices of χ_{ij} that are related by (8.16) describe the same physical quantum spin liquid, so the mean-field decoupling only fixes a class of gauge-equivalent solutions and not a unique set of hoppings. This redundancy is the origin of an **emergent gauge theory** for quantum spin liquids.

Once we see this gauge structure, χ_{ij} becomes a dynamical field on the links of the lattice, and it can have its own topological defects. This is in direct analogy with the superconducting order parameter, whose gauge degree of freedom can have vortices. Let us write the mean-field Hamiltonian as $\mathcal{H} = \sum_{ij,s} |\chi_{ij}| e^{i\theta_{ij}} f^\dagger_{i,s} f_{j,s}$, and let us define the flux through a plaquette as the product of the phases that we pick up when we go around its boundary, $\Phi = \prod e^{i\theta_{ij}}$. For concreteness, let us take a square lattice where the hoppings around every elementary loop pick up the phase $\Phi = +1$, which is a ground state, except for one loop, where the phase is $\Phi = -1$. This single flipped plaquette is an excited state. It is a topological defect in the emergent gauge field, and it is called a **vison**. The vison is the excitation of the gauge sector, and in general it is coupled to the spinon excitations of the matter sector that we built in Section 8.9. In this language, a quantum spin liquid is described not only by its spinon dispersion but also by an emergent gauge theory and by the vison excitations of that theory. This is a new organizing principle, and nothing like it exists in the symmetry-broken magnets of Session 6. There is no notebook exercise for this construction, which is a purely conceptual and formal part of the QSL picture. Figure 8.5 shows both parts of it, why the flux and not the link variable is the physical quantity, and what a single flipped flux looks like on the lattice.

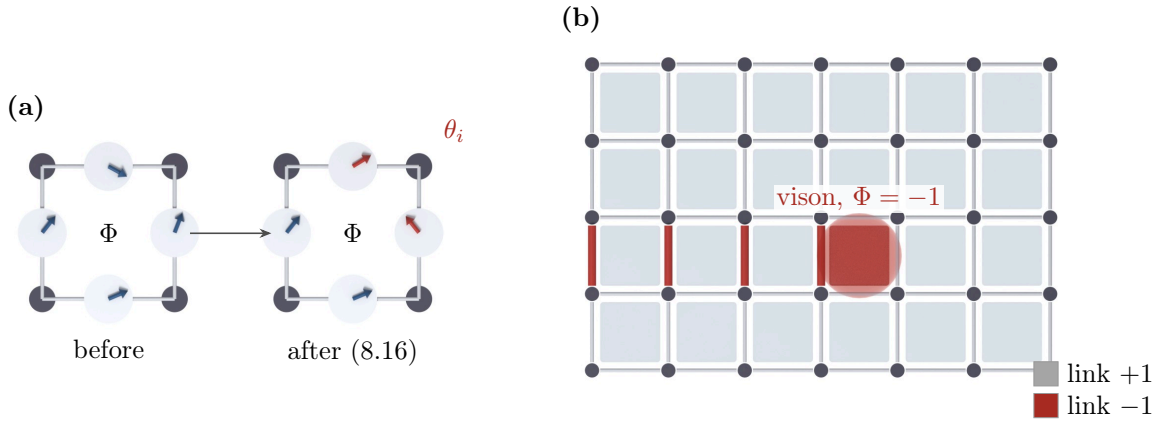


Figure 8.5: The emergent gauge theory of the parton construction, and its excitation, seen from above. (a) The mean-field bond variable is defined on the links, $\chi_{ij} = |\chi_{ij}|e^{i\theta_{ij}}$, drawn as a dial at the phase θ_{ij}, and the local phase rotation (8.16) on one site shifts every link phase by $\theta_i - \theta_j$, which changes the two red needles. When we go once around a plaquette, each site phase appears twice with opposite signs, so the flux $\Phi = \prod e^{i\theta_{ij}}$ does not change. The link variables are pure gauge, and the flux is physical. This is the reason why two mean-field solutions related by (8.16) describe the same spin liquid. (b) As a result, the flux is what labels the state. With $\Phi = +1$ on every plaquette as the ground state, flipping a single plaquette to $\Phi = -1$ costs energy, and no gauge transformation can remove it. This localized defect of the gauge field is a **vison**. Which links have the minus sign is a gauge choice, here a string from the edge, and it can be moved freely, while the shaded plaquette cannot be moved.

8.11 From a single Kondo impurity to the Kondo lattice

Now that we have the method for quantum spin liquids, the lecture moves to a second physical system, which is quite different but which can be solved with the same method. This is the **Kondo lattice model**, which appears whenever a metal and a correlated system of local moments exist together.

Let us start with the simplest case, which is a single spin-1/2 magnetic impurity in a non-interacting two-dimensional electron gas. The total Hamiltonian is a free kinetic term with a dispersion for the conduction electrons, plus a Heisenberg-like coupling between the spin of the impurity and the local spin density of the conduction electrons at the impurity site,

$$H = -t \sum_{\langle ij \rangle, \sigma} \left(c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.} \right) + H_K, \qquad H_K = J_K \sum_{\alpha\beta} \left(c^\dagger_{0\beta} \, \boldsymbol{\sigma}_{\beta\alpha} \, c_{0\alpha} \right) \cdot \mathbf{S}, \tag{8.17}$$

where H_K is the **Kondo coupling**. To solve this problem exactly is hard. However, we can describe the ground state in a simple way, even without the full solution. The spin of the impurity forms a singlet, which has the same structure as (8.3), with an effective conduction electron close to it,

$$|GS\rangle \sim \frac{1}{\sqrt{2}} \Big(|\Uparrow\downarrow\rangle - |\Downarrow\uparrow\rangle \Big),$$

and the rest of the electron gas behaves as a Fermi sea without interactions. This is a good approximation to the many-body ground state. The resulting state has total spin zero and does not break time-reversal symmetry, so even if it has a real local moment, it is already a quantum magnet in the sense of Section 8.1.

Let us now generalize this to a whole **lattice** of magnetic impurities, with one impurity per site, where each impurity is coupled to the same bath of conduction electrons,

$$H = -t \sum_{\langle ij \rangle, \sigma} \left(c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.} \right) + J_K \sum_{j,\alpha\beta} \left(c^\dagger_{j\beta}\, \boldsymbol{\sigma}_{\beta\alpha}\, c_{j\alpha} \right) \cdot \mathbf{S}_j. \tag{8.18}$$

Every impurity again tends to form a singlet with a conduction electron close to it. Since many impurities compete for the same sea of electrons, the electrons that screen the impurities are shared, in a fractional way, between the singlets of different impurities. The result is a state with a large entanglement over the whole lattice, between all the magnetic sites and the electron gas, while the total spin stays exactly zero.

Notebook check-in: `session8.ipynb`, "Magnetization of a large Heisenberg model with a magnetic impurity" (Challenge) and "Magnetization of a dimerized Heisenberg model with a magnetic impurity" (Challenge)

These exercises show, numerically and in real space, how the local field of a single magnetic impurity decays with the distance in a quantum magnet that is otherwise gapless or dimerized. The gapless chain shows an algebraic and staggered decay, which follows the correlation profile of a critical spin liquid. The dimerized chain of Section 8.5, which is a valence bond solid, shows instead an exponential decay over a finite correlation length, because its ground state is gapped. This is the same physics of the screening cloud as in (8.17), seen from the side of a single chain in real space and not from the side of itinerant electrons.

8.12 Solving the Kondo lattice with the parton method and a hybridization

The lecture makes the parallel with Section 8.8 explicit, since we solve this problem in the same way as we solved the QSL. We replace every impurity spin by an Abrikosov fermion, $\mathbf{S}_j = \frac{1}{2} f^\dagger_{j\alpha} \boldsymbol{\sigma}_{\alpha\beta} f_{j\beta}$, with the same single-occupancy constraint (8.11), which we write here in the equivalent traceless form

$$S_{\alpha\beta}(j) = f^\dagger_{j\alpha} f_{j\beta} - \frac{n_f(j)}{2} \delta_{\alpha\beta}. \tag{8.19}$$

When we insert this into (8.18), the Kondo coupling becomes a real quartic interaction. Now it does not couple the auxiliary f fermions to the f fermion of another site, as in (8.13), but to the *real* conduction electrons $c_{j\sigma}$ on the same site,

$$H \sim \sum_{\mathbf{k}\alpha} \varepsilon_{\mathbf{k}}\, c^\dagger_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} \;-\; J_K \sum_{j,\alpha\beta} \left(c^\dagger_{j\beta} f_{j\beta} \right) \left(f^\dagger_{j\alpha} c_{j\alpha} \right). \tag{8.20}$$

As in (8.14), we decouple this quartic term at the mean-field level, with the expectation value $\langle c^\dagger_j f_j \rangle$, and we do a Fourier transform to momentum space. This gives a bilinear Hamiltonian that mixes real electrons and auxiliary fermions,

$$H \sim \sum_{\mathbf{k}\alpha} \varepsilon_{\mathbf{k}}\, c^\dagger_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} \;-\; \gamma_K \sum_{\mathbf{k},\alpha} f^\dagger_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} + \text{h.c.}, \tag{8.21}$$

where $\varepsilon_{\mathbf{k}}$ is the **conduction band dispersion** and γ_K is the **Kondo hybridization**. The hybridization is the only mean-field parameter, it is determined self-consistently, and it contains all the information about the strength of the original Kondo coupling J_K.

8.13 Electronic structure of the Kondo lattice and the Kondo insulator

For each spin, Equation (8.21) is an ordinary two-band Bloch Hamiltonian in the basis $(c_{\mathbf{k}\alpha}, f_{\mathbf{k}\alpha})$. When $\gamma_K = 0$, there is no mixing at all. The spectrum is then the ordinary dispersive conduction band $\varepsilon_{\mathbf{k}}$, together with a completely **flat** band of f-fermion levels at zero energy. This band is flat because the auxiliary fermions have no hopping of their own when the Kondo coupling is zero. When we switch the Kondo coupling on again, γ_K becomes finite, and the flat band and the dispersive band have an **anti-crossing**. In every place where the flat spinon band would cross the conduction band, a **gap** opens instead, and the gap is larger when the Kondo coupling is larger. We see this directly when we compare the spectra for increasing J_K. At $J_K = 0$ the bands cross. At $J_K = 0.25$ there is already a visible gap at the crossing. At $J_K = 0.5$ the gap is much larger. Figure 8.6 shows the two ends of this comparison, and where the Fermi level is in each case.

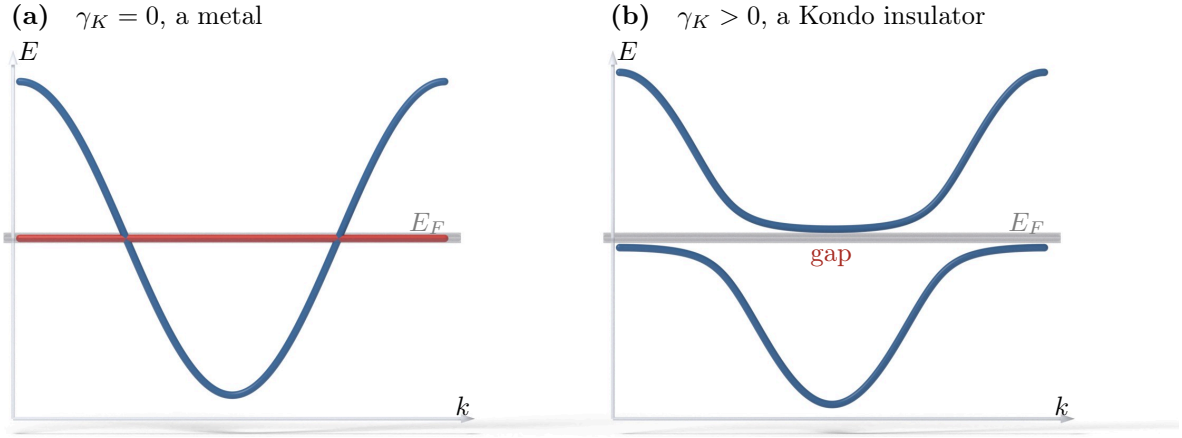


Figure 8.6: Why a lattice of local moments is an insulator, drawn from (8.21) per spin in one dimension. (a) With the Kondo coupling switched off, (8.21) is a dispersive conduction band $\varepsilon_{\mathbf{k}}$ together with a completely flat band of f levels, the red line. This band is flat because the auxiliary fermions have no hopping of their own, and the two bands cross at the Fermi level, which gives a metal. (b) A finite hybridization γ_K removes the crossing. The bands repel each other, the states close to the crossing become mixtures of conduction electron and local moment, and a gap opens exactly where the crossing was. With one f fermion per site from the constraint (8.11) and a half-filled conduction band, the lower hybridized band is exactly full. So the Fermi level is inside the gap, and the metal has become a **Kondo insulator**. The gap grows with γ_K, and so with the Kondo coupling J_K, and the nearly flat band edges that remain are the reason why the remaining quasiparticles are heavy.

Physically, the ground state of this hybridized system is non-magnetic by construction. It has total spin zero and it is time-reversal symmetric, in the same way as for the single impurity (8.17) and for the quantum spin liquid of Section 8.9. The consequence for the electronic structure, however, is new. A lattice of Kondo impurities turns a system that would otherwise be an ordinary metal into an **insulator**, because every electron at the Fermi surface forms a bound state with a magnetic impurity. This is the **Kondo insulator**, which is also called a heavy-fermion insulator, and SmB_6, which we already introduced in Section 8.1, is the standard real material example.

Note that there is a similarity with a quite different construction of Session 5. Superconductors also open a gap in their quasiparticle spectrum through a hybridization, but there it is between electrons and holes and not between electrons and auxiliary fermions. The two systems are physically very different, but the *mechanism* that opens a gap in the single-particle spectrum is very similar. The way the effect appears is also similar. The temperature scale

at which the gap of the Kondo insulator becomes visible depends on the Kondo coupling in a strongly non-linear way, which is the same kind of non-linear dependence that we know from the dependence of the superconducting gap on the pairing interaction.

Notebook check-in: `session8.ipynb`, "Quantum magnetism in the Kondo lattice" (Core) and "Spinon dispersion in the Kondo lattice" (Challenge)

The first exercise builds (8.21) directly on a lattice of van der Waals type. You identify the flat band of the Kondo sites and the anti-crossing with the conduction band, and you check what happens to the gap when you change the chemical potential. The second exercise adds a direct exchange coupling between the Kondo sites, which gives a dispersion to the spinon band that would otherwise be flat. This is the lattice version of the Doniach picture. The Kondo coupling J_K favors a local screened singlet, while the direct exchange J favors magnetic correlations between the local moments, and the competition between them decides which regime is stable, the screened one or the magnetic one.

8.14 Closing summary

This is the final recap of the lecture, with the equations that it refers to.

- Magnetism, in all the forms that we saw in Sessions 1 and 6 and in this session, comes from strong repulsive interactions between electrons.
- Frustrated magnetic models, such as those on triangular and kagome lattices (Section 8.4), can show **quantum spin liquid** behavior. This is a maximally entangled ground state with zero local spin expectation value on every site (8.5). It generalizes the singlet of the Heisenberg dimer (8.3), through the valence bond solid state (8.5), to a resonating and frustrated ground state (8.6)-(8.7), and conventional mean-field theory cannot describe it by construction.
- The zero magnetization of a QSL is the opposite regime from the one that Holstein-Primakoff needs (8.9). For this reason, its fundamental excitations are not magnons but **spinons**, with $S = 1/2$. These are emergent fermions without charge that we obtain from the Abrikosov or parton construction, (8.10)-(8.15), which is the mean-field method for electrons of Sessions 1 and 6 applied to the spin sector. Spinon bands can be gapless, gapped, or of Dirac type (Section 8.9).
- The parton construction has an emergent gauge redundancy (8.16). The mean-field bond variable is a link field, and its topological defects are gauge excitations called **visons**, which are in general coupled to the spinons (Section 8.10).
- Quantum entanglement that mixes spin and fermion degrees of freedom also appears in **Kondo systems**. The same parton method, applied to a lattice of local moments inside a metal, (8.17)-(8.21), turns the Kondo exchange coupling into a mean-field **hybridization** γ_K. This hybridization opens a gap in the conduction band at the anti-crossing and turns a metal into a **Kondo insulator**, also called a heavy-fermion insulator, and the gap grows monotonically with the strength of the Kondo coupling (Section 8.13).

Session 9 continues directly from this construction. The method of Abrikosov fermions and spinons that we built here is used again, now for a single magnetic impurity in a metal that is otherwise ordinary, with an explicit reference to a result on Friedel oscillations that is derived earlier in that same session.

Session 9

Excitations and Defects in Quantum Materials

In Session 8 we built a small set of new ground states. These were quantum spin liquids described by mean-field spinons, and Kondo lattices, where the local moments are screened by the conduction electrons with the same Abrikosov-fermion construction, so that a metal becomes a heavy-fermion insulator. Up to now every session has followed the same procedure. We take a many-body Hamiltonian, we write it in a controlled or approximate way, and we read the resulting ground state. In this session we use all of these methods in a different way. Instead of building one more ground state, we will *probe* the ground states that we already have. The idea is easy to state. We take a quantum material, which can be a metal, a superconductor, a topological insulator or a quantum magnet, we disturb it a little, and we look at how it responds. The response itself is computed only from the undisturbed ground state, and it tells us a lot about what that ground state is. We will use this one idea, linear response, as a probe of almost every class of systems that we have built in the course so far. These are ordinary metals, the superconductors of session 5, the topological matter of sessions 3 and 4, and the quantum magnets of session 8. In each case we will see that the response function works as a diagnostic, and it gives us Fermi surfaces, superconducting pairing symmetries, topological invariants and spin gaps, all from the same formalism. We will also see that the Stoner criterion for itinerant magnetism follows directly from the same language of response functions. This connects again to the magnetism of session 1 and session 6, but now we see magnetism as an *instability* of the paramagnetic response and not as a mean-field ansatz for the ground state that we impose by hand. In this sense, this is the diagnostic and synthesis session of the course, which gives a new way to look at everything that we have built.

9.1 Response functions as a probe of the ground state

Let us first set up the general experiment, since it will come back in every example of this session. We take a quantum material with a ground state that we understand, or that we think we understand, and we make a small perturbation at position x and time t. Then we measure some observable at a different position x' and at a different time t', which is typically later. The relation between what we did at (x, t) and what we observe at (x', t') is contained in a **response function**. Once we know that function completely, we know how the material reacts to a perturbation applied at any place and at any time.

This point of view does not depend on the observable that we choose, and the same formalism describes the following cases.

- A perturbation in the local electrostatic potential, measured as a change in the local

charge density, which gives the **dielectric function**.

- A local magnetic moment or field, measured as a change in the local magnetization at another place, which gives the **magnetic susceptibility**, also called the spin-fluctuation response.
- A local change of temperature, measured as a change of temperature at another place, which gives the **thermal conductance**.

There are also two quite different ways to do this experiment in the lab, and we will use both in this session. The first way is a real physical perturbation. If we replace one atom of a metal by an atom of a different element, we have created a real, localized impurity. As we will see in detail in Section 9.6, the **Friedel oscillations** that appear in the electronic density show us directly the Fermi surface of the material. In the second way, the measurement itself is the perturbation. We send a neutron, whose magnetic moment couples to the local moments in a magnetic material, and the scattering cross-section that we record gives directly the magnetic fluctuations of the material, which is called inelastic neutron scattering. These two experiments look very different, but the formalism behind them, and the kind of information that we obtain, is the same in both cases.

Finally, a simplification makes almost everything below possible to compute. When the unperturbed Hamiltonian has translational symmetry in space and does not depend on time, the response function can only depend on $x' - x$ and $t' - t$, and not on x, t, x', t' separately. In this situation the natural language is to Fourier transform to reciprocal space and frequency, (q, ω), and this is the language that we will use for almost all the rest of this session.

9.2 The response of a classical magnet as a first example

Before we use any quantum many-body method, let us see how the response predicts the ground state in a setting that we can solve exactly. This setting is a classical Ising magnet at finite temperature, treated in mean-field theory, and Section 1.12 gives the mean-field method itself. At a temperature well above any ordering transition, is there a measurement that we can do *at that temperature*, in the disordered phase, which tells us what kind of order the system will develop when we cool it down?

The relevant response is the **magnetic susceptibility**, $\chi = \partial M / \partial H$, which is the linear-response coefficient that relates an applied field H to the magnetization M that it induces. If we solve the mean-field Ising model at high temperature, we find three different behaviors, which depend on the sign and the strength of the exchange coupling.

$$\chi(T) \propto \begin{cases} \dfrac{1}{T} & \text{paramagnet (no exchange coupling), diverges only at } T = 0, \\ \dfrac{1}{T - T_c} & \text{ferromagnet, diverges at a positive } T_c, \\ \dfrac{1}{T + T_c} & \text{antiferromagnet, apparent divergence at a negative temperature.} \end{cases} \tag{9.1}$$

The physical picture behind (9.1) is important, because we will find it again in a fully quantum form in Section 9.5. In the paramagnetic regime the thermal fluctuations dominate, and the susceptibility stays finite at any $T > 0$, since nothing in the system tends to order. In the ferromagnetic case the fluctuations *reinforce* the applied field, so as $T \to T_c^+$ the system becomes polarized more and more easily, and χ really diverges at the transition. In the antiferromagnetic case the fluctuations *oppose* the applied field, since they favor staggered order and not uniform order, and this appears as the opposite sign in the denominator. The

susceptibility then seems to go towards a divergence at a temperature below absolute zero. Of course it never reaches it, but the sign alone already tells us that the system tends to antiferromagnetic order and not to ferromagnetic order. In experiments this is a practical diagnostic. We plot $\chi^{-1}(T)$, and if the straight line extrapolates to a positive intercept, the system is close to a ferromagnet, while a negative intercept means that the system is close to an antiferromagnet. All of this is measured in the disordered phase at high temperature. This is the model for the whole session, which is that **the response of the disordered/uniform state, computed without ever solving for the ordered state itself, already contains the information about which ordered state is coming.**

9.3 Linear response and perturbation theory in detail

Let us now build the quantum version of this idea, since everything that follows is based on it. We split the full Hamiltonian into an unperturbed part H_0, which can be a metal, a superconductor, a quantum magnet or any system that we study, and a perturbation V. We also add a coupling constant λ that helps us keep track of the orders, and at the end we will set it to one.

$$H(\lambda) = H_0 + \lambda V, \qquad \lambda = 0 \text{ (pristine)}, \quad \lambda = 1 \text{ (full perturbation)}. \tag{9.2}$$

Every observable of the system, such as energies, densities, magnetizations and the wavefunctions themselves, becomes a function of λ once we fix $H(\lambda)$. If this function is smooth enough to make a Taylor expansion around $\lambda = 0$, and later we will see when this assumption fails, then for any observable A we have

$$A(\lambda) = A^{(0)} + \lambda\, \chi_A + O(\lambda^2), \qquad \chi_A \equiv \left.\frac{\partial A}{\partial \lambda}\right|_{\lambda=0},$$

and by definition χ_A, the first-order coefficient, *is* the response function that we have been talking about in this session. To compute χ_A for a generic observable, we first need the perturbed wavefunctions. We expand an unperturbed eigenstate $|n\rangle$ of H_0, with energy $E_n^{(0)}$, in the full basis $\{|m\rangle\}$ of H_0. Standard Rayleigh-Schrödinger perturbation theory then gives, to first order in λ and away from any degeneracy,

$$|n(\lambda)\rangle = |n\rangle + \lambda \sum_{m \neq n} \frac{\langle m|V|n\rangle}{E_n^{(0)} - E_m^{(0)}}\, |m\rangle + O(\lambda^2). \tag{9.3}$$

The structure of (9.3) is the basis of every derivation in the rest of this session. A perturbation mixes a state $|n\rangle$ with every other state $|m\rangle$. The weight of the mixing is the matrix element of V between the two states, and it is suppressed by the inverse of the energy gap between them. States that are close in energy mix strongly, and states that are far apart in energy almost do not mix. Once we have (9.3), we can write any observable, in particular any density or magnetization, to first order in λ by using the perturbed states.

When does this break down? The construction in (9.3) assumes that $A(\lambda)$ is an analytic function of λ close to $\lambda = 0$, and this assumption is not always true. There are three paradigmatic quantum states that are known to be intrinsically non-perturbative in this sense, and we mention them explicitly since they mark the limit of everything that we derive below.

- **An impurity in a metal, the orthogonality catastrophe.** In the thermodynamic limit, the full many-body wavefunction of a metal with the impurity switched on becomes exactly orthogonal to the wavefunction with the impurity switched off, for any strength of the impurity potential, even a very weak one.

- **A quantum spin in a metal, the Kondo effect.** The emergent Kondo temperature is a non-perturbative function of the Kondo coupling, since it contains an essential singularity in the coupling constant and not a power series.
- **Superconductivity in the BCS limit.** In the same way, the superconducting gap is non-perturbative in the attractive coupling constant.

In the rest of this session we stay in the regime where perturbation theory *does* work, which is the regime where (9.2) and (9.3) give a controlled and systematic way to compute responses.

9.4 The Fermi sea and the Lindhard function

To make (9.3) concrete, let us take the case that we will use for most of this session, which is a non-interacting electronic Hamiltonian in diagonal form, $H_0 = \sum_k \varepsilon(k)\, c_k^\dagger c_k$. Its many-body ground state is the Fermi sea, where every single-particle state with $\varepsilon(k)$ below the Fermi energy is filled and every state above it is empty. In the same way, $\langle c_k^\dagger c_k \rangle = n_F(\varepsilon(k))$, which is equal to 1 for occupied states and 0 for unoccupied states at $T = 0$, and more generally is the Fermi-Dirac function. The set of k-points that are exactly at the Fermi energy defines the **Fermi surface**. This surface will be the most important geometric object in everything that follows, because a small perturbation can only change the states that are close to it. To move an electron from deep inside the Fermi sea to far above it costs too much energy to be relevant at first order.

Now we use (9.3) with a specific perturbation that has a clear physical meaning, which is a local shift of the onsite energy. This is a local change of the chemical potential at one point of the material. The observable that we follow is the change in the local electronic density, $n(x) - n_0(x)$. We use translational symmetry, we Fourier transform to q-space, and we insert the perturbed wavefunctions (9.3) into the density operator. After some algebra, where each term comes from a matrix element $\langle k+q|V|k\rangle$ divided by the energy denominator $\varepsilon(k+q) - \varepsilon(k)$, with the same structure as (9.3), we obtain the **static Lindhard function**,

$$\chi_0(q) = \frac{1}{N} \sum_k \frac{n_F(\varepsilon(k)) - n_F(\varepsilon(k+q))}{\varepsilon(k+q) - \varepsilon(k)}. \tag{9.4}$$

Let us first check (9.4). As $q \to 0$, we expand both the numerator and the denominator to first order in q and we obtain $\chi_0(q \to 0) \to -\sum_k n_F'(\varepsilon(k))/N$. At $T = 0$ this is exactly $D(E_F)$, the density of states per site at the Fermi level, which is a positive and finite number, as expected. Equation (9.4) is different from zero only when k is occupied and $k + q$ is unoccupied, or the opposite. If both are occupied or both are empty, the numerator vanishes. So the response is built only from states on the two sides of the Fermi surface, as the general argument above predicted.

Instabilities and nesting. Equation (9.4) has a strong feature in one dimension. If there is a wavevector $q^\star$ that connects an occupied state at k to an empty state at $k + q^\star$ with the *same* energy, the denominator in (9.4) vanishes for that term, and $\chi_0(q^\star)$ diverges. For a generic 1D band the Fermi surface is only the two points $\pm k_F$, and the single nesting vector that connects them is $q^\star = 2k_F$ (figure 9.1a,b). When we sum (9.4) over the whole band we obtain a logarithmic divergence, $\chi_0(2k_F) \sim \ln(W/T)$, where W is a cutoff given by the bandwidth. This divergence is the Peierls instability, which drives one-dimensional metals to dimerize into a charge density wave. If a response function diverges, an arbitrarily weak perturbation at that wavevector gives an arbitrarily large change of the ground state, meaning that the ground state that we started from is unstable to that specific fluctuation. This is the

central strategy of this session. To predict a new phase of matter, we look for the wavevector where the response function of the old phase diverges.

Everything above can be generalized directly to finite frequency. If we repeat the derivation with a perturbation that depends on time as $\propto e^{-i\omega t}$, we obtain the retarded **dynamical Lindhard function**,

$$\chi_0(q,\omega) = \frac{1}{N}\sum_k \frac{n_F(\varepsilon(k)) - n_F(\varepsilon(k+q))}{\varepsilon(k+q) - \varepsilon(k) - \hbar\omega - i0^+}, \tag{9.5}$$

which becomes (9.4) at $\omega = 0$. For a one-dimensional chain, if we plot $\operatorname{Im}\chi_0(q,\omega)$ over the whole (q,ω) plane, we obtain a characteristic surface of divergences that follows the particle-hole continuum of the band structure. This ω-q plot is only on the slides. In the spoken lecture its shape is described in words, as "a whole collection of ω versus q values where there are divergences", but it is not drawn, so it is useful to open the slides to see the figure.

The same logic extends directly to two dimensions. For a square lattice with dispersion $\varepsilon(k) = -2t(\cos k_x + \cos k_y)$, a low filling gives a small Fermi surface with a shape close to a circle. Then $\chi_0(q,0)$ is large for q inside a circle whose size is set by the diameter of the Fermi surface, and small outside it, so the response function is a direct image of the Fermi surface. This is the physical basis of **quasiparticle interference (QPI)**. In real STM experiments, impurities are put into a 2D metal, the modulation of the local density of states that they create is imaged, and the observed wavelength is inverted to obtain the Fermi surface. At half filling, however, the specific dispersion of the square lattice gives $\varepsilon(k+Q) = -\varepsilon(k)$ for $Q = (\pi,\pi)$. This is a *perfect* nesting condition, which connects very large parts of the Fermi surface with the same wavevector at the same time, and at half filling the Fermi surface itself is a square made of straight lines and not a circle. Equation (9.4) then has a very large divergence at $q = (\pi,\pi)$, which signals a strong instability there. Figure 9.1 shows the whole argument in one place. It shows the single 1D nesting vector, the logarithmic divergence that it gives in $\chi_0(q)$, and the 2D Fermi surface with straight edges that $Q = (\pi,\pi)$ nests all at once. The shape of the Fermi surface at half filling and the explicit nesting vector are shown on the slides in a more concrete way than in the spoken lecture. The figure shows the straight parallel edges that make the (π,π) nesting so effective.

Notebook check-in: `session9.ipynb`**, "The charge-charge response function" (Challenge)**

Plot $\operatorname{Im}\chi_0(q,\omega)$ for a band structure and a chemical potential μ that you can change, as in (9.5). Look at how the divergence follows the Fermi surface when you change μ, and try to predict from the band structure alone at which q-vector the instability will appear, before you run the calculation.

9.5 Interactions as a perturbation, with RPA, plasmons and the Stoner instability

So far the perturbation has been an external impurity. However, we can also treat the electron-electron interaction *itself* as the perturbation. We use χ_0, the non-interacting response that we just built, as input, and we ask for the response χ of the full interacting system. For this we need two things. The first is a rule that connects χ to χ_0, and the second is a criterion to find instabilities from χ.

From χ_0 to χ with the random phase approximation. Let us take a contact interaction, of Hubbard type, with strength U, which couples to the density. The physical picture is a diagrammatic one. The response to a probe is the bare response χ_0, plus the bare response followed by one interaction event and another bare response, plus two interaction events, and

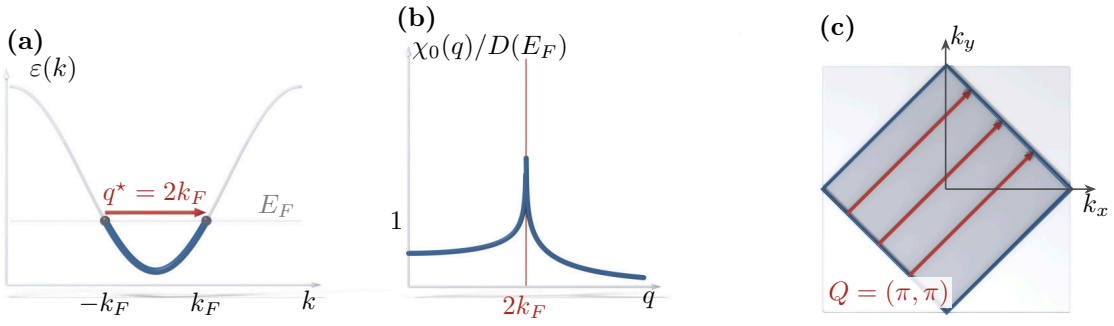


Figure 9.1: Nesting is what makes a response function diverge. (a) A one-dimensional band $\varepsilon(k) = -2t\cos k$ at partial filling. The occupied states (thick) end at $\pm k_F$, and the single wavevector $q^\star = 2k_F$ connects one Fermi point to the other, so the energy denominator of (9.4) vanishes for that term. (b) The static Lindhard function that results, with the closed form given in Section 9.6. It is finite and equal to $D(E_F)$ as $q \to 0$, as the $q \to 0$ check of (9.4) requires, and it diverges logarithmically at $q^\star = 2k_F$, which is the Peierls instability. (c) In two dimensions the criterion becomes geometric. At half filling the Fermi surface of the square lattice is the diamond with straight edges $|k_x| + |k_y| = \pi$, and since $\varepsilon(k + Q) = -\varepsilon(k)$ for $Q = (\pi, \pi)$, this one vector, the red arrows, maps *every* point of one edge onto the parallel opposite edge at the same time. For this reason the divergence at (π, π) is much stronger than the one-dimensional one, since whole segments of the Fermi surface nest, and not isolated points. At low filling the same Fermi surface is a small circle, which no single q can nest in this way.

so on, which is an infinite ladder of terms. The most efficient way to include all of them is not to sum the series term by term, but to write the self-consistent equation that the full response must satisfy,

$$\chi(q,\omega) = \chi_0(q,\omega) - \chi_0(q,\omega)\, U\, \chi(q,\omega),$$

and its algebraic solution gives the **random phase approximation (RPA)**,

$$\chi(q,\omega) = \frac{\chi_0(q,\omega)}{1 + U\,\chi_0(q,\omega)}. \tag{9.6}$$

Equation (9.6) has a mixed character. It is derived order by order in U, so it is perturbative, but since we sum the whole geometric series it becomes non-perturbative. In this way it includes physics that no finite order in U can give, which is a real pole, meaning a real phase transition.

Two different sources of divergence. The interacting response (9.6) can diverge for two quite different reasons. It is useful to separate them, because they correspond to two different emergent quasiparticles.

From χ_0 itself, as in Section 9.4. This is an electron-hole excitation, which is a single-particle process that moves one electron from an occupied state to an empty state.

From the RPA denominator, $1 + U\chi_0 = 0$, which is a new collective phenomenon that comes from the interaction. For a long-range Coulomb interaction, $U \to v(q)$, which depends strongly on q and diverges as $q \to 0$. The pole of (9.6) as a function of ω at fixed q then defines a new quasiparticle, the **plasmon**, which is a collective charge oscillation of the whole electron gas. Its dispersion depends on the screening and on the dimensionality of the system. In the standard RPA treatment of the three-dimensional electron gas, which is the Bohm-Pines treatment, at long wavelength the dispersion has the form

$$\omega_{\mathrm{pl}}(q)^2 = \omega_p^2 + \tfrac{3}{5} v_F^2\, q^2 + O(q^4), \tag{9.7}$$

where ω_p is the classical plasma frequency. The plasmon exists even at $q = 0$, which is different from the particle-hole continuum, because it is a collective mode of the whole charge density and not a single-particle excitation. Figure 9.2 shows both kinds of divergence in the same (q, ω) plane, and it shows explicitly the window at small q where the pole exists and no single-particle excitation exists.

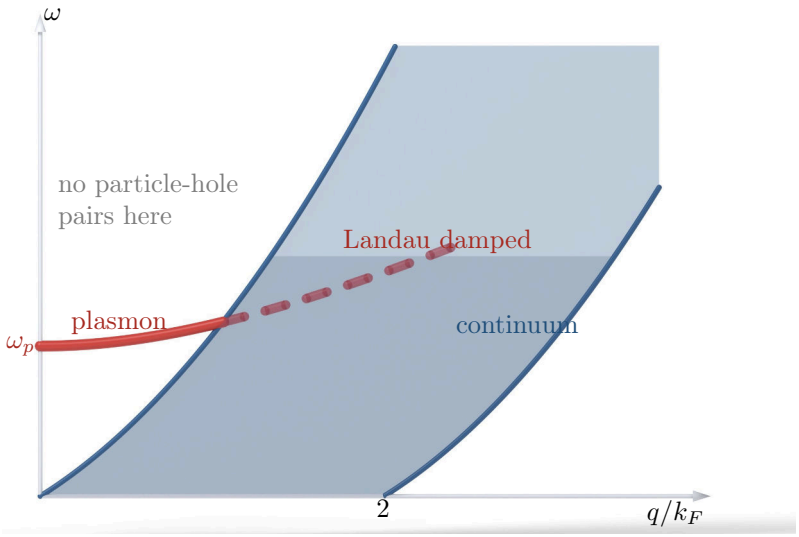


Figure 9.2: The two divergences of the RPA response (9.6), drawn in the (q, ω) plane for the three-dimensional electron gas. The shaded region is where $\mathrm{Im}\,\chi_0(q,\omega) \neq 0$, which is the *particle-hole continuum*, the set of all (q, ω) at which a single electron can be moved from an occupied state k to an empty state $k + q$. Its upper boundary is $\hbar\omega_+ = \hbar v_F q + \hbar^2 q^2/2m$, and beyond $2k_F$ a lower boundary $\hbar\omega_- = \hbar^2 q^2/2m - \hbar v_F q$ appears. Both follow from the energy denominator of (9.5), and for $q < 2k_F$ the continuum goes all the way down to $\omega = 0$, which is the same statement as saying that χ_0 is built only from states on the two sides of the Fermi surface. The plasmon (9.7) is a different object. It is a zero of the RPA *denominator* $1 - v(q)\chi_0$, so it exists at ω_p even as $q \to 0$, in a window where no single-particle excitation exists at all. This window, marked in the figure, is the full meaning of the phrase "collective mode, not a particle-hole pair". Once the branch meets the continuum it can decay into particle-hole pairs, and it is Landau damped (dashed).

The Stoner criterion as an instability of the response function. Now we use the same RPA logic in the *spin* channel of an interaction of Hubbard type, so that χ_0 in (9.6) becomes the non-interacting spin susceptibility. The sign is the opposite of the density channel, because $U n_\uparrow n_\downarrow = \frac{U}{4}(n^2 - m^2)$ is repulsive for the density and attractive for the magnetization, so the same U screens the charge response and enhances the spin response, and the denominator of (9.6) becomes $1 - U\chi_0$. The interacting spin response diverges when

$$U\,\chi_0(q, 0) = 1. \tag{9.8}$$

At $q = 0$ we use the limit $q \to 0$ of (9.4), $\chi_0(0, 0) = D(E_F)$, so (9.8) becomes $U\,D(E_F) = 1$. This is exactly the **Stoner criterion** for itinerant ferromagnetism, which says that an interacting metal becomes magnetic once the product of the density of states at the Fermi level and the interaction strength is larger than one. However, (9.8) is more general than the special case $q = 0$. For a Fermi surface with strong nesting, such as the half-filled square lattice of Section 9.4, $\chi_0(q, 0)$ has its *maximum* not at $q = 0$ but at the nesting vector $Q = (\pi, \pi)$, where it is already logarithmically enhanced. The instability threshold in (9.8) is then reached first at $Q = (\pi, \pi)$ and not at $q = 0$. As a result, the system develops a *staggered* antiferromagnetic instability and not a uniform ferromagnetic one, and this happens at a smaller value of U than the simple Stoner estimate predicts.

This is the connection that the course has been preparing. In Sessions 1 and 6 we obtained magnetism by writing the Hubbard model, applying mean-field theory directly to the *ordered* state, and solving self-consistently for a non-zero magnetization m, as in Equation 1.7 of Section 1.14. In this session we have obtained the *same physics*, with the same threshold $U\,D(E_F) = 1$ and the same preference for antiferromagnetism at half filling, without writing any ordered mean-field ansatz. Instead, we asked when the *paramagnetic*, disordered response function develops a divergence, which is the same logic as the classical susceptibility of Section 9.2. Mean-field theory and the response function are two ways to reach the same instability. The first one computes it from the side where the symmetry is broken, and the second one computes it only from the symmetric side, by following how this state becomes unstable.

The same method, applied to different response channels and to different signs of the interaction, classifies a whole family of instabilities, which we write here explicitly. On the slides this classification into three cases is presented as a clear summary table, in a more systematic way than in the spoken lecture, which mainly worked out the magnetic case in detail.

Instability	Response channel	Interaction sign
Magnetism	spin response	repulsive
Dimerization / charge density wave	charge-phonon response	repulsive
Superconductivity	electron-electron scattering response	attractive

Notebook check-in: `session9.ipynb`, "The charge-charge response function" (Challenge)

The same notebook section that we used above for nesting is extended, in its second part, to the interacting case. You add a Hubbard interaction U and an external staggered, antiferromagnetic field to a dimerized chain, and you look at what happens to the response as U increases. This is the version of equation (9.8) on the scale of a notebook, where the Stoner instability appears when U goes above the threshold set by χ_0.

9.6 Impurities as diagnostics I, Friedel oscillations in a metal

We now go to the most physical realization of a perturbation, which is a real local impurity. Four case studies follow, one for each broad class of ground state that we have built in the course. These are metals, superconductors, topological matter and quantum magnets. In each case the diagnostic logic is the same. We perturb the system locally, and we obtain the nature of the ground state from how the effect of the perturbation decays in space.

Let us start with a metal. An impurity is a local shift of the onsite energy. If we add one to a Hamiltonian with translational symmetry, this symmetry is necessarily broken. Note, however, that perturbation theory itself never needed translational symmetry, since equations (9.2) and (9.3) are valid for any H_0 and any V. If we apply them directly to the local density of states, we obtain a response that is proportional to matrix elements between states at the Fermi surface. For a generic band, close to a parabolic one, the oscillation that results is set only by the Fermi wavevector.

To see this in an exact way, note that the static Lindhard function of a one-dimensional free-electron gas has the closed form

$$\chi_0(q) = \frac{m}{\pi\hbar^2 q} \ln \left| \frac{2k_F + q}{2k_F - q} \right|,$$

which is smooth everywhere except for a logarithmic non-analyticity exactly at $q = 2k_F$. This is the same nesting singularity that we found in Section 9.4. A standard result of Fourier

analysis is that a non-analyticity of a function $\chi_0(q)$ at a wavevector $q^\star$ gives an oscillating tail with an algebraic decay in its real-space Fourier transform. The period of this tail is fixed by $q^\star$, and its envelope is set by the *type* of the singularity. We apply this to the change of density induced by the impurity, $\delta n(x) = \int \frac{dq}{2\pi}\, e^{iqx}\, \chi_0(q)\, v_0$, in the Born approximation with impurity strength v_0, and we obtain the **Friedel oscillation**,

$$\delta n(x) \xrightarrow{|x|\to\infty} A(v_0)\, \frac{\cos\bigl(2k_F|x| + \varphi_0\bigr)}{|x|^{\,d}} \qquad (d = 1, 2, 3), \tag{9.9}$$

The oscillation has a universal wavevector that is fixed only by the Fermi surface, which is $2k_F$ in 1D, or more generally the diameter of the Fermi surface. It has an algebraic envelope, whose power depends on the dimensionality d, and it has an amplitude and a phase that are not universal and are set by the impurity strength and by microscopic details. This closes the argument that we started in Section 9.4. The same singularity at $q = 2k_F$ in $\chi_0(q)$, which we found there as an instability channel, gives here a long-ranged real-space oscillation with a power-law decay, and not a real instability. The reason is that a single local impurity is a perturbation that is much too weak to reorganize the whole Fermi sea. The divergence in $\chi_0(q)$ only makes its Fourier partner decay slowly instead of exponentially.

For a concrete example, let us take a one-dimensional tight-binding chain with an impurity at site 0. The local density of states oscillates around the impurity with wavelength π/k_F, which is half of the Fermi wavelength $\lambda_F = 2\pi/k_F$, since the modulation is set by $2k_F$ and not by k_F. If we raise the chemical potential, k_F^{-1} becomes shorter and the oscillation becomes faster in the same way. The same phenomenon, with the same physical content, appears in a two-dimensional square lattice. This real-space LDOS modulation around impurities is what an STM measures in practice, and it is quasiparticle interference again, seen this time in real space and not in reciprocal space. The explicit plots of the LDOS as a function of position at two different fillings, for both the 1D and the 2D case, are clearer on the slides than in words. The comparison that shows that the "oscillations get faster as k_F grows" is a visual one, and it is better seen than read from the formula. As a complementary check, the slides and the notebook also show the *insulating* case. An impurity placed in a gapped 1D chain gives an LDOS that converges very quickly to its uniform value, instead of oscillating with a slow power-law envelope. This is a first view of the diagnostic that separates gapped and gapless systems, which we will make fully general in Section 9.9 and draw in figure 9.5.

Notebook check-in: `session9.ipynb`, "Impurity in a 1d metal" (Core), "Impurities in a 2d metal" (Core), and "Impurity in an electronic insulator" (Core)

Place a single impurity in a 1D chain, and then in a 2D square lattice, and look at how the LDOS oscillates as in (9.9). Change the chemical potential and check that the wavelength of the oscillation follows π/k_F, as predicted. This is the clearest place in the whole session to see a response formula and a numerical experiment agree digit by digit. The third section repeats the exercise with the chemical potential inside a gap, instead of in a partially filled band. Check that the LDOS now converges to its bulk value within one or two sites, without an oscillating tail that decays slowly. This is the contrast between gapped and gapless systems that Section 9.9 makes fully general.

9.7 Impurities as diagnostics II, magnetic impurities in a superconductor

Superconductors behave in a completely different way under a local perturbation, because for single-particle excitations a superconductor is an insulator in the bulk. There is a finite gap Δ, so there is no Fermi surface that can give oscillations of the Friedel type. The interesting

physics is instead the difference between magnetic and non-magnetic impurities.

Let us remember that an s-wave Cooper pair binds one spin-up electron and one spin-down electron. A **non-magnetic** impurity, which is an ordinary local potential, does not distinguish spin up from spin down. So it does not break time-reversal symmetry, and it cannot lower the energy needed to break any specific Cooper pair. In practice, the density of states of a conventional s-wave superconductor is almost the same with and without a non-magnetic impurity, including the gap, and the slides show the two spectra on top of each other.

A **magnetic** impurity is completely different. It couples with strength J to the local spin, it distinguishes up from down, and so it *does* break time-reversal symmetry. This lowers the energy needed to break one specific Cooper pair and gives a real bound state inside the gap. In the approximation of a classical spin, in the Born approximation, this is the **Yu-Shiba-Rusinov** problem, which is not named in the transcript, but the setup is the same. With $\alpha \equiv \pi\nu_0 JS$, where ν_0 is the density of states of the normal state at the Fermi level and S is the impurity spin, the energy of the bound state in the gap is

$$E_b(J) = \Delta \frac{1-\alpha^2}{1+\alpha^2}. \tag{9.10}$$

Equation (9.10) gives the qualitative curve that the slides show as a function of J. At $J = 0$ ($\alpha = 0$), $E_b = \Delta$, which is at the gap edge, meaning that there is no real bound state yet. As J increases, α grows and E_b moves down through the gap. At $\alpha = 1$, which is $\pi\nu_0 JS = 1$, the state crosses zero energy exactly, and for $\alpha > 1$ the numerator of (9.10) becomes negative, so the bound state appears again on the other side of zero. This crossing is a real quantum phase transition, where the fermion parity of the electronic ground state changes. Below it the electronic ground state is even, with total electron spin $S_{\text{tot}} = 0$, where every Cooper pair is intact and the impurity spin is not screened. Above it the electronic ground state has $S_{\text{tot}} = 1/2$, since one quasiparticle is bound to the impurity and partially screens its spin. So a superconductor with a single magnetic impurity that we can tune is a minimal, exactly solvable model of a quantum phase transition, which is driven only by a local perturbation. Both parts of this picture are drawn in figure 9.3. It shows the in-gap states that a magnetic impurity adds to a BCS spectrum that is otherwise unchanged, and the path that (9.10) follows across the gap as α grows.

Notebook check-in: `session9.ipynb`, "Magnetic impurities in a conventional superconductor" (Core) and "Single magnetic impurity in a superconductor" (Challenge)

Build a 1D superconducting chain with a magnetic impurity with an exchange J that you can change, and follow the energy of the in-gap state as in (9.10). Check the change of sign at the crossing. Check also that a non-magnetic impurity, where you set $J = 0$ and change only the local potential, leaves the gap edges almost unchanged, as the qualitative argument above predicts.

9.8 Impurities as diagnostics III, vacancies in topological matter

The strongest local impurity that is possible is a **vacancy**, where we remove a site completely, which is the same as sending its onsite potential to infinity. Here the bulk-boundary correspondence that we built in sessions 3 and 4 gives a direct result. This correspondence includes the Chern number, the Zak phase, and the guarantee of gapless modes at any boundary of a topologically non-trivial phase. It is known that topological boundary modes survive around a large hole cut into a topological material. If we make that hole smaller, down to a single missing site, the argument still works. A vacancy is in practice a boundary of zero size, and the bulk-boundary correspondence still applies. When the bulk has a non-trivial topological

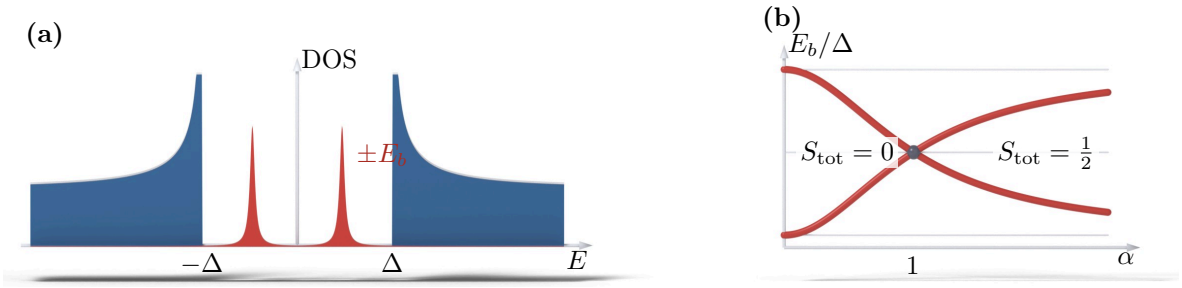


Figure 9.3: A magnetic impurity in an s-wave superconductor. (a) The single-particle density of states. The BCS continuum, with a gap Δ and coherence peaks at the gap edges, is almost unchanged by a *non-magnetic* impurity. Such an impurity does not distinguish up from down, so it cannot lower the energy needed to break any specific Cooper pair. A *magnetic* impurity distinguishes them, breaks time-reversal symmetry, and creates a pair of bound states inside the gap at $\pm E_b$, the red peaks, drawn for $\alpha = 0.6$. (b) The energy of these states, from (9.10), as a function of $\alpha = \pi\nu_0 JS$. At $\alpha = 0$ they are exactly at the gap edges, meaning that there is no bound state yet. As the exchange grows they move across the gap and cross at $\alpha = 1$. The crossing is not a level repulsion that did not open. It is a real quantum phase transition of the electronic ground state, with a change of fermion parity, and the electron spin goes from $S_{\text{tot}} = 0$, where every Cooper pair is intact and the impurity spin is not screened, to $S_{\text{tot}} = 1/2$, where one quasiparticle is bound to the impurity and partially screens its spin. So a single impurity that we can tune is an exactly solvable quantum phase transition.

invariant, a vacancy is guaranteed to host an in-gap bound state, as long as the vacancy itself preserves the symmetry that protects that invariant.

This gives a useful diagnostic, because a fully gapped bulk spectrum alone cannot distinguish a topological phase from a trivial one. Let us take a **chiral, spin-triplet topological superconductor**, which is the class that we studied with Majorana modes and the Chern-number classification in session 5. Without impurities, its bulk spectrum is fully gapped, and by itself it looks the same as the spectrum of a trivial s-wave superconductor. But a vacancy is a non-magnetic perturbation. It is a local change of the potential that does not distinguish spin, even in the extreme limit where the local potential diverges, so it does not change the topological invariant of the bulk, and once we add it, an in-gap mode appears. The same logic holds for a **Chern insulator**. A vacancy in a system with a finite Chern number gives an in-gap mode, while the same vacancy in a topologically trivial insulator gives no such mode. In both cases the observable signature is the same and easy to read, which is that *if introducing vacancies produces in-gap modes, the underlying phase is topological, and if it does not, the phase is trivial*, regardless of what the clean bulk spectrum alone would tell us. Figure 9.4 draws both the geometric argument, where a hole becomes smaller down to a single site, and the three spectra that make the diagnostic possible to use.

Notebook check-in: `session9.ipynb`, "Non-magnetic impurities in a topological superconductor" (Challenge), "DOS of non-magnetic impurities in a topological superconductor" (Challenge), "Single vacancy in a topological superconductor" (Challenge), "Vacancies in a topological insulator" (Challenge), and "Single vacancy in a topological insulator" (Challenge)

Build a chiral p-wave superconductor and a Chern insulator of the Haldane model, each with a single vacancy, and check that the in-gap mode appears in both. The check that the notebook itself suggests is a good one. Change the pairing symmetry to conventional s-wave, or change the topological gap to a trivial gap from a sublattice imbalance, and look at how the in-gap mode disappears. It disappears because the bulk invariant, which guaranteed the mode through the bulk-boundary correspondence, is now zero.

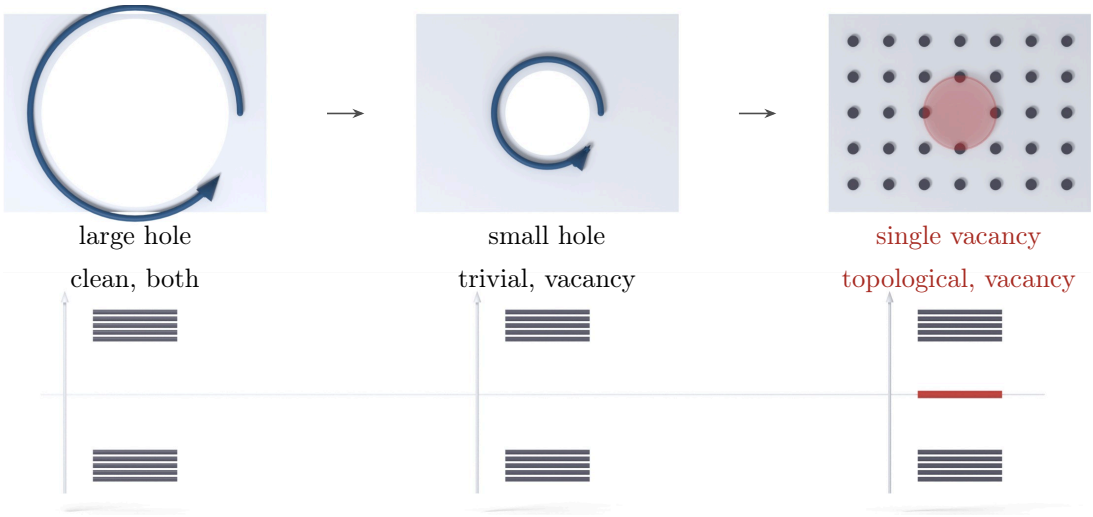


Figure 9.4: Why a single missing atom detects topology. Top, seen from above. If we cut a large hole into a topologically non-trivial insulator or superconductor, the bulk-boundary correspondence guarantees gapless modes that run around its edge. Nothing in that argument depends on the size of the hole, so making it smaller does not remove these modes. A vacancy is a boundary of zero size, and it binds an in-gap mode. Bottom, schematic spectra. This is the diagnostic that a clean spectrum cannot give. A trivial and a topological insulator with the same gap have bulk spectra that look the same, so nothing there distinguishes them (left). If we add a vacancy, the trivial spectrum stays as it was (middle), and only the topological one develops a state inside the gap (right). The rule is as direct as it looks. Vacancies produce in-gap modes if and only if the bulk invariant is non-zero, provided that the vacancy itself respects the symmetry that protects that invariant.

9.9 Impurities as diagnostics IV, screening in quantum magnets

The last case study brings us directly back to the construction of session 8. We take a frustrated Heisenberg model with a ground state similar to a quantum spin liquid. This ground state has zero net magnetization on every site, and it does not break time-reversal symmetry. We add one local magnetic perturbation, and we ask how the spins around it screen it.

For $S = 1/2$ we do not need to repeat the construction from the beginning, since session 8 already built the tool that we need. We write each spin with a pair of auxiliary (Abrikosov) fermions, $S_i = \frac{1}{2}\sum_{\alpha\beta} f^\dagger_{i\alpha}\boldsymbol{\sigma}_{\alpha\beta}f_{i\beta}$, with the single-occupancy constraint $\sum_\sigma f^\dagger_{i\sigma}f_{i\sigma} = 1$ at every site, and we make a mean-field decoupling of the quartic Hamiltonian that we obtain. As we derived in session 8, this turns the Heisenberg model with a local magnetic perturbation into a bilinear Hamiltonian for the spinons, similar to a tight-binding one, with an effective onsite term at the impurity site that mixes with the spinon band. We will not derive this mapping again here, and the notes of session 8, in the part on the Abrikosov-fermion construction, give the full mean-field decoupling.

What matters for us in this session is what this mapping gives us. It turns the problem of screening a magnetic moment inside a quantum spin liquid into *exactly* the same mathematical problem as the impurity in a one-dimensional electron gas of Section 9.6, where the electrons are replaced by spinons. The induced spin correlation $\langle S_0 \cdot S_n\rangle$, as a function of the distance n from the impurity, therefore has the same structure as (9.9). It has an oscillation set by the antiferromagnetic exchange, and it decays with distance in a way that depends on whether the spinon band is gapped or gapless.

Here the comparison between the $S = 1/2$ and $S = 1$ Heisenberg chains becomes clear. Both have a ground state similar to a spin liquid, with zero onsite magnetization when there is no impurity, and both develop an oscillating induced moment once the impurity is switched

on. However, the decay is qualitatively different in the two cases.

$$\langle S_0 \cdot S_n \rangle \;\sim\; \begin{cases} \dfrac{(-1)^n}{n} & S = 1/2 \quad \text{(power-law, gapless)}, \\ (-1)^n \, e^{-n/\xi} & S = 1 \quad \text{(exponential, gapped)}. \end{cases} \tag{9.11}$$

The reason is the same reasoning as for (9.9). Through the Abrikosov-fermion construction, the spin-1/2 Heisenberg chain maps onto a gapless tight-binding chain of spinons, with a real Fermi surface of two points, as for the 1D metal of Section 9.4. So the impurity response must decay algebraically, in the same way as the Friedel oscillations of a metal. The spin-1 chain, on the other hand, has an excitation gap. This is the Haldane gap, which is not named explicitly in the transcript, but it is the gap that opens for antiferromagnetic chains with integer spin. This gap sets a finite correlation length ξ, and it forces the impurity response to decay exponentially, in the same way as charge screening inside a band insulator.

This gives us a completely general diagnostic, which connects all the case studies of this session, namely that *a power-law decay of an impurity response signals a gapless ground state, and an exponential decay signals a gapped one, where the decay length in the exponent is the correlation length.* The metal of Section 9.6, which has a power law, the insulating chain at the end of that section, which has an exponential, and the two quantum-magnet cases here are all examples of the same statement, and they are drawn side by side in figure 9.5.

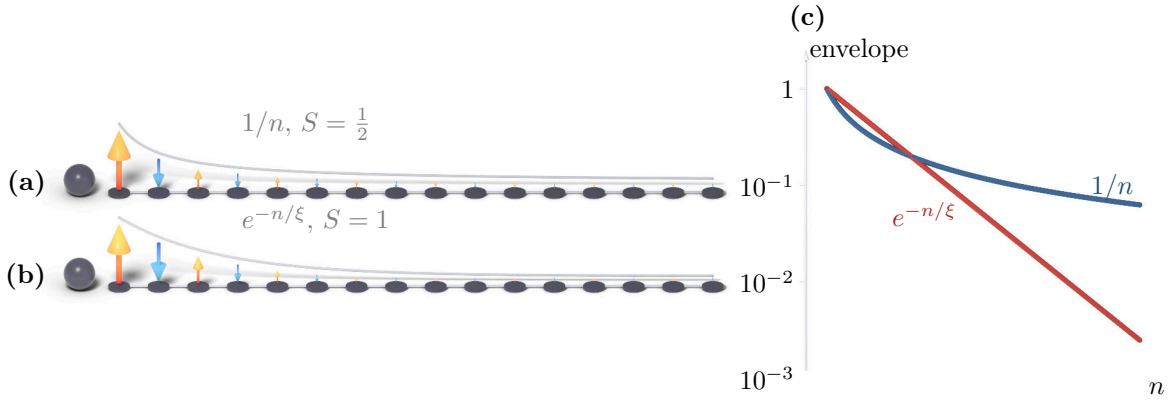


Figure 9.5: The diagnostic that connects the whole session, drawn for the two Heisenberg chains of (9.11). Close to the impurity both look the same, with an induced moment that changes sign from site to site, because the exchange is antiferromagnetic. The only difference between them is how the envelope decays. (a) The $S = 1/2$ chain maps, through the Abrikosov-fermion construction of session 8, onto a *gapless* spinon band with a real Fermi surface of two points, so its envelope decays as the power law $1/n$. This is the same mathematics, and the same reason, as for the Friedel oscillations (9.9) of a metal. (b) The $S = 1$ chain has the Haldane gap, which fixes a finite correlation length ξ and gives an exponential envelope $e^{-n/\xi}$, drawn for $\xi = 2.5$, in the same way as charge screening decays inside a band insulator. (c) On a logarithmic vertical scale the two are easy to distinguish, since an exponential is a straight line of slope $-1/\xi$ and a power law is not. The statement can be read in both directions, so a measured decay law is a measurement of whether the ground state is gapped.

This specific topic, the screening in quantum magnets, is only in the slides and in the lecture, and it has no direct counterpart in `session9.ipynb`. None of the sections of the notebook build a chain of spinons or Abrikosov fermions, so there is no hands-on exercise that reproduces (9.11) itself.

Notebook check-in: `session9.ipynb`, "The impact of a strong impurity" (Core)

This is not a spin model, but it is the closest exercise in its logic. It compares how a dimerized chain, with a stronger or a weaker gap, responds to a strong local impurity. Run it with (9.11) in mind, and check the same principle in the fermionic language, where the chain with the stronger gap screens the impurity over a shorter length.

9.10 Disorder and Anderson localization

So far we have only discussed a *single* impurity. What happens when we put many impurities in the whole material? A single impurity changes the ground state only a little. Many impurities, in particular when they are placed at random, can change it completely, because they break translational symmetry in the whole system and not only locally. It is useful to list all the possibilities, since there are three different regimes of translational symmetry to consider, and not only the symmetric and the non-symmetric ones.

1. **Periodic.** There is full translational symmetry. Every eigenstate is extended, with weight $\sim 1/N$ on each of N sites.

2. **Quasiperiodic.** The system is not periodic, since there is no repeating unit cell, but it is not random either, since the potential follows a fixed and deterministic rule that never repeats. Here the eigenstates can be extended, critical, or fully localized, depending on the strength of the potential. Critical means localized with a power law, so that the state is not even normalizable in the strict sense.

3. **Strongly (randomly) disordered.** The eigenstates are localized, and each of them has a finite localization length.

Quasiperiodic case. The standard model here is a 1D tight-binding chain with an onsite modulation at an irrational wavevector. It belongs to the Aubry-André-Harper family, although it is not named in this way in the transcript.

$$H = -t \sum_n \left(c_n^\dagger c_{n+1} + \text{h.c.} \right) + V_0 \sum_n \cos(2\pi q\, n + \phi)\, c_n^\dagger c_n, \qquad q \notin \mathbb{Q}.$$

Since q is irrational, the potential never repeats exactly, so the chain has no translational symmetry at all. However, it is not random either, because the value of the potential at every site is completely fixed once we fix q and ϕ. As we increase V_0 from zero, the eigenstates change from extended to a marginal, critical behavior with a power law close to the self-dual point of the model, which is $V_0 = 2t$ in the plain Aubry-André model, and then to exponentially localized states for V_0 well above it.

The tool that we use to distinguish these regimes is the **inverse participation ratio (IPR)**,

$$\text{IPR}(\psi) = \sum_i |\psi_i|^4, \qquad \sum_i |\psi_i|^2 = 1. \tag{9.12}$$

An extended state with weight $\sim 1/N$ per site gives IPR $\sim N \cdot (1/N)^2 = 1/N$, which vanishes as the system grows. A localized state, with $O(1)$ weight on $O(1)$ sites, gives IPR $\sim O(1)$, which does not depend on the system size. A critical state is between the two, and its IPR follows a non-trivial power law in N. This is a multifractal signature, which is different from both of the two clean limits. Figure 9.6 shows the three eigenstates side by side, together with the scaling of (9.12) with N, which separates them in a quantitative way.

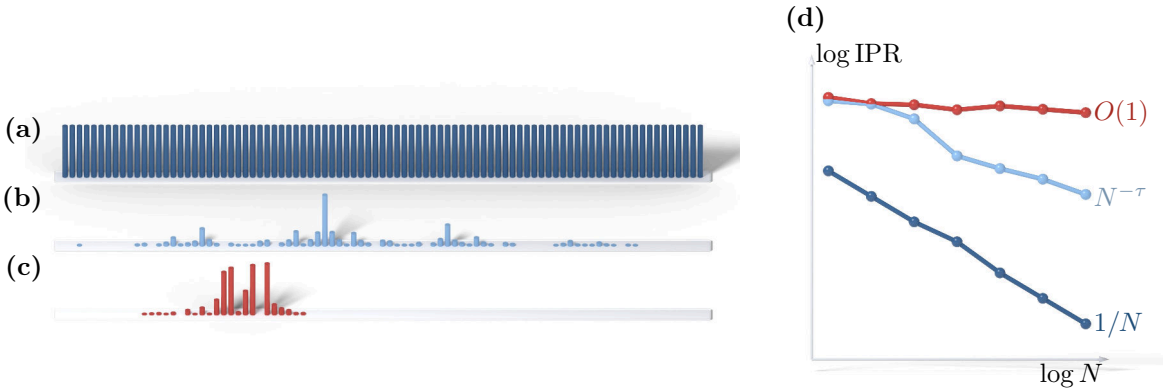


Figure 9.6: The three regimes of Section 9.10, shown through their effect on an eigenstate of a chain of $N = 89$ sites, with the weight $|\psi_i|^2$ of each site as a column. (a) With a periodic potential every eigenstate is a Bloch wave, with weight exactly $1/N$ on every site, and this height becomes smaller as the chain grows. (b) At the self-dual point of the quasiperiodic model the state is neither extended nor localized. It has structure on every length scale at the same time, with large peaks and smaller copies of the same pattern between them, and this state is called critical, or multifractal. (c) With really random disorder the state is concentrated on a few sites and it is strictly zero, to numerical precision, over most of the chain. *Note the three different vertical scales.* For this reason we need a number and not a visual check, and this number is the inverse participation ratio (9.12). (d) The IPR separates the three regimes clearly by how it scales with system size, with IPR $\sim 1/N$ for extended states, IPR $\sim O(1)$ for localized ones, and an anomalous power $N^{-\tau}$ with $0 < \tau < 1$ for the critical state between them, here computed for $N = 34$ to 610. On a log-log plot these are three straight lines with three different slopes.

Fully disordered case. Now we take an onsite potential that is really random, with strength W, $H = -t\sum_n (c_n^\dagger c_{n+1} + \text{h.c.}) + \sum_n w_n\, c_n^\dagger c_n$, where w_n is taken uniformly from $[-W, W]$. At $W = 0$ this is an ordinary clean metal, with the band structure of the periodic hopping alone, and as $W \to \infty$ every eigenstate becomes concentrated on a single site. In dimension $d \geq 3$ there is a real **Anderson localization transition** at a finite critical disorder W_c. There is a real mobility edge that separates extended metallic states from localized states, which is the kind of "critical point" described on the slides. Below the transition, or at any energy on the extended side of a mobility edge, the system conducts. Above it, there are states at every energy, so there is *no spectral gap* in the density of states, but these states cannot carry current, so there *is* a transport gap. This means that the **Anderson insulator** is an insulator with no gap at all in its spectrum, which is a very unusual state of matter compared with every band insulator or Mott insulator that we have seen so far in this course.

The special role of one dimension. In $d = 1$, and in a marginal way also in $d = 2$ according to the scaling theory of localization, no real metallic phase survives at any non-zero disorder in the thermodynamic limit, since an arbitrarily small $W > 0$ localizes every state at long enough distances. What survives is a finite localization length $\xi(W)$. For weak disorder close to the band center of the 1D model, it scales as $\xi(W) \propto (t/W)^2$, which diverges as $W \to 0$. A finite sample that is shorter than $\xi(W)$ still looks metallic in practice, and only when the length of the sample is larger than $\xi(W)$ do we see the localized, insulating behavior in transport. This is the exact sense in which "weak disorder in 1D" and "genuinely metallic" are never the same statement, although they can look the same at short length scales.

Notebook check-in: `session9.ipynb`, "The impact of disorder in different systems" (Core), "Extended and localized states in quasicrystals" (Challenge), and "Localization due to random disorder" (Core)

Change the strength of the quasiperiodic modulation, and separately the strength W of the random disorder, and compute (9.12) for the eigenstates that you obtain in each case. Look at how the IPR changes from a behavior like $1/N$ to a behavior like $O(1)$ as you increase the strength of the potential. Try to find by eye where the change from extended to localized states happens in each of the two models, and compare it with the language of critical points and mobility edges above.

9.11 Closing summary

- A perturbation that is applied to a quantum material and measured at another place defines a **response function**. With translational symmetry and time-translation symmetry, it is most natural to write it in (q, ω)-space (Section 9.1).
- A classical first example already contains the logic of the whole session. The sign of the divergence in the magnetic susceptibility at high temperature (9.1) predicts, from the disordered phase alone, whether the order at low temperature will be ferromagnetic or antiferromagnetic.
- Linear response theory (9.2)-(9.3) computes the leading effect of any perturbation only from the unperturbed ground state. It fails for the orthogonality catastrophe, the Kondo effect and the BCS gap, since none of them are connected to the unperturbed state in a perturbative way.
- For a Fermi sea, the static and dynamical Lindhard functions (9.4)-(9.5) are built only from matrix elements between occupied and empty states close to the Fermi surface. They diverge when a nesting vector connects large parts of that surface, which signals an instability (Section 9.4).
- If we sum the interaction to all orders with the random phase approximation (9.6), we find two different sources of divergence. These are electron-hole excitations from χ_0 itself, and collective plasmon excitations (9.7) from the RPA denominator.
- The Stoner criterion (9.8), $U\chi_0(q, 0) = 1$, gives again the magnetism of session 1 and session 6 as an *instability of the paramagnetic response function*, including the preference for antiferromagnetism at a nested (half-filled) Fermi surface, and not as an ordered mean-field ansatz imposed by hand.
- A local impurity in a metal gives Friedel oscillations (9.9). These are a density modulation with period set by $2k_F$ and a power-law decay, and they give a direct image of the Fermi surface, which is the same physics that an STM measures with quasiparticle interference.
- A magnetic impurity in a superconductor creates a Yu-Shiba-Rusinov in-gap bound state (9.10). Its energy crosses zero at a real quantum phase transition, where the fermion parity of the electrons changes and their spin goes as $S_{\text{tot}} = 0 \to 1/2$, while a non-magnetic impurity leaves the spectrum almost unchanged.
- A vacancy in a topological superconductor or in a Chern insulator gives an in-gap mode only because of the bulk-boundary correspondence. This is a diagnostic for topology that a clean and fully gapped bulk spectrum alone cannot give (Section 9.8).

- The screening of a magnetic impurity in a quantum spin liquid uses directly the Abrikosov-fermion mean-field construction of session 8, and it has the same mathematics as the impurity in a metal of Section 9.6. The induced spin correlation (9.11) decays as a power law for the gapless spin-1/2 chain and exponentially for the gapped spin-1 chain. This gives a fully general diagnostic of gapped and gapless states from the impurity response alone.

- Many impurities at the same time can break translational symmetry in the whole system, which gives rise to Anderson localization. In $d \geq 3$ there is a real mobility edge, which we diagnose with the inverse participation ratio (9.12), and an Anderson insulator, which has no spectral gap but has a transport gap. In one dimension an arbitrarily weak disorder localizes every state, and the localization length diverges as the disorder strength goes to zero (Section 9.10).

Every session so far has gone in the same direction. We start from a Hamiltonian, and we compute what it predicts, which can be an observable, a response function or a ground state. Session 10 goes in the opposite direction. In this whole course we computed observables from Hamiltonians, and in the next session we ask the inverse question. Given only the observed data, can we find the Hamiltonian that produced it? We will see that machine learning is the right tool to make this question precise.

Session 10

Machine Learning Quantum Materials

At the end of Session 9 we used linear response as a diagnostic tool, and we applied it to almost every class of system that we built in the course, which are metals, superconductors, topological matter and quantum magnets. That session was a synthesis in a narrow sense, since one formalism was applied to all the systems. In this session we do a synthesis in a much broader sense. We will leave condensed matter for a moment and ask what happens when machine learning meets everything that we built in the last nine sessions, in both directions. In some cases, a machine-learning architecture will be the right tool for a hard many-body problem. In other cases the reverse will be true, and a many-body technique that we already know will be the right architecture for a machine-learning problem. This is also, on purpose, the last session of the course, and before we start we should say clearly that there is one practical difference with all the sessions before. **There is no companion Jupyter notebook for this topic**. The `jupyter-notebooks/` folder stops at `session9.ipynb`, so in these notes there are no "try it in the notebook" boxes. The reason is not that the material is less practical, but that this part of the infrastructure was not built for this session. For this reason, we should study the derivations of this session directly on the page.

The session closes the arc of the course in two ways, and we name both of them before we start. First, we will build neural network quantum states from the beginning. This is the first time that the course meets this idea in full. A neural network takes the place of the exponentially many coefficients of a many-body wavefunction, and it is trained by using the variational principle as a loss function. It has its own place in the hierarchy of variational many-body states, together with matrix product states and projected entangled pair states, and we will use it a second time to reconstruct wavefunctions from a small number of measurements. Second, and this is where the course comes back to its starting point, we will see Hamiltonian learning, which is the inverse of the very first step that this course made. Session 1 started by saying that, if we are given a Hamiltonian, we can integrate the Schrödinger equation forward in time to get observables. The last topic of this session starts from the observables and asks for the Hamiltonian. Ten sessions after we first wrote that forward arrow, we will reverse it.

10.1 Machine learning and many-body physics in two directions

Let us first give the plan of this session, since the session goes in two directions and it is useful to know which direction we are following at each point. In one direction, we use machine-learning architectures, mostly neural networks, to solve many-body problems. Examples are to represent wavefunctions that are too large to store, to reconstruct states from a small number of measurements, to classify phases of matter and to learn Hamiltonians from experimental observables. In the other direction, we use many-body architectures, in particular tensor networks, as machine-learning tools, and we apply them to ordinary tasks such as image

classification. Under both directions there is a third theme, which is probably the most important one in practice. This is to combine machine learning with the condensed-matter methods that we already have, such as density functional theory, phase classification and the extraction of Hamiltonians, to solve problems that are hard for each of the two approaches alone. We will follow the same order as the lecture. We start with neural networks and how they learn, then neural network quantum states and tomography, then tensor networks used as machine learning, then machine learning inside density functional theory, then phase classification, and at the end Hamiltonian learning.

10.2 Neural networks as function approximators

Machine learning is a large field, and we should say clearly which part of it we use in this session. There are three broad types of task. **Supervised learning** maps labeled inputs to outputs, and the standard example is image classification. This is the type that we will use in the rest of the session. **Generative learning** models a probability distribution directly, so that one can *sample* new instances from it. For instance, a generative adversarial network that is trained on photographs of cats can produce realistic images of cats that never existed. **Reinforcement learning** trains an agent to choose the actions that maximize a cumulative reward, given the state of its environment. In this session we will not develop generative or reinforcement learning further, even though both of them have real uses in other parts of quantum materials. We keep the GAN example with cat faces in mind only to remember that supervised learning is not the whole field, even if it is all that we use in this session.

We can think of a deep neural network as a very flexible function with parameters. It takes an input, which is an array of numbers, passes it through a sequence of layers, and gives an output, which is another array of numbers. Inside a layer, the value of each node is a nonlinear function of the values of all the nodes in the previous layer, combined with a set of free parameters. These parameters are the *weights*, one for each connection, and the *biases*, one for each node. The architecture, which is the number and the width of the layers, fixes how complex a function the network *can* represent. The specific values of the weights and biases, which we call together θ, fix which function it *does* represent at a given moment. To train a network means to search over θ for the values that make the network give the function that we want. In practice, we define a **loss function** $\mathcal{L}(\theta)$ that measures how far the output of the network is from the desired one, and we minimize it. This is usually done with backpropagation, which is an efficient and exact way to compute $\partial\mathcal{L}/\partial\theta$ with the chain rule, together with gradient descent,

$$\mathcal{L}(\theta) = \sum_n \left\| f_\theta(x_n) - y_n \right\|^2, \qquad \theta \longrightarrow \theta - \eta\, \nabla_\theta \mathcal{L}(\theta), \tag{10.1}$$

where $\{(x_n, y_n)\}$ is a set of labeled training examples and η is the learning rate. Every use of a neural network in this session is a variation of (10.1), even when the physics attached to it is exotic. We have to choose the right input, the right output and the right loss, and then the training procedure is always the same gradient descent.

The concrete example that we keep in mind is the classification of images of cats and dogs, since we will use the same architecture two more times in this session. An image is an $N \times N$ matrix of pixel values. We flatten it into the input layer of the network, and the output layer has only two nodes, with the target $(1, 0)$ for "cat" and $(0, 1)$ for "dog." We train on a labeled set of images by minimizing (10.1), with f_θ the output of the network and y_n the one-hot label. Once the network is trained, it classifies new images that have no label. This template can be used in many cases, since nothing in it depends on the input being a photograph. It only needs the input to be an array of numbers with a categorical label. We will see the

same structure, an input matrix and a one-hot output, again in Section 10.6, where a spin configuration takes the place of the photograph. Figure 10.1 shows the template once, and the four later constructions of this session use it without change except for the choice of input, output and loss.



Figure 10.1: The construction that the rest of this session uses again and again. **Left.** A deep network is a parametric function f_θ. The architecture, which is the number of layers and their width, fixes which functions it *can* represent, and the weights and biases θ fix which one it *does* represent at a given moment. Here the input is a small picture and the output is a one-hot label. **Right.** Training is the loop of (10.1). We evaluate the output, compare it with the label through a loss, obtain $\partial\mathcal{L}/\partial\theta$ exactly with backpropagation, take a gradient step and repeat.

10.3 Neural network quantum states

In this section we meet neural network quantum states (NNQS) for the first time in the course, and we treat them directly with the machine-learning methods of Section 10.2. The idea in one sentence is that a neural network can take the place of the exponentially many coefficients of a many-body wavefunction. In this way it has its own place in the hierarchy of variational many-body states that this course has been building since the matrix product states of session 7. In this section we make this idea precise. We will see what the network computes, which loss function is used to train it, and how it is related to the ansätze that we already know.

We start from a difficulty that we have met before, most recently in the discussion of entanglement growth in session 7. A wavefunction for L spins,

$$|\psi\rangle = \sum_{s_1,\ldots,s_L} C(s_1,\ldots,s_L)\,|s_1,\ldots,s_L\rangle, \tag{10.2}$$

is given by 2^L coefficients, and it is not possible to store them explicitly once L is much bigger than 30 or 40. The idea of a neural network quantum state (NNQS) is to not store $C(s_1,\ldots,s_L)$, and instead to *generate* it. We build a neural network C_θ that takes a spin configuration $(s_1,\ldots,s_L)$ as input, which is formally an L-pixel "image," and returns the coefficient. For a wavefunction that is complex, it returns its real and imaginary parts as two output numbers, or in an equivalent way its modulus and phase. This is the same idea as the matrix product state of Section 10.4 below, where a compact generator takes the place of exponentially many numbers, but here it is built from the parameters of a neural network and not from a chain of matrices.

The question is then how to find the right θ for the ground state, and here the methods of Section 10.2 appear again with almost no change. The variational principle tells us that

$$E[\theta] \;=\; \frac{\langle\psi_\theta|H|\psi_\theta\rangle}{\langle\psi_\theta|\psi_\theta\rangle} \;=\; \frac{\sum_{s,s'} C_\theta^*(s)\,H_{ss'}\,C_\theta(s')}{\sum_s |C_\theta(s)|^2} \;\geq\; E_0, \tag{10.3}$$

where the equality holds only when $|\psi_\theta\rangle$ is the true ground state, or in practice the closest state that the ansatz can reach. Equation (10.3) *is* the loss function of (10.1), meaning that

the "observable output" that the network is trained to minimize is the energy of the system. To compute (10.3) without a sum over all the 2^L configurations, one samples s from the Born distribution $p(s) = |C_\theta(s)|^2 / \sum_{s'} |C_\theta(s')|^2$ and evaluates the energy as an average of the sampled local energy,

$$E_{\mathrm{loc}}(s) = \sum_{s'} H_{ss'} \frac{C_\theta(s')}{C_\theta(s)}, \qquad E[\theta] = \big\langle E_{\mathrm{loc}}(s) \big\rangle_{s\sim p}, \tag{10.4}$$

which only needs H between the configurations that are connected by the Hamiltonian, which is usually sparse, and not the full $2^L \times 2^L$ matrix. The training is then done in the same way as in (10.1). We compute the energy and its gradient with respect to the parameters of the network,

$$\frac{\partial E}{\partial \theta_k} = 2\,\mathrm{Re}\Big[\big\langle E_{\mathrm{loc}}(s)\, O_k^*(s)\big\rangle - \big\langle E_{\mathrm{loc}}(s)\big\rangle\big\langle O_k^*(s)\big\rangle\Big], \qquad O_k(s) \equiv \frac{\partial \ln C_\theta(s)}{\partial \theta_k}, \tag{10.5}$$

and we update $\theta_k \to \theta_k - \eta\, \partial E/\partial \theta_k$ many times, with a learning rate η as in any other training of a neural network, until the gradient vanishes and the energy has converged to its variational minimum.

We should now say clearly how NNQS is related to the other variational ansätze of this course. The slides show this as a set of nested regions. The full many-body Hilbert space is one region, and inside it there is a smaller region with all the states that an NNQS with a given architecture can reach. Projected entangled pair states are inside this NNQS region, and matrix product states, which we built in session 7, are inside the PEPS region. The parameterizations of gapped one-dimensional systems are in the innermost region, since a matrix product state is a particular and highly structured choice of "network." NNQS is therefore the most expressive of the variational families of this course, because it imposes the least structure on how the coefficients depend on the configuration. Equation (10.3) is the loss function, and (10.4)-(10.5) are the training procedure, and together they make this picture a working method and not only a structure. Figure 10.2 shows all three parts, which are what the network computes, how the sampled estimate of the energy is built, and the nested picture.

10.3.1 Quantum state tomography

A second and different use of the same NNQS ansatz is **quantum state tomography**. Let us take a real experimental system in some unknown state $|\psi\rangle$, and let us say that the only access that we have to it is through projective measurements. We measure the state of every spin, and the system collapses to one configuration s with probability $|C(s)|^2$. If we repeat the measurement many times, we obtain a set of observed configurations that are distributed according to $|C(s)|^2$. However, we never observe $C(s)$ itself but only its modulus squared, and to do this for all the 2^L configurations one by one is the same exponential problem that we started with.

The solution has the same structure as the ground-state search above. We parameterize the unknown amplitude and phase with a network, for instance a restricted Boltzmann machine, or with two separate networks, one for $|C_\theta(s)|$ and one for its phase. Then we fit θ by **maximum likelihood** to the observed measurement outcomes, instead of minimizing an energy. In practice, one maximizes $\sum_s n(s) \ln p_\theta(s)$ over θ, where $n(s)$ counts how many times the configuration s was observed and $p_\theta(s) = |C_\theta(s)|^2 / \sum_{s'} |C_\theta(s')|^2$ is the probability predicted by the network. This reconstructs a full approximation to the unknown wavefunction with far fewer than 2^L measurements, and this is what makes it useful as an experimental tool, since nobody can run a tomography experiment that needs an exponential number of measurements.

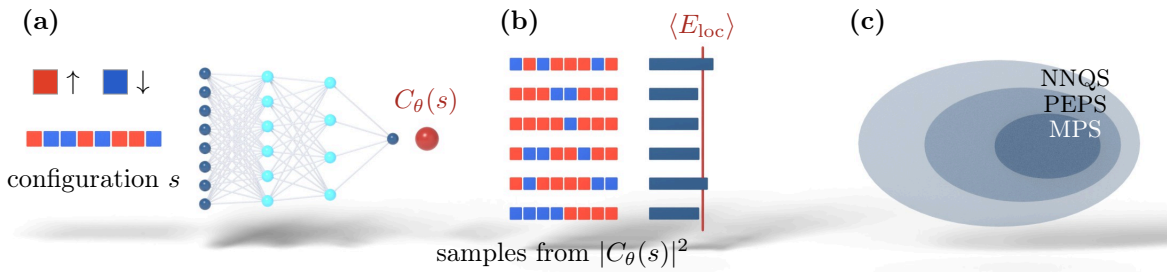


Figure 10.2: Neural network quantum states. **(a)** What the ansatz is. The exponentially long table of coefficients in (10.2) is not compressed, it is never written down at all. The configuration s is given as an L-pixel "image," here a row of up and down spins, and the network returns the one coefficient $C_\theta(s)$ that we ask for. This is the same structure as in figure 10.1, where only the input and the output have a new label. **(b)** Training. The variational energy (10.3) is the loss, and we estimate it by sampling configurations from the Born distribution and averaging the local energy (10.4), the bars and their mean. This is what makes the method possible, since $E_{\rm loc}(s)$ only needs the few matrix elements $H_{ss'}$ that a sparse Hamiltonian connects to s, and never the full $2^L \times 2^L$ matrix. **(c)** The nested regions drawn on the slides. A matrix product state is a highly structured special case of a PEPS, a PEPS is in turn a state that a large enough NNQS can represent, and NNQS is the outer region because it imposes the least structure of all on how C depends on s. This picture organizes the lecture and is not a theorem, since the outer inclusion is a statement about networks that are expressive enough and not about every architecture. Note also that expressiveness has a cost, because the order is the opposite one for how easy each family is to optimize.

This part comes from the slides and the transcript and has no notebook counterpart, in line with the fact that this session has no companion notebook. There is no `sessionN.ipynb` cell to point you to here, so (10.3)-(10.5) and the tomography likelihood above are the complete and self-contained derivation.

10.4 Tensor networks for machine learning and the many-body wavefunction as a big-data problem

We now follow the logic of the previous section in the reverse direction. Instead of using a machine-learning architecture to solve a many-body problem, we treat the many-body wavefunction itself as a big-data problem, and we use a many-body compression technique, which is tensor networks, as a machine-learning architecture.

Let us recall the matrix product state (MPS) ansatz, which we already know from session 7. Instead of storing all the 2^L coefficients of (10.2) independently, we assume that they factorize into a product of matrices, one for each site,

$$C(s_1, \ldots, s_L) = \mathrm{Tr}\left[A^{s_1} A^{s_2} \cdots A^{s_L}\right], \tag{10.6}$$

where for each physical value of s_i, for example spin up or spin down, there is a different $D \times D$ matrix A^{s_i}, and D, which is called the *bond dimension*, is a free parameter that we choose. The number of free parameters in (10.6) scales as $L \times d \times D^2$, with d the dimension of the local Hilbert space, so it is linear in the system size and not exponential. This is the same compression idea as the NNQS of Section 10.3, but the parameters are matrix elements instead of network weights. The same compression can be generalized beyond the one-dimensional geometry of the MPS. A *tree tensor network* puts the same matrices in a tree instead of a chain. The *multiscale entanglement renormalization ansatz* (MERA) adds extra layers of coarse-graining on top of a tree structure. *Projected entangled pair states* (PEPS) generalize the whole construction to a two-dimensional grid. For a ground-state search these architectures are not equivalent, since an MPS is much easier to optimize than a PEPS, and

this is the reason why MPS works well for one-dimensional problems while two-dimensional problems are often still hard. However, if we only want a flexible function of a high-dimensional input with an efficient parameterization, as in machine learning, the difference between the architectures matters much less, and any of them can be used.

This is a direct reuse of the architectures of session 7, and not only an analogy. The same matrix product states, tree tensor networks, MERA and PEPS that we built there to compress ground-state wavefunctions are used here without change as function approximators. This approach has a name, which is "quantum-many-body-inspired machine learning," and a clear example of it is image classification with tensor networks. We map each pixel of an image with N pixels to a two-dimensional vector, for example white $\to$ spin up and black $\to$ some other reference direction. Then we take the tensor product of all the N pixel vectors and obtain a single vector in a 2^N-dimensional space, which is the same exponentially large space where a many-body wavefunction with N spins lives. The classification task then becomes the search for a *linear* operator on this 2^N-dimensional space that maps each image to its one-hot label. A tensor network, and here a tree tensor network works well, is the efficient and structured parameterization of such a linear operator that (10.6) and its generalizations already gave us. We apply the tensor network to the tensor-product image, and then a final small tensor that projects down to two output numbers. This gives the same structure as the classification of cats and dogs in Section 10.2, where now the worked example is "purse versus shirt" and the tensor product of the pixels takes the place of the flattened input matrix. Figure 10.3 shows the four geometries as tensor diagrams, and then the classifier built from one of them.

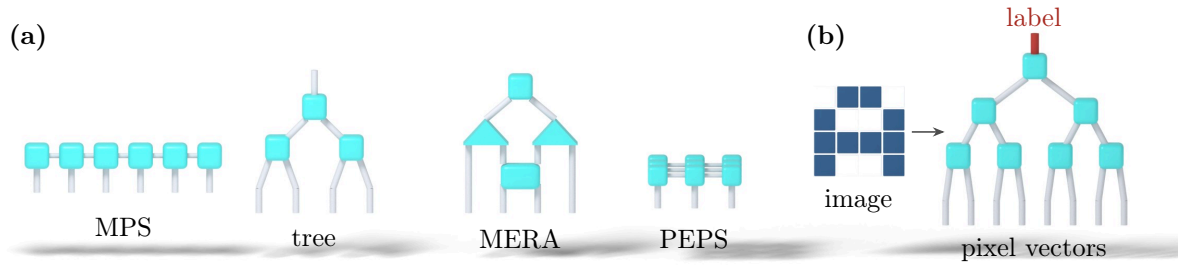


Figure 10.3: Tensor networks used in the reverse direction. **(a)** The four geometries that the text names, drawn as tensor diagrams. Every block is a tensor, every rod is an index that is summed over, and the open legs are the physical indices. The bond dimension D of the internal lines is what (10.6) calls a free parameter. For a ground-state search these geometries are not equivalent, since an MPS is easy to optimize and a PEPS is not. However, as flexible functions of a high-dimensional input with parameters, which is all that a classifier needs, any of them can be used. **(b)** Image classification with a tree tensor network. Each pixel becomes a vector with two components, and their tensor product is a single vector in a 2^N-dimensional space, which is the same space where an N-spin wavefunction of (10.2) lives. Classification is then a *linear* map from this space down to two numbers, the red leg, and the tensor network is the efficient and structured parameterization of such a map that (10.6) already gave us. Note what changes and what does not change with respect to figure 10.1. The input is still an array with a categorical label and the output is still one-hot, and only the function between them is now built from matrices instead of layers of nodes.

This is not a toy result. On the Fashion-MNIST dataset, which is a standard benchmark of images of clothes, this tensor-network classifier reaches **95.38% accuracy on the training set and 88.97% on the test set**, which is comparable with conventional neural network architectures on the same task. These numbers appear only on the slides, and in the recording they are only described as "very high accuracy." They are the kind of concrete result that we want to have in the written notes and not leave implicit. The general point is the same as in Section 10.3, in the other direction. A variational architecture that was built to compress

many-body wavefunctions is, in mathematical terms, one more kind of neural network, and we can use it directly as one.

10.5 Machine learning in density functional theory

We now turn to a different problem, and to a much older method of condensed matter, which is density functional theory (DFT). The motivation is the same as in Section 10.3, but now for real materials instead of lattice models. Let us take the Hamiltonian of a material,

$$H = \sum_i \left(-\frac{\hbar^2 \nabla_i^2}{2m} \right) + \sum_i v_{\text{ext}}(\mathbf{r}_i) + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|},$$

with v_{ext} the potential of the nuclei. For a realistic number of electrons, it is not possible to obtain the many-body ground state $\psi_0(\mathbf{r}_1, \ldots, \mathbf{r}_N)$ directly. DFT avoids this problem by asking a smaller question. It does not ask for the full wavefunction, but for the ground-state electronic density $n(\mathbf{r}) = N \int |\psi_0|^2 \, d^3 r_2 \cdots d^3 r_N$, which has a much lower dimension.

The **Hohenberg-Kohn theorem** is what makes this approach exact and not only convenient. For a given external potential $v_{\text{ext}}(\mathbf{r})$, the ground-state density $n(\mathbf{r})$ determines $v_{\text{ext}}(\mathbf{r})$ uniquely, up to an additive constant, and therefore it determines the full Hamiltonian and the full many-body wavefunction. The arrow $\psi_0 \to n$ is trivial, since it is an integral. The arrow $n \to \psi_0$ is the nontrivial content of the theorem, and the theorem says that this arrow is *guaranteed to exist*. As a result, the ground-state energy is a functional of the density alone,

$$E[n] = F[n] + \int d^3 r \, v_{\text{ext}}(\mathbf{r}) \, n(\mathbf{r}), \tag{10.7}$$

where $F[n] = T[n] + V_{ee}[n]$ is a *universal* functional, meaning that it is the same for every system and does not depend on v_{ext}, and nobody knows its exact mathematical form.

The **Kohn-Sham construction** makes (10.7) useful in practice. We introduce a fictitious system of non-interacting electrons, which is chosen so that it has the *same* density $n(\mathbf{r})$ as the real interacting system, and which is described by an effective single-particle Hamiltonian

$$H_{\text{KS}} = -\frac{\hbar^2 \nabla^2}{2m} + v_{\text{eff}}(\mathbf{r}), \qquad v_{\text{eff}}(\mathbf{r}) = v_{\text{ext}}(\mathbf{r}) + \underbrace{e^2 \int d^3 r' \, \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}}_{\text{Hartree (mean-field) term}} + \underbrace{\frac{\delta E_{\text{xc}}[n]}{\delta n(\mathbf{r})}}_{v_{\text{xc}}(\mathbf{r})}. \tag{10.8}$$

The first part of v_{eff} is the external potential of the nuclei. The second part, the Hartree term, is a mean-field decoupling of the interaction, with the same structure as the mean-field theory of session 1, but applied here to the Coulomb interaction instead of a Hubbard term on a lattice. The last part, the exchange-correlation potential v_{xc}, is defined so that it contains everything that the mean-field approximation misses, and it is as unknown as $F[n]$ was. In practice one uses an approximation for $E_{\text{xc}}[n]$, such as LDA, GGA, meta-GGA and further refinements, and solves (10.8) self-consistently. We start from a guess for the density, build H_{KS} and diagonalize it to obtain the Kohn-Sham orbitals. Then we compute the density again from these orbitals, and we repeat until convergence.

In the lecture we discussed two different places where machine learning can enter this loop. We keep them separate, because they solve different problems.

- **Learning the functional itself.** We replace or improve the LDA, GGA or meta-GGA approximation to $E_{\text{xc}}[n]$, or to its functional derivative v_{xc}, with a neural network, which is trained to reproduce reference calculations with a higher accuracy. This can improve the accuracy, but it does *not* remove the self-consistency loop. We still have to

diagonalize H_{KS} at every iteration, and this remains the expensive step for large systems, independently of how good v_{xc} is.

- **Learning the whole map.** A more ambitious use of the Hohenberg-Kohn theorem is to remove the self-consistency loop completely. We train a neural network to map the atomic positions, or equivalently $v_{\mathrm{ext}}(\mathbf{r})$, directly to the ground-state density $n(\mathbf{r})$, with no diagonalization at all when the network is used. This is the idea behind machine-learned density functionals, which are related to the machine learning interatomic potentials that map atomic positions to energies and forces, an established technique in quantum chemistry and condensed matter. We train the network on conventional Kohn-Sham calculations for small systems that we can solve, and then we apply it to much larger systems, where a direct Kohn-Sham calculation would be too expensive.

The Hohenberg-Kohn reduction is very asymmetric on purpose, since going from the many-body ground state to the density is a $3N \to 3$ integral that anyone can do, while going back is guaranteed to be possible but has no formula, and this missing formula is where machine learning can enter. Figure 10.4 places both routes on the self-consistency loop, which is the fastest way to see why only the second one changes which systems we can compute.

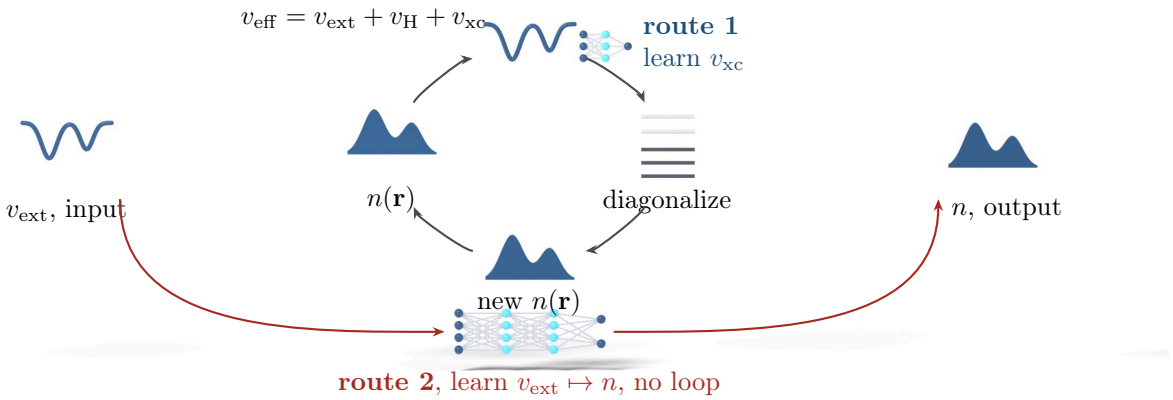


Figure 10.4: Where machine learning enters density functional theory, and why the two entry points are different ideas. The Kohn-Sham loop (10.8), the ring in the middle, turns the problem into a self-consistent single-particle problem. **Route 1** replaces the unknown exchange-correlation part by a trained network. This is more accurate, but the diagonalization stays inside the loop, and the diagonalization, not v_{xc}, is what limits the system size. **Route 2** learns the Hohenberg-Kohn map $v_{\mathrm{ext}} \mapsto n$ directly, trained on conventional Kohn-Sham results for small systems, and then does no diagonalization at all, so the whole loop is skipped. Only route 2 changes which systems we can compute, while route 1 only changes how good the answer is.

This part also comes from the slides and the transcript and has no notebook counterpart. DFT calculations were never part of the numerical tools of this course, since `pyqula` is a tight-binding and mean-field library and not a DFT code, so there is nothing to point you to beyond the derivation above.

10.6 Phase classification with local order parameters and topological order

We now come to a section that is, in a real sense, a review of the classification scheme of the whole course, seen from the point of view of machine learning. In this course we have used two quite different tools to distinguish phases. Sessions 1, 5, and 6 used a **local order parameter**, which is a quantity built from a simple local expectation value that is zero in

one phase and nonzero in the other. Examples are the charge imbalance for the CDW, the magnetization for the magnet and the pairing amplitude for the superconductor. Sessions 3 and 4 instead used a **topological invariant**, such as the Chern number or the Zak phase, because the phases there are *not* distinguished by any local quantity at all, and the difference can only be seen in a non-local way. The phase-classification example of this session uses this contrast as an "easy" and a "hard" worked example of machine learning, and the lecture itself presents it in this way.

The easy case, the Ising model. Let us take classical Ising spins $\sigma_i = \pm 1$ on a square lattice, with the standard nearest-neighbor Hamiltonian and a local order parameter,

$$H_{\text{Ising}} = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j, \qquad m = \frac{1}{N} \sum_i \sigma_i, \tag{10.9}$$

which has ferromagnetic order ($|m| \to 1$) below a critical temperature T_c and is disordered ($m \to 0$) above it. If we only have a single snapshot of the spin configuration, at some unknown temperature, it is easy to see the phase even by eye. We sum the spins and check whether the result is close to $\pm N$ or close to 0. Even so, it is a clean and controlled test case for machine learning. We train the same cat-versus-dog architecture of Section 10.2 on labeled snapshots below and above T_c, where the input is the $N \times N$ spin snapshot and the output is a one-hot label for ordered or disordered. The classifier then gives a transition that becomes sharper as the system size grows, where the sweep on the slides goes from $L = 10$ up to $L = 60$, and this reproduces what we already know analytically.

The hard case, the Ising gauge theory. Now we replace the coupling between pairs with a **plaquette** interaction, which is the product of the four spins around each elementary square of the lattice,

$$H_{\text{IGT}} = -J \sum_p \prod_{i \in \partial p} \sigma_i, \tag{10.10}$$

where the sum runs over the plaquettes p and the product runs over the four spins on the four bonds around that plaquette. Note that here the spins live on the *links* of the lattice and not on its sites. At zero temperature the ground state satisfies $\prod_{i \in \partial p} \sigma_i = +1$ on *every* plaquette, but this constraint is satisfied by a macroscopic number of different spin configurations. For instance, if we flip the four spins on the bonds that meet at a single vertex, every plaquette product stays the same, because each of the four plaquettes that touch this vertex contains exactly two of these four bonds and every other plaquette contains none of them. This local gauge transformation is what gives the model its name. At infinite temperature, instead, plaquettes with product $+1$ and -1 appear with equal probability. Both the ordered configurations, which satisfy the constraint, and the disordered ones can look, one by one, like a featureless 50/50 mix of up and down spins. As a result, the naive order parameter (10.9), which worked perfectly for the Ising model, is *useless* here. The reason is that no local quantity can distinguish the two regimes, which is the same obstruction that forced sessions 3 and 4 to leave local order parameters and to use Chern numbers and Zak phases. What (10.10) reproduces, in a simple classical model, is this feature of those earlier examples. Note that in two dimensions this model has no true transition and no topological order at finite temperature, and only its ground state at zero temperature has topological order. On a finite system, however, the low-temperature and high-temperature regimes are clearly different, and no local quantity can distinguish them, which is what makes this model a good test for the classifier.

A convolutional neural network, trained on labeled configurations at low T and high T with the same supervised architecture as before, succeeds where the naive summary statistic fails. In practice, it learns a nonlocal function of the spin configuration, which plays the role

that a topological invariant would play analytically. We should state one more physical point carefully, since it is easy to mix up. The two regimes here are separated by a **size-dependent crossover** and not by a true phase transition at finite temperature, so there is no sharp T_c that the result converges to as the system grows, unlike the true Ising transition of (10.9). What makes the machine-learning result useful is that the trained network has only seen one system size during training, and it still captures this crossover correctly for *other* sizes and temperatures on which it was never trained. The slides show it trained at a single size and evaluated from $L = 4$ up to $L = 28$. Figure 10.5 shows the easy and the hard case next to each other in real space, which is where the difference between them can be seen.

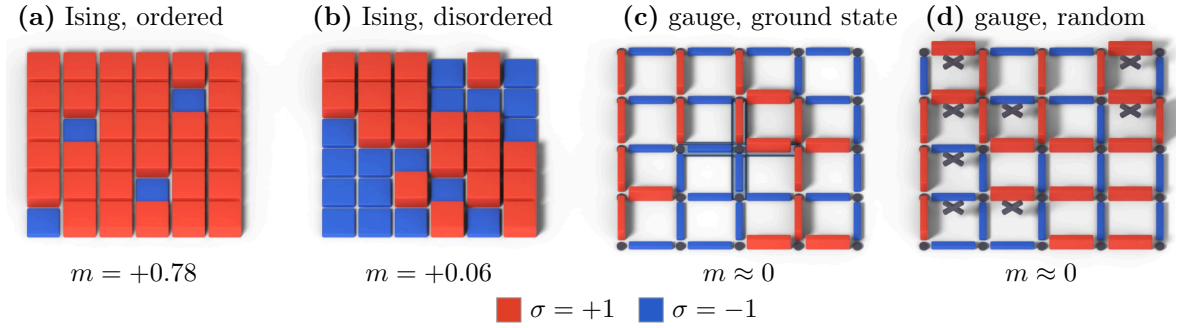


Figure 10.5: Why one phase-classification problem is easy and the other is hard, seen from above with $\sigma = +1$ raised and $\sigma = -1$ sunk. **(a,b)** For the Ising model (10.9) the magnetization of a single snapshot already separates the phases, and a network trained on such snapshots only finds again the logic of a local order parameter of sessions 1, 5 and 6. **(c,d)** For the Ising gauge theory (10.10) the spins live on the *bonds*, and the $T = 0$ condition $\prod_{i\in\partial p}\sigma_i = +1$ leaves an exponentially large degenerate manifold. A typical ground state is random from bond to bond, so the magnetization is ≈ 0 in both columns and no quantity on a single bond can distinguish them. The difference is in the four-spin structure of the plaquettes, which is marked by a cross where the product is -1, and the convolutional network has to find by itself a function of the configuration that is sensitive to it. The shaded star in (c) is a gauge flip of the four bonds at one vertex, which keeps every plaquette product. This is the same obstruction that forced sessions 3 and 4 to go from local order parameters to Chern numbers and Zak phases.

10.7 Hamiltonian learning as the inverse of the first step of session 1

We come at the end to the part of this session that closes the loop in the most explicit way. Let us go back to session 1, Section 1.4. Every quantum mechanical system is described by the Schrödinger equation, and if we have a Hamiltonian and an initial state we can integrate forward in time and obtain, in principle, every observable of the system, such as the density, the magnetization, the response functions and so on. This forward arrow, Hamiltonian $\rightarrow$ observable, is what almost this entire course has been built to compute, one physical setting at a time.

However, this is not the arrow that an experimentalist can use. A real measurement never gives you the Hamiltonian. It gives you an observable directly, such as a density, a conductivity, an ARPES spectrum or a neutron-scattering structure factor, and the Hamiltonian is the thing that you do *not* have and would like to infer. **Hamiltonian learning** writes this explicitly as the inverse problem. On the slides the Hamiltonian is written as $H(\lambda_i, w_i)$, where its dependence is split into two parts. The first part are the physical parameters of interest λ, such as magnetic or electric fields, hopping amplitudes and many-body interaction strengths. The second part is a set of "hidden," or nuisance, variables w that one does *not* want to fit,

such as smaller parameters, disorder realizations, artifacts of the computational method that generated the training data, and plain randomness. If we write $\mu \equiv w$ for this vector of hidden variables, the forward direction is the map

$$\mathcal{O} = \mathcal{O}\big[H(\lambda, \mu)\big], \tag{10.11}$$

and Hamiltonian learning asks for the inverse map, $\mathcal{O} \mapsto \lambda$, where we marginalize over μ as well as we can instead of trying to fit it. In general this is a difficult inverse problem, since many different λ can be compatible with the same measured $\mathcal{O}$, and there is little real experimental data to train on. The practical strategy is, once again, to use theory to compensate for the lack of experimental data. We simulate the observable $\mathcal{O}$ from many different and *known* Hamiltonians $H(\lambda_n)$ that cover the range of parameters of physical interest. These are the same forward calculations, such as the density, the magnetization and the spectroscopy, that we learned to compute directly in the earlier sessions. Then we train a neural network $\mathcal{N}_\theta$ on this synthetic data to invert (10.11), and we do this by minimizing

$$\mathcal{L}(\theta) = \sum_n \big\| \mathcal{N}_\theta\big(\mathcal{O}[H(\lambda_n)]\big) - \lambda_n \big\|^2 . \tag{10.12}$$

Once the network is trained on simulated pairs, where the answer λ_n is known by construction, the same network is applied to a real experimental observable $\mathcal{O}_{\text{exp}}$. The concrete example of the lecture was a two-dimensional spectroscopy of a quantum magnet. Then $\hat{\lambda} = \mathcal{N}_\theta(\mathcal{O}_{\text{exp}})$ gives an estimate of the physical parameters of the Hamiltonian, for instance the first-, second-, and third-neighbor exchange couplings of that magnet, extracted directly from a measurement for which there is no closed-form inversion formula at all. Figure 10.6 shows both stages, the synthetic training set and the use on real data, together with the forward arrow that the two stages invert.

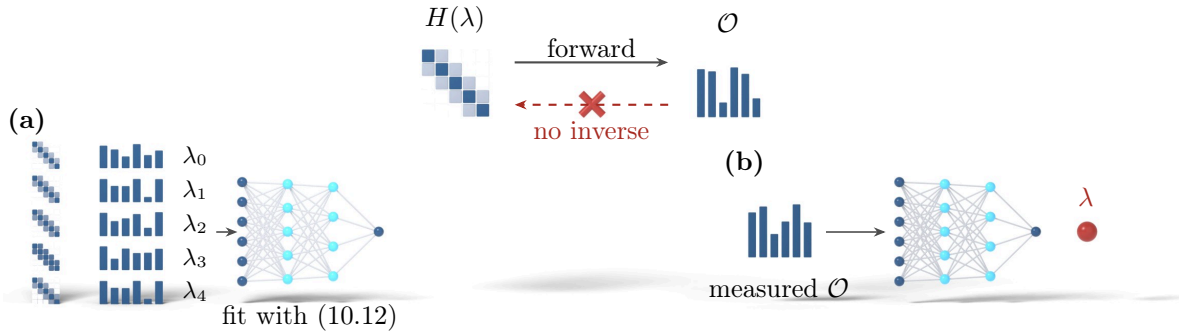


Figure 10.6: Hamiltonian learning, and why it closes the loop of the course instead of going outside it. **Top.** The forward arrow $H \mapsto \mathcal{O}$ of (10.11) is what sessions 1 to 9 built during all their time. The arrow that an experiment needs goes in the other direction, and there is no closed-form inversion of it. **(a)** The solution to the lack of experimental training data is to produce the data. We simulate $\mathcal{O}$ from many *known* Hamiltonians, so that every training pair comes with its answer λ_n by construction, and we fit the network with (10.12). **(b)** Only after this step the trained network is applied to a real measurement. Note what the training set is made of, which is the forward methods of the whole course, run many times. The final method of the course and its first method are the same method, applied in opposite directions.

This is the mirror image of the starting point of session 1, and we want to stop for a moment on this symmetry. In nine sessions we built more and more sophisticated ways to go from a Hamiltonian to an observable. These are mean-field decoupling, Bloch's theorem, topological invariants, Landau levels, Nambu spinors, magnon spectra, tensor-network ground states, parton constructions and response functions. Hamiltonian learning does not leave any

of these behind. Instead, it uses the same forward methods, run many times on synthetic data, as the training set to learn how to run the process in reverse. In this sense, the method of the course and the final method of the course are the same method, applied in opposite directions.

10.8 Closing summary

- A deep neural network is a parametric function that is trained by minimizing a loss function with gradient descent, equation (10.1). Every other construction in this session is a specific choice of input, output and loss built on top of this one template.
- **Neural network quantum states** replace the explicit storage of the 2^L coefficients of a wavefunction with a network $C_\theta(s)$, equation (10.2), which is trained by using the variational energy as the loss function, equations (10.3)-(10.5). NNQS is above matrix product states and PEPS in the variational hierarchy, and it imposes the least structure of all the ansätze of this course. The same ansatz, fitted by maximum likelihood instead of energy minimization, solves quantum state tomography with far fewer than 2^L measurements.
- **Tensor networks**, built from the matrix product state ansatz (10.6) and its generalizations to trees, MERA and PEPS, can be used in the reverse direction as machine-learning architectures. For image classification on Fashion-MNIST, this reaches 95.38% training accuracy and 88.97% test accuracy, which is comparable with conventional neural networks.
- **Density functional theory** is based on the Hohenberg-Kohn functional (10.7) and the Kohn-Sham self-consistency loop (10.8). Machine learning enters in two ways. It can learn the unknown exchange-correlation part of (10.8), and then the self-consistency loop stays the same, or it can learn the whole map from potential to density directly, and then the self-consistency loop is removed completely.
- **Phase classification** uses one more time the split of this course between local order parameters and topological order. The Ising model (10.9) is the easy case, which can be detected locally and follows the logic of sessions 1, 5 and 6. The Ising gauge theory (10.10) is the hard case, where no local order parameter distinguishes the low-temperature and high-temperature regimes, which are separated only by a crossover, and it follows the logic of sessions 3 and 4. A trained network succeeds where the naive order parameter fails.
- **Hamiltonian learning** treats the inference of a Hamiltonian from measured observables, equation (10.11), as the inverse of the first step of session 1. We train a network on synthetic forward data, equation (10.12), and apply it to real experimental observables, and in this way we close the loop in practice and not only in principle.

We started this course by asking how a small number of identical atoms, arranged in a lattice, can give rise to a phenomenology as rich as the one that we studied in ten sessions. This includes charge density waves, magnets, superconductors, topological insulators, fractional excitations, spin liquids and more, and none of it can be seen in a single atom alone. To answer this question we went through second quantization and mean-field theory, through Bloch's theorem and topological invariants, through Nambu spinors and magnon spectra, through tensor networks and parton constructions, and through linear response as a diagnostic for all of them. This session closed the arc in two ways. First, it showed that a neural network can take the place of the many-body wavefunction itself. Second, it showed that all the forward

methods of the course, with a Hamiltonian as input and an observable as output, can be run backward to infer the Hamiltonians of real materials. This is a good place to end the course, not with a final answer, but with a general method to ask the question again, for the next quantum material that we want to understand.